Cloud Basics

Foreword

- This chapter describes the cloud trend of enterprise IT facilities, overview and features of cloud computing, background, definition, and technical features of public cloud development, and basic architecture, basic concepts, delivery modes, and ecosystem construction of HUAWEI CLOUD .

Objectives

- On completion of this course, you will be able to:
  - Understand the background of cloud computing and the trend of enterprise IT cloudification.
  - Understand the definition and technical features of public cloud.
  - Understand the basic architecture, basic concepts, and ecosystem construction of HUAWEI CLOUD.

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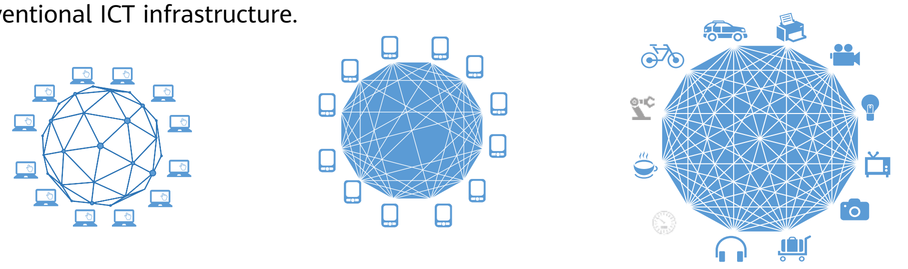
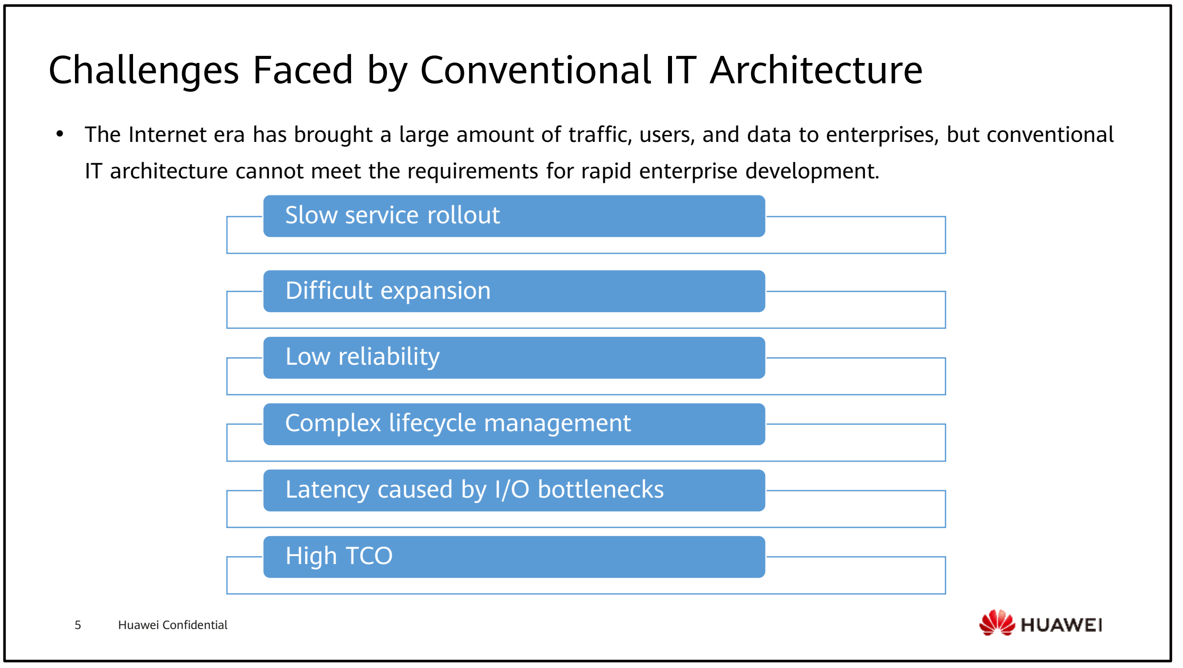
Network era transformation, information and data growth

- With the prevalence of the mobile Internet and fully connected era, more terminal devices are being used and data is exploding every day, posing unprecedented challenges on conventional ICT infrastructure.



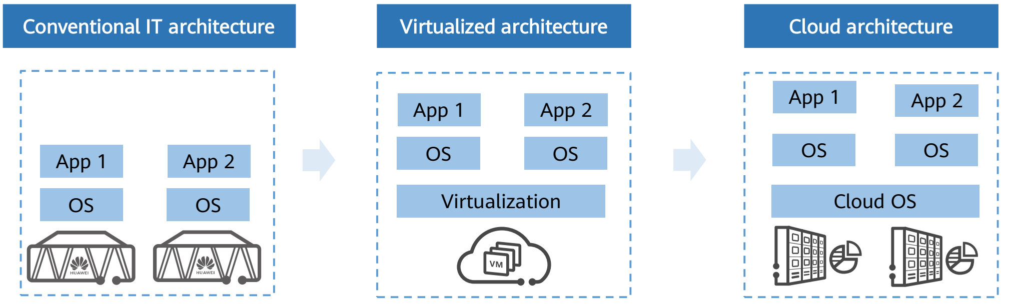
PC era	Mobile Internet era	IoT era
Computers of the x86 architecture	Mobile phones of the ARM architecture	x86, ARM, DSP, MIPS, FPGA, etc.
Windows/Linux	**Android/iOS**	**IoT OS**

- The PC era is essentially in which computers are networked, and personal computers are connected through servers. Now, in the mobile era, we can assess the Internet through mobile phones. With the advent of 5G, all computers, mobile phones, and intelligent terminals can be connected, and we can enter an era of Internet of Everything (IoE).
- In the IoE era, the entire industry will compete for ecosystem. From the PC era to the mobile era, and to the IoE era, the ecosystem experiences fast changes at the beginning, then tends to relatively stable, and rarely changed when it is stable. In the PC era, a large number of applications run on Windows, Intel chips, and x86 architecture. Then, browsers come with the Internet. In the mobile era, applications run on iOS and Android systems that use the ARM architecture.
- Compared with the previous generation, the number of devices and the market scale of each generation increase greatly, presenting future opportunity. As the Intel and Microsoft in the PC era and the ARM and Google in the mobile era, each Internet generation has its leading enterprises who master the industry chain. In the future, those who have a good command of core chips and operating systems will dominate the industry.



- The Internet brings a large amount of traffic, users, and data, so enterprises need to continually purchase traditional IT devices to keep pace with their rapid development. Therefore, the disadvantages of traditional IT devices gradually emerge.
  - Long procurement period causes slow rollout of new business systems.
  - The centralized architecture has poor scalability and can only increase the processing performance of a single node.
  - Traditional hardware devices exist independently, and their reliability depends only on software.
  - Devices and vendors are heterogeneous and hard to manage.
  - The performance of a single device is limited.
  - The utilization of devices is low, while the total cost remains high.

Enterprises Are Migrating To the Cloud Architecture



- The traditional IT architecture consists of hardware and software, including infrastructure, data centers, servers, network hardware, desktop computers, and enterprise application software solutions. This architecture requires more power, physical space, and capital, and is usually installed locally for enterprises or private use.
- With the virtualization technology, computer components run on the virtualization environment, not on the physical environment. Virtualization enables maximum utilization of the physical hardware and simplifies software reconfiguration.
- With cloud transformation, enterprise data centers are transformed from resource silos to resource pooling, from centralized architecture to distributed architecture, from dedicated hardware to software-defined storage (SDS) mode, from manual handling to self-service and automatic service, and from distributed statistics to unified metering.

Definition and Features of Cloud Computing

- Definition

  Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.

  --National Institute of Standards and Technology (NIST)

- Features
  - On-demand self-service
  - Broad network access
  - Resource pooling
  - Quick deployment and auto scaling
  - Measured service

- On-demand self-service: Customers can deploy processing services based on actual requirements on the server running time, network, and storage, and do not need to communicate with each service provider.
- Broad network access: Various capabilities can be obtained over the Internet, and the Internet can be accessed in standard mode from various clients, such as mobile phones, laptops, and PADs.
- Resource pooling: Computing resources of the service provider are centralized so that customers can rent services. In addition, different physical and virtual resources can be dynamically allocated and reallocated based on the customer requirements. Customers generally cannot control or know the exact location of the resources. The resources include the storage devices, processors, memory, network bandwidth, and virtual machines.
- Quick deployment and auto scaling: Cloud computing can rapidly and elastically provide computing capabilities. A customer can rent unlimited resources and purchase required resources at any time.
- Measured services: Cloud services are billed based on the actual resource usage, such as the CPU, memory, storage capacity, and the bandwidth consumption of cloud servers. Cloud services provide two billing modes: pay-per-use and yearly/monthly.

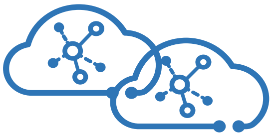
Key Cloud Computing Technologies

- **Virtualization Technology**
  - Server virtualization is an important cornerstone of the underlying architecture of cloud computing. In server virtualization, virtualization software needs to abstract hardware and allocate, schedule, and manage resources.
- **Data storage technology**
  - The cloud computing system needs to meet the requirements of a large number of users and provide services for a large number of users in parallel. Therefore, the data storage technology of cloud computing must have the characteristics of distributed, high throughput and high transmission rate.
- **Massive Data Management Technology**
  - Cloud computing is characterized by massive data storage and analysis after reading. How to improve the data update rate and further improve the random read rate is a problem that must be solved by future data management technology.

- In addition to the preceding key technologies, there are two important technologies:
  - Programming mode: Cloud computing provides a distributed computing mode, which objectively requires a distributed programming mode. Cloud computing adopts a simple distributed parallel programming model Map-Reduce. Map-Reduce is a programming model and task scheduling model. It is mainly used for parallel computing of data sets and scheduling of parallel tasks. In this mode, Users only need to compile Map and Reduce functions to perform parallel computing.
  - Cloud computing platform management technology: Cloud computing resources are huge, servers are numerous and distributed in different places, and hundreds of applications are running at the same time. How to effectively manage these servers and ensure that the entire system provides uninterrupted services is a huge challenge.
- Virtualization Technology
  - A Virtual machine(VM) consists of disk files and description files, which are encapsulated in the same folder.
  - Multiple VMs running on the server are encapsulated and isolated from each other. That is, multiple folders exist.
  - The essence of virtualization is to logically convert a physical device into a folder or file to decouple software and hardware.

Eight Common Characteristics of Cloud Computing

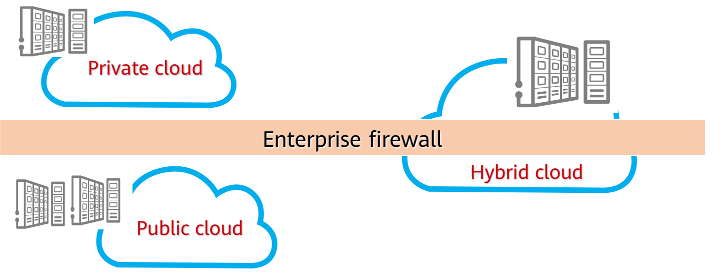
- Massive scale
- Homogeneity
- Virtualization
- Resilient computing
- Low-cost software
- Advanced security technologies
- Geographical distribution
- Service orientation



- Massive scale: Cloud computing service is in large scale as it centralizes IT resource supply. This makes cloud computing different from conventional IT.
- Homogeneity: Homogeneity can also be understood as standardization, which is similar to power utilization. Voltage and socket interface should be the same for various electrical appliances and devices.
- Virtualization: Virtualization has two meanings. One is accurate computing units. If a cake is too large for one person, it is better to divide it into small pieces to share. That is, with smaller computing units, IT resources can be fully used. The other meaning is the separation of software and hardware. Before virtualization, software and specified hardware are bound together, and after virtualization, software can be freely migrated on all hardware, which is like renting a house instead of buying one.
- Elastic computing: Elastic computing means that IT resources can be elastically provided.
- Low-cost software: Low-cost software is provided to meet the market competition and requirements. Cloud computing, with low individual technical skill and financial requirements, makes IT easy to use. Small and micro startups are always willing to enjoy the more IT services at the lowest cost. Based on this situation, low-cost software is required to earn money at small profits but quick turnover.

- Geographic distribution: As the broad access mentioned above, IT services can be provided anytime and anywhere. From the perspective of users, cloud computing data centers, are geographically distributed and the performance of network bandwidth varies by regions. Large public cloud service providers have dozens or even hundreds of data centers or service nodes to provide cloud computing services to global customers.
- Service orientation: Cloud computing is a service model, and the overall design is service-oriented.
- Advanced security technology: Public cloud has a large number of users with different requirement. Therefore, advanced security technologies must be adopted to protect cloud computing.

Deployment Models for Cloud Computing



Private cloud: The cloud infrastructure is provisioned for exclusive use by a single organization.
Public cloud: The cloud infrastructure is owned and managed by a third-party provider and shared with multiple organizations using the public Internet.
Hybrid cloud: This is a combination of public and private clouds, viewed as a single cloud externally.

- Private cloud is a cloud infrastructure operated solely for a single organization. All data of the private cloud is kept within the organization's data center. Attempts to access such data will be controlled by ingress firewalls deployed for the data center, offering maximum data protection.
- Public cloud service provider owns and operates the cloud infrastructure and provides cloud services open to the public or enterprise customers. This model gives users access to convenient, on-demand IT services, comparable to how they would access utilities like water and electricity.
- A hybrid cloud is a combination of a public cloud and a private cloud or on-premises resources, that remain distinct entities but are bound together, offering the benefits of multiple deployment models. Users can migrate workloads across these cloud environments as needed.

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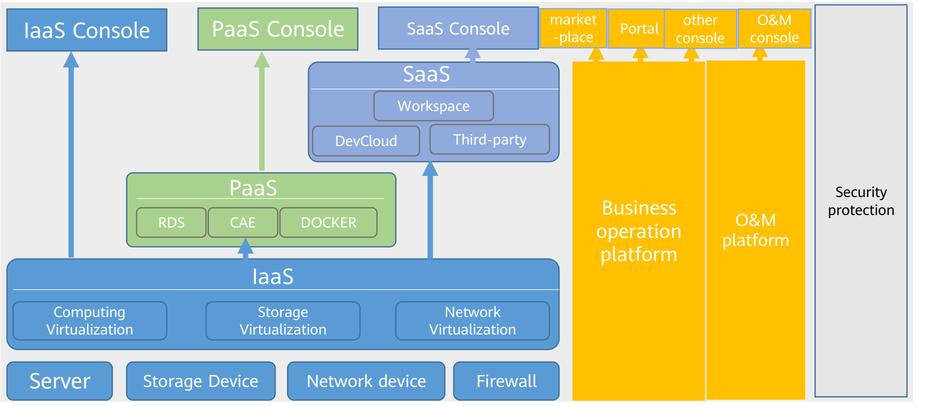
What is Public Cloud?

- **Concepts**
  - Public cloud refers to cloud services provided by third-party providers over the public Internet. Users can access the cloud and enjoy various services, including but not limited to computing, storage, and network services. Public cloud services can be free or pay-per-use.
- **Features and Values**
  - The core attribute of the public cloud is the shared resource service. Third-party providers provide shared computing, storage, and network resources to users on demand. Users can enjoy IT services on a pay-per-use basis without initial IT infrastructure investment, greatly reducing digital barriers and IT costs.
  - For most SMBs or startups, public cloud is the best choice:
    - From the perspective of operation, the public cloud can provide users with required resources on demand and charge fees to reduce the TCO and reduce costs.
    - From the perspective of O&M, traditional enterprises build their own data centers to support their services. The workload brought by self-construction includes infrastructure. (including wind, fire, water and electricity, servers, storage, switches, firewalls, etc.) , systems, middleware services, etc. Maintenance is complex and costly.
    - From the perspective of services, the public cloud provides a wide variety of services, enabling users to enjoy the convenience brought by the cloud.
    - From the perspective of security, the security level of mainstream public cloud service providers is beyond the reach of most enterprises.

- From the perspective of operation, especially for small- and medium-sized enterprises, the public cloud can meet the requirements of devices that do not have sufficient budget to purchase, use and release devices in a short period of time (testing and verification), and require ultra-large computing capabilities.
- From the perspective of O&M: By using the public cloud, users only need to focus on their own services. This greatly reduces maintenance complexity and costs and focuses on continuous service innovation.
- From the perspective of security: Mainstream public cloud providers provide services that have passed most security and privacy certifications, effectively ensuring user data and privacy security.

Public Cloud Architecture

- The following figure shows the common public cloud architecture, including Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS). Software as a Service (SaaS), O&M, operation, and security.

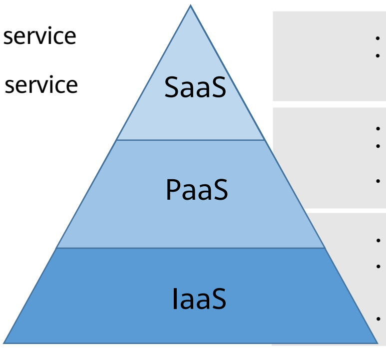


- The IaaS layer abstracts computing, storage, and network resources for users to use and provides corresponding services based on actual application requirements.
- The PaaS layer provides container services and microservice development services for users based on the IaaS layer. That is, an open platform is provided for users.
- The SaaS layer mainly provides scenario-based applications, that is, provides applications as services for users.
- At the O&M layer, the public cloud provides user- and platform-oriented O&M capabilities. The public cloud provides O&M capabilities for users using cloud services, such as permission control, performance monitoring, status monitoring, and fault alarm reporting. On the platform side, the public cloud assurance team performs O&M to ensure high reliability, high availability, and security of the platform.
- At the operation layer, the public cloud provides user- and platform-oriented operation capabilities. Users have operation capabilities such as submitting work orders, orders, and charging to help users understand operation costs and analyze service trends. The public cloud operation team processes and manages users' work orders and investments, and performs visualized management on the overall revenue of the public cloud.
- At the security level, the public cloud needs to meet requirements on system security, platform security, O&M security, and network security to ensure the data and property security of users and cloud service providers.

- RDS: Relational Database Service
- CAE: Cloud Application Engine (CAE) is a serverless hosting service oriented to web and microservice applications. It provides a one-stop application hosting solution that features fast deployment, low cost, and simplified O&M.
- CodeArts : is a one-stop cloud DevSecOps platform for developers. It can be used out of the box and deliver the entire software lifecycle on the cloud anytime and anywhere, covering requirement delivery, code submission, code check, code compilation, verification, deployment, and release. Streamline the complete software delivery path and provide end-to-end support for the software R&D process.

Three Service Modes of the Public Cloud

- The public cloud service mode is being developed and improved. In the industry, the public cloud service mode is classified into the following three types:
  - **IaaS:** Infrastructure as a service
  - **PaaS:** Platform as a service
  - **SaaS**: Software as a service



- Target : enterprise users/individual users
- Examples: enterprise application services, email, IM, and microblogging.

- Target: Developers
- Provides database middleware, MySQL, Mangodb, and Java.
- Examples: CloudFoundry and OpenShift

- Target: enterprises (or a few individual users)
- Provides basic equipment services such as computers, storage devices, and networks.
- Examples: AWS, Google Cloud, OpenStack

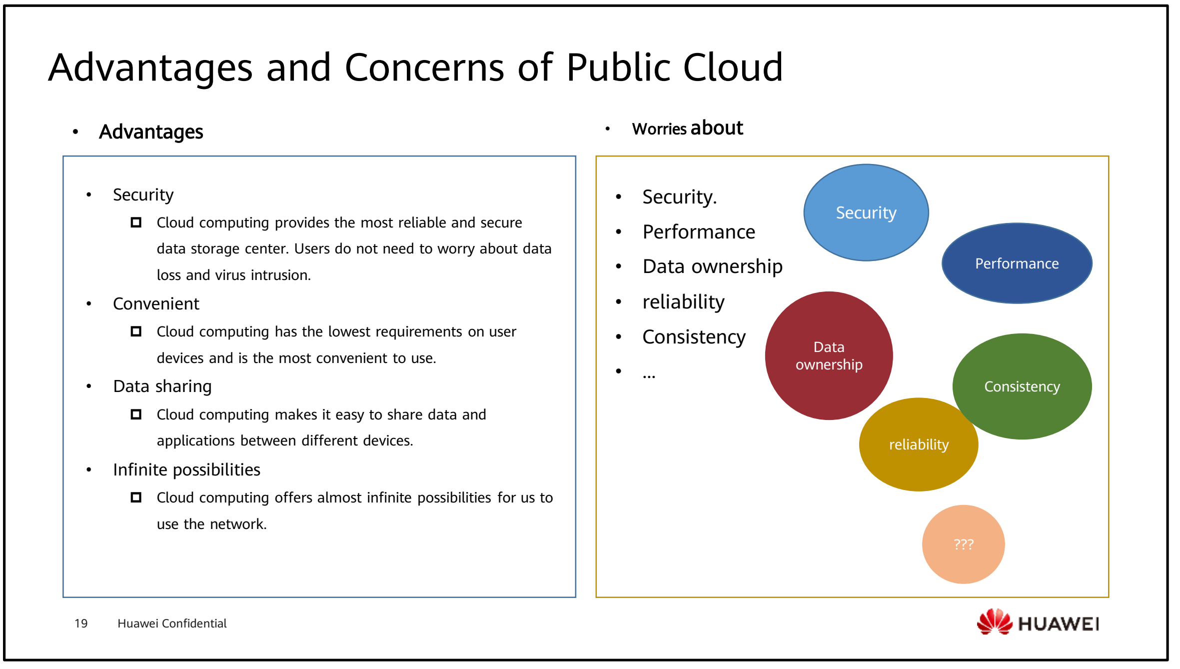
Features of the Three Service Modes

- **IaaS is infrastructure as a service. IT infrastructure is provided as a service through the network.**
  - Users do not need to build data centers. Instead, they rent infrastructure services, including servers, storage devices, and networks.
  - In terms of usage, IaaS is similar to traditional host hosting, but IaaS has strong advantages in service flexibility, scalability, and cost.
- **PaaS is Platform as a Service. A software platform has been built on the cloud, and the customer rents the required software platform.**
  - When users use the cloud, the operating system, database, middleware, and runtime library have been set up.
  - Compared with IaaS, PaaS has low freedom and flexibility and is not suitable for highly professional IT technical professionals.
- **SaaS is software as a service. The operating system, middleware, database, runtime library, and software applications required by the customer have been deployed on the cloud. Most SaaS applications can run directly through the browser without the need for client installation.**
- Summary: For users, the relationship between the three service models is independent because the user groups are different. Technically, the three are not simply inherited. SaaS is based on PaaS, and PaaS is based on IaaS.

- A simple example, convenient more intuitive understanding of the three modes, if Users want to develop a small program mall system.
- The first solution is: buy servers, buy databases, buy domain names, develop small program mall, that this model is IaaS model.
- The second solution is that applets provide cloud development services, eliminating the need for servers, storage, and domain names. I can only develop programs. This mode is the PaaS mode.
- The third solution is: Huawei provides the mall applet. Users only need to enable it. This mode is the SaaS mode.

Advantages of the Public Cloud over Traditional IT Systems

Item	Traditional IT	Public cloud
Resource utilization	Low resource utilization • The resource usage of traditional servers is unbalanced, ranging from 30% to 40% in some cases to 10% in most cases. The IT resources put into production are not effectively used.	High resource utilization • Select cloud services of different specifications and models as required to make full use of resources.
Cost	Expensive • It is expensive to prepare network, computing, and storage resources. As the business grows, the cost increases.	Savings • With the elastic computing capabilities of the public cloud, resources can be added or released at any time when services increase. • Various computing modes, including yearly/monthly and pay-per-use.
Scalability	Poor scalability • Architectures typically require a long time and a lot of resources to upgrade or downgrade.	Good scalability • Scalability allows organizations to better control costs and resources. • Businesses can upgrade or downgrade their computing and storage capacity as needed.
Service rollout duration	Long service rollout time • Deploying new services takes months to years.	Quick service rollout • Users can quickly purchase, start, and update resources on the GUI. • Deploy services by using various applications, high-availability templates, and solutions.
Maintenance Interval	Long maintenance period • The fault rectification period is long. • The maintenance labor cost is high.	Short maintenance time • Select resource specifications and perform quick upgrade in one-click mode. • Multi-channel after-sales ensures continuous and efficient business operation.



- **Security:** Who is allowed to view the enterprise's proprietary data?
- **Performance:** Will the application system perform as expected at peak processing times?
- **Data Ownership:** Is ownership of the "cloud" the ownership of the data on the system platform?
- **Reliability:** An enterprise can deploy many data centers and redundant systems to meet the need for uptime. Will companies that offer "cloud" services offer the same services?
- **Consistency:** A growing number of companies in the public enterprise, financial services and health sectors are facing strict regulations; They need to be able to prove who accessed the data, when or where it was processed, and what software and hardware is required when it is processed. In an enterprise's internal database, this is very difficult to do. Can they allow the same work in the cloud? More likely, for important applications, the enterprise will deploy web-based access mechanisms that allow those applications to run in the current host location. As for application updates, the enterprise may create an on-premise cloud.
- Think about any other concerns? If Users are a user, what other concerns do Users have?

Cloud customers are generally concerned about cloud security

- As with many emerging technologies, the security of cloud services has attracted much attention, and the emerging security and compliance issues will challenge the widespread deployment and development of cloud services.
  - Security vendor Ermetic surveyed more than 300 information security executives. Nearly 80% of enterprises have experienced at least one cloud data breach in the past 18 months, and 43% have reported more than 10 times.
  - According to a survey conducted by security vendor Barracuda, 70 percent of respondents said security concerns were limiting their organization's adoption of public cloud. These security concerns include the security of the public cloud infrastructure, the impact of cyber attacks, and the security of applications deployed in the public cloud.
- When providing services, cloud service providers may face both internal and external security threats. For example:
  - **In terms of internal threats,** there may be unknown or uncontrolled assets and devices. Data centers may be damaged by extreme natural disasters. Cloud service products may have security vulnerabilities caused by design defects. Ineffective access control may cause data leakage, malicious use of data, and abuse of access rights.
  - **In terms of external threats**, organizations may face hacker attacks, third-party vendors' products may have defects, and business processes may have vulnerabilities and be exploited for fraud.

- According to the cloud security report released in 2021, up to 96% of enterprises are worried about public cloud security. According to the anxiety level, the security concerns are 23% (moderate), 41% (very), and 32% (extremely).
- In September 2020, the Cloud Security Alliance (CSA) released 11 types of top cloud computing threats. Compared to the 12 threats released in the previous 2016 release, CSA noted a decline in the ranking of traditional cloud security issues due to efforts by cloud service providers. Concerns such as denial of service, sharing technology vulnerabilities, and cloud service provider data loss and system vulnerabilities (All of the previous top 12 potential risks) Now the rating is so low that it is no longer on the list of top threats. This suggests that the traditional security problems that are the responsibility of cloud service providers seem to have been effectively mitigated.

Cloud Service Providers Improve Security Management Capabilities

- How to solve the cloud service security problems and challenges faced by cloud service providers and reduce customers' concerns is a key issue for all cloud service providers to continuously provide services. In order to better address the potential security risks posed by public cloud services, internal and external security threats, cloud security compliance risks, and enhance the understanding of the shared security responsibility model among stakeholders, cloud service providers must adopt appropriate management and technical means. Gradually improve cloud security and privacy management capabilities. For example:
  - Integrates security services from third-party security vendors to quickly integrate more and more updated security products and capabilities into the cloud platform.
  - Strengthen measures such as access management, log review, and security training for internal personnel to mitigate internal security risks.
  - Strengthen vulnerability management and in-depth protection measures to defend against external threats.
  - Deeply understand compliance requirements and improve compliance capabilities,
  - Avoid fines, lawsuits, and damage to the reputation of the enterprise caused by violations and regulations.

Cloud service customers can leverage security services and products provided by cloud service providers

- Because the responsibility for cloud service security is shared between the cloud service provider and the cloud service customer, the cloud service customer also needs to think about how to manage security in the cloud computing environment. To meet the increasing cloud security management requirements, cloud service customers can use the service products provided by cloud service providers to improve their cloud security management capabilities.
  - **Visible advanced security capabilities**. Cloud service providers can provide visualized security monitoring and protection capabilities for cloud computing environments to help cloud service customers discover and block security vulnerabilities, detect suspicious behaviors, and respond to possible intrusion attacks in a timely manner.
  - **Security solutions applicable to multiple scenarios**. Cloud service providers use innovative capabilities to integrate multiple mature products and the latest technologies to design network security solutions for cloud service customers in various business scenarios, escorting customers' digital transformation and enabling customers to invest in new technology changes with confidence. For example, the Content Moderation service automatically detects content violation, helping customers reduce service violation risks.
  - **Rich cloud security ecosystem**. The rich cloud security ecosystem greatly expands the variety of cloud security services, enables cloud service customers to have more autonomy in product selection, and helps cloud service customers flexibly select services and products based on different scenario requirements, improving the security of their IT systems.
  - **Other cloud security services**. Cloud service providers can also provide security and compliance consulting and security hosting services to customers, so that cloud service customers can quickly obtain high-level security management capabilities by leveraging the capabilities and experience of cloud service providers.

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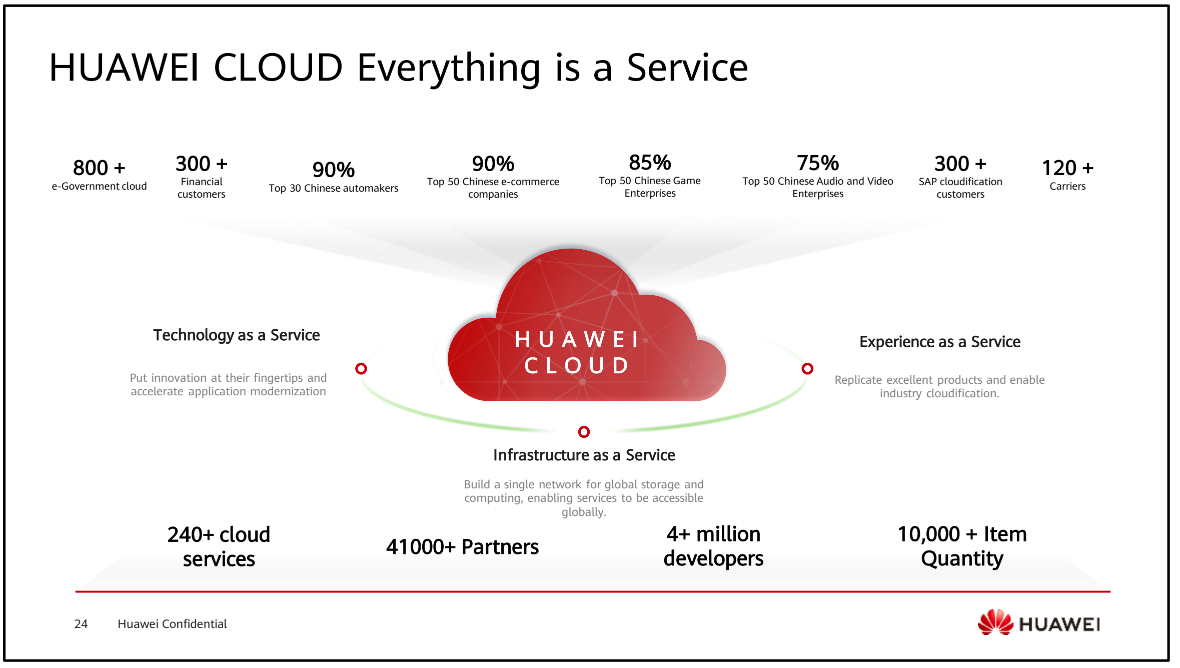


- In 2017, Huawei officially launched the HUAWEI CLOUD brand, which opens Huawei's 30-year-old technology accumulation and product solutions in the ICT field to customers. Through infrastructure as a service, technology as a service, and experience as a service, we realize "everything is a service". Provides stable, reliable, secure, reliable, and sustainable cloud services for customers, partners, and developers.
- According to Gartner's Market Share: IT Services, Worldwide 2021 research report released in April 2022, HUAWEI CLOUD ranks top 5 in the global IaaS market, second in China, third in Thailand, and fourth in emerging Asia Pacific.
- HUAWEI CLOUD has launched 248 cloud services and more than 78,000 APIs, has joined more than 40 million partners around the world, and has developed more than 4 million developers. More than 10,000 applications have been released to the market.
- In China, HUAWEI CLOUD has served more than 700 government cloud projects and has worked with more than 150 cities to build "one city, one cloud". Serves six major banks, 12 joint-stock commercial banks, top 5 insurance institutions and 7 top 10 traditional securities firms in China. Serves more than 30 smart airports, more than 30 urban rail, and 29 provincial highways; It serves 14 provincial companies of State Grid Group, more than 30 automobile manufacturing enterprises, more than 20 top building materials & mining enterprises, and more than 15 top household appliance enterprises.

- HUAWEI CLOUD Enablement Cloud has deployed more than 160 innovation centers and built more than 60 industrial Internet innovation centers across the country, helping 23,000 manufacturing enterprises with digital transformation. 80% of the top 50 Internet enterprises have chosen HUAWEI CLOUD. 90% of China's top 30 automobile enterprises have chosen HUAWEI CLOUD. HUAWEI CLOUD opens the autonomous driving ecosystem, and 80% of the enterprises in the autonomous driving industry chain conduct R&D on HUAWEI CLOUD.
- In the Asia-Pacific region, HUAWEI CLOUD is the fastest growing mainstream public cloud provider. It ranks top 3 in Thailand and top 4 in emerging markets. HUAWEI CLOUD has served more than 20 financial customers, more than 100 government customers, and more than 170 Internet and cloud-native valued customers in the Asia-Pacific region. In 2021, the number of valued customers will increase by more than 150%, and the revenue of partners will increase by more than 150%. HUAWEI CLOUD has become one of the best partners in enterprise digital transformation.

Everything as a Service - Infrastructure as a Service

- HUAWEI CLOUD is deployed globally.
  - 93 AZs are operated in 33 regions (self-operated and jointly operated), covering Asia Pacific, Latin America, Africa, Europe, and Middle East. Currently, more than 220 cloud services and 210 solutions have been launched, meeting the service requirements of various users.
- Huawei Cloud Global Infrastructure
  - Product categories cover infrastructures such as computing, container, storage, network, CDN and intelligent edge, database, AI, big data, IoT, application middleware, development and O&M, enterprise applications, video, security and compliance, management and supervision, migration, and blockchain.

- Based on the operation status, the regions, AZs, cloud services, and solutions deployed by HUAWEI CLOUD will be adjusted based on the actual situation.

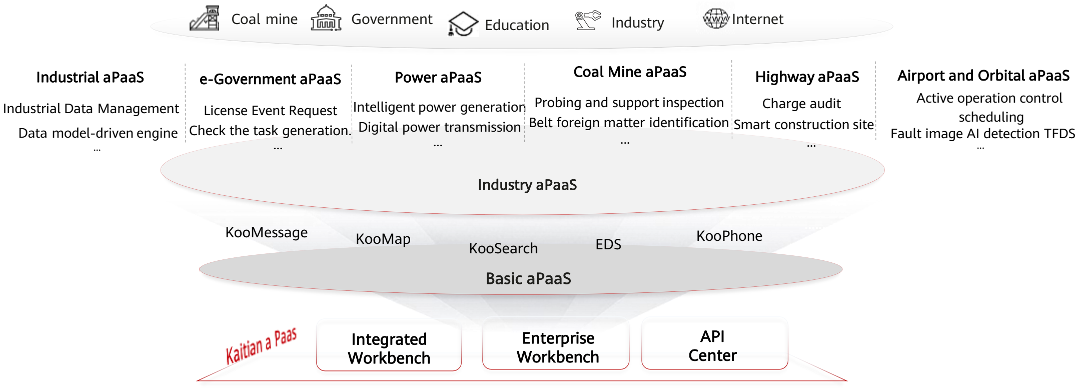
Everything as a Service - Technology as a Service

- Technology-as-a-Service, bringing innovation within reach
  - Huawei's more than 30 years of ICT technology accumulation will be translated into various cloud services on HUAWEI CLOUD, which will be applied by more enterprises. Instead of creating wheels repeatedly, we will focus on customers' own service innovation.
- 100,000 R&D engineers invest tens of billions of dollars in R&D every year, covering four tPaaS development production lines.
  - MetaStudio, a digital content development line. Help thousands of industries to achieve the seamless integration of virtual world and real world.
  - DataArts Studio, a data governance production line. Help enterprises quickly build data operation capabilities and implement integrated governance of batch, stream, and interactive data.
  - Software development production line CodeArts. A one-stop, end-to-end, secure, and reliable software development production line, which is ready to use out of the box and has years of Huawei's best R&D practices built in, facilitating efficiency double and digital transformation.
  - AI development production line, AI platform ModelArts. Helps users quickly create and deploy models and manage full-cycle AI workflows.

- MetaStudio, a digital content production line, provides platform capabilities such as 3D model creation (Creator), asset management (Store), content editing (Editor), and cloud rendering (Rendering) based on two media engines: graphics engine and space engine. Joint partners will build production lines such as digital human production, virtual live broadcast, enterprise 3D space, and virtual and real integration to help thousands of industries seamlessly integrate the virtual world and the real world.

Everything as a Service - Experience as a Service aPaaS

- HUAWEI CLOUD aPaaS integrates industry capabilities and industry experience, provides a one-stop open platform for unified application distribution and operation, improves application construction, development, and use experience, and supports rapid innovation of applications in various industries.

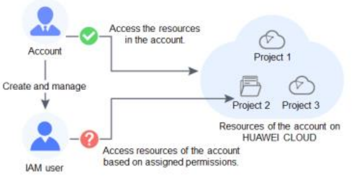


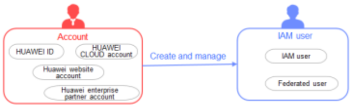
Basic aPaaS, accelerating enterprise digital upgrade

Basic aPaaS	Definition and Function
Cloud Message Service KooMessage	• Integrate multiple customer access channels, including intelligent information, service numbers, PUSH messages, and 5G messages, provide one-stop industry services and user growth services for industry customers, achieve all-scenario and all-end customers, and improve final consumer service satisfaction and marketing conversion rate. • This feature is available only to enterprise certified customers.
Cloud Map Service KooMap	• The KooMap satellite image processing service converges high-quality satellite sources and provides global satellite image processing, supporting application transformation and innovation for government and enterprise customers. • Precipitate industry assets, build an open platform, and provide one-stop out-of-the-box space-time information services, such as space-time processing, analysis, and visualization.
Cloud Phone Service KooPhone	• KooPhone is a cloud mobile phone service that features excellent experience and high security based on Huawei Kunpeng ARM servers, introduces Huawei core technologies such as audio and video codec and real-time transmission, and the rich application ecosystem of HUAWEI CLOUD. It provides new application scenarios for customers in industries such as government, enterprise, and Internet. • Breaks through physical resource restrictions and enables on-demand scaling and flexible conversion of mobile phone instance specifications. Based on cloudification advantages, tens of thousands of mobile phones are provisioned in minutes, and massive resources are centrally managed and controlled.
Enterprise search service KooSearch	• KooSearch is a fully managed search service. It provides search services for Huawei internal office and customer search services. • With built-in capabilities such as industry word segmentation, semantic understanding, and industry sorting algorithms, Huawei provides customers with simpler, more accurate, and faster search services. • Huawei cloud provides enterprise-level data security, permission control, and global deployment capabilities to meet enterprise-level application requirements.
Exchange data space EDS	• The Exchange Data Space (EDS) is an exchange and sharing platform designed to protect enterprise data sovereignty, promote efficient data circulation, and maximize data value. • The platform provides 21 policies, which are based on policies during data use. (e.g. "validity period, number of viewing times, downloading, etc.) Implement corresponding use control to ensure that data is used in compliance with the rules and regulations on the basis of data sovereignty and control.

HUAWEI CLOUD Basic Concepts – Account

- The HUAWEI CLOUD account system consists of two types of accounts:
  - Accounts: registered or created on HUAWEI CLOUD. An account has the highest permissions on HUAWEI CLOUD. It can access all of its resources and pays for the use of these resources. Accounts include HUAWEI IDs and HUAWEI CLOUD accounts.
  - IAM users: created and managed using an account in IAM. The account administrator grants permissions to IAM users and makes payment for the resources they use. IAM users use resources as specified by the permissions.
- Users can log in to HUAWEI CLOUD using a HUAWEI ID, Huawei website account, Huawei enterprise partner account, or HUAWEI CLOUD account, and use their resources and cloud services.
- If Users are an IAM user created by an account or a user of a third-party system that has established a trust relationship with HUAWEI CLOUD, log in to HUAWEI CLOUD through the corresponding page and then use resources and cloud services as specified by the permissions granted by the account.

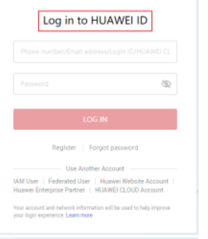


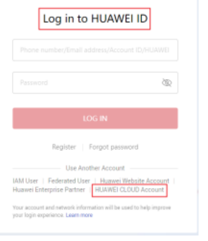


HUAWEI

Huawei ID and HUAWEI CLOUD Account

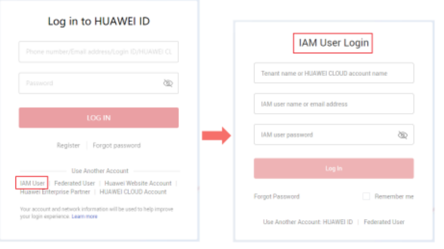
- You can register a HUAWEI ID to access all Huawei services, such as HUAWEI CLOUD and Vmall.
  - **Registration:** Register a HUAWEI ID on any Huawei service website, such as the HUAWEI ID website.
  - **HUAWEI CLOUD login**: Log in to HUAWEI CLOUD by clicking HUAWEI ID. If this is the first time you log in to HUAWEI CLOUD with a HUAWEI ID, enable HUAWEI CLOUD services or bind the HUAWEI ID to your HUAWEI CLOUD account by following the on-screen prompts.
- HUAWEI CLOUD accounts can only be used to log in to HUAWEI CLOUD.
  - **Registration**: To improve login experience, we have unified our account system. You can only register HUAWEI IDs on HUAWEI CLOUD from October 30, 2021.
  - **HUAWEI CLOUD login**: Log in to HUAWEI CLOUD by clicking HUAWEI ID or HUAWEI CLOUD Account.





IAM User

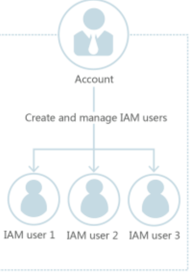
- Huawei Cloud Identity and Access Management (IAM) provides permissions management to help you securely control access to your cloud services and resources. If you want to share resources with others but do not want to share your own account and password, you can create an IAM user.
  - You can use your account to create IAM users and assign permissions for specific resources. Each IAM user has their own identity credentials (passwords or access keys) and uses cloud resources based on assigned permissions. IAM users cannot make payments themselves.
  - IAM users do not own resources and cannot make payments. Any activities performed by IAM users in your account are billed to your account.



- Identity and Access Management (IAM) is a unified identity authentication service that helps users more securely control access to cloud services and resources. The IAM user is used to prevent multiple users from sharing the passwords of the accounts. This section describes IAM in detail in the following sections.

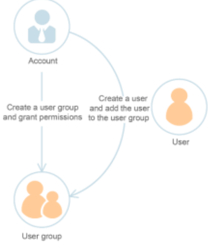
Relationship between accounts and IAM users

- An account and its IAM users share a parent-child relationship. The account owns the resources and makes payments for the resources used by IAM users. It has full permissions for these resources.
  - IAM users are created by the account administrator, and only have the permissions granted by the administrator. The administrator can modify or revoke the IAM users' permissions at any time.
  - Fees generated by IAM users' use of resources are paid by the account.



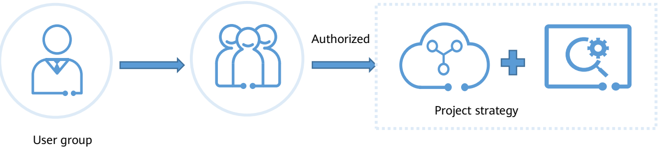
User Group

- You can use user groups to assign permissions to IAM users.
  - After an IAM user is added to a user group, the user has the permissions of the group and can perform operations on cloud services as specified by the permissions.
  - If a user is added to multiple user groups, the user inherits the permissions assigned to all these groups.
  - The default user group **admin** has all permissions required to use all of the cloud resources. Users in this group can perform operations on all the resources, including but not limited to creating user groups and users, modifying permissions, and managing resources.



Permission

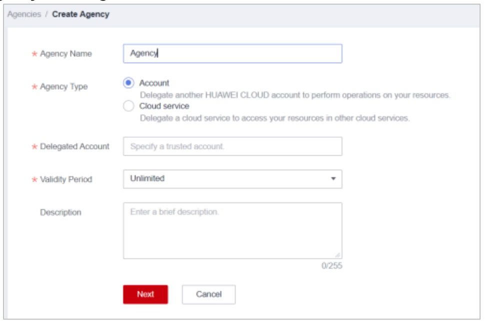
- You can grant permissions by using roles and policies.
  - Roles: A coarse-grained authorization strategy provided by IAM to assign permissions based on users' job responsibilities. Only a limited number of service-level roles are available for authorization.
  - Policies: A fine-grained authorization strategy that defines permissions required to perform operations on specific cloud resources under certain conditions. IAM supports both system-defined and custom policies.
    - system-defined policy defines the common actions of a cloud service. System-defined policies can be used to assign permissions to user groups, and cannot be modified.
    - Custom policies function as a supplement to system-defined policies. You can create custom policies using the actions supported by cloud services for more refined access control. You can create custom policies in the visual editor or in JSON view.



- A fine-grained authorization strategy that defines permissions required to perform operations on specific cloud resources under certain conditions. This type of authorization is more flexible and is ideal for least privilege access. For example, you can grant users only permission to manage ECSs of a certain type.
- If you need to assign permissions for a specific service to a user group or agency on the IAM console but cannot find corresponding policies, it indicates that the service does not support permissions management through IAM. You can submit a service ticket to request that permissions for the service be made available in IAM.

Agency

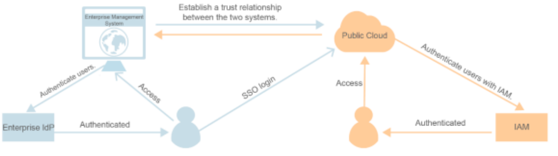
- A trust relationship that you can establish between your account and another account or a cloud service to delegate resource access.
  - Account delegation: You can delegate another account to implement O&M on your resources based on assigned permissions.
  - Cloud service delegation: Huawei Cloud services interwork with each other, and some cloud services are dependent on other services. You can create an agency to delegate a cloud service to access other services.



- The IAM. Agency element is used to create agencies on IAM, specify entrusted accounts, and grant rights. After an administrator assigns agent operator permissions to an entrusted account user, the user can manage corresponding resources.

Advantages of IAM

- Fine-grained access control for Huawei Cloud resources
  - If you purchase multiple Huawei Cloud resources for different teams or applications in your enterprise, you can use your account to create IAM users for the team members or applications and grant them permissions required to complete specific tasks.
  - The IAM users use their own usernames and passwords to log in to Huawei Cloud and access resources in your account.
- Cross-account resource access delegation
  - If you purchase multiple Huawei Cloud resources, you can delegate another account to manage some of your resources for efficient O&M
- Federated access to Huawei Cloud with existing enterprise accounts (identity federation)
  - If your enterprise has an identity system, you can create an identity provider (IdP) in IAM to provide single sign-on (SSO) access to Huawei Cloud for employees in your enterprise. The identity provider establishes a trust relationship between your enterprise and Huawei Cloud, allowing the employees to access Huawei Cloud using their existing accounts.



- In addition to IAM, you can use Enterprise Management to control access to cloud resources. Enterprise Management supports more fine-grained permissions management and enterprise project management. You can choose either IAM or Enterprise Management to suit your requirements.
- For example, you can create an agency for a professional O&M company to enable the company to manage specific resources with the company's own account. If the delegation changes, you can modify or revoke the delegated permissions at any time. In the following figure, account A is the delegating party, and account B is the delegated party.

Huawei Cloud-Security Cloud Platform

- 100+ global security compliance certifications
  - Currently, HUAWEI CLOUD has passed various international authoritative certifications and practice standards. The following are some examples:
    - Security-related certifications include ISO 27001, ISO 27017, CSA STAR Gold Certification, China Ministry of Public Security Information Security Level 3/Level 4 Certification, PCI DSS for the payment card industry, and NIST CSF Cyber Security Framework.
    - The following privacy-related specifications are ISO 27018, ISO 27701, BS 10012, ISO 29151, and ISO 27799.
  - "3CS" is a new security governance system for the entire process of cloud services.
    - HUAWEI CLOUD has developed a governance system that covers mainstream cloud security standards in the industry and security management requirements of HUAWEI CLOUD. It is called Cloud Service Cybersecurity & Compliance Standard (3CS for short).
    - This governance system provides valuable reference solutions for enterprises or partners who are willing to learn from Huawei's practical experience.
  - DevSecOps, covering the entire lifecycle of services from development, deployment, to operation.
    - HUAWEI CLOUD seamlessly embeds the security lifecycle (SDL) into the new DevOps process with fast iteration, combining security R&D and O&M, ensuring cloud service security activities without affecting rapid continuous integration, release, and deployment.

Enterprise Solutions

- HUAWEI CLOUD provide comprehensive cloud solutions to help you accelerate growth, from startup to management and expansion.

Website Building Solution	• Build your enterprise website with ease, flexibility, and speed, and at low costs.
Enterprise Cloud Box	• Content management powered by AI and cloud computing for efficiency, security, and ease of use
Marketing Automation	• Marketing Automation helps you streamline data, manage leads, identify and incubate quality potential customers
Cross-border Enterprise Business	• HUAWEI CLOUD help you expand your business internationally and help you enter and thrive in Chinese market
On-premises to On-cloud	• Free cloud resources and professional migration services

Solutions by Industry

- HUAWEI CLOUD provides solutions for a wide range of industries, so you can always find the cloud services you need.

Smart City	Telecom	Automotive
Campus	E-Commerce	Education
Financial Services	Gaming	Healthcare and Life Sciences
Manufacturing	Media & Entertainment	Retail

- Huawei cloud industry solutions are as follows (for the latest classification, see the HUAWEI CLOUD official website):
  - Smart City: Facilitating the upgrade of city infrastructure, management, and services. Serves the needs of four types of users — residents, legal persons, government employees, and decision-makers, and helps refactor or optimize public service processes for better user experience, improving people's livelihoods, satisfaction, and sense of security.
  - Telecom: Enable carriers/operators to achieve network monetization, innovate services, and improve operation efficiency. Huawei Cloud provides powerful solutions representing accumulation of more than 30 years of ICT expertise, solutions that help you go cloud and help your customers go digital.
  - Automotive: Cut operations cost, improve quality and efficiency, and enhance sales support. The global automotive industry is witnessing a CASE (connected, autonomous, shared, and electric) transformation. Automotive enterprises are going digital with intelligent upgrades. Huawei Cloud offers these enterprises tailored solutions that leverage cloud computing, big data, AI, IoT, and 5G
  - Campus: Empowers your industrial park and campus with innovative AI, IoT, big data, and cloud computing. Campuses contain a large number of facilities that must all be monitored to ensure security. There were more than 1.2 million campuses spread across China in 2017. Seeing the great potential, HUAWEI CLOUD geared up its solution so partners can build smart campuses with higher safety standards at lower cost.
  - E-Commerce: Build and host your e-commerce websites on your highly scalable and available cloud infrastructure.

- Education: Cost-effectively upgrade communications and learning systems. This solution ensures stable and efficient resource allocation. It provides quality services for customers in the education sector. Designed for scenarios such as talent cultivation, scientific research and innovation, smart campus, and online education, this solution uses cloud computing, big data, IoT, and artificial intelligence to accelerate education modernization, promoting education equity and improving the quality of education available to all.
- Financial Services: Get the agility while maintaining your FSI safe, stable, and secure. By combining industrial features and Huawei's cloud services, this solution provides end-to-end cloud services for financial customers such as banks, insurance agents, security companies, or Internet finance enterprises. It helps customers quickly migrate their services to the cloud, promoting fast growth and improving their competitiveness.
- Gaming: Deliver a flawless, lag-free gaming experience. . Cloud, AI, and 5G technologies are transforming the industry with a brand new gaming experience. With its powerful cloud infrastructure capabilities and innovative technical advantages, HUAWEI CLOUD provides professional, fast, stable, and secure one-stop cloud service solutions for gaming enterprises to build high-quality, comprehensive cloud gaming platforms.
- Manufacturing: Increase production, perform preventive maintenance, and accurately predict customer demands.
- Healthcare and Life Sciences: Accelerate research, scale telemedicine services, and improve health outcomes. Leveraging core cloud services such as cloud-network synergy, Big Data, and artificial intelligence of HUAWEI CLOUD and its partners, the Healthcare and Life Sciences solution provides high-performance, reliable, and secure resources and technologies and a full portfolio of applications and services for the medical and healthcare industry.

Practical Application of Huawei Cloud Solutions

- The Practical Application of Huawei Cloud Solutions describes the architecture and deployment of Huawei Cloud solutions in specific scenarios. The source codes have been technically verified by experts for one-click deployment. Technical support is also available to assist you in resolving problems that arise during the deployment.

Source Code Compilation with Jenkins	CDN for Download Acceleration	CSS-based SQL Acceleration
Application Containerization on the Cloud	Quickly Deploying a High-Availability RabbitMQ Cluster	Quick Deployment of an MHY MySQL Cluster
Serverless Real-Time Log Analysis	Website O&M Analysis Based on LTS	Scheduled ECS Specification Modification
Scheduled ECS Startup/Shutdown		

- Source Code Compilation with Jenkins: Quickly deploy source code compilation environments on ECS.
- CDN for Download Acceleration: Use CDN and OBS to provide turnkey download acceleration for static resources.
- CSS-based SQL Acceleration; Use CSS to quickly build SQL acceleration solutions.
- Application Containerization on the Cloud: Quickly deploy a cross-AZ HZ container cluster environment and containerize service systems.
- Quickly Deploying a High-Availability RabbitMQ Cluster: Set up a high-availability RabbitMQ cluster.
- Quick Deployment of an MHY MySQL Cluster: Use MHA to deploy highly available MySQL clusters on ECSs.
- Serverless Real-Time Log Analysis: Collect, analyze, and archive ECS logs with a serverless architecture.
- Website O&M Analysis Based on LTS: Quickly interconnect LTS with ELB for routine website O&M analysis.
- Scheduled ECS Specification Modification: Use a FuctionGraph timer trigger to periodically modify ECS specifications.
- Scheduled ECS Startup/Shutdown: Use a FunctionGraph timer trigger to periodically start and stop ECSs.

Create, share, and win-win results to build a new industry ecosystem

co-creation

Application enable, data enable, and AI enable
Help enterprises explore and practice innovative solutions

Sharing

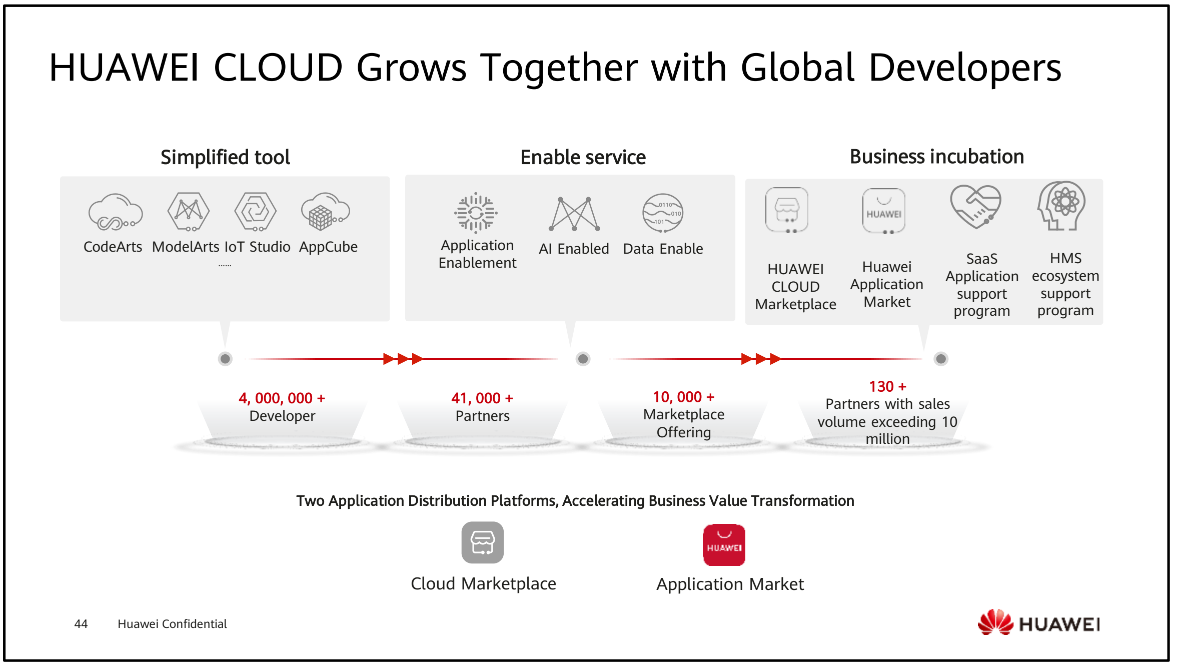
Public cloud + hybrid cloud + edge cloud
Share innovation capabilities and multi-cloud application ecosystem

Win-win

Marketplace, reseller channel
Enable excellent software to serve more enterprises

Build a Black Land for Ecosystem Development with HUAWEI CLOUD as the Foundation

- HUAWEI CLOUD adheres to the concept of joint creation, sharing, and win-win ecosystem. With HUAWEI CLOUD as the foundation, we build a black land for ecosystem development. Our colleagues and partners work together to facilitate digital transformation and intelligent upgrade of industries.
- Co-creation: Continuous technological innovation enables industry innovation. HUAWEI CLOUD builds three enablement platforms: application enablement, data enablement, and AI enablement to help ecosystem partners realize cloudification, SaaS, and intelligence of applications.
- Sharing: Industry applications are evolving towards cloud-edge-device synergy. HUAWEI CLOUD uses the Optimus architecture to streamline the public cloud, hybrid cloud, and edge cloud to build a unified application ecosystem and share innovation capabilities in multiple industries, application scenarios, and deployment forms.
- Win-win: HUAWEI CLOUD works with partners to create value for customers, enable excellent software to serve more enterprises, and achieve win-win results with customers and partners in the digital era.
- Currently, we have aggregated 1800000 developers, more than 13,000 consulting partners, more than 7,000 technical partners, and more than 100000 paid users. We have released more than 4000 applications on the cloud market. The annual transaction amount exceeds 1 billion RMB, and the number of paid users exceeds 100,000. We sincerely invite more excellent enterprises to join the HUAWEI CLOUD ecosystem.



- Specifically, HUAWEI CLOUD provides a series of simplified tools and templates to improve development efficiency. In addition, Huawei provides application, data, and AI enabling services, and builds multiple industry knowledge and asset models to flexibly respond to market requirements.
- More importantly, HUAWEI CLOUD should help developers monetize their business. HUAWEI CLOUD provides powerful application distribution capabilities and the most potential business support program. Developers can obtain rich cloud resources and traffic support and have the opportunity to communicate and cooperate with top enterprise accelerators and incubators. We hope that developers can grow, succeed, and achieve success on HUAWEI CLOUD.
- HUAWEI CLOUD Marketplace is an application distribution platform for government and enterprise users. Over the past two years, we have developed rapidly. Currently, the annual transaction volume of HUAWEI CLOUD Marketplace has exceeded 1 billion, and the number of orders has exceeded 100,000. The sales volume of 30 partners has exceeded 10 million.
- AppGallery, the familiar Huawei terminal application distribution platform, is the third largest application market in the world. It has 530 million active users worldwide, and the total number of distributed applications has exceeded 384.4 billion. HUAWEI CLOUD cooperates with AppGallery Connect to build a one-stop solution for mobile applications, providing more technical and resource support for innovative applications that integrate HMS Core.
- HUAWEI CLOUD hopes that the two application distribution platforms can help developers accelerate business value transformation.

Quiz

1. In the cloud computing deployment mode, the infrastructure is owned by a single organization and runs only for that organization. Which of the following deployment modes is the cloud computing deployment mode?
   A. Private cloud
   B. Public cloud
   C. Hybrid cloud
2. (True or false) Huawei Cloud uses Identity and Access Management (IAM) projects to group and isolate resources in different regions.
   A. True
   B. False

- Answer:
- 1, A
- 2, False B. IAM can restrict the permissions of IAM users and user groups to use resources in different regions, but cannot isolate resources and groups in different regions.

Summary

After reviewing this chapter, we have a preliminary understanding of cloud computing and public cloud, and have a basic understanding of their development background, future trends, and technical characteristics. This chapter described the basic architecture, basic concepts, technical features, ecosystem construction, and future market trends of HUAWEI CLOUD, and have a preliminary impression on HUAWEI CLOUD.

Recommendations

- Huawei Talent
  - https://e.huawei.com/en/talent/cert/#/careerCert
- Huawei Technical Support Website
  - https://support.huaweicloud.com/intl/en-us/help-novicedocument.html
- HUAWEI CLOUD Academy
  - https://edu.huaweicloud.com/intl/en-us/

Acronyms and Abbreviations

AI: Artificial intelligence

AS: Auto Scaling

APM: Application Performance Management

AOM: Application Operations Management

AZ: Availability Zone

API: Application Programming Interface

BMS: Bare Metal Server

BCS: Hyperledger Fabric

CCE: Cloud Container Engine

CDN: Content Delivery Network

Acronyms and Abbreviations

CBH: Cloud Bastion Host

CPTS: Cloud Performance Test Service

CAE: Computer Aided Engineering

CES: Cloud Eye Service

CTS: Cloud Trace Service

CCS: Cloud Catalog Service

CRS: Cloud Record Service

CDM: Cloud Data Migration

CMC: Cloud Migration Center

DES: Data Express Service

Acronyms and Abbreviations

DNS: Domain Name Service

DDS: Document Database Service

DDM: Distributed Database Middleware

DAS: Data Admin Service

DBSS: Database Security Service

DMS: Distributed Message Service

DWS: Data Warehouse Service

DevOps: Development and Operations

ECS: Elastic Cloud Server

EVS: Elastic Volume Service

Acronyms and Abbreviations

ELB: Elastic Load Balance

EI: enterprise intelligence

ERP: Enterprise Resource Planning

GES: Graph Engine Service

HMS: Huawei Mobile Service

HSS: Host Security Service

ICT: Information and Communications Technology

IMS: Image Management Service

IAM: Identity and Access Management

IOPS: Input/Output Operations Per Second

Acronyms and Abbreviations

I/O: Input/Output

LTS: Log Tank Service

MVP: Most Valuable Player

MRS: MapReduce Service

NIC: Network Interface Controller

OBS: Object Storage Service

OCR: Optical Character Recognition

OMS: Object Storage Migration Service

OVS: Open Virtual Switch

QoS: Quality of Service

Acronyms and Abbreviations

RoCE: RDMA over Converged Ethernet, a network protocol that allows remote direct memory access (RDMA) over Ethernet

RDS: Relational Database Service

SDK: Software Development Kit

SFS: Scalable File Service

SA: Situation Awareness, situational awareness

SMN: Simple Message Notification

SWR: SoftWare Repository for Container

SMS: Server Migration Service

SSD: Solid State Disk

Acronyms and Abbreviations

SAP: System Applications and Products

TMS: Tag Management Service

TTS: Text-To-Speech

VBS: Volume Backup Service

VPC: Virtual Private Cloud

VPN: Virtual Private Network

VXLAN: Virtual Extensible Local Area Network

WAF: web application firewall

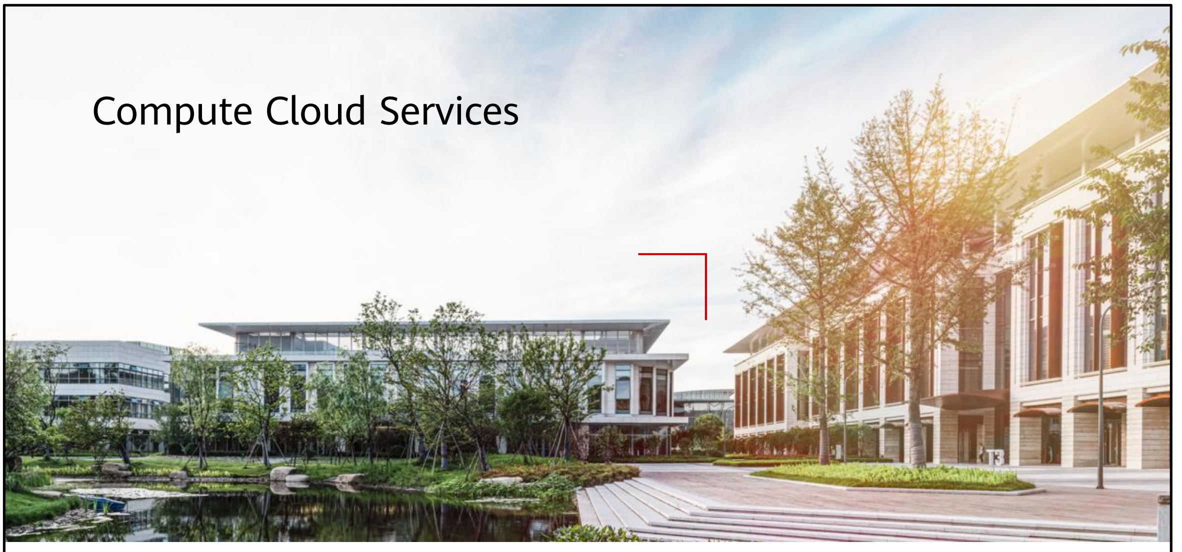
Thank Users.

把数字世界带入每个人、每个家庭、
每个组织，构建万物互联的智能世界。

Bring digital to every person, home, and organization for a fully connected, intelligent world.

HUAWEI

Compute Cloud Services



Foreword

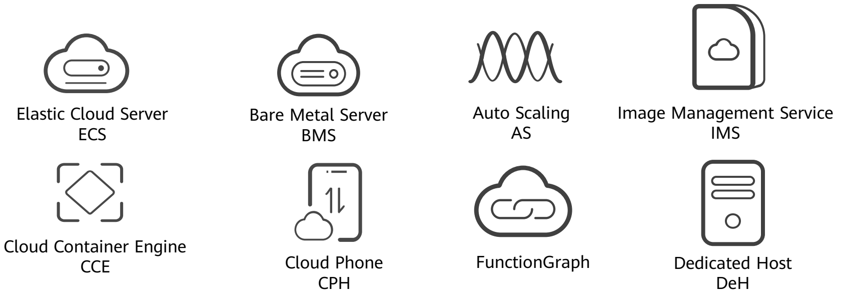
- Computing resources have always been the main artery for the development of the entire enterprise service system. Without computing resources, enterprise services cannot run properly. In the cloud computing era, computing services are also the first type of cloud services. Therefore, the importance of computing resources can be seen.
- This chapter describes the computing services on HUAWEI CLOUD.

Objectives

- On completion of this course, you will be able to:
  - Understand common computing services on HUAWEI CLOUD.
  - Understand the basic concepts, features, usage methods, and application scenarios of ECSs.
  - Have a good command of the basic concepts, features, usage methods, and application scenarios of IMS.
  - Understand the basic concepts, features, usage methods, and application scenarios of the AS service.
  - Have a good command of the basic concepts and features of the BMS service.
  - Learn about CCE and other computing cloud services.

Computing cloud service

- A compute resource is a measurable amount of computing power that can be requested, allocated, and used for a compute activity. Common computing resources include the CPU and memory.
- A computing cloud service is a service or product that can provide computing resources on the cloud.

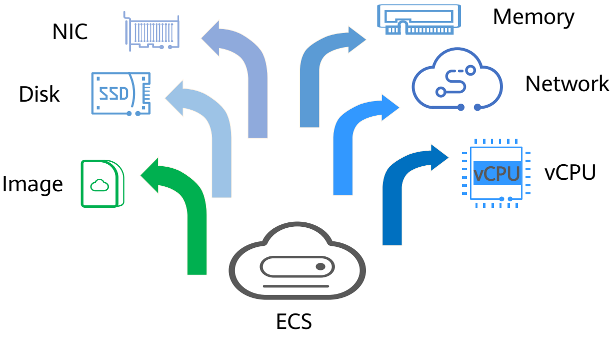


HUAWEI

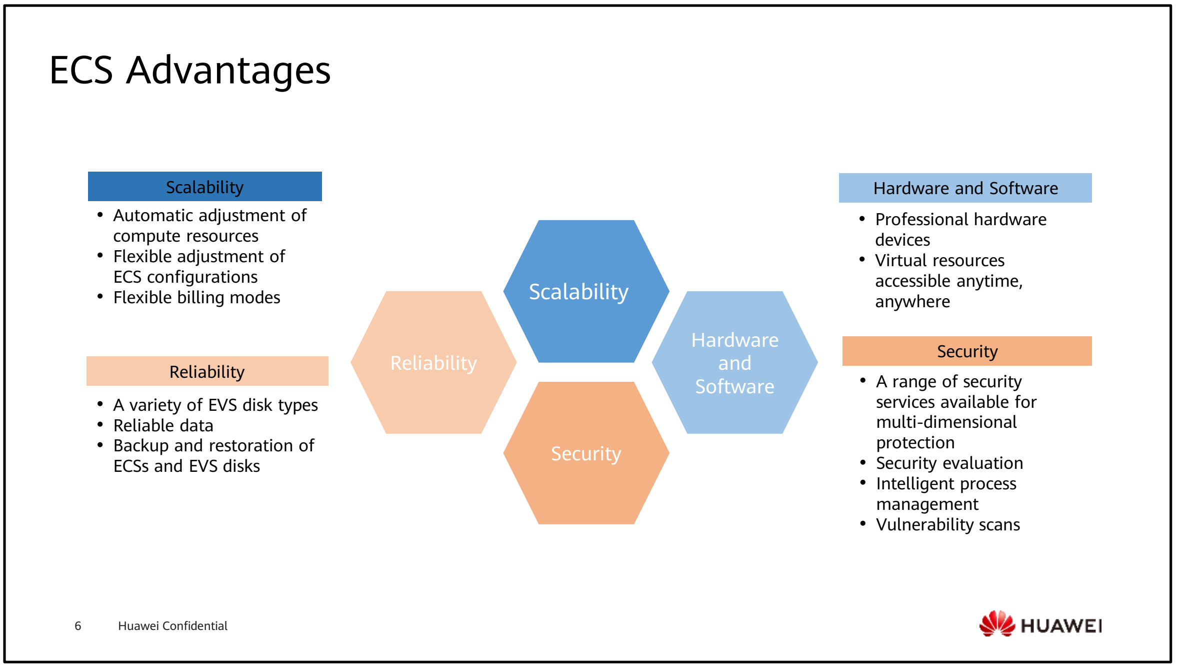
Contents

What Is Elastic Cloud Server (ECS)?

- An ECS is a basic computing unit that consists of vCPUs, memory, an OS, and Elastic Volume Service (EVS) disks. After an ECS is created, you can use it on the cloud similarly to how you would use your local computer or physical server.



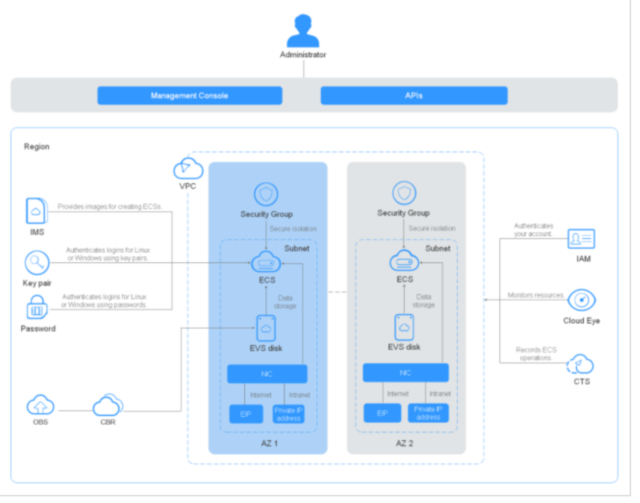
- An ECS is a computer system that has complete hardware, operating system, and network functions and runs in a completely isolated environment.
- ECS has the following advantages:
  - A variety of specifications to choose from: Different ECS types are available for different applications. There are multiple, customizable specifications for each type.
  - A wide range of available images: Public, private, and shared images can be selected.
  - Different types of EVS disks: Common I/O, high I/O, general-purpose SSD, and ultra-high I/O EVS disks are available for different service requirements.
  - Flexible billing: Yearly/monthly and pay-per-use billing modes are available for different applications. You can purchase and release resources as service levels fluctuate.
  - Reliable data: Virtual block storage based on a distributed architecture provides robust throughput that is scalable and reliable.
  - Security: The network is isolated from viruses and Trojans by security group rules. Security services, such as Anti-DDoS, Web Application Firewall and Vulnerability Scan Service are also available to protect your ECSs.
  - Flexible, easy-to-use: Elastic computing resources are automatically adjusted based on service requirements and policies to efficiently meet service requirements.
  - Highly efficient O&M: Multi-choice management via the management console, remote access, and APIs with full management permissions.
  - Cloud monitoring: Cloud Eye monitors your ECSs in real time, generating alarms and sending notifications when it detects abnormal metrics.
  - Load balancing: Elastic Load Balance automatically distributes traffic to multiple ECSs to keep the loads on the servers balanced. It improves the fault tolerance of your applications and enhances application capabilities.



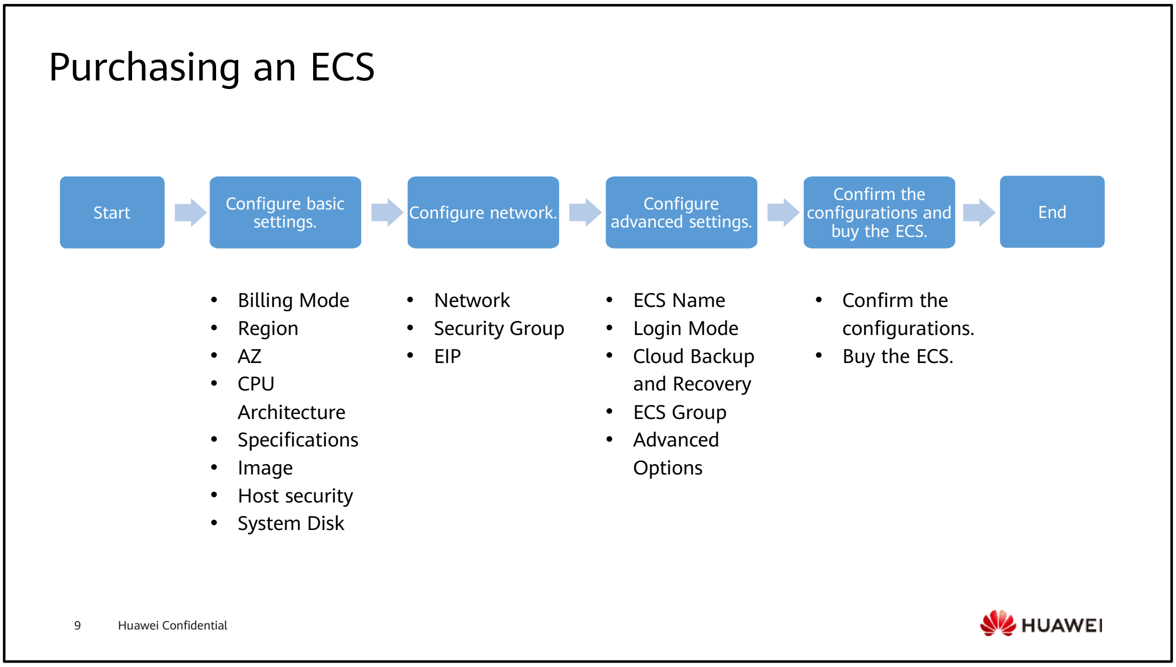
- Reliability
  - A variety of EVS disk types: Common I/O, high I/O, ultra-high I/O, general purpose SSD, and extreme SSD disks are available for different service requirements.
  - Reliable data: Scalable, reliable, high-throughput virtual block storage is based on a distributed architecture. This architecture ensures that data can be quickly migrated or restored, if necessary, which means you will not lose your data as the result of a single hardware fault.
  - Backup and restoration of ECSs and EVS disks: You can configure backup policies on the management console or use an API to back up ECSs and EVS disks periodically or at a specified time.
- Security Protection
  - A range of security services provide multi-dimensional protection: Security services, such as Web Application Firewall and Vulnerability Scan Service, are available to protect your ECSs.
  - Security evaluation: The security of cloud environments is evaluated to help you quickly detect security vulnerabilities and threats. Security configurations are reviewed and suggestions provided on how to improve system security. Actions will be recommended to reduce or avoid altogether potential losses resulting from viruses or other malicious attacks.

- Hardware and Software
    - Professional hardware devices ECSs are equipped with professional hardware devices and can be optimized for virtualization. Users do not need to build their own equipment rooms.
    - Virtual resources can be obtained at any time from the virtual resource pool and exclusively used. Elastic cloud servers can be used on the cloud like local PCs, ensuring reliable, secure, flexible, and efficient application environments.
- Scalability
    - Automatic scaling of computing resources: Based on the monitoring data of the scaling group, ECSs are dynamically added or deleted based on the running status of applications.
    - Scheduled scaling: Based on service expectations and operation plans, you can customize scheduled and periodic policies to automatically add or delete ECS instances on time.
    - ECS configuration specifications and bandwidth can be flexibly adjusted based on service requirements, efficiently matching service requirements.
    - Flexible billing modes: yearly/monthly, pay-per-use, and bidding ECSs can be purchased and released at any time based on service fluctuation.

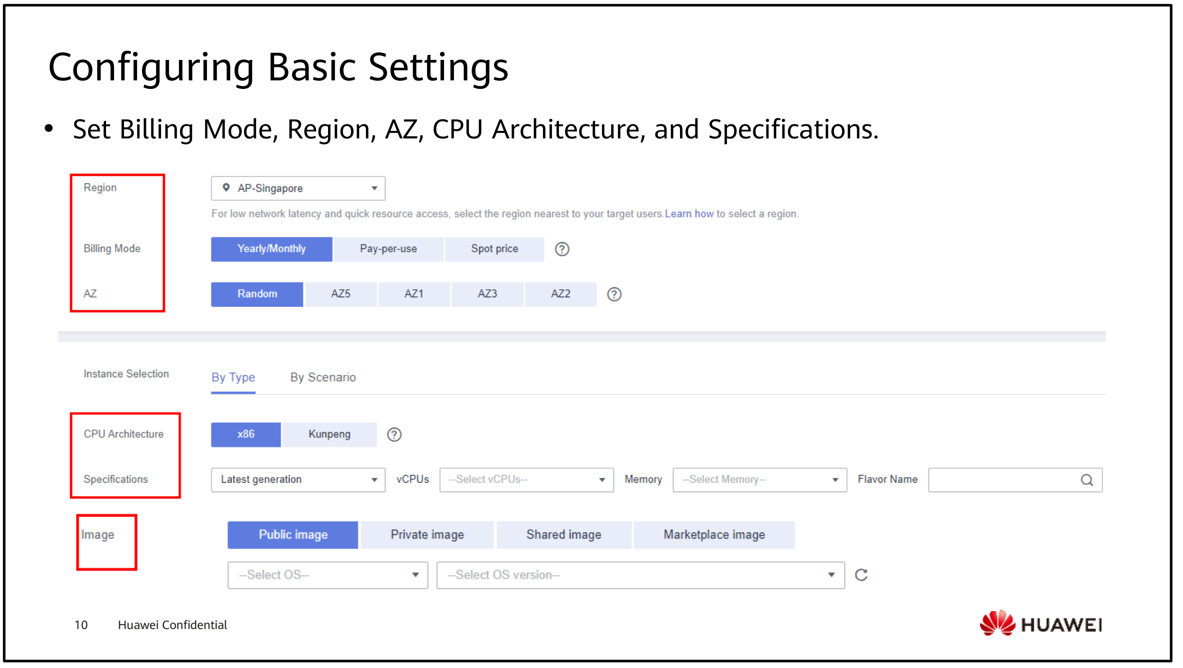
ECS Architecture



- ECS works with other products and services to provide computing, storage, network, and image installation functions.
  - ECSs are deployed in multiple Availability Zones (AZs) connected with each other through an intranet. If an AZ becomes faulty, other AZs in the same region will not be affected.
  - With the Virtual Private Cloud (VPC) service, you can build a dedicated network, configure subnets and security groups, and allow the VPC to communicate with the external network through an EIP with bandwidth assigned.
  - With the Image Management Service (IMS), you can create images for ECSs, or create ECSs using private images for rapid service deployment.
  - EVS provides storage and Volume Backup Service (VBS) provides data backup and recovery functions.
  - Cloud Eye is a key service to help ensure ECS performance, reliability, and availability. You can use Cloud Eye to monitor ECS resource usage.
  - Cloud Backup and Recovery (CBR) backs up data for EVS disks and ECSs and creates snapshots in case you need to restore them.



- Select a billing mode, yearly/monthly or pay-per-use.
    - You can purchase a yearly/monthly ECS subscription and enter your required duration. Yearly/monthly subscriptions are pre-paid, using a single, lump sum payment.
    - If you choose pay-per-use billing, you do not need to choose a required duration. Pay-per-use usage is postpaid.
- Select required specifications: HUAWEI CLOUD provides various ECS types for you to select based on different applications. You can view the available ECS types and specifications in the list. Alternatively, you can enter a flavor (such as c3) or search for a flavor by vCPU and memory.
- Set Network by selecting an available VPC and subnet from the drop-down list, and specifying a private IP address assignment mode. You can also create a VPC if needed. VPC provides a network, including subnets and security groups, for an ECS.
- Set EIP. If you want the ECS to connect to the Internet, it needs to have an EIP bound.
- Set Login Mode. Key pair is recommended because key pair authentication is more secure than using a password.



- Configure basic settings
  - . **Billing Mode**: An ECS can be billed on a pay-per-use, yearly/monthly, or spot price basis. For yearly/monthly subscriptions, the longer the subscription, the more you save.
  - **Region** and **AZ**: ECSs in different regions cannot communicate with each other over an intranet. Select a region closest to your target users to ensure low network latency and quick access.
  - **CPU Architecture**: x86-based CPUs use Complex Instruction Set Computing (CISC). Kunpeng CPUs use Reduced Instruction Set Computing (RISC).
  - **Specifications**: Select a flavor and image based on service requirements.
- The central processing unit (CPU) is mainly composed of three parts: calculator, controller and register. The calculator plays the role of operation. The controller is responsible for issuing the information required by each instruction of the CPU. The register stores the temporary files of the operation or instruction, which can ensure a higher speed.
- The CPU architecture is a specification defined by the CPU vendor for the CPU products of the same series. The main purpose is to distinguish different types of CPUs.
- Complex instruction set computer (CISC): Intel, AMD
- Reduced instruction set computer (RISC) CPU: IBM, ARM
- CISC has strong compatibility, a variety of instructions, variable length, and is implemented by microprogramming. RISC, on the other hand, has few instructions and similar frequency, and is mainly implemented by hardware (general-purpose registers, hard-wired logic control).

Billing Mode

- Yearly/Monthly
  - prepaid billing mode and is cost-effective for long-term use.
- Pay-per-Use
  - A postpaid billing mode in which an ECS will be billed based on usage frequency and duration.
- Spot price
  - Spot price ECSs are billed based on the market price, which varies according to the changes in supply and demand.

- Yearly/Monthly: The ECS will be billed based on the service duration. This cost-effective mode is ideal when the duration of ECS usage is predictable.
- Pay-per-use: The ECS will be billed based on usage frequency and duration. This mode is ideal when you want more flexibility and control on ECS usage.
- Spot price: The ECS will be billed based on the price that is effective for the time it is being used. This mode is more cost-effective than pay-per-use, and the spot price will be adjusted based on supply-and-demand changes.

Region

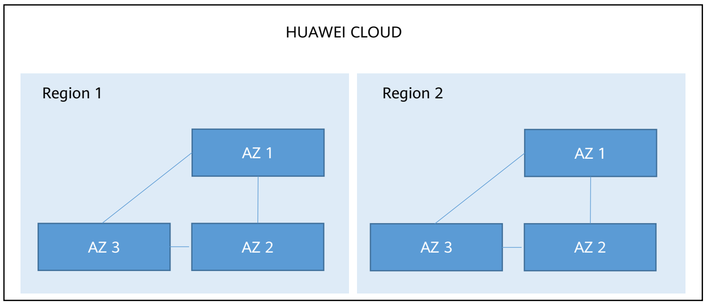
- Regions are divided based on geographical location and network latency. Public services, such as ECS, EVS, OBS, VPC, EIP, IMS, are shared within the same region.
- It is recommended that you select the closest region for lower network latency and quick access.

- A region can be regarded as a large independent data center, which is divided by geographical location. Intranets in different areas are not connected
- Huawei Cloud provides services in many regions around the world. You can select a region and an AZ based on requirements. For more information, see https://www.huaweicloud.com/intl/en-us/global/.
- Regions are classified into universal regions and dedicated regions. A universal region provides universal cloud services for common tenants. A dedicated region provides specific services for specific tenants.
- If your target users are in Asia Pacific (excluding the Chinese mainland), select the CN-Hong Kong, AP-Bangkok, or AP-Singapore region.
- If your target users are in Africa, select the AF-Johannesburg region.
- If your target users are in Latin America, select the LA-Santiago region. The **LA-Santiago** region is located in Chile.
- Resource prices may vary in different regions. .

- https://www.huaweicloud.com/intl/en-us/

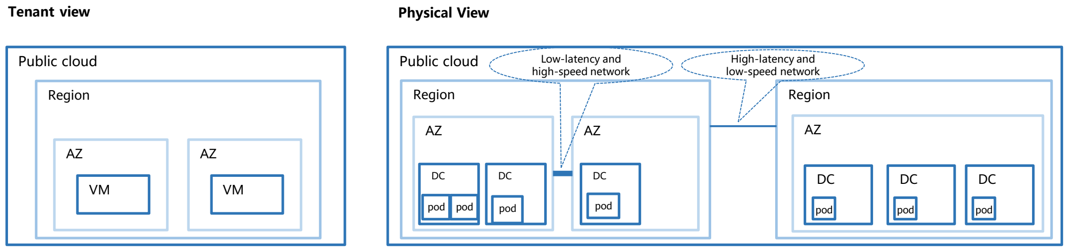
Availability Zone

- An AZ contains one or more physical data centers. Each AZ has independent cooling, fire extinguishing, moisture-proof, and electricity facilities. Within an AZ, computing, network, storage, and other resources are logically divided into multiple clusters.



- AZs within a region are interconnected using high-speed optical fibers, to support cross-AZ high-availability systems.
- When deploying resources, consider your applications' requirements on disaster recovery (DR) and network latency.
    - For high DR capability, deploy resources in different AZs within the same region.
    - For lower network latency, deploy resources in the same AZ.
- A region has multiple equipment rooms, and each equipment room is an AZ. A region can have multiple AZs, and an AZ can belong to only one region. Each AZ is independent of each other, for example, an independent network and a separate power supply system.
- In addition, AZs in each region can communicate with each other. Although each AZ has its own independent network (in the HA layer), they can communicate with each other at the network layer.

Relationship Between Regions and AZs

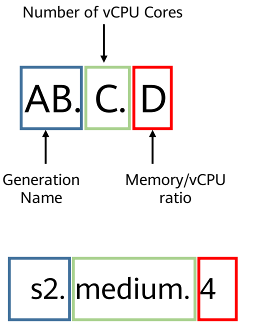


Name	target	Description
Region	Users can select low-latency cloud data centers nearby, avoiding long-distance physical transmission delay.	The network latency to the nearest region is less than or equal to 100 ms. Inter-region network latency is greater than 10 ms, and intra-region network latency is 1–10 ms.
AZ	Physically isolated resource areas. Different AZs have independent wind, fire, and water resources.	The network latency in the AZ is less than 1 ms. The storage in the AZ can be shared. The distance between AZs ranges from 30 km to 200 km.
DC	Data center with the concept of physical location; A single DC can carry one or more sites. A single DC can have one or more Layer 2 networks.	Not visible to tenants. Currently, an AZ has only one DC and supports multiple DCs. The layer 2 network latency in a DC is equivalent to that in an AZ. The layer-3 network latency in a DC is equivalent to the latency in a region.

- A DC is the infrastructure of the public cloud domain.
- POD, virtualization platform management resource pool, and independent cloud platform software instance. Invisible to tenants. A POD contains 512 servers, which will be expanded to 1024 servers in the future. If a POD is located on a Layer 2 network, the network latency is equivalent to that of the AZ.

Specifications

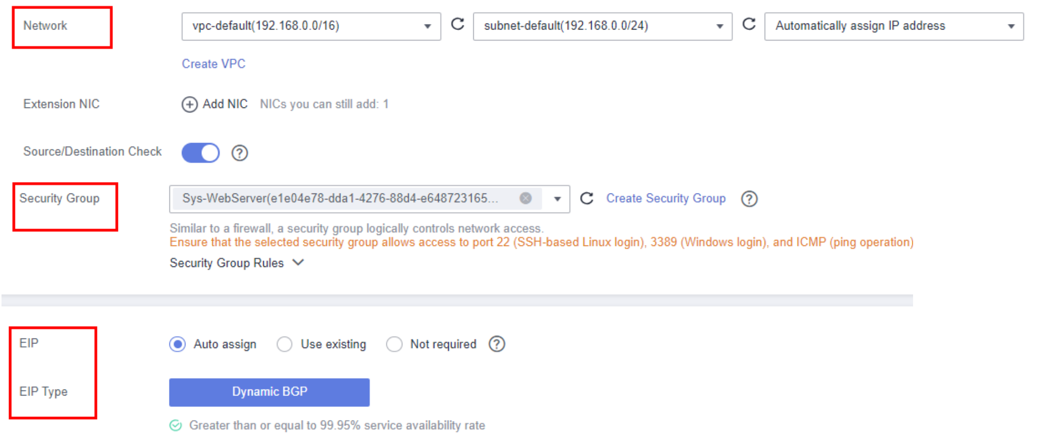
- ECS specifications refer to ECS configurations, including the CPU, memory, bandwidth, disk, and OS.
- AB indicates the ECS type and type ID.
  - A specifies the ECS type.
    - Kunpeng flavor names start with letter k. For example, kc indicates Kunpeng general computing-plus.
  - B specifies the type ID.
- C specifies the flavor size (the number of vCPUs), such as small, medium, large, xlarge, 2xlarge, 4xlarge, and 8xlarge.
- D specifies the ratio of memory to vCPUs and is expressed in a digit. For example, value 4 indicates that the ratio of memory to vCPUs is 4.



- A: For example, s indicates a general-computing ECS, c indicates a general computing-plus ECS, and m indicates a memory-optimized ECS.
- B: For example, 1 in s1 indicates the first-generation general-computing ECS, and 2 in s2 indicates the second-generation general-computing ECS. Generally, a larger number indicates a newer generation, which is more cost-effective. For example, compared with s1 and s2, s6 is more cost-effective.

Configuring Network

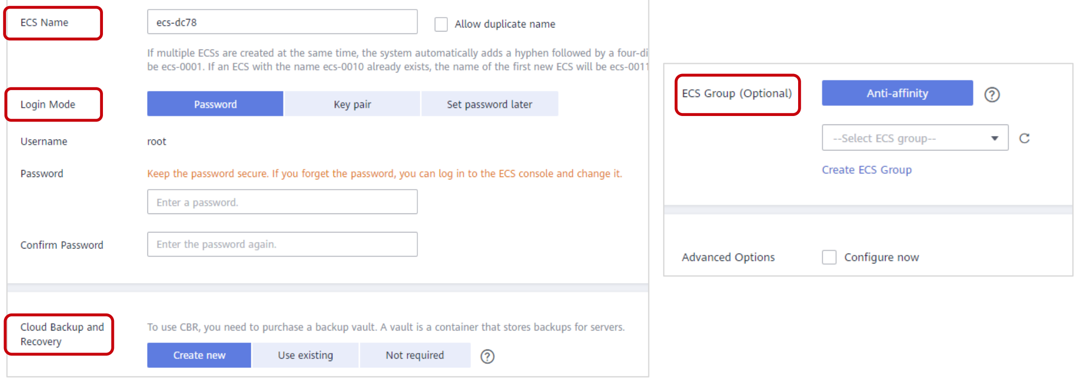
- Select a VPC, subnet, and security groups for the ECS.



- VPC: A Virtual Private Cloud (VPC) is a logically isolated virtual network. You can create subnets, configure route tables, assign EIPs, and implement access control through security groups and network ACLs.
- Subnet: Subnets are logical subdivisions in your VPC. Each subnet is a unique CIDR block with a range of IP addresses.
- Network Interfaces: A network interface is a virtual network card. You can create network interfaces and attach them to your ECSs for flexible and highly available network configurations.
- Source/destination check ensures that the ECS processes only traffic that is destined specifically for it. This function is enabled by default but should be disabled if the ECS functions as a SNAT server or has a virtual IP address bound to it.Source/destination check is not used for existing NICs. It is only used for NICs created together with the ECS.
- Create a security group with inbound and outbound rules to control traffic to and from the ECSs in the security group.
- An EIP bound to an ECS enables the ECS to access the Internet. EIP bandwidth can be modified at any time.

Configuring Advanced Settings

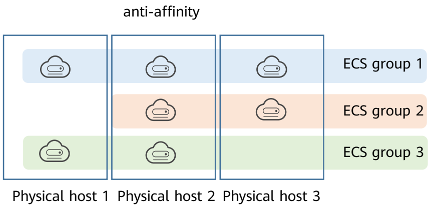
- Set ECS Name, Login Mode, Cloud Backup and Recovery, ECS Group, and Advanced Options.



- The advanced configurations of the ECS include:
  - **ECS Name**: It can be customized but must comply with the naming rules. If multiple ECSs are purchased at a time, the system automatically sequences these ECSs.
  - **Login Mode**:
    - **Key pair**: You use a key pair for login authentication.
    - **Password**: You use a username and its initial password for ECS authentication. For Linux ECSs, the initial password is the root password. For Windows ECSs, it is the Administrator password.
  - **Cloud Backup and Recovery**: With CBR, you can back up data for EVS disks and ECSs, and use backups to restore the EVS disks and ECSs if something happens.
  - **ECS Group (Optional)**: An ECS group applies the anti-affinity policy to the ECSs in it so that the ECSs are automatically allocated to different hosts.
  - **Advanced Options** is optional.

ECS group

- An ECS group allows ECSs within the group to be automatically allocated to different hosts.



- An ECS group logically groups ECSs. ECSs in an ECS group comply with the same policy associated with the ECS group.
- If an ECS is associated with an anti-affinity policy, ECSs added to the ECS group are deployed on different hosts for higher reliability.

Access Methods

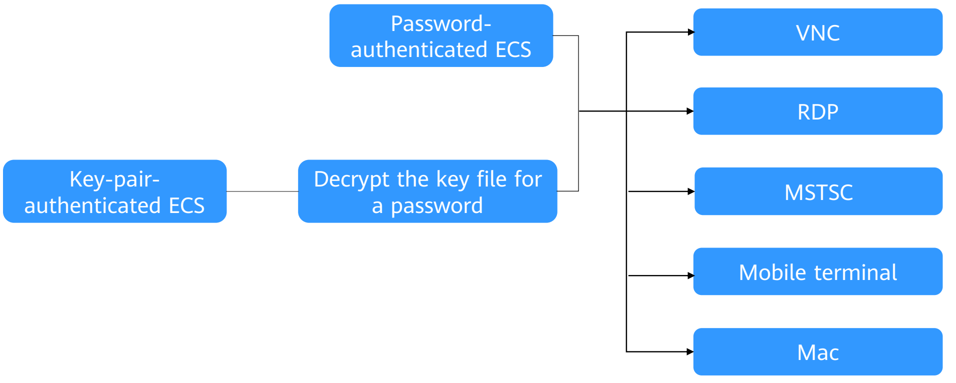
- HUAWEI CLOUD provides a web-based management platform. You can access ECSs through the management console or HTTPS-based REST APIs.

API	Management Console
Use an API if you need to integrate the ECSs into a third-party system for secondary development.	After registering on HUAWEI CLOUD, log in to the management console and click Elastic Cloud Server under Compute on the homepage.

- You can also access your ECSs using SDKs.

Logging In to a Windows ECS

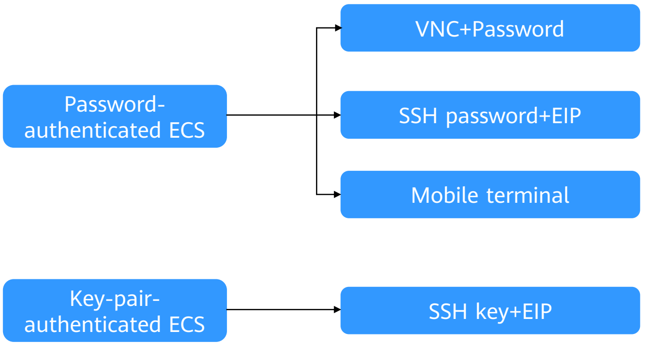
- Select a login method and log in to the ECS.



- Select a login method and log in to the Windows ECS.
  - Through the management console (VNC): The login username is Administrator.
  - Using the RDP file provided on the management console: The login username is Administrator, and the ECS must have an EIP bound.
  - Using MSTSC: The login username is Administrator, and the ECS must be bound with an EIP.
  - From a mobile terminal: The login username is Administrator, and the ECS must have an EIP bound.
  - From a Mac: The login username is Administrator, and the ECS must have an EIP bound.
- For more login methods, visit https://support.huaweicloud.com/intl/en-us/usermanual-ecs/en-us_topic_0092494943.html.

Logging In to a Linux ECS

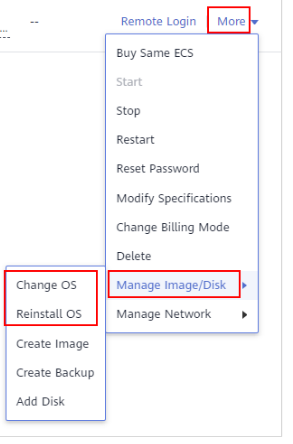
- The method of logging in to an ECS varies depending on the login authentication configured when you purchased the ECS.



- To log in to Linux ECS using a password for the first time, you can log in as root:
  - Through the management console (VNC).
  - Using an SSH password, as long as the ECS has an EIP bound.
  - From a mobile terminal, as long as the ECS has an EIP bound.

Reinstalling/Changing an ECS OS

- Scenarios: If the OS of an ECS fails to start, requires optimization, or cannot meet service requirements, reinstall or change the OS.



Notes

- Only the original image of the ECS can be used to reinstall the OS.
- Changing the OS will change the system disk of the ECS. After the change, there will be a new system disk ID, and the original system disk will be gone.

- Procedure
  - Log in to the management console.
  - Click the map icon in the upper left corner and select the desired region and project.
  - Under **Compute**, select **Elastic Cloud Server**.
  - Locate the row containing the target ECS. Click **More** in the **Operation** column and select **Manage Image/Disk** > **Reinstall OS**. Before reinstalling the OS, stop the ECS or select **Automatically stop the ECSs and then reinstall OSs**.
  - Configure the login mode. If the target ECS used key pair authentication, you can replace the original key pair.

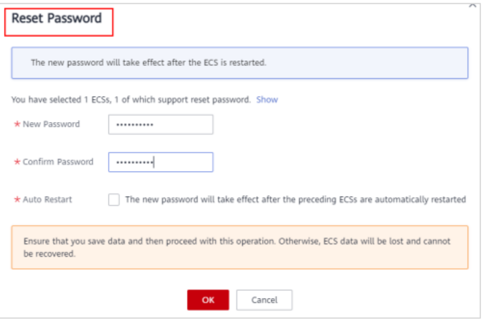
Modifying ECS Specifications

- If the specifications of an existing ECS cannot meet service requirements, modify the ECS specifications as needed, for example, by increasing the number of vCPUs or adding memory.
- Notes
  - To modify the specifications of a yearly/monthly ECS, select the target specification, pay the difference in price or claim the refund, and restart the ECS.
  - There is no need to make an additional up front payment and there are no refunds if you modify the specifications of a pay-per-use ECS.

- When changing the ECS Specification, you cannot select the CPU and memory resources that have been sold out.
- If the ECS specification (CPU or memory) decreases, the ECS performance will be affected.
- If the EVS disk status is Expanding, the ECS specifications cannot be modified.

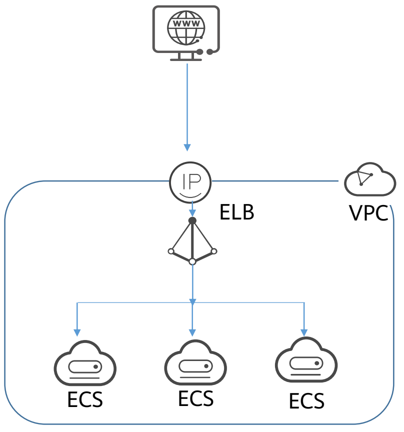
Resetting the ECS Login Password

- Scenarios: The ECS password is lost or has expired.
- Prerequisites: One-click password reset plug-ins have been installed on the ECS.
- Notes: ECSs created using a public image have the one-click password reset plug-in installed by default.



- Plug-in name: CloudResetPwdAgent and CloudResetPwdUpdateAgent.
- After installing the one-click password reset plug-ins, do not delete the CloudResetPwdAgent or CloudResetPwdUpdateAgent process, or one-click password reset will not be supported.

Scenarios – Internet



Application Scenarios

Website R&D and testing, and small-scale databases

Recommended ECS

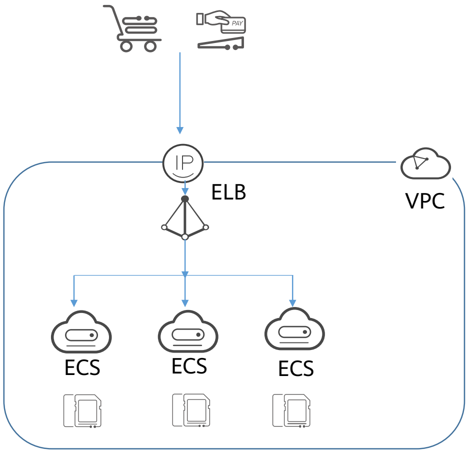
General-computing ECSs and general computing-plus ECSs

Recommendation Reasons

- Requirements: To minimize upfront deployment and O&M costs, applications need to be deployed on only one or just a few servers, but there are no special requirements for CPU performance, memory, disk capacity, or bandwidth, strong security and reliability.
- Solution: General-computing ECSs provide a balance of compute, memory, and network resources. They are appropriate for medium-workload applications and meet the cloud service needs of both enterprises and individuals.

- General-computing ECSs provide a balance of compute, memory, and network resources and a baseline level of vCPU performance with the ability to burst above the baseline. These ECSs are suitable for many applications, such as web servers, enterprise development, and small databases.
- General computing-plus ECSs use dedicated vCPUs to deliver powerful performance. In addition, the ECSs use latest-generation network acceleration engines and Data Plane Development Kit (DPDK) to provide high network performance.

Scenarios – E-Commerce



Application Scenarios

Precision marketing, E-Commerce, and mobile apps

Recommended ECS

Memory-optimized ECSs

Recommendation Reasons

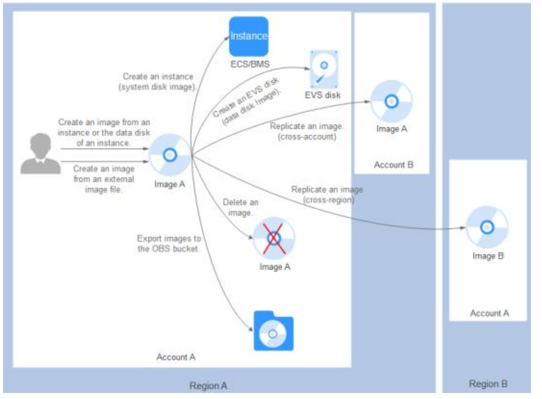
- Requirements: large amount of memory, rapid processing of large volumes of data, and fast network access
- Solution: memory-optimized ECSs, which feature a large amount of memory, ultra-high I/O EVS disks, and appropriate bandwidths

- Memory-optimized ECSs have a large memory size and provide high memory performance. They are designed for memory-intensive applications that involve a large amount of data, such as precision advertising, e-commerce big data analysis, and IoV big data analysis.
- E-commerce presents special challenges.
  - Sudden Traffic Surges: Access traffic can surge to hundreds of times normal levels during promotions, flash sales, and sweepstakes. Servers become overloaded and e-commerce platforms may even crash.
  - Poor User Experience: Massive amounts of static data, such as product pictures and videos content, is usually stored on servers, resulting in slow loading, time-consuming and costly. Users in different network environments may experience delayed access to such data, resulting in poor user experience.
  - Lack of Proper Analytics: Due to the lack of big data platforms and analysis tools, existing customers, financial products, and transaction data cannot be effectively analyzed. As a result, there are problems such as high promotion investment and low second-order rate.
  - Security: E-commerce enterprises have to deal with risks in various processes, such as traffic diversion, registration and login, browsing and comparison, preference obtaining, ordering, payment, delivery, and evaluation. The vulnerabilities may come from credential stuffing, scalpers, web page tampering, DDoS attacks, data breaches, and Trojans.

Contents

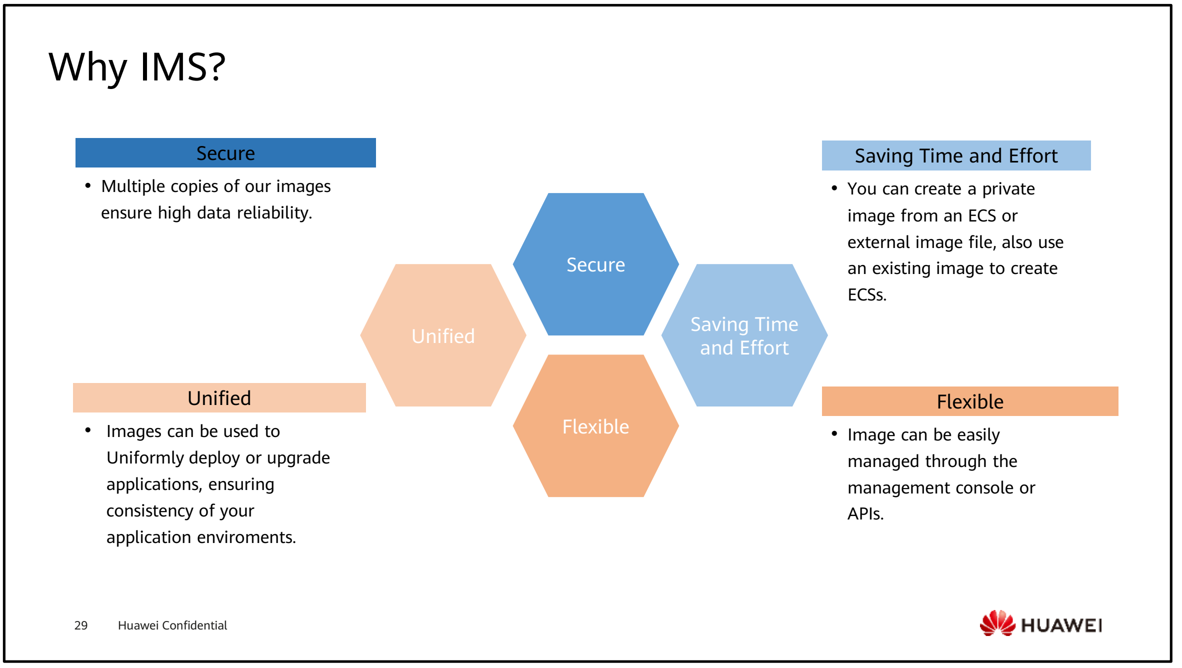
What Is IMS?

- Image Management Service (IMS) allows you to manage the entire lifecycle of your images. You can create ECSs or BMSs from public, private, or shared images. You can also create a private image from a cloud server or an external image file to make it easier to migrate workloads to the cloud or on the cloud.



HUAWEI

- An image is a server or disk template that contains an operating system (OS), service data, and necessary application software, such as database software. IMS provides public, private, Marketplace, and shared images.

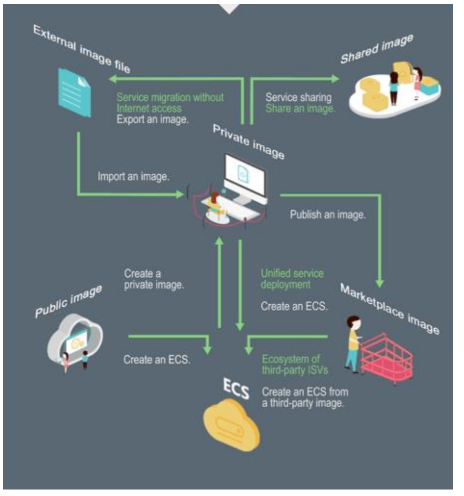


- **Saving Time and Effort**
  - Deploying services on cloud servers is much faster and easier when you use images.
  - A private image can be created from an ECS, a BMS, or an external image file. It can be a system, disk, or full-ECS image that suites your different needs.
  - Private images can be transferred between accounts, regions, or cloud platforms through image sharing, replication, and export.
- **Secure**
  - Public images use Huawei EulerOS and mainstream OSs such as Ubuntu, Windows Server, and CentOS. These OSs have been thoroughly tested to provide secure and stable services.
  - Multiple copies of image files are stored on Object Storage Service (OBS), which provides excellent data reliability and durability.
  - Private images can be encrypted for data security by using envelope encryption provided by Key Management Service (KMS).

- **Flexible**
  - You can manage images through the management console or using APIs.
  - You can use a public image to deploy a general-purpose environment, or use a private image or Marketplace image to deploy a custom environment.
  - You can use IMS to migrate servers to the cloud or on the cloud, and back up server running environments.
- **Unified**
  - IMS provides a self-service platform to simplify image management and maintenance.
  - IMS allows you to batch deploy and upgrade application systems, improving O&M efficiency and ensuring consistency.
  - Public images comply with industry standards. Preinstalled components only include clean installs, and only kernels from well-known third-party vendors are used to make it easier to transfer images from or to other cloud platforms.

Image Types

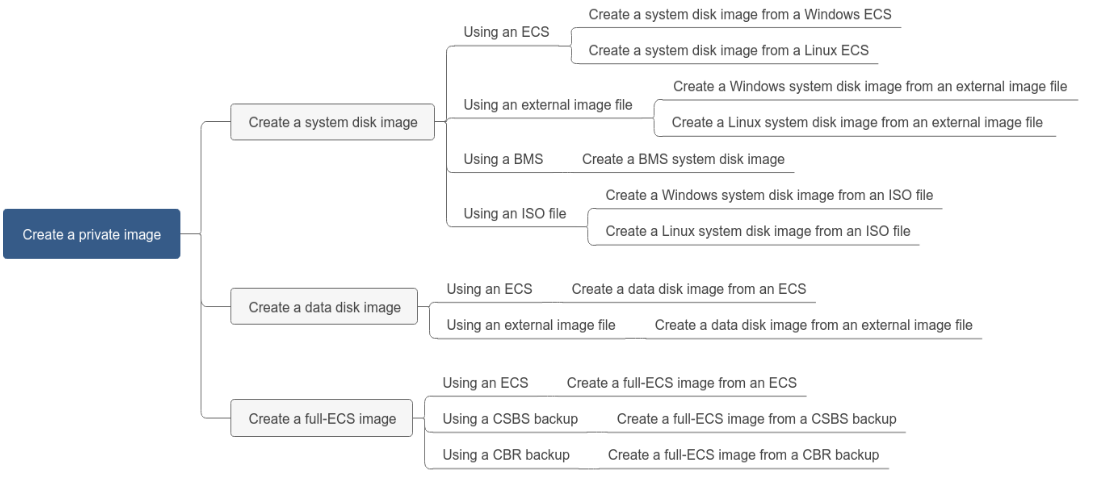
- A public image is a standard image provided by the cloud platform. It contains an OS and various preinstalled applications, and is available to all users.
- A private image is created by users and is visible only to the user who created it.
- A shared image is a private image another user has shared with you.
- A Marketplace image is a third-party image published in the Marketplace. It has an OS, various applications, and custom software preinstalled.



- Public image: A public image is a standard image provided by the cloud platform and is available to all users. It contains an OS and various preinstalled public applications. If a public image does not contain the application environment or software you need, you can use a public image to create an ECS and then install the software you need. Public images include the following OSs to choose from: Windows, CentOS, Debian, openSUSE, Fedora, Ubuntu, EulerOS, and CoreOS. When you use certain public images, the system recommends the Host Security Service (HSS) and server monitoring. HSS supports two-factor authentication for logins, defense against account cracking, and weak password detection to protect your ECSs against brute force attacks.
- Private image: A private image is only available to the user who created it. It contains an OS, service data, preinstalled public applications, and custom applications that the image creator added. A private image can be a system disk image, data disk image, or full-ECS image.
  - A system disk image contains an OS and pre-installed software for various services. You can use a system disk image to create ECSs and migrate your services to the cloud.
  - A data disk image contains only service data. You can use a data disk image to create EVS disks and use them to migrate your service data to the cloud.

    - An ISO image is created from an external ISO image file. It is a special image that is not available on the ECS console.
    - A full-ECS image contains an OS, pre-installed software, and service data.
- Shared image: A shared image is a private image another user has shared with you.
- Marketplace image: A Marketplace image is a third-party image published in the Marketplace. It has an OS, application environment, and software pre-installed. You can use these images to deploy websites and application development environments in just a few clicks. No additional configuration is required. Marketplace images are provided by service providers who have extensive experience configuring and maintaining cloud servers. All the images are thoroughly tested and have been approved by HUAWEI CLOUD before being published.

Creating a Private Image



- You can use an ECS or external image file to create an ECS private image.
- You can also:
  - Use an ISO file to create an ECS system disk image.
  - Use a CBR backup to create a full-ECS image.
  - Use a BMS to create a system disk image.

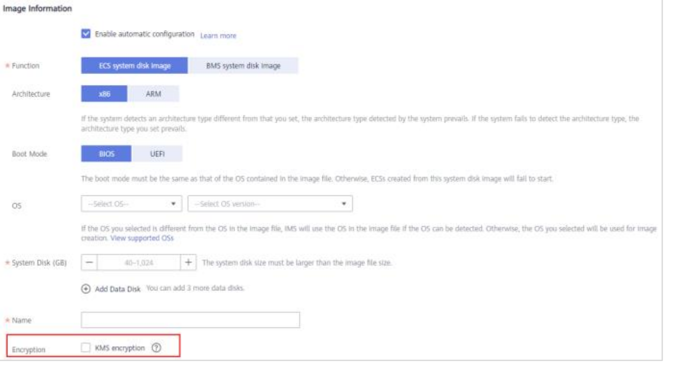
Common IMS Operations

- Modifying an Image
  - You can modify the following information of an image: name, description, minimum memory, maximum memory, NIC multi-queue, and SR-IOV driver.
- Sharing images
  - You can share your images with other tenants. The tenants can use the shared images to quickly create identical ECSs or EVS disks.
- Exporting Images
  - You can export private images to your OBS bucket and download them to your local PC for backup.
  - By exporting an image of a cloud server from the cloud platform, you can reproduce the cloud server and its running environments in on-promises clusters or private clouds. The following figure shows the process of exporting an image.
- image replication
  - In-region replication: This is used for conversion between encrypted and unencrypted images or for enabling advanced features (such as fast ECS creation) for images.
  - Cross-region replication: This is used for replicating a private image in the current region to the same account in another region. You can use this private image to deploy the same application environment in the two regions.

- You can modify the following attributes of a private image: Name/Description/Minimum Memory/Maximum Memory/NIC Multi-Queue(NIC multi-queue enables multiple CPUs to process NIC interruptions for load balancing.)/Boot Mode/SR-IOV driver(After the SR-IOV driver is installed on an image, the network processing performance of the ECS is greatly improved.)

Encrypting an Image

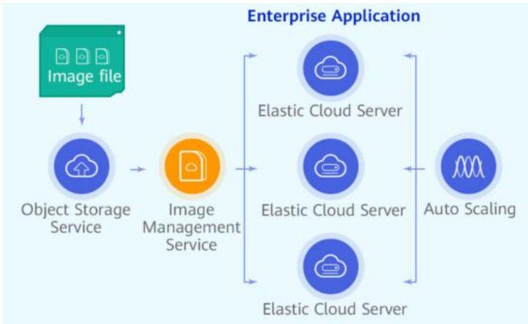
- You can create an encrypted image to securely store data.
- Encrypted images cannot be shared with other users or published in the Marketplace.
- The system disk of an ECS created from an encrypted image is also encrypted, and its key is the same as the image key.
- If an ECS has an encrypted system disk, private images created from the ECS are also encrypted.

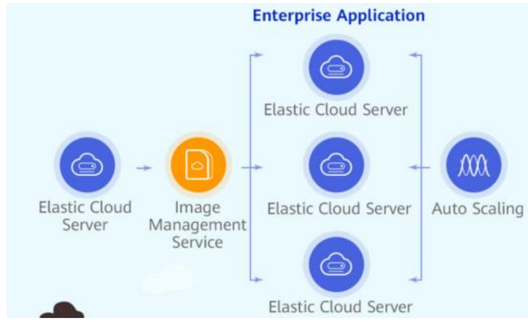


Scenarios - Migrating Servers to the Cloud or in the Cloud

Recommendation Reasons

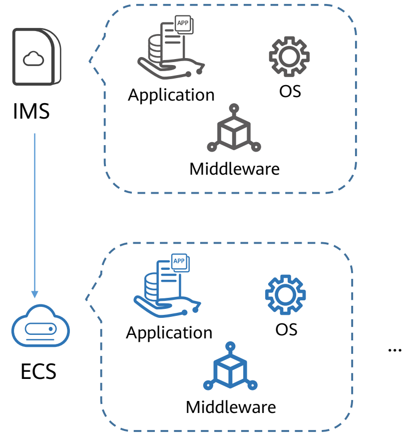
You can import local images to the cloud platform and use the images to quickly create cloud servers for service migration to the cloud. You can also share or replicate images across regions to migrate ECSs between accounts and regions.





- A variety of image formats can be imported, including VMDK, VHD, QCOW2, RAW, VHDX, QED, VDI, QCOW, ZVHD2, and ZVHD. Image files in other formats need to be converted to one of these formats before being imported. You can use the open-source tool **qemu-img** or the Huawei tool **qemu-img-hw** to convert the image.
- https://support.huaweicloud.com/intl/en-us/productdesc-ims/ims_01_0001.html

Scenarios - Deploying a Specific Software Environment



Application Scenarios

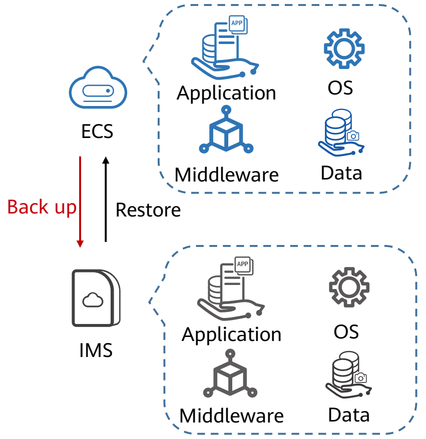
Deploying a specific software environment

Recommendation Reasons

You can use shared or Marketplace images to quickly build custom software environments without having to manually configure environments or install any software. This is especially useful for Internet startups.

- In traditional batch service deployment, you need to evaluate different service scenarios, select an OS, database, and software, and install them. The deployment quality depends on the skills of R&D and O&M personnel.
- On the cloud platform, you can quickly create ECSs by using public, private, Marketplace, or shared images. You only need to identify sources of shared images. Public, private, and Marketplace images have been thoroughly tested to ensure security and stability.

Scenarios - Backing Up Server Environments



Application Scenarios

Backing up server environments

Recommendation Reasons

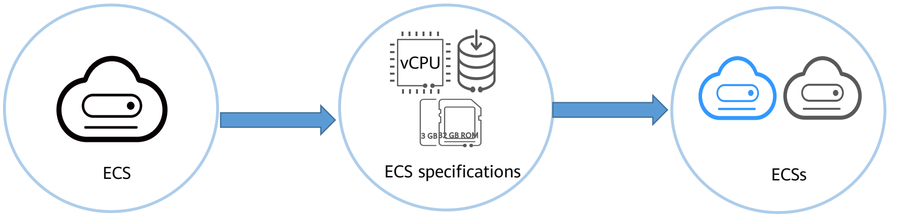
You can create an image from an ECS to back up the ECS. If the ECS breaks down for some reason, you can use the image to restore it.

- This is similar to system restoration with Ghost. You can create a Ghost recovery point for your PC. If the PC is infected with a virus or the system breaks down for some reason, you can restore it to the recovery point you created.
- On the public cloud, you can create a private image to back up an ECS. If periodic backup is required, you are advised to use cloud services such as Cloud Server Backup Service (CSBS) and Volume Backup Service (VBS) for the backup.

Contents

What Is AS?

- Auto Scaling (AS) automatically adjusts resources to keep up with changes in demand based on pre-configured AS policies. You can specify AS configurations and policies based on service requirements. These configurations and policies free you from having to repeatedly adjust resources to keep up with service changes and spikes in demand, helping you reduce the resources and manpower required.

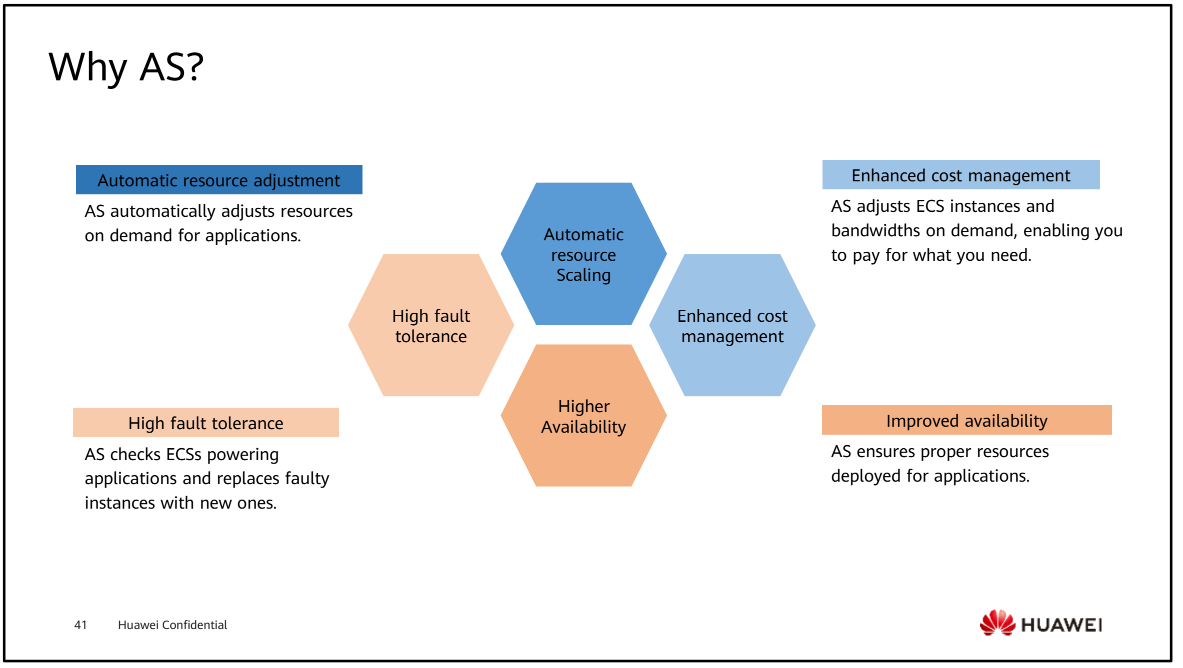


Create an AS group.

Define AS policies and configurations.

AS automatically scales ECSs.

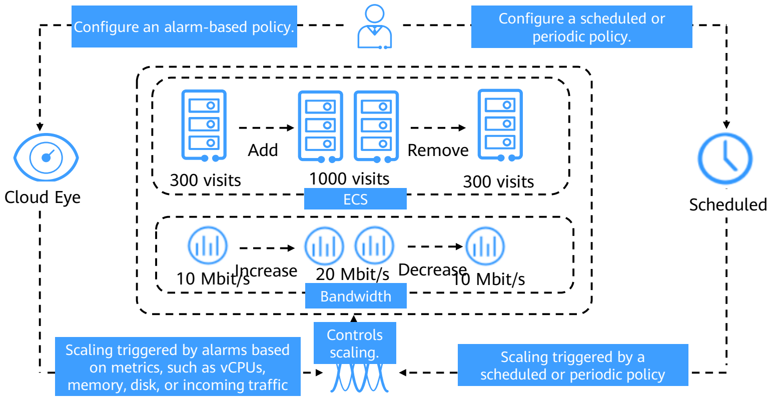
- Auto Scaling (AS) helps you automatically scale Elastic Cloud Server (ECS) and bandwidth resources to keep up with changes in demand based on pre-configured AS policies. It allows you to add ECS instances or increase bandwidths to handle load increases and also save money by removing resources that are sitting idle.



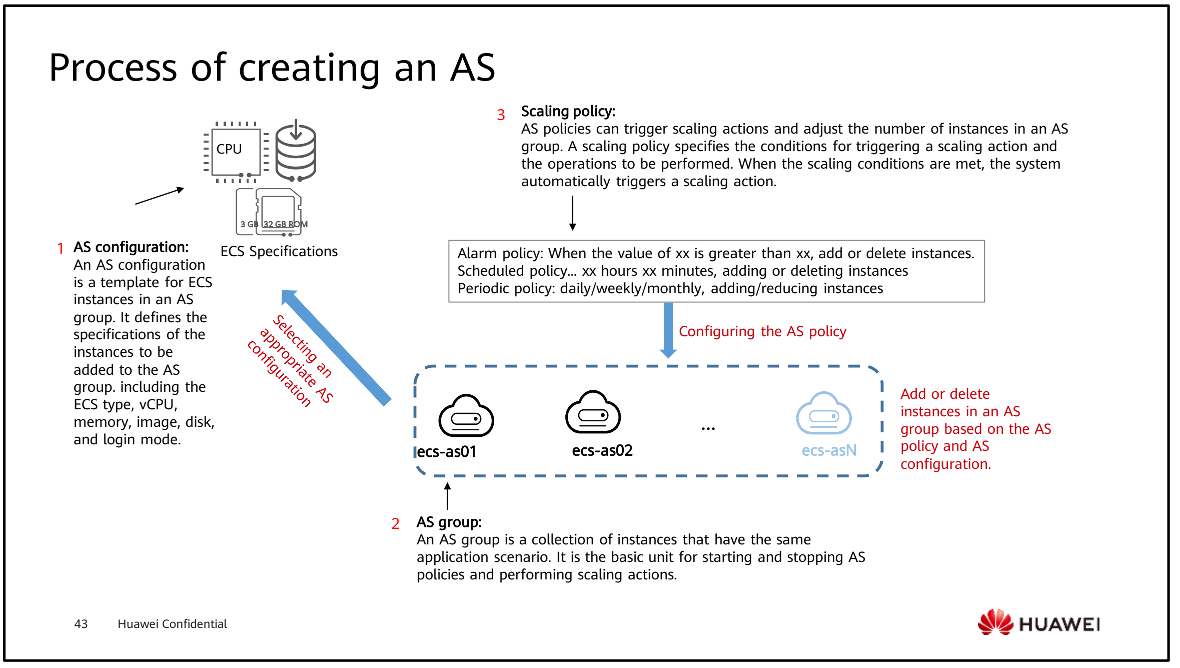
- AS advantages:
  - Automatic resource Scaling: AS adds ECS instances and increases bandwidths for your applications when the access volume increases and removes unneeded resources when the access volume drops, ensuring system stability and availability.
  - Enhanced cost management: AS enables you to use ECS instances and bandwidths on demand by automatically scaling resources for your applications, eliminating waste of resources and reducing costs.
  - Higher Availability: AS ensures that you always have the right amount of resources available to handle the fluctuating load of your applications. When working with ELB, AS automatically associates a load balancing listener with any instances newly added to the AS group. Then, ELB automatically distributes access traffic to all instances in the AS group through the listener, which improves system availability.
  - High fault tolerance: AS monitors instances in an AS group, and replaces any unhealthy instances it detects with new ones.

AS Architecture

- AS automatically adjusts compute resources based on service demands and configured AS policies. The number of ECS instances changes to match service demands, ensuring service availability.



- AS allows you to adjust the number of ECSs in an AS group and EIP bandwidths bound to the ECSs.
    - Scaling control: You can specify thresholds and schedule when different scaling actions are taken. AS will trigger scaling actions on a repeating schedule, at a specific time, or when configured thresholds are reached.
    - Policy configuration: You can configure alarm-based, scheduled, and periodic policies as needed.
    - Alarm-based: You can configure alarm metrics such as vCPU, memory, disk, and inbound traffic.
    - Scheduled: You can schedule actions to be taken at a specific time.
    - Periodic: You can configure scaling actions to be taken at scheduled intervals, a specific time, or within a particular time range.
    - When Cloud Eye generates an alarm for a monitoring metric, for example, CPU usage, AS automatically increases or decreases the number of instances in the AS group or the EIP bandwidth.
    - When the configured triggering time arrives, a scaling action is triggered to increase or decrease the number of ECS instances or the bandwidth.



- AS Basic Concepts
  - AS group: An AS group consists of a collection of instances and AS policies that have similar attributes and apply to the same scenario. It is the basis for enabling or disabling AS policies and performing scaling actions.
  - AS configuration: An AS configuration is a template specifying specifications for the instances to be added to an AS group. The specifications include the ECS type, vCPUs, memory, image, disk, and login mode.
  - AS policy: An AS policy can trigger scaling actions to adjust the number of instances in an AS group. An AS policy defines the condition to trigger a scaling action and the operations to be performed. When the triggering condition is met, the system automatically triggers a scaling action.
  - Scaling action: A scaling action adds instances to or removes instances from an AS group. It ensures that the number of instances in an application system is the same as the expected number of instances by adding or removing instances when the triggering condition is met, which improves system stability.
  - Cooldown period: To prevent an alarm policy from being repeatedly triggered for the same event, we use a cooldown period. The cooldown period specifies how long any alarm-triggered scaling action will be disallowed after a previous scaling action is complete. The cooldown period is not used for scheduled or periodic scaling actions.
  - Bandwidth scaling: AS automatically adjusts a bandwidth based on the configured bandwidth scaling policy. AS can only adjust the bandwidth of pay-per-use EIPs and shared bandwidths. It cannot adjust the bandwidth of yearly/monthly EIPs.

Creating an AS Configuration

- Configuration Template options

Create a specifications template

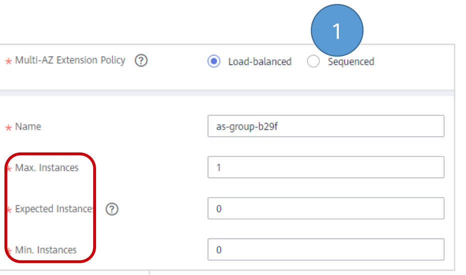
If you have special requirements on the specifications of the ECSs used for capacity expansion, specify the specifications in a template and use it to create an AS configuration. Then, the specifications will be applied to the ECSs added to the AS group in scaling actions.

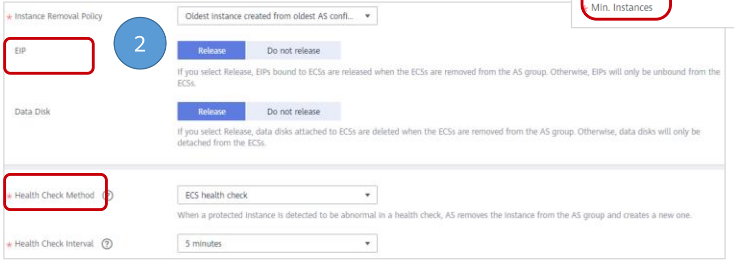
Use specifications of an existing ECS

You can use an existing ECS to quickly create an AS configuration. Then, the specifications of this ECS, such as the vCPUs, memory, image, disk, and ECS type, will be applied to ECSs added to the AS group in scaling actions.

Creating an AS Group

- An AS group consists of a collection of instances and AS policies that have similar attributes and apply to the same scenario. It is the basis for enabling or disabling AS policies and performing scaling actions.
- AS automatically scales in or out instances or maintains a fixed number of instances in an AS group through scaling actions triggered by configured AS policies.
- When creating an AS group, you need to configure parameters, such as Max. Instances, Min. Instances, Expected Instances, and Load Balancing.

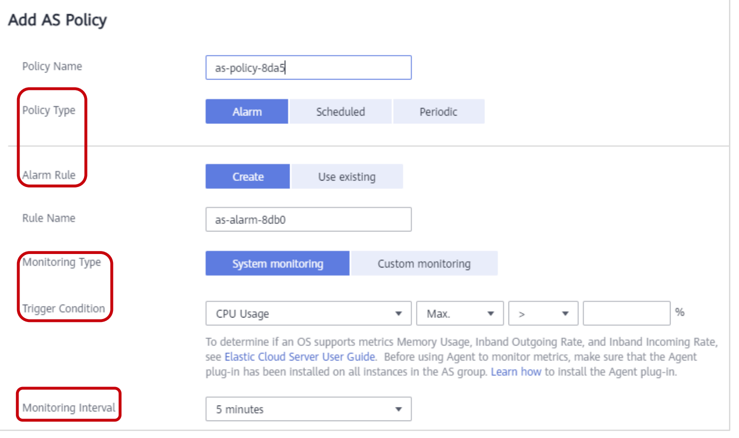




- Main parameters for creating an AS group
  - **Multi-AZ Expansion Policy**: This parameter is required only when two or more AZs are selected.
  - **Max./Min. Instances**: Specifies the minimum or maximum number of ECS instances in an AS group.
  - **Expected Instances**: Specifies the number of ECSs that are expected to run in an AS group. It is between the minimum and maximum numbers of instances. Generally, when the service peak is about to arrive, **Expected Instances** enables you to quickly provision a large number of ECS instances.
  - **Instance Removal Policy**: When instances are automatically removed from your AS group, the instances that are not in the currently used AZs will be removed first. Additionally, AS will check whether instances are evenly distributed in the currently used AZs. If the load among AZs is unbalanced, AS balances the load among AZs when removing instances. If the load among AZs is balanced, AS removes instances following the instance removal policy you configured here.

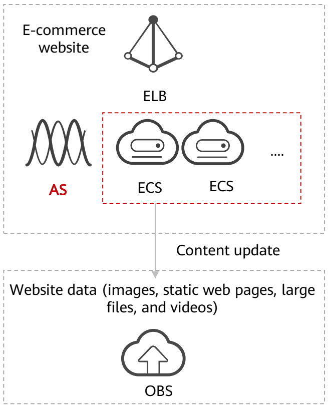
Creating an AS Policy

- Main parameters: Policy Type and Cooldown Period



- If the service workloads are unpredictable, you can configure alarm-based AS policies. These policies are used to trigger scaling actions based on real-time monitoring data (such as CPU usage) to dynamically adjust the number of instances in the AS group. AS restarts the cooldown period after a scaling action is complete. During the cooldown period, scaling actions triggered by alarms will be denied. Scheduled and periodic scaling actions are not affected.

Scenarios – Web Applications



Application Scenarios

- E-commerce websites
- Heavy-traffic web portals

Recommendation Reasons

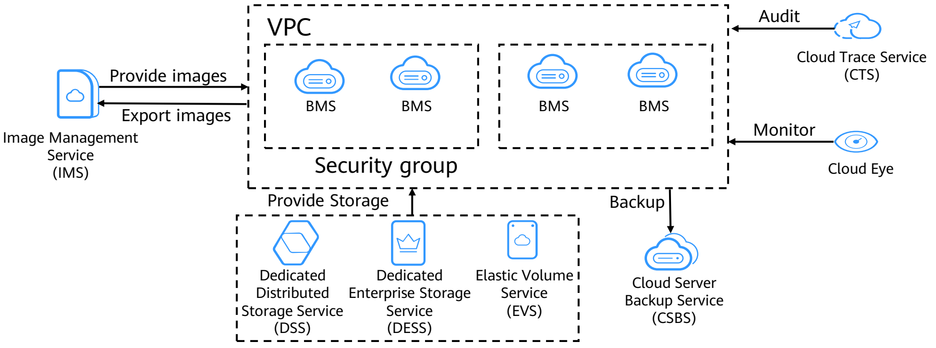
- E-commerce: During big promotions, E-commerce websites need more resources. AS automatically scales out ECS instances and bandwidth within minutes to ensure that promotions go smoothly.
- Heavy-traffic portals: Service load changes are difficult to predict for heavy-traffic web portals. AS dynamically scales in or out ECS instances based on monitored ECS metrics, such as vCPU usage and memory usage.

- Using ELB with AS
  - Working with ELB, AS automatically increases or decreases resources based on changes in demand while ensuring that the load of all the ECS instances in the AS group stays balanced.
  - After ELB is enabled in an AS group, AS automatically associates a load balancing listener with instances newly added to the AS group. Then, ELB automatically distributes access traffic to all instances in the AS group through the listener, which improves system availability. If the instances in the AS group are running a range of different types of applications, you can bind multiple load balancing listeners to the AS group to listen to each of these applications, improving scalability.

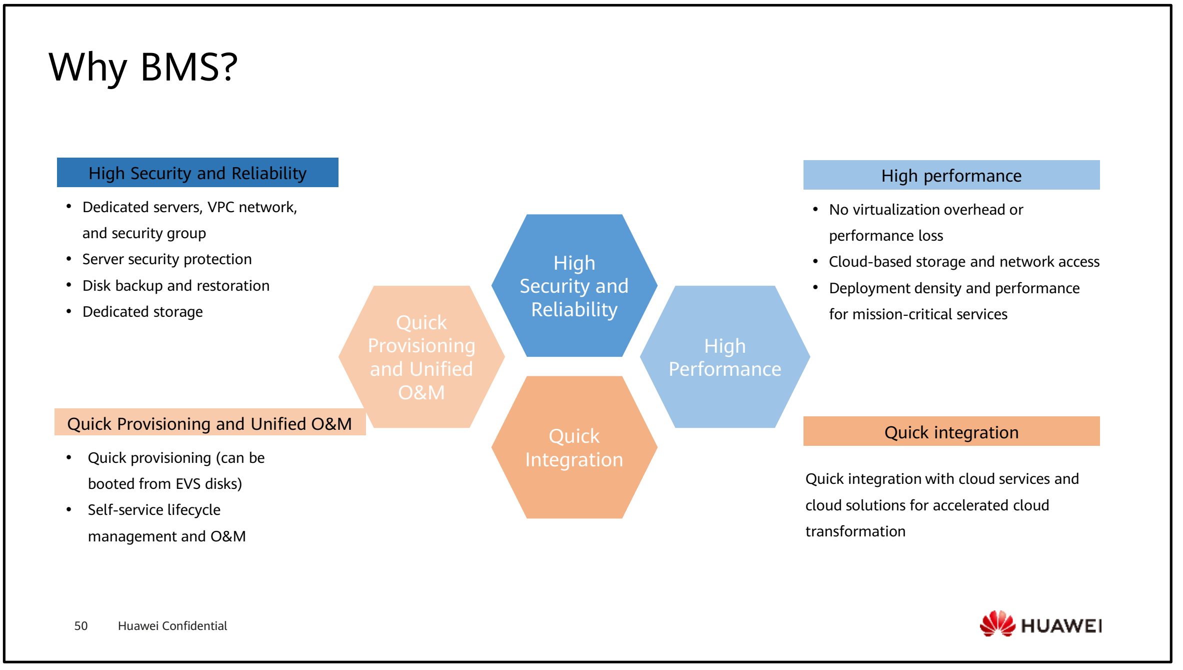
Contents

What Is BMS?

- Bare Metal Server (BMS) provides tenants with dedicated servers featuring excellent computing performance equivalent to physical servers as well as high security and reliability. You can obtain BMSs as easily and quickly as ECS and also use the service together with IMS, EVS, and VPC. The BMS service offers both the stability of traditional hosted servers and the high scalability of cloud-based services.

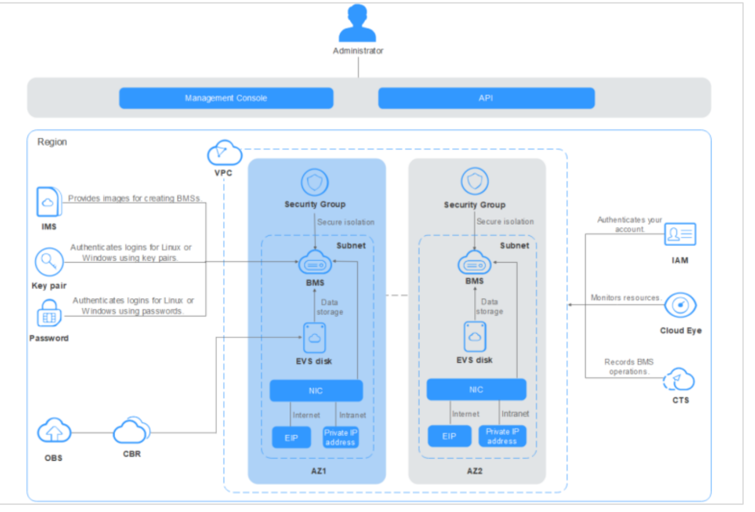


- Essentially, a BMS is a physical server. The difference is that BMSs can be easily configured and purchased on the cloud platform, but traditional physical servers can only be configured and purchased in person.
- BMSs support automatic provisioning, automatic O&M, VPC connection, and interconnection with shared storage. You can provision and use BMSs as easily as ECSs and enjoy excellent computing, storage, and network performance of physical servers.
- A Bare Metal Server (BMS) features both the scalability of Elastic Cloud Servers (ECSs) and high performance of physical servers. It provides dedicated servers on the cloud, delivering the performance and security required by core databases, critical applications, high-performance computing (HPC), and Big Data.



- Advantages of BMS:
  - High Security and Reliability: BMS allows you to use dedicated compute resources, add servers to VPCs and security groups for network isolation, and integrate related components for server security. The BMSs running on the QingTian architecture can use EVS disks, which can be backed up for restoration. BMS interconnects with Dedicated Storage Service (DSS) to ensure the data security and reliability required by enterprise services.
  - High performance: BMS has no virtualization overhead, allowing compute resources to be dedicated to running services. Running on QingTian, an architecture from Huawei that is designed with hardware-software synergy in mind, BMS supports high-bandwidth, low-latency storage and networks on the cloud, meeting the deployment density and performance requirements of mission-critical services such as enterprise databases, big data, containers, HPC, and AI.
  - Quick Provisioning and Unified O&M: Hardware-based acceleration provided by the QingTian architecture enables EVS disks to be used as system disks. The required BMSs can be provisioned within minutes after you submit an order. You can manage your BMSs through their lifecycle from the management console or using open APIs with SDKs.
  - Quick integration of cloud services and solutions: Based on the unified VPC model, cloud services and cloud solutions (such as database, big data, container, HPC, and AI solutions) can be quickly integrated to run on BMSs. This accelerates cloud transformation.

BMS Architecture



HUAWEI

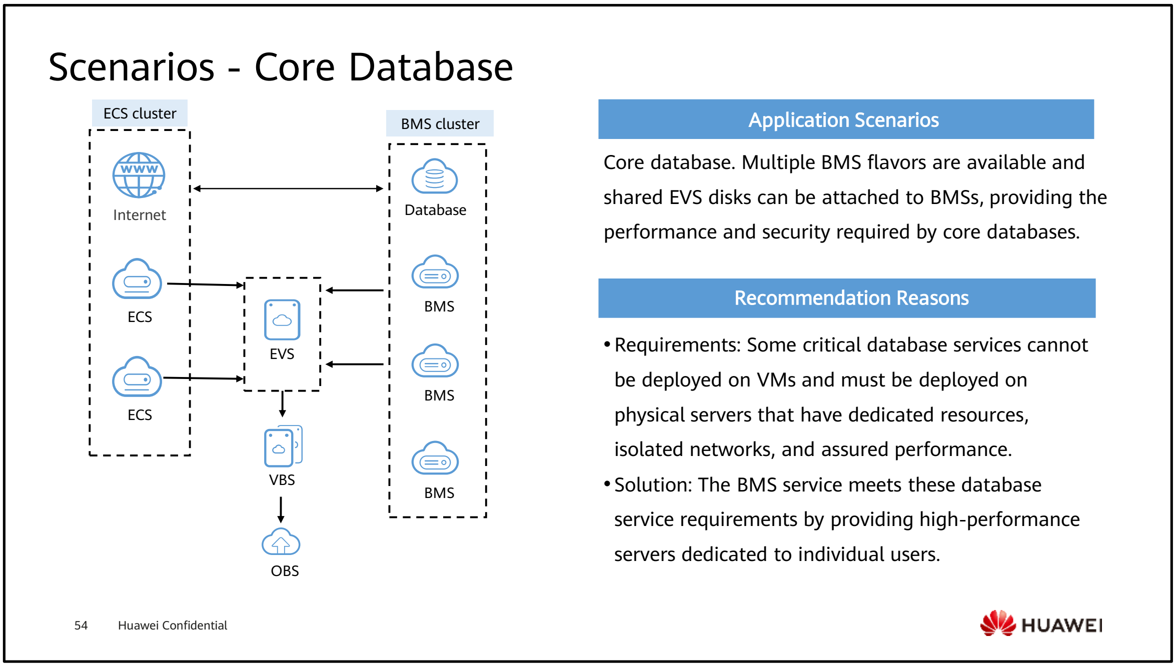
- BMS works together with other cloud services to provide compute, storage, network, and imaging.
  - BMSs are deployed in multiple availability zones (AZs) connected with each other through an internal network. If an AZ becomes faulty, other AZs in the same region will not be affected.
  - With the Virtual Private Cloud (VPC) service, you can build a dedicated network for BMS, configure subnets and security groups, and allow resources deployed in the VPC to communicate with the Internet through an EIP (with bandwidth assigned).
  - With the Image Management Service (IMS), you can install OSs on BMSs or create BMSs using private images for rapid service deployment.
  - The Elastic Volume Service (EVS) provides storage, and Volume Backup Service (VBS) provides data backup and restoration.
  - Cloud Eye is a key tool to monitor BMS performance, reliability, and availability. Using Cloud Eye, you can monitor BMS resource usage in real time.
  - Cloud Backup and Recovery (CBR) backs up data for EVS disks and BMSs, and uses snapshot backups to restore the EVS disks and BMSs when necessary.

Comparisons Between a BMS, ECS, and Physical Server

Item	BMS	ECS	Physical Server
Physical resources	Exclusive	Shared	Exclusive
Application scenarios	Mission-critical applications or services that require high performance	General-purpose and specific services	Traditional services
Provisioning	Flexible	Flexible	Inflexible
Advanced features	Automatic provisioning, automatic O&M, VPC interconnection, and interconnection with shared storage	Automatic provisioning, automatic O&M, VPC interconnection, and interconnection with shared storage	Traditional features

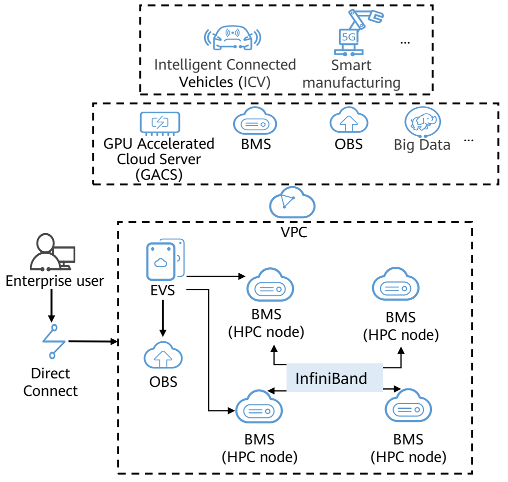
- A lack of flexibility is the main problem with physical servers. Although cloud computing is super popular right now, some enterprises may still choose physical servers for absolute best possible performance. The only reason is that physical servers do not have performance loss due to no virtualization overhead.
- However, it takes a long time to deploy physical servers, the O&M is complex, and the architecture cannot be reconstructed easily. When physical servers break down, it takes a lot of time, effort, and money to fix them.
- When Enterprises choose to avoid VMs (ECSs), it is typically because VMs are not able to provide the performance required by their core databases. Additionally, they do not want to adjust their core applications to adapt to VM deployment. These enterprises are faced with a dilemma.
- BMS is designed to address this dilemma. It provides physical servers exclusive to a particular enterprise's use, so they do not have to compromise on performance or resource isolation.
- Meanwhile, it delivers cloud capabilities such as online delivery, automatic O&M, VPC interconnection, and interconnection with shared storage. You can provision and use BMSs as easily as ECSs and enjoy excellent computing, storage, and network performance of physical servers.

- BMS can also offer services that ECSs cannot provide due to various architecture restrictions, such as virtualization services, high-performance computing services, services that have high requirements on I/O performance, and services that have high requirements on core data control and resource isolation. In addition, HUAWEI CLOUD provides O&M for BMSs, which helps keep your costs down.



- Cluster Deployment: An EVS disk can be attached to multiple BMSs, enabling cluster-based application deployment.
- Large Capacity: A BMS can have multiple EVS disks attached, each as large as 32 TB. EVS disk capacity can be expanded as needed and you only pay for what you use.
- High Reliability: Three-copy backup ensures high data durability.

Scenarios - High Performance Computing (HPC)



Application Scenarios

Supercomputing centers and DNA sequencing. For high performance and high throughput scenarios, BMSs with the latest CPUs, coupled with a 100 Gbit/s network, provide low latency and high performance services.

Recommendation Reasons

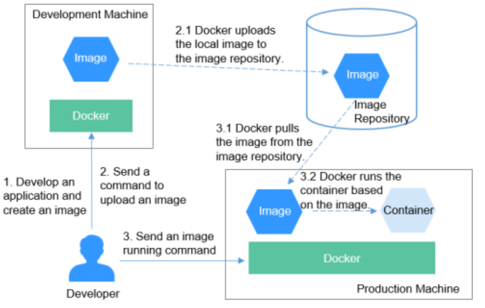
- Requirements: In HPC scenarios, such as supercomputer centers and DNA sequencing, massive volumes of data need to be processed and the computing performance, stability, and real-time responsiveness need to be stellar.
- Solution: HPC node (BMS)
  - Low latency: 100 Gbit/s, isolated, microsecond-level latency network
  - High performance: the latest Intel CPUs
  - Convenient scale-up: open APIs for easy ecosystem integration

- High-performance ECS: Compute-intensive ECSs, such as general computing-plus (C6) and memory-optimized (M6) ECSs, use 2nd Gen Intel® Xeon® scalable processors to provide robust, stable computing performance, and Huawei-developed intelligent high-speed NICs to provide networks with ultra-high bandwidth and ultra-low latency.
- High-performance BMS: High-performance computing (H2) BMSs with 100 Git/s EDR InfiniBand NICs provide excellent computing performance with no virtualization overhead. You can apply for BMSs on demand through the management console.
- Excellent network performance: Secure, isolated virtual networks are provided for HPC users on the public cloud. The networks communicate with each other through intelligent high-speed NICs that deliver excellent bandwidth.
- Other application scenarios:
  - Database scenario: Key database services for government, enterprise, and financial services must be carried on physical servers with dedicated resources, isolated networks, and guaranteed performance. The bare metal server provides dedicated high-performance physical servers to meet service requirements.
  - Container scenario: Internet elastic service load. Compared with VMs, BMS containers provide higher deployment density, lower resource overhead, and more agile deployment efficiency. Cloud native technologies help customers achieve the goal of reducing cloudification costs.

Contents

Docker and Kubernetes

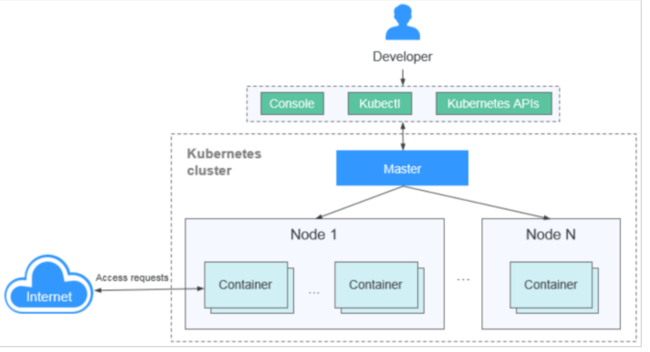
- Container is a lightweight virtualization technology. It can pack applications and their dependencies together to form an independent running environment, implementing quick deployment and migration of applications.
- The core of the container technology is the namespace and CGroups functions of the Linux kernel. The namespace and CGroups functions can isolate different processes, file systems, networks, and resources, thereby implementing application isolation and secure running.
- Docker is the first system that allows containers to be portable in different machines. It simplifies both the application packaging and the application library and dependency packaging.
- Kubernetes is a containerized application software system that can be easily deployed and managed. It facilitates container scheduling and orchestration.



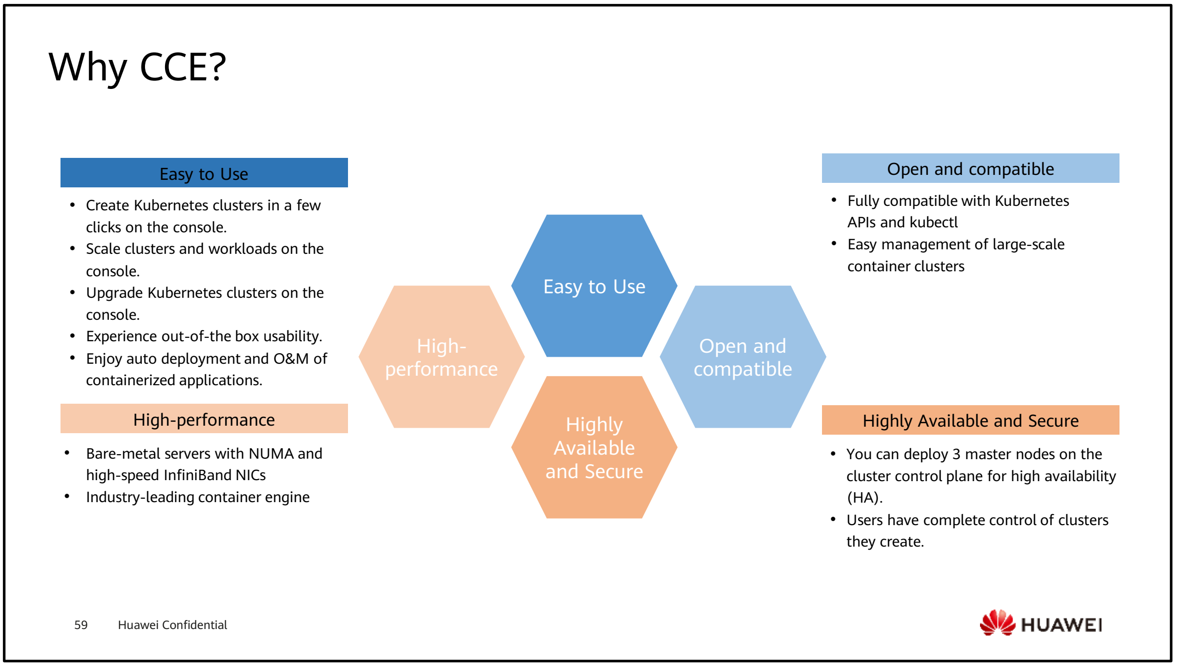
- Containers have become popular since the emergence of Docker.
- Even the OS file system can be packaged into a simple portable package, which can be used on any other machine that runs Docker.
- Docker is developed and implemented by using the Go language launched by Google. Docker encapsulates and isolates processes based on technologies such as cgroup and namespace of the Linux kernel and Union FS of the AUFS type. Docker is an operating system virtualization technology. Because an isolated process is independent of the host and other isolated processes, it is also called a container.
- Docker further encapsulates the file system, network interconnection, and process isolation based on containers, greatly simplifying container creation and maintenance.
- The "8" in the Kubernetes abbreviation K8S represents the eight letters of the word "ubernete".
- For application developers, Kubernetes can be regarded as a cluster operating system. Kubernetes provides functions such as service discovery, scaling, load balancing, self-healing, and even leader election, freeing developers from infrastructure-related configurations.When using Kubernetes, it's like you run a large number of servers as one on which your applications run. Regardless of the number of servers in a Kubernetes cluster, the method for deploying applications in Kubernetes is always the same.
- A Kubernetes cluster consists of master nodes (Masters) and worker nodes (Nodes). Applications are deployed on worker nodes, and you can specify the nodes for deployment.

What Is CCE?

- Cloud Container Engine (CCE) is a highly scalable, high-performance, enterprise-class Kubernetes service for you to run containers and applications. With CCE, you can easily deploy, manage, and scale containerized applications on HUAWEI CLOUD.



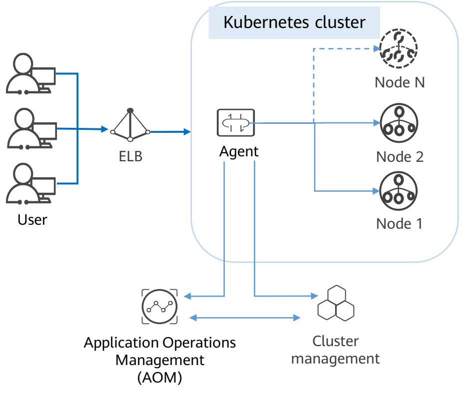
- CCE is a one-stop platform integrating compute, networking, storage, and many other services. It supports heterogeneous computing architectures such as GPU, NPU, and Arm. Supporting multi-AZ and multi-region disaster recovery, CCE ensures high availability of Kubernetes clusters.



- Easy to Use:
  - Creating a Kubernetes cluster is as easy as a few clicks on the console. You can create either VM nodes or bare-metal nodes, or both, in a cluster.
  - From auto deployment to O&M, you can manage your containerized applications all in one place throughout their lifecycle.
  - You can also scale your clusters and workloads in just a few clicks on the console. Auto scaling policies can be flexibly combined to deal with in-the-moment load spikes.
  - The console enables you to easily upgrade your clusters.
  - Application Service Mesh (ASM) and Helm charts are pre-integrated, delivering out-of-the-box usability.
- High-performance:
  - CCE draws on Huawei's years of field experience in computing, network, storage, and heterogeneous infrastructure. You can concurrently launch containers at scale.
  - The bare-metal NUMA architecture and high-speed InfiniBand network cards yield a three- to five-fold improvement in computing performance.

- Highly Available and Secure:
  - You can create 3 master nodes for your cluster control plane to avoid single points of failure. Faults in one or two of the master nodes do not interrupt the whole cluster. CCE allows you to deploy nodes and workloads in a cluster across AZs. Such a multi-active architecture ensures service continuity against host faults, data center outages, and natural disasters.
  - Clusters are private and completely controlled by users. With deeply integrated Kubernetes RBAC capabilities, CCE allows you to set different RBAC permissions for sub-users on the console.
- Open and compatible
  - CCE streamlines deployment, resource scheduling, service discovery, and dynamic scaling of applications that run in Docker containers.
  - CCE is built on Kubernetes and compatible with Kubernetes native APIs and kubectl (a command line tool). CCE provides full support for the most recent Kubernetes and Docker releases.

Scenario - Auto Scaling in Second



Function Description

CCE adjusts compute resources based on auto scaling policies to handle fluctuating service loads. Specifically, CCE automatically adds or reduces cloud servers for your cluster or containers for your workload.

Benefits

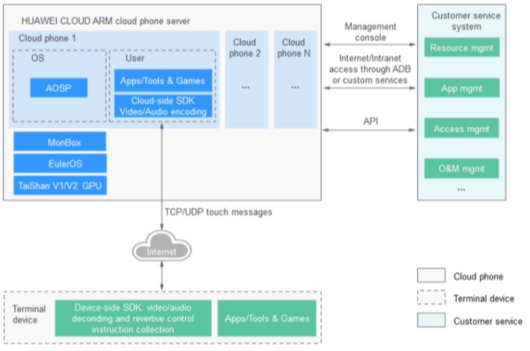
- Flexible: Multiple scaling policies are supported and containers can be provisioned within seconds when specific conditions are met.
- Highly available: Pods are automatically monitored and unhealthy pods will be replaced with new ones to ensure high service availability.
- Low cost: You are billed only for the cloud servers you use.

- Application scenarios:
  - Shopping apps and websites, especially during promotions and flash sales
  - Live streaming, where service loads often fluctuate
  - Games, where many players may go online in certain time periods

Contents

Cloud Phone Service (CPH)

- Cloud Phone Host (CPH) provides you with cloud servers virtualized from Huawei Cloud BMSs and running native Android. Just one of these cloud servers can virtualize up to 60 cloud phones with the functions of virtual phones. You can remotely control cloud phones in real time and run Android applications on the cloud. Cloud phone compute lets you build and test phone applications more efficiently.



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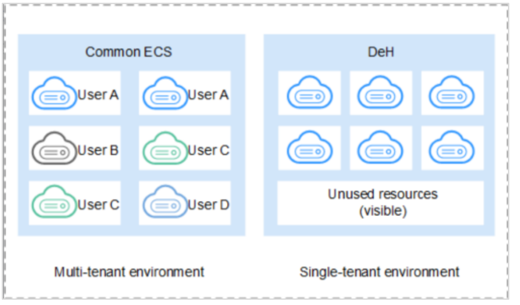
- **Scenarios**
  - Cloud gaming is a popular trend of the game industry. It provides players with a download-free game service that is independent of mobile phone performance. The video streaming modes it uses include PC game streaming and mobile game streaming. As a cloud-based emulation phone, a cloud phone server can take the advantage of instruction isomorphism of mobile games and carry game applications on the cloud.
  - With the popularization of mobile apps, more and more enterprises are starting to allow work from mobile terminals. However, they are faced with the challenge of data security. Although purchasing customized secure mobile phones can enhance security, leakage of sensitive data cannot be prevented. As an alternative solution, cloud phones store core enterprise data on the cloud and control access to mobile phone screens only within authorized employees.
  - Generally, mobile phones provide services for individuals. As the type and number of mobile applications increase, enterprises may need to run a large number of mobile applications on mobile phones in specific scenarios to implement automation or intelligence functions. To run these applications, a large number of simulation mobile phones are needed.
  - Streaming interaction is another CPH scenario. It allows the host to stream the mobile phone screen to audiences and interact with them to bring an enjoyable interaction experience.

What Is DeH?

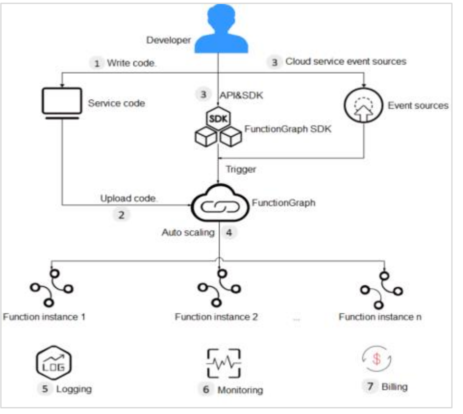
- Dedicated Host (DeH) provides dedicated physical hosts to ensure isolation, security, and performance for your ECSs. You can bring your own license (BYOL) to DeH to reduce the costs on software licenses and facilitate the independent management of ECSs..



- Application scenarios:
  - Industries that have high requirements for regulation compliance and security: You can exclusively use a physically isolated host to meet compliance and security requirements.
  - Tenants that need to use their existing licenses (BYOL): If you have a licensed OS or software (licensed based on the number of physical sockets or cores), you can bring your own license and migrate your services to the cloud platform.
  - Industries that are extremely sensitive to performance and stability: DeH is ideal for service scenarios with higher requirements on server performance and stability such as finance, securities and gaming applications. DeH guarantees the stability of CPUs and network I/O, ensuring smooth running of applications.
  - Independent resource deployment and flexible management: You can create ECSs on a specified DeH and specify your ECS specifications based on the type of DeH. You can also migrate ECSs between DeHs or migrate ECSs from public resource pool to a specified DeH.

What Is FunctionGraph?

- FunctionGraph allows you to run your code without provisioning or managing servers, while ensuring high availability and scalability. All you need to do is upload your code and set execution conditions, and FunctionGraph will take care of the rest. You pay only for what you use and you are not charged when your code is not running.



- FunctionGraph is designed for real-time file and data stream processing, web and mobile app backends, and artificial intelligence (AI) applications.
  - FunctionGraph processes files in real time by triggering a function once a client uploads a file to OBS. Functions can generate image thumbnails, convert video formats, and aggregate and filter data files.
  - FunctionGraph also works with Data Ingestion Service (DIS) to process data streams in real time. It supports application activity tracking, sequential transaction processing, data stream analysis, data sorting, metric generation, log filtering, indexing, social media analysis, and IoT device data telemetry and metering.
  - FunctionGraph also interconnects with your VMs or other services to build highly available and scalable web and mobile app backends.
  - Finally FunctionGraph also works with Enterprise Intelligence (EI) services for text recognition and illicit image identification. For example, build a function to identify pornographic and terrorism-related images.

Quiz

1. (True or False) There is a hypervisor layer in containerization, just like the traditional virtualization featuring VMs.
   A. True
   B. False
2. (True or False) The functions of an IMS image are the same as those of an ISO image.
   A. True
   B. False

- 1.B False. Containerization has no virtualization layer.
- 2.B False. An ISO image is used to install an OS. An IMS image is more like a template that is generated after an ISO image is modified. It is mainly used to batch create cloud servers instead of just installing cloud server OSs.

Summary

After reviewing this chapter, we have learned about the features, usage methods, and application scenarios of Elastic Cloud Server (ECS), Auto Scaling (AS), Image Management (IMS), bare metal server, and Cloud Container Engine (CCE). Based on the knowledge in this chapter and the lab manual, you can use HUAWEI CLOUD to deploy your own compute instances and services.

Recommendations

- Huawei Talent
  - https://e.huawei.com/en/talent/cert/#/careerCert
- Huawei Technical Support Website
  - https://support.huaweicloud.com/intl/en-us/help-novicedocument.html
- HUAWEI CLOUD Academy
  - https://edu.huaweicloud.com/intl/en-us/

Acronyms and Abbreviations

- AI: Artificial intelligence
- API: Application Programming Interface
- AS: Auto Scaling
- BMS: Bare Metal Server
- CCE: Cloud Container Engine
- CI/CD: Continuous Integration/Continuous Delivery
- CISC: Complex Instruction Set Computer
- CPH: Cloud Phone
- CPU: Central Processing Unit
- DeH: Dedicated Host

Acronyms and Abbreviations

- DevOps: Development and Operations
- DHCP: Dynamic Host Configuration Protocol
- ECS: Elastic Cloud Server
- EI: Enterprise Intelligence
- GPU: Graphics Processing Unit
- HPC: High Performance Computing
- HTTPS: Hypertext Transfer Protocol over Secure Sockets Layer
- IB: InfiniBand
- IMS: Image Management Service
- K8s: Kubernetes

Acronyms and Abbreviations

- IPoIB: Internet Protocol over Infiniband
- NUMA: Non-Uniform Memory Access
- RDMA: Remote Direct Memory Access
- RISC: Reduced Instruction Set Computer
- SR-IOV: Single Root Input/Output Virtualization
- VLAN: Virtual Local Area Network
- VPC: Virtual Private Cloud

Thank Users.

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每个组织，构建万物互联的智能世界。

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Network Cloud Services

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Foreword

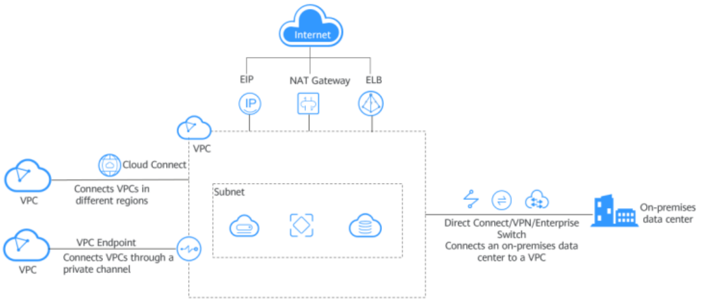
- Network resources are essential to the development of the ICT infrastructure. With network resources, devices and systems can communicate with each other so that enterprises can provide better services to their end users.
- This chapter describes the network services provided by HUAWEI CLOUD.

Objectives

- On completion of this course, you will be able to:
  - Understand what network services are and the application scenarios of different services.
  - Understand the concepts and usage of VPC.
  - Understand the concepts and usage of EIP.
  - Understand the concepts and usage of VPN and NAT gateway.
  - Learn about other Huawei cloud network services.

Network Services

- A network is a system that connects multiple computers or other devices together so that they can communicate and share resources with each other. A network can be a different type, such as a local area network (LAN), a wide area network (WAN), or the Internet.
- Huawei Cloud provides various network services to help you build secure and scalable networks on the cloud, connect cloud and on-premises networks in a high-speed and reliable way, and connect your on-premises data center to the Internet.



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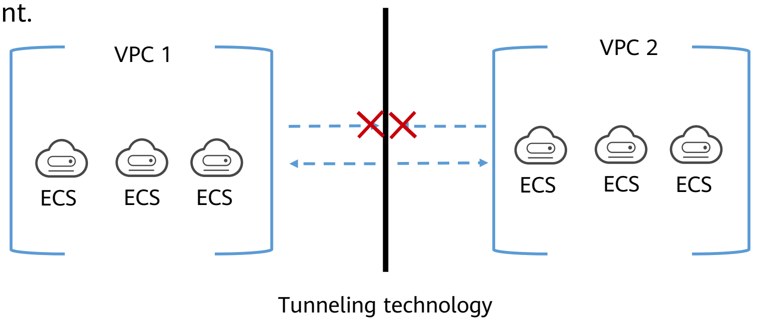
- VPC provides an isolated network environment on HUAWEI CLOUD.
- VPCEP provides secure access to cloud services and private services hosted on HUAWEI CLOUD.
- ELB automatically distributes incoming traffic across multiple backend servers.
- NAT Gateway provides network address translation (NAT) for cloud servers.
- EIP provides independent public IP addresses for accessing the Internet.
- Direct Connect establishes a dedicated channel between an on-premises data center and the cloud.
- VPN establishes an IPsec encrypted channel between an on-premises data center and the cloud.
- CC connects VPCs in multiple regions and allows one or more on-premises data centers to access multiple VPCs.
- DNS provides authoritative DNS services and domain name management services.

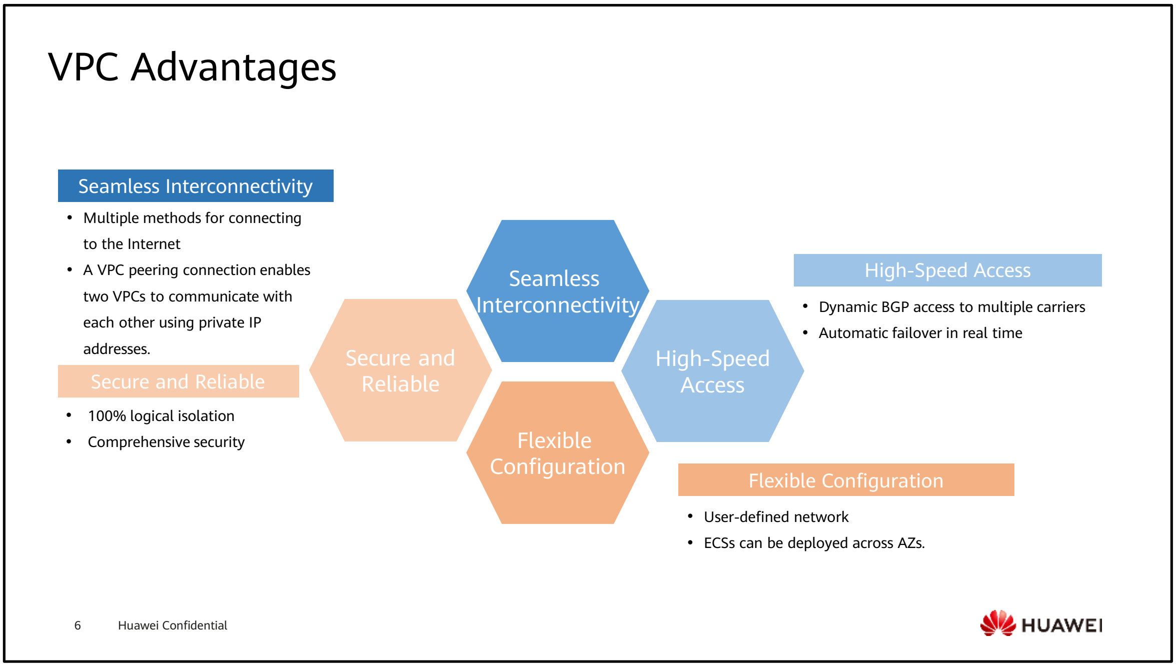
Contents

What Is a VPC?

- The Virtual Private Cloud (VPC) service enables you to provision logically isolated, configurable, and manageable virtual networks for cloud servers, cloud containers, and cloud databases, improving cloud service security and simplifying network deployment.

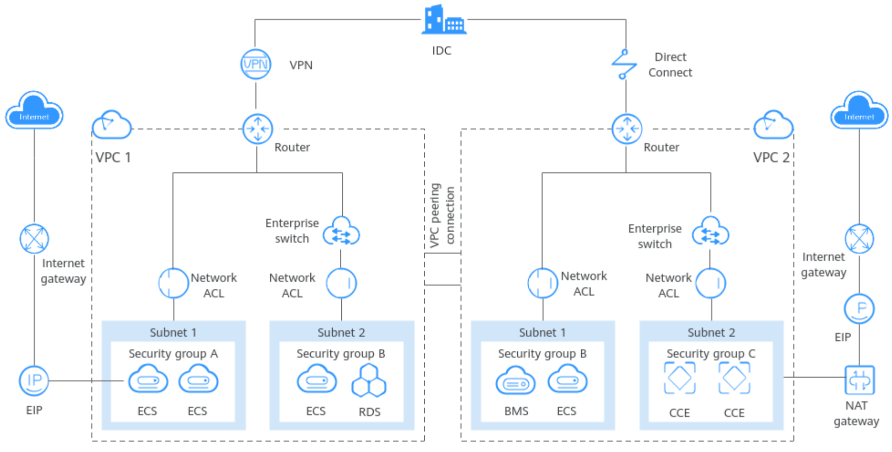


- A Virtual Private Cloud (VPC) is a logically isolated virtual network. Within your own VPC, you can create subnets, configure route tables, assign EIPs and bandwidths, and configure security groups to manage access control.
- VPC is the basis of HUAWEI CLOUD networks. VPC provides secure and isolated networks based on tunneling technology. You can customize your own VPCs, including dividing subnets, configuring route tables, specifying IP addresses, and configuring network ACLs and security groups.



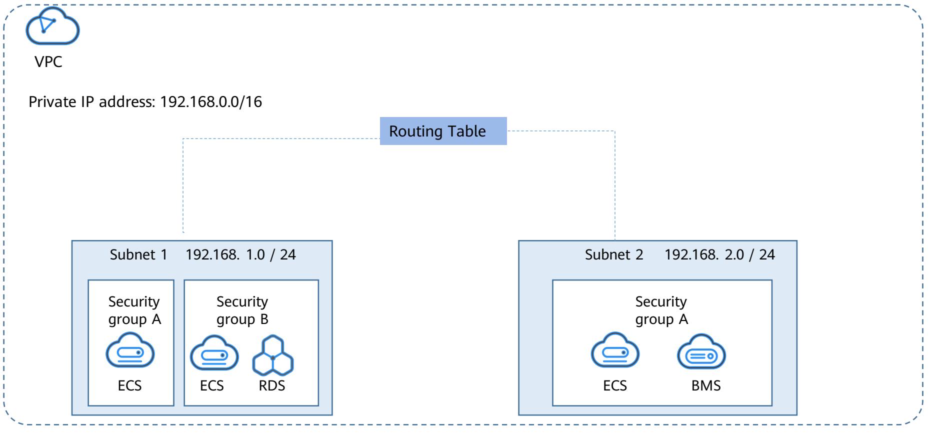
- A VPC has many advantages.
  - You can create VPCs, add subnets, specify IP address ranges, and configure DHCP and route tables. You can configure the same VPC for ECSs that are in different availability zones (AZs).
  - Secure and Reliable: VPCs are logically isolated from each other. By default, different VPCs cannot communicate with each other. Network ACLs protect subnets, and security groups protect ECSs.
  - Seamless Interconnectivity: By default, a VPC cannot communicate with the Internet. You can use EIP, ELB, NAT Gateway, VPN, and Direct Connect to enable access to or from the Internet. By default, two VPCs in the same region cannot communicate with each other. You can create a VPC peering connection to enable them to communicate with each other using private IP addresses.
  - High-speed access: Up to 21 dynamic BGP connections are established to multiple carriers. Dynamic BGP provides automatic failover in real time and chooses the optimal path when a network connection fails.

VPC Architecture

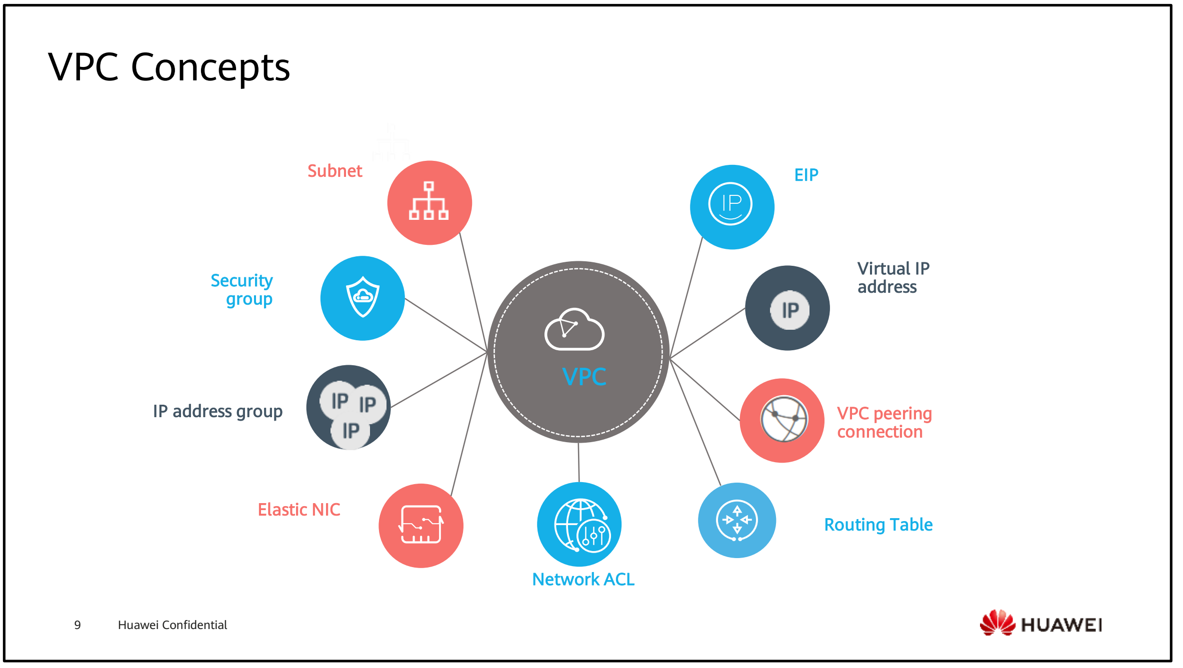


VPC Components

- Each VPC consists of a private CIDR block, route tables, and at least one subnet.



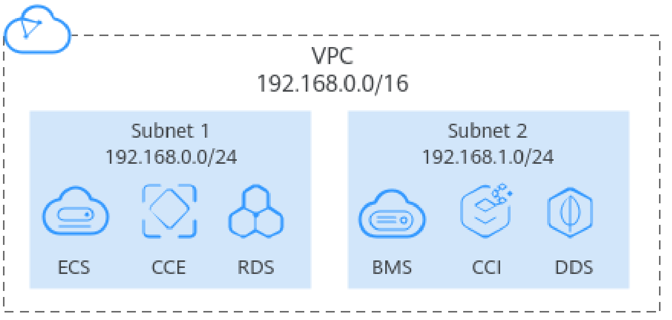
- Private CIDR blocks: When creating a VPC, you need to specify the private CIDR block used by the VPC. The VPC service supports the following CIDR blocks: 10.0.0.0 – 10.255.255.255, 172.16.0.0 – 172.31.255.255, and 192.168.0.0 – 192.168.255.255
- Subnets: Cloud resources, such as cloud servers and databases, must be deployed in subnets. After you create a VPC, you can divide the VPC into one or more subnets. Each subnet must be within the VPC.
- Route tables: When you create a VPC, the system automatically generates a default route table. The route table ensures that all subnets in the VPC can communicate with each other. If the routes in the default route table cannot meet application requirements (for example, if there is an ECS without an elastic IP address (EIP) bound that needs to access the Internet), you can create a custom route table.



- An elastic network interface (NIC) is a virtual network card. You can create and configure network interfaces and attach them to your instances (ECSs and BMSs) to create flexible and high availability network configurations.
- An IP address group is a collection of IP addresses that use the same security group rules. You can use an IP address group to manage IP addresses that have the same security requirements or whose security requirements change frequently. An IP address group frees you from repeatedly modifying security group rules and simplifies security group rule management.

Subnet

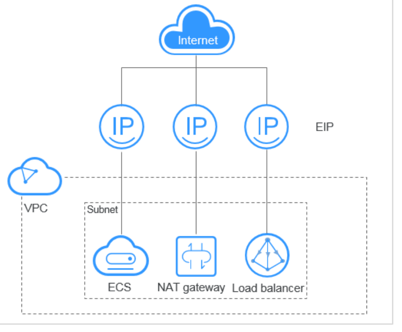
- A subnet is a unique CIDR block, a range of IP addresses, in your VPC.
- All resources in a VPC must be deployed on subnets.
- Once a subnet has been created, its CIDR block cannot be modified.



- By default, ECSs in all subnets of the same VPC can communicate with one another, but ECSs in different VPCs cannot.
- You can create VPC peering connections to enable ECSs in different VPCs to communicate with one another.
- The subnets used to deploy your resources must reside within your VPC, and the subnet masks used to define them can be between the netmask of its VPC CIDR block and /28 netmask.
  - 10.0.0.0 – 10.255.255.255
  - 172.16.0.0 – 172.31.255.255
  - 192.168.0.0 – 192.168.255.255
- https://support.huaweicloud.com/intl/en-us/productdesc-vpc/en-us_topic_0030969424.html

Elastic IP

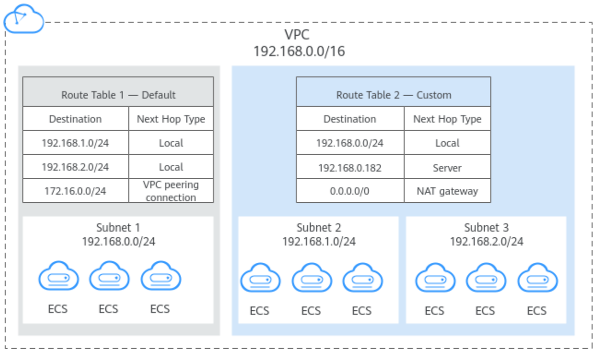
- The Elastic IP (EIP) service enables your cloud resources to communicate with the Internet using static public IP addresses and scalable bandwidths. EIPs can be bound to or unbound from ECSs, BMSs, virtual IP addresses, NAT gateways, or load balancers.



- Each EIP can be used by only one cloud resource at a time.

Route Table

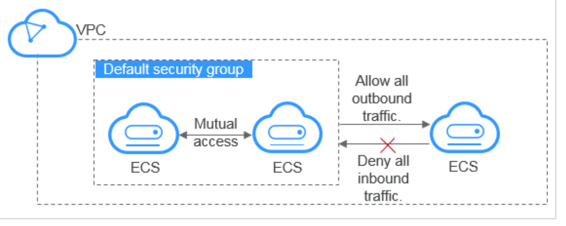
- A route table contains a set of routes that are used to determine where network traffic from your subnets in a VPC is directed. Each subnet must be associated with a route table. You can associate a subnet with only one route table at a time, but you can associate multiple subnets with the same route table.



- You can add, query, modify, and delete routes.
- When you create a VPC, the system automatically generates a default route table for the VPC. If you create a subnet in the VPC, the subnet automatically associates with the default route table. You can add, delete, and modify routes in the default route table, but you cannot delete the route table. When you create a VPN connection, the default route table automatically delivers a route that cannot be deleted or modified. If you want to modify or delete the route, you can associate your subnet with a custom route table and replicate the route to the custom route table to modify or delete it.
- You can also create a custom route table and associate subnets that have the same routing requirements with this table. Custom route tables can be deleted if they are no longer required.
- The way you can access the route table module varies by region.
  - If the route table module is not decoupled from the VPC module in your selected region, access the route table module by clicking the **Route Tables** tab on the VPC details page.
  - If the route table module is decoupled from the VPC module in your selected region, access the route table module by clicking **Route Tables** in the left navigation pane of the VPC console.

Security Group

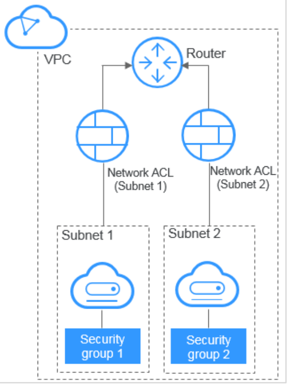
- A security group is a collection of access control rules for ECSs that have the same security requirements and are mutually trusted within a VPC. After you create a security group, you can create different access rules for the security group, and the rules will apply to any ECS that the security group contains.



- Your account automatically comes with a security group by default. The default security group allows all outbound traffic and denies all inbound traffic. Your ECSs in this security group can communicate with each other without the need to add rules.

Network ACL

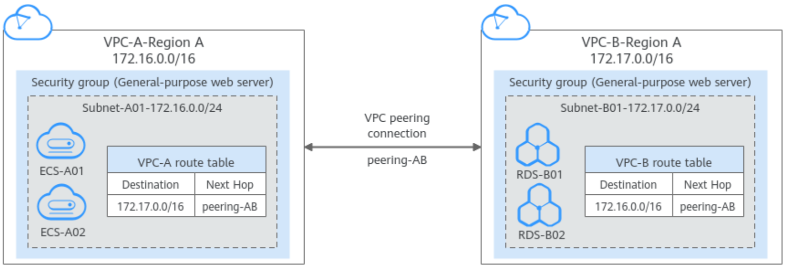
- A network ACL is an optional layer of security for your subnets. After you associate one or more subnets with a network ACL, you can control traffic in and out of the subnets.



- Similar to security groups, network ACLs control access to subnets and add an additional layer of defense to your subnets. Security groups only have the "allow" rules, but network ACLs have both "allow" and "deny" rules. You can use network ACLs together with security groups to implement comprehensive and fine-grained access control.
- You can associate a network ACL with multiple subnets. However, a subnet can only be associated with one network ACL at a time.
- Each newly created network ACL is in the **Inactive** state until you associate subnets with it.
- Network ACLs are stateful. If the network ACL allows outbound traffic and you send a request from your instance, the response traffic for that request is allowed to flow in regardless of inbound network ACL rules. Similarly, if inbound traffic is allowed, responses to allowed inbound traffic are allowed to flow out, regardless of outbound rules.

VPC Peering Connection

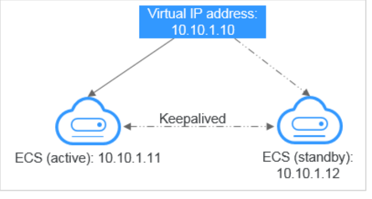
- A VPC peering connection is a network connection between two VPCs in the same region. It enables you to route traffic between them using private IP addresses. You can create a VPC peering connection between your own VPCs, or between your VPC and a VPC of another account within the same region. However, you cannot create a VPC peering connection between VPCs in different regions.

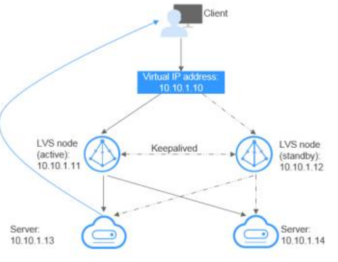


- The VPCs to be peered can be in the same account or different accounts, but must be in the same region.
- If you create a VPC peering connection between two VPCs in your account, the system accepts the connection by default. To enable communication between the two VPCs, you need to add routes for the local and peer VPCs.
- If you request a VPC peering connection with a VPC in another account in the same region, the VPC peering connection will be in the **Awaiting acceptance** state. After the owner of the peer account accepts the connection, the connection status changes to **Accepted**. The owners of both the local and peer accounts must configure the routes required by the VPC peering connection to enable communication between the two VPCs.

Virtual IP Address

- A virtual IP address can be shared among multiple ECSs. An ECS can have both private and virtual IP addresses, and you can access the ECS through either IP address. A virtual IP address has the same network access capability as a private IP address. Virtual IP addresses are used for high availability as they make active/standby ECS switchover possible.





- Networking mode 1: high availability of ECSs
  - If you want to improve service availability and avoid single points of failure, you can deploy ECSs in the active/standby mode or deploy one active ECS and multiple standby ECSs. In this arrangement, the ECSs all use the same virtual IP address. If the active ECS becomes faulty, a standby ECS takes over services from the active ECS and services continue uninterrupted.
- Networking mode 2: high availability of a load balancing cluster
  - If you want to build a high-availability load balancing cluster, use Keepalived and configure LVS nodes as direct routers.

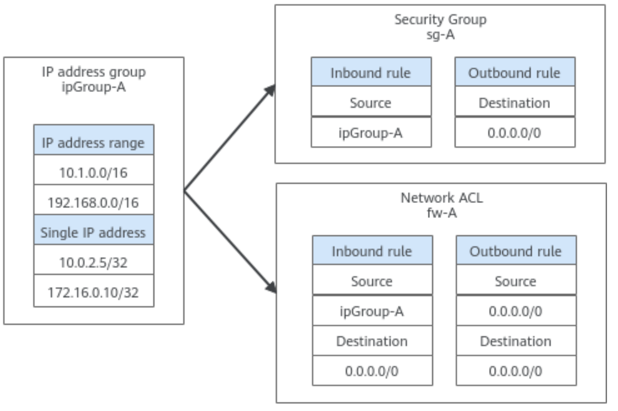
Elastic Network Interface

- An elastic network interface (Elastic NIC) is a virtual network card. You can create and configure network interfaces and attach them to your instances (ECSs and BMSs) to obtain flexible and highly available network configurations.
  - A **primary network interface** is created together with an instance by default, which cannot be detached from its instance.
  - You can create **extension network interfaces**, attach them to an instance, and detach them from the instance. The number of extension network interfaces that you can attach to an ECS varies by ECS flavor.

- A primary network interface cannot be detached from its instance.
- The number of extension network interfaces that you can attach to an instance varies by instance flavor.
- Elastic network interfaces and extension NICs cannot be used to directly access Huawei Cloud services, such as DNS. You can use VPCEP to access these services.

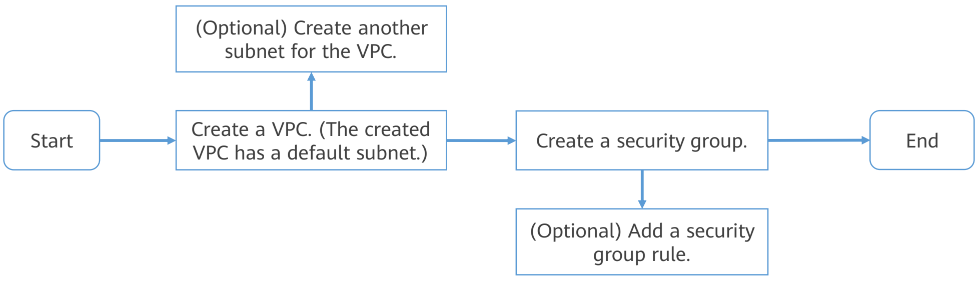
IP address group

- An IP address group is a collection of IP addresses. It can be associated with security groups and network ACLs to simplify IP address configuration and management.



- You can add IP address ranges and IP addresses that need to be managed in a unified manner to an IP address group. An IP address group can work together with different cloud resources.

VPC Configuration Process



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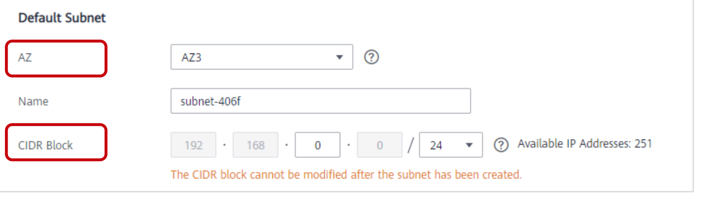
- Before creating your VPCs, determine how many VPCs, the number of subnets, and what IP address ranges you will need. Ensure that the subnets do not overlap with those of the end of VPN or Direct Connect connections.
- For network security, define access control policies based on specific services and minimize the access permissions. For example, a security group allows access from only certain source IP addresses on certain ports.

VPC Configuration - Subnet

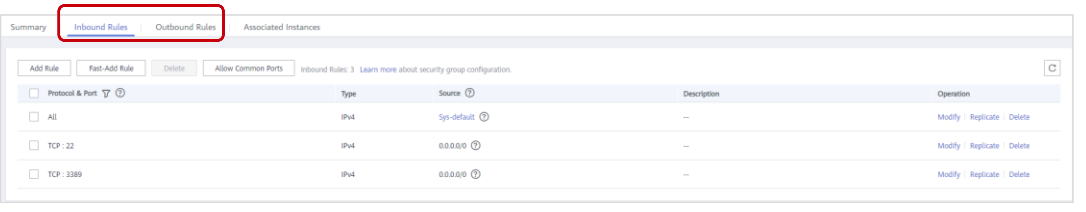
- Each VPC comes with a default subnet. If the default subnet cannot meet your requirements, create one.
- The subnet is configured with DHCP by default. When an ECS in this subnet starts, the ECS automatically obtains an IP address using DHCP.
- An AZ is a physical location where resources use independent power supplies and networks within a given region.



- The CIDR block of a subnet can be within the CIDR block for the VPC. The supported CIDR blocks are 10.0.0.0/8-24, 172.16.0.0/12-24, and 192.168.0.0/16-24.
- An external DNS server address is used by default. If you need to change the DNS server address, ensure that the configured DNS server address is available.
- Dynamic Host Configuration Protocol (DHCP) is a network protocol for local area networks. It means that the server controls a range of IP addresses, and the client can automatically obtain the IP address and subnet mask assigned by the server when logging in to the server.

VPC Configuration - Security Group

- Your account automatically comes with a default security group. You can add inbound and outbound rules to the default security group or create a new security group.
- Inbound rules control incoming traffic to ECSs in the security group.
- Outbound rules control outgoing traffic from ECSs in the security group.
- Default security group rules



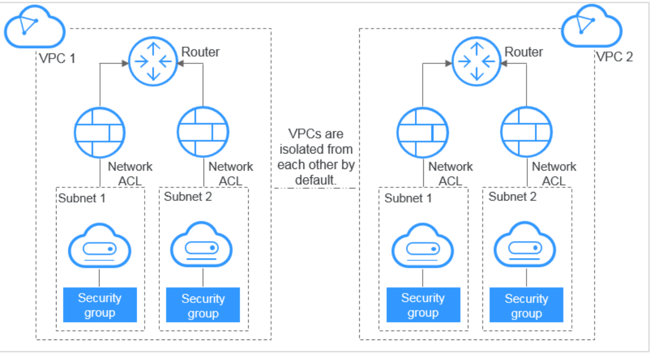
- The default security group cannot be deleted, but you can modify the rules in the default security group.
- If two ECSs are in the same security group but in different VPCs, the ECSs cannot communicate with each other. To enable communications between the ECSs, use a VPC peering connection to connect the two VPCs.
- In a VPC, if you want to copy resources from an ECS in a security group to another ECS in another security group, you can add rules to enable internal network communication between the ECSs and then copy resources. Within a given VPC, ECSs in the same security group can communicate with one another by default. However, ECSs in different security groups cannot communicate with each other by default. To enable these ECSs to communicate with each other, you need to add certain security group rules.

VPC vs. Traditional IDC

Comparison Item	Virtual private cloud	Traditional IDC
Deploymentcycle	You do not need to perform complex engineering deployment, including engineering planning and cabling. You can determine your networks, subnets, and routes on Huawei Cloud based on service requirements.	You need to set up networks and perform tests. The entire process takes a long time and requires professional technical support.
Total cost	Huawei Cloud provides flexible billing modes for network services. You can select whichever one best fits your business needs. There are no upfront costs and network O&M costs, reducing the total cost of ownership (TCO).	You need to invest heavily in equipment rooms, power supply, construction, and hardware materials. You also need professional O&M teams to ensure network security. Asset management costs increase with any change in business requirements.
Flexibility	Huawei Cloud provides a variety of network services for you to choose from. If you need more network resources (for instance, if you need more bandwidth), you can expand resources on the fly.	You have to strictly comply with the network plan to complete the service deployment. If there are changes in your service requirements, it is difficult to dynamically adjust the network.
Security	VPCs are logically isolated from each other. You can use security features such as network ACLs and security groups, and even security services like Advanced Anti-DDoS (AAD) to protect your cloud resources.	The network is insecure and difficult to maintain. You need professional technical personnel to ensure network security.

Application Scenario - Dedicated Networks on Cloud

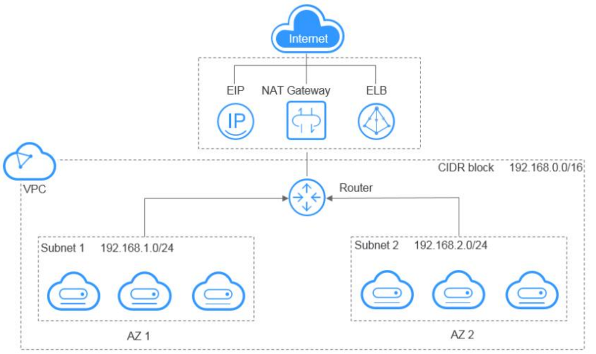
- Each VPC represents a private network and is logically isolated from other VPCs. You can deploy your service systems in a private network on the cloud. If you have multiple service systems, for example, a production system and a test system, you can keep them isolated by deploying them in two different VPCs.



- To enable two VPCs in the same region to communicate with each other, you can create a VPC peering connection between them.

Application Scenario - Web Application/Website Hosting

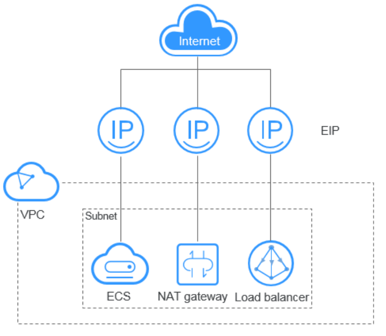
- You can host web applications and websites in a VPC and use the VPC as a regular network. With EIPs or NAT gateways, you can connect ECSs running your web applications to the Internet. You can then use load balancers provided by the ELB service to evenly distribute traffic across multiple ECSs.



Contents

What Is Elastic IP

- An EIP is a public IP address that can be accessed directly over the Internet. An EIP consists of a public IP address and some amount of public network egress bandwidth. EIPs can be bound to or unbound from ECSs, BMSs, virtual IP addresses, NAT gateways, and load balancers.



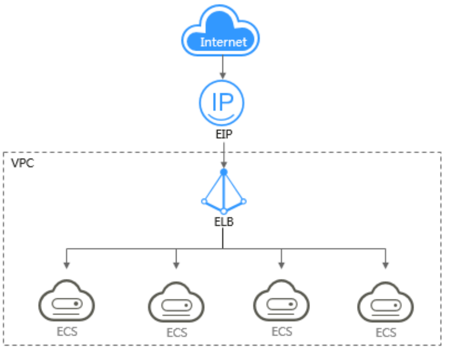
EIP Types

Comparison Dimension	Static BGP	Dynamic BGP	Preferred BGP
Definitions	Static routes are manually configured and must be manually reconfigured anytime the network topology or link status changes.	Dynamic BGP provides automatic failover and chooses the optimal path based on the real-time network conditions as well as preset policies.	Premium BGP chooses the optimal path and ensures low-latency and high-quality networks. BGP is used to interconnect with lines of multiple mainstream carriers. Public network connections that feature low latency and high quality are directly established between Chinese mainland and Hong Kong (China).
Assurance	When changes occur on a network that uses static BGP, the manual configuration takes some time and high availability cannot be guaranteed.	When a fault occurs on a carrier's link, dynamic BGP will quickly select another optimal path to take over services, ensuring service availability.	Premium BGP has the same assurance capability as that of dynamic BGP. In addition, premium BGP ensures higher network quality and lower latency.
Service availability	99%	99.95%	99.95%

- If you select static BGP, your application system must have disaster recovery setups in place.
- Currently, mainstream Dynamic BGP carriers in Hong Kong (China) are supported.
- Premium BGP is now available only in the CN-Hong Kong region.

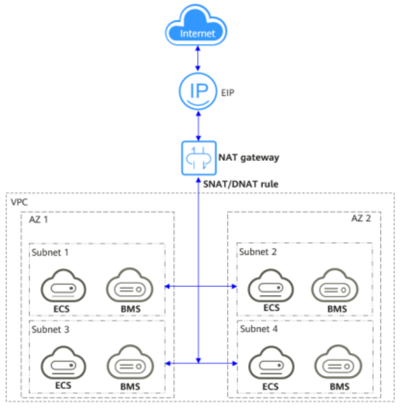
Application Scenario - Binding an EIP to a Load Balancer

- After you attach an EIP to a load balancer, the load balancer can distribute requests from the Internet to backend servers.



Application Scenario - Binding an EIP to a NAT Gateway

- After an EIP is bound to a NAT gateway and SNAT and DNAT rules are added, multiple servers (such as ECSs and BMSs) can use the same EIP to access the Internet and provide services accessible from the Internet.



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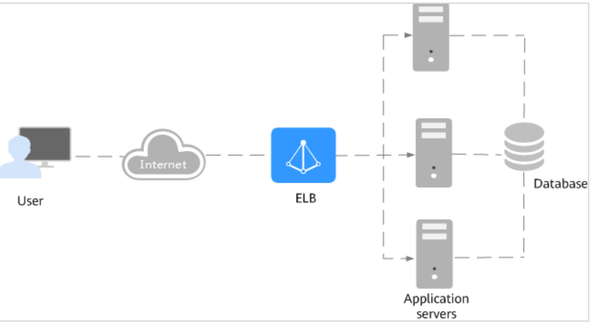
- An SNAT rule allows servers in a specific VPC subnet to use the same EIP to access the Internet.
- A DNAT rule enables servers in a VPC to provide services accessible from the Internet.

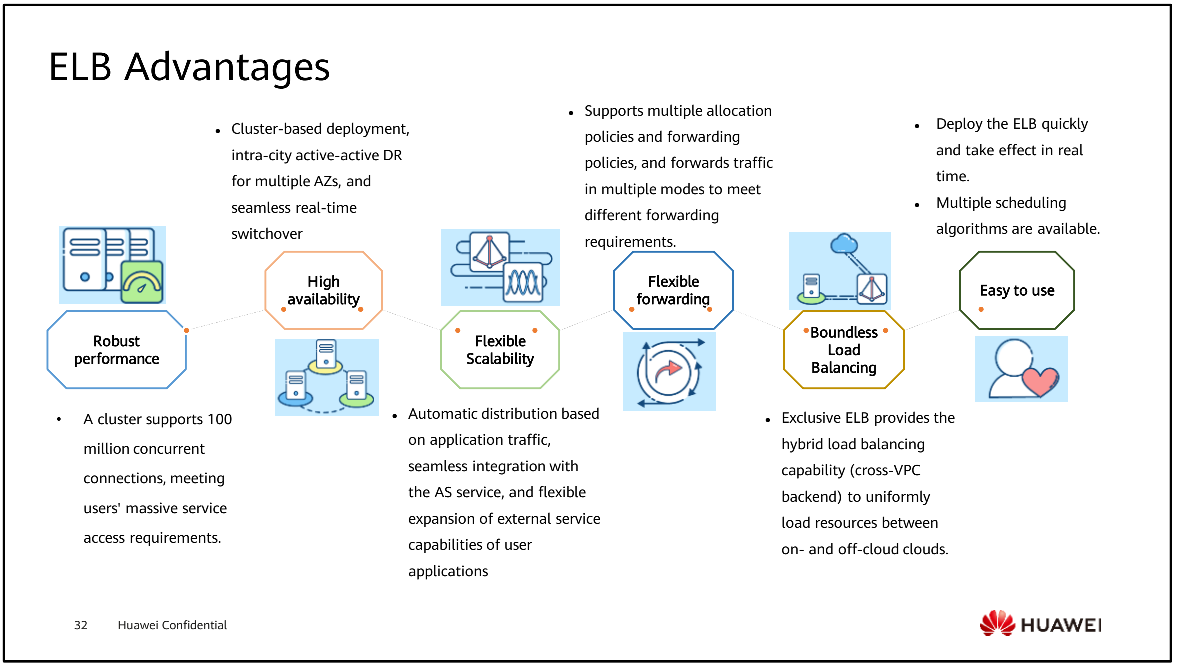
Contents

What Is ELB?

- Elastic Load Balance (ELB) automatically distributes incoming traffic across multiple backend servers based on the listening rules you configure. ELB expands the service capabilities of your applications and improves their availability by eliminating single points of failure (SPOFs).



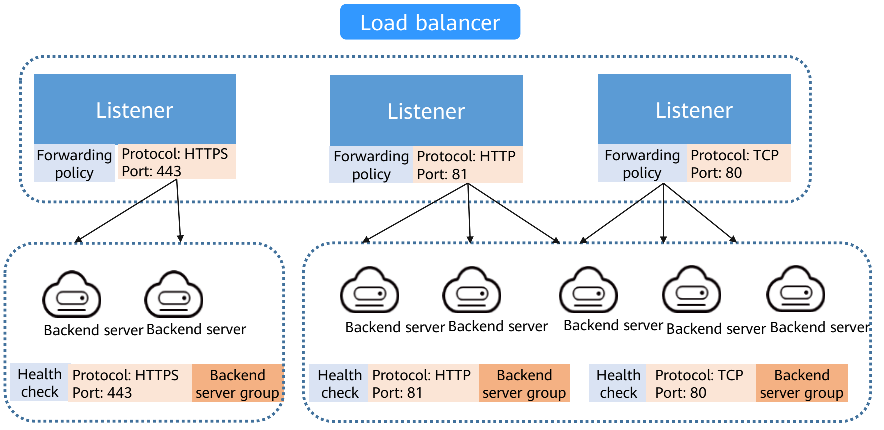
- ELB provides shared load balancers and dedicated load balancers.
  - Dedicated load balancers have exclusive use of underlying resources, so that the performance of a dedicated load balancer is not affected by other load balancers. In addition, there are a wide range of specifications available for selection.
  - Shared load balancers are deployed in clusters and share underlying resources, so that the performance of a load balancer is affected by other load balancers. Shared load balancers were previously named enhanced load balancers.



- ELB has the following advantages:
  - Robust performance: Each load balancer has exclusive use of isolated resources, meeting your requirements for handling a massive number of requests. A single load balancer deployed in one AZ can handle up to 20 million concurrent connections.
  - High availability: ELB can route traffic uninterruptedly. If your servers in one AZ are unhealthy, it automatically routes traffic to healthy servers in other AZs. ELB provides a comprehensive health check system to ensure that incoming traffic is routed only to healthy backend servers, improving the availability of your applications..
  - Flexible Scalability: ELB ensure that your applications are always ready for any size of workloads. It works with Auto Scaling (AS) to flexibly adjust the number of backend servers and intelligently distribute incoming traffic across them.
  - Flexible Forwarding: ELB allows you to create custom forwarding policies to meet your requirements in different scenarios.
  - Boundless Load Balancing: Dedicated load balancers can route traffic to both cloud and on-premises servers, improving the scalability of your hybrid cloud network.
  - Ease-of-use: A diverse set of protocols and algorithms enable you to customize traffic routing policies to your needs while keeping deployments simple.

ELB Architecture

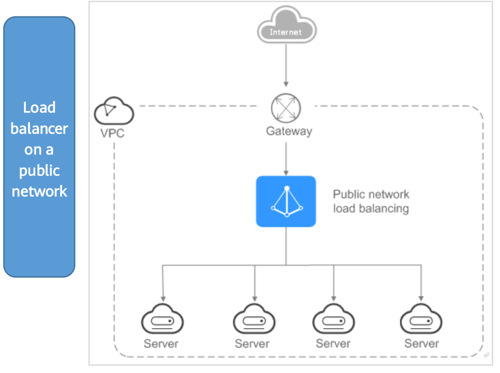
- ELB consists of three components: load balancers, listeners, and backend server groups.

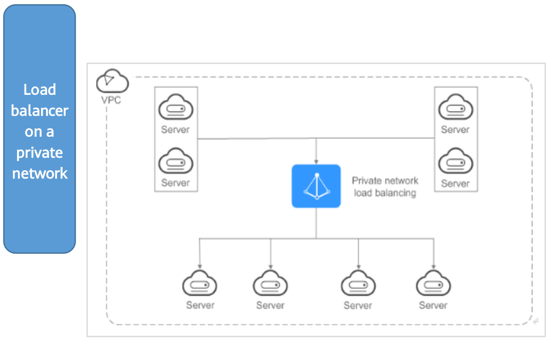


- ELB consists of load balancers, listeners, and backend server groups.
  - A load balancer is an instance that distributes incoming traffic across backend servers in one or more availability zones (AZs).
  - A listener uses the protocol and port you specify to check for requests from clients and route the requests to associated backend servers based on the listening rules you define. You can add one or more listeners to a load balancer.
  - A backend server group uses the protocol and port you specify to receive the requests from the load balancer and route them to one or more backend servers. You need to add at least one backend server to a backend server group. You can set a weight for each backend server so that the load balancer can route requests based on their performance. You can also configure health checks for a backend server group to check the health of backend servers in the group. If a backend server is unhealthy, the load balancer stops routing new requests to this server until it recovers.

ELB - Load Balancer

- A load balancer distributes incoming traffic across multiple backend servers. Load balancers can work on both public and private networks.

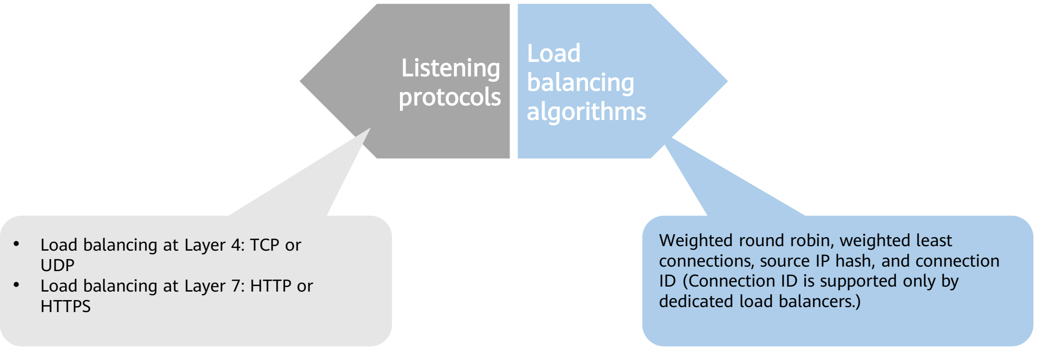




- Each load balancer on a public network has an EIP bound to it and routes requests from clients to backend servers over the Internet.
- Load balancers on a private network work within a VPC and route requests to backend servers in the same VPC as the clients.

ELB - Listener

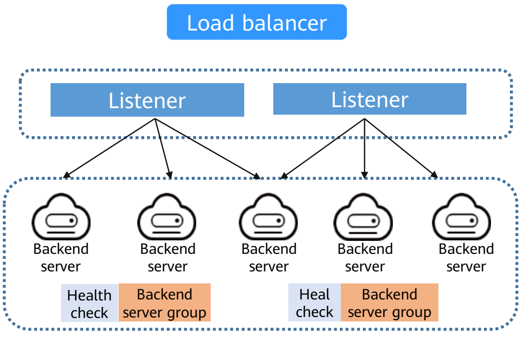
- A listener listens on requests from clients and routes the requests to backend servers based on the settings that you configure when you add the listener.



- A listener specifies the protocol and port used to receive requests from the clients, and the protocol, the port, and the load balancing algorithm to forward the requests to one or more backend servers. A listener also defines the health check configuration, which the load balancer uses to continually check the statuses of backend servers. If a backend server is unhealthy, the load balancer routes traffic to the healthy ones. Traffic routing to this server resumes after it recovers.
- The OSI model consists of the application layer, presentation layer, session layer, transport layer, network layer, data link layer, and physical layer.
    - Protocols at the application layer: HTTP, SNMP, FTP, NFS, Telnet, and SMTP
    - Protocols at the presentation layer: none
    - Protocols at the session layer: none
    - Protocols at the transport layer: TCP and UDP
    - Protocols at the network layer: IP and ICMP
    - Protocols at the data link layer: FDDI, Ethernet, ARPANET, PDN, SLIP, and PPP
    - Protocols at the physical layer: IEEE 802.1A, IEEE 802.2 to IEEE 802.11

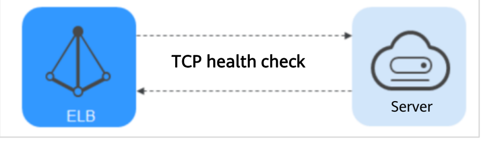
ELB-Backend Server Group

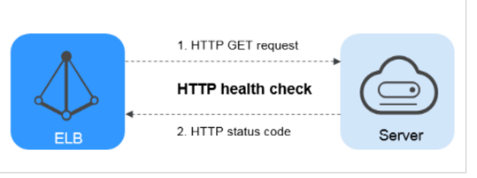
- A backend server group is a group of cloud servers that have same features. When you add a listener, you select a load balancing algorithm and create or select a backend server group. Incoming traffic is routed to the corresponding backend server group based on the listener's configuration.



ELB - Health Check

- ELB periodically sends heartbeat messages to associated backend servers to check their health and ensure that traffic is distributed only to healthy servers. This can improve the availability of your applications. If a backend server is unhealthy, the load balancer stops routing traffic to it. The load balancer will resume routing requests to the backend server after it recovers.





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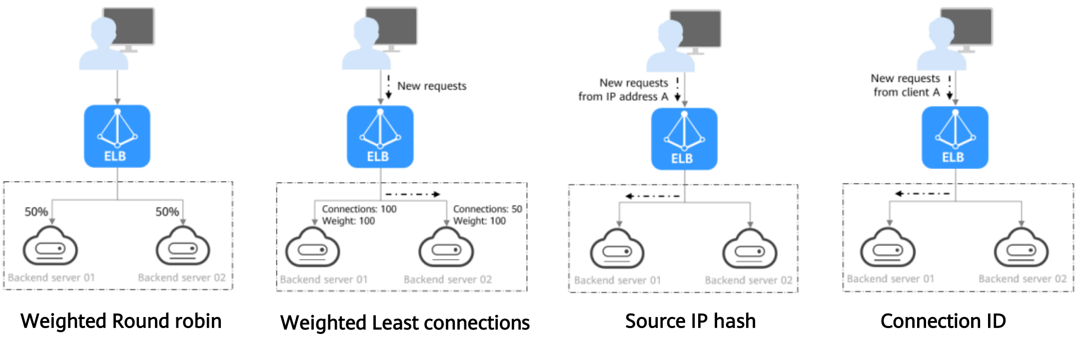
- How does a health check work?
  - UDP listeners: UDP is used for health checks by default, and UDP probe packets are sent to backend servers to obtain their health results.
  - TCP, HTTP, or HTTPS listeners: HTTP can be used for health checks. ELB sends HTTP GET requests to backend servers to check their health.
- If the health check result of a backend server is **Unhealthy**, you need to check its configuration.
- The security group that contains the backend servers must allow access from 100.125.0.0/16. Otherwise, health checks cannot be performed.
- If UDP is used for health checks, the backend server group's protocol must be UDP.

ELB - Load Balancing Algorithms

- The load balancer forwards the request from the client to the backend server for processing. You can add an ECS instance as the backend server of the load balancer. The listener uses the configured protocol and port to check connection requests from clients and forwards the requests to backend ECSs in the backend server group based on the user-defined allocation policy. The specific policies are as follows:



- Each backend server can be given a numeral value from 0 to 100 to indicate the proportion of requests the backend server can receive. The higher the weight, the more requests the backend server receives. You can set a weight for each backend server when you select one of the following algorithms:
    - Weighted round robin: Requests will not be routed to a backend server whose weight is 0, even if the backend server is considered healthy. If none of the servers have a weight of 0, the load balancer routes requests to these servers using the round robin algorithm based on their weights. If two backend servers have the same weights, they receive the same number of requests.
    - Weighted least connections: Requests will not be routed to a backend server whose weight is 0. If none of the servers have a weight of 0, the load balancer calculates each server's overhead using the formula: Overhead = Number of current connections/Server weight. The load balancer routes requests to the backend server with the lowest overhead.
    - Source IP hash: If a backend server's weight is 0, requests will not be routed to this server. If the server weights are not 0, they will not take effect, and requests from the same IP address will be routed to the same backend server.
    - Connection ID: If a backend server's weight is 0, requests will not be routed to this server. If the server weights are not 0, they will not take effect, and requests from the same client and with the same connection ID will be routed to the same backend server.
- Currently, only dedicated load balancers support the Connection ID algorithm.
- In addition to the load balancing algorithm, fa
- Assume that there are two backend servers with the same weight (not zero), the weighted least connections algorithm is selected, sticky sessions are not enabled,

and 100 connections have been established with backend server 01, and 50 connections with backend server 02.

- When client A wants to access backend server 01, the load balancer establishes a persistent connection with backend server 01 and continuously routes requests from client A to backend server 01 before the persistent connection is disconnected. When other clients access backend servers, the load balancer routes the requests to backend server 02 using the weighted least connects algorithm.
- In addition to the load balancing algorithm, factors that affect load balancing generally include connection type, session stickiness, and server weights.

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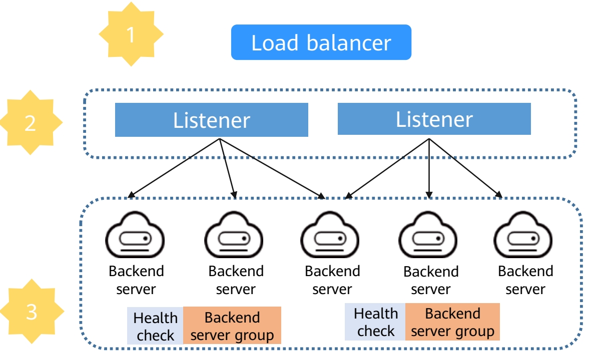
- In addition to the load balancing algorithm, factors that affect load balancing generally include connection type, session stickiness, and server weights.

ELB Configuration Process

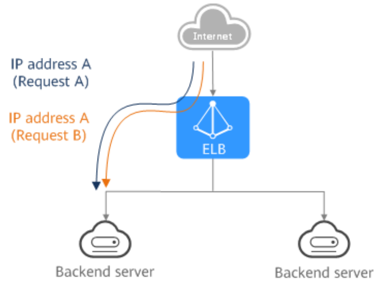
1. **Creating a Load Balancer**
   - Click **Buy Elastic Load Balancer**.
   - Select the load balancer type.
   - Configure the network.
2. **Adding a Listener**
   - Locate the created load balancer.
   - Configure the protocol and port.
3. **Adding a Backend Server Group**
   - Select a load balancing algorithm.
   - Configure a health check.



- A listener specifies the protocol and port used to receive requests from the clients, and the protocol, the port, and the load balancing algorithm to forward the requests to one or more backend servers. A listener also defines the health check configuration, which the load balancer uses to continually check the statuses of backend servers. If a backend server is unhealthy, the load balancer routes traffic to the healthy ones. Traffic routing to this server resumes after it recovers.

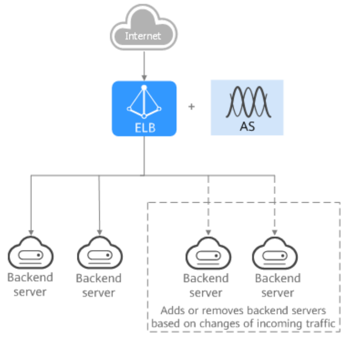
Application Scenario: Heavy-Traffic Applications

- For an application with heavy traffic, such as a large web portal or mobile app store, ELB evenly distributes incoming traffic to multiple backend servers, balancing the load while ensuring stable performance. Sticky sessions ensure that requests from one client are always forwarded to the same backend server.



Application Scenario: Applications with different Traffic

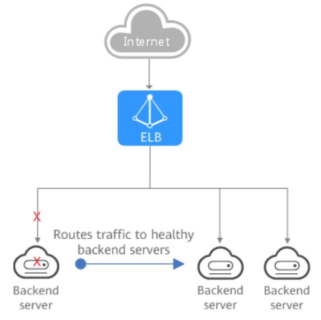
- For an application that has predictable peaks and troughs in traffic volumes, ELB works with AS to add or remove backend servers to keep up with changing demands. One example is flash sales, during which there are predictable traffic spikes that only last a short while. ELB can work with AS to run only the required number of backend servers needed to handle the load of your application.



- Traffic to some applications may fluctuate significantly at different time periods.

Application Scenario: Zero SPOFs

- ELB routinely performs health checks on backend servers. If any backend server is unhealthy, ELB will not route requests to this server until it recovers. This makes ELB a good choice for running applications that require high reliability.



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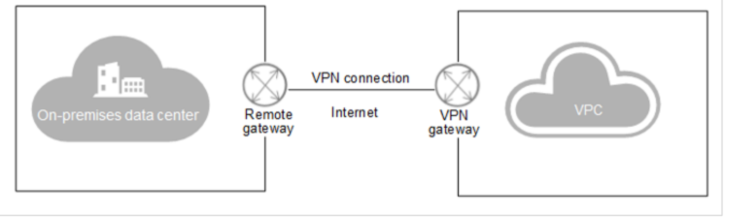
- A single point of failure (SPOF) is a part of a system that, if it fails, will stop the entire system from working. SPOFs are undesirable in any system with a goal of high availability or reliability, such as a business system, software application, or other industrial system.

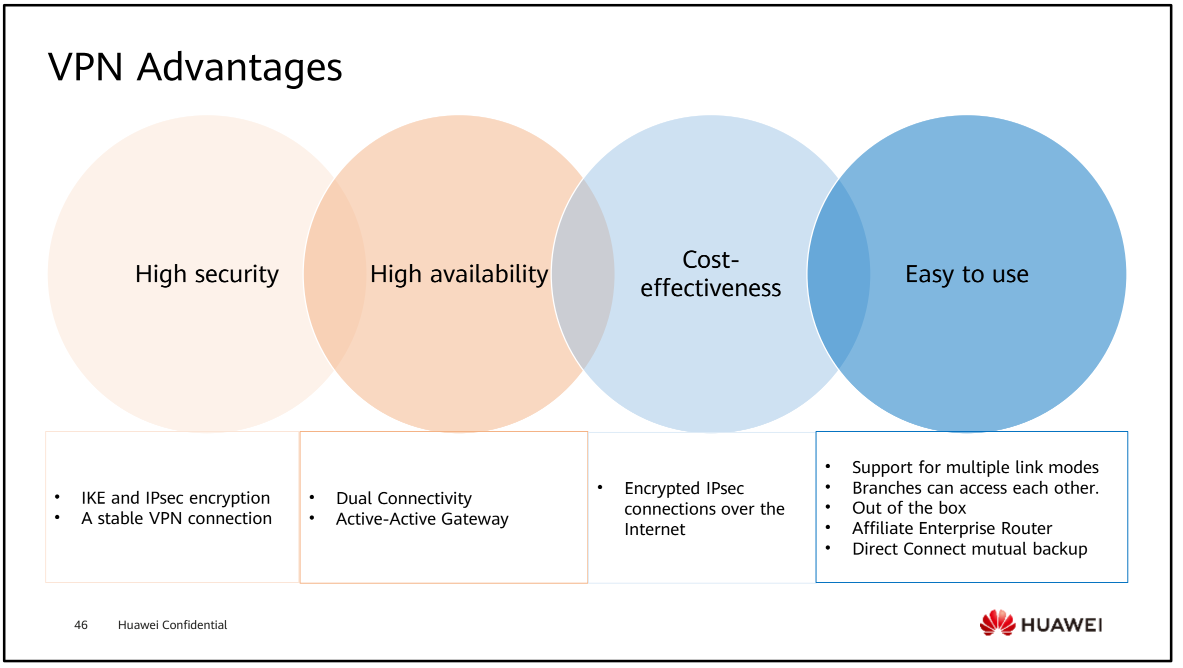
Contents

Virtual Private Network

- Virtual Private Network (VPN) allows you to establish an encrypted, Internet-based communications tunnel between your on-premises data center and a VPC, so you can access resources in the VPC remotely.



- VPN tunnels support three protocols: PPTP, L2TP, and IPsec.
- Virtual Private Network (VPN) establishes a secure, encrypted communications tunnel between your local data center and your VPC on HUAWEI CLOUD. With VPN, you can build a flexible and scalable hybrid cloud environment.

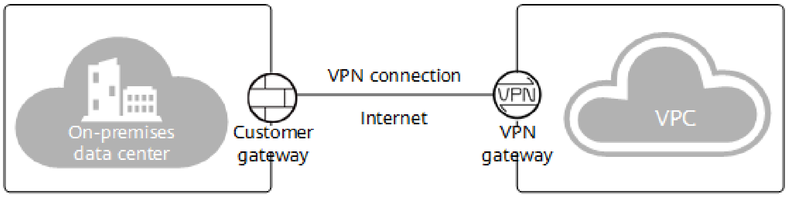


- VPN advantages:
- **High security**
  - Data is encrypted using IKE and IPsec, ensuring high data security.
  - A VPN gateway is exclusive to a tenant, isolating tenants from each other.
  - VPN gateways of the GM specification are supported, in compliance with the GB/T 36968-2018 Information security technology —Technical specification for IPSec VPN.
- **High availability**
  - A VPN gateway provides two elastic IP addresses (EIPs) to establish dual independent VPN connections with a customer gateway. If one VPN connection fails, traffic can be quickly switched to the other VPN connection.
  - Active-active gateways are deployed in different availability zones (AZs) to ensure AZ-level high availability.
  - Active/Standby mode: In normal cases, a VPN gateway communicates with a customer gateway through the active connection. If the active connection fails, traffic is automatically switched to the standby VPN connection. After the fault is rectified, traffic is switched back to the active VPN connection.

- **Cost-effectiveness**
  - IPsec connections over the Internet provide a cost-effective alternative to Direct Connect.
  - A VPN gateway can be bound to EIPs that share bandwidth, reducing bandwidth costs.
  - The bandwidth can be adjusted when an EIP instance is created.
- **Easy to use**
  - A VPN gateway supports multiple connection modes, including policy-based, static routing, and BGP routing, to meet different access requirements of customer gateways.
  - A VPN gateway on the cloud can function as a VPN hub, enabling on-premises branch sites to access each other.
  - A VPN connection can be created in a few simple steps on the VPN device in an on-premises data center and on the VPN console, and is ready to use immediately after being created.
  - VPN can be used together with the enterprise router service, allowing enterprises to build more flexible cloud-based networks.
  - Backup between VPN and Direct Connect is supported, and automatic failover is supported.
  - Private VPN gateways are supported to encrypt traffic transmitted over Direct Connect connections, improving data transmission security.

VPN Networking

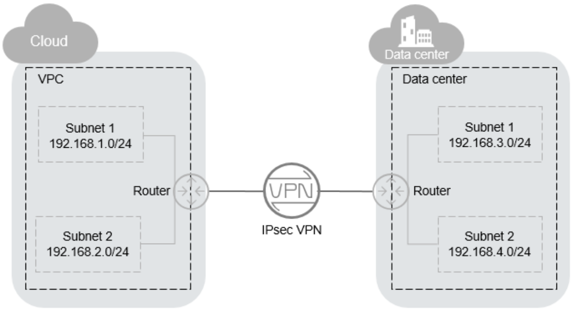
- A VPN consists of a VPN gateway and one or more VPN connections.
  - A VPN gateway provides an Internet egress for a VPC and works together with the gateway in your on-premises data center.
  - A VPN connection connects a VPN gateway to a customer gateway through encrypted tunnels, enabling communication between a VPC and your on-premises data center. This helps quickly establish a secure hybrid cloud environment.



- VPN components:
  - A VPN gateway is an egress gateway for a VPC. With a VPN gateway, you can create a secure, reliable, and encrypted connection between a VPC and your on-premises data center or between two VPCs in different regions. Each data center must have a gateway, which works as the remote gateway. Each VPC must have a VPN gateway. A VPN gateway needs to be paired with a remote gateway. Each VPN gateway can connect to one or more remote gateways, so you can set up point-to-point or hub-and-spoke VPN connections.
  - **Customer gateway**: is a resource that provides information to Huawei Cloud about your customer gateway device, which can be a physical device or software application in your on-premises data center.
  - A VPN connection is a secure and reliable communications tunnel established between a VPN gateway and a gateway in your on-premises data center. Only IPsec VPNs are supported. VPN connections use IKE and IPsec protocols to cost-effectively and securely encrypt data transmitted over the Internet.

IPSec VPN

- Internet Protocol Security (IPsec) VPN uses a secure network protocol suite that authenticates and encrypts the packets of data to provide secure encrypted communication between different networks.



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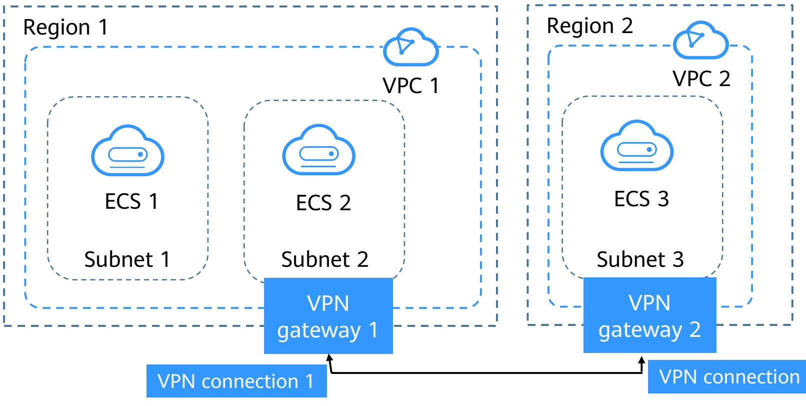
- Assume that you have created a VPC with two subnets (192.168.1.0/24 and 192.168.2.0/24) on the cloud, and the router in your on-premises data center also has two subnets (192.168.3.0/24 and 192.168.4.0/24). In this case, you can create a VPN to connect the VPC subnets and the data center subnets.

VPN Configuration Process(Classic)

- You can create a VPN gateway and a VPN connection on the management console.



- A VPN enables communications between VPCs in different regions.
- In this example, ECS 2 in region 1 needs to communicate with ECS 3 in region 2. A VPN connection linking region 1 and region 2 can make this possible.
- Step 1: Create a VPN gateway in region 1 and configure parameters such as Billing Mode, Region (Region 1), VPC (VPC 1), Billed by, Bandwidth, and encryption policies.
- Step 2: Create a VPN connection in region 1. Select subnet 2 for Local Subnet and subnet 3 for Remote Subnet. Configure the remote gateway. (VPN gateway 2 has not been created. Just enter a random address. You can change it later.)
- Step 3: Create a VPN gateway in region 2 and configure parameters such as Billing Mode, Region (Region 2), VPC (VPC 2), Billed by, Bandwidth, and encryption policies.
- Step 4: Create a VPN connection in region 2. Select subnet 3 for Local Subnet and subnet 2 for Remote Subnet. Configure the remote gateway by entering the IP address of VPN gateway 1.
- Step 5: Change the remote gateway address of VPN connection 1 to the address of VPN gateway 2.
- Step 6: Test the connectivity between ECS 2 and ECS 3 and check the VPN connection status.

VPN Configuration: VPN Gateway

- To allow your ECSs in a VPC to access your on-premises network, you must first create a VPN gateway.

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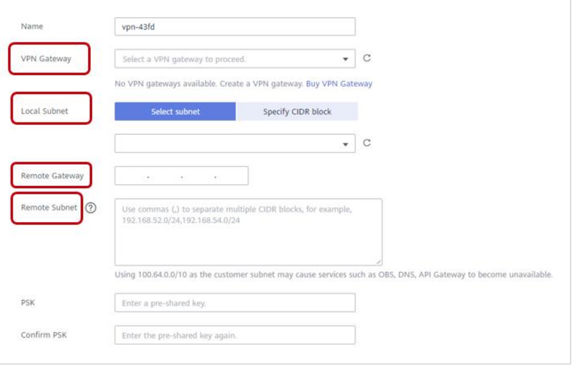
- **VPN Gateway**
  - You can modify the name and description of a VPN gateway if needed. If the bandwidth of a VPN gateway cannot meet your requirements, you can modify the bandwidth, too. If the number of VPN connections associated with a VPN gateway cannot meet your requirements, you can modify the VPN gateway specifications. You can change the billing mode of a VPN gateway billed by bandwidth from pay-per-use to yearly/monthly.
  - If a VPN gateway is no longer required, you can delete it to release network resources as long as it has no VPN connections configured. If it has any connections configured, they have to be deleted before you can delete the gateway.
- **VPC**: the name of the VPC that the VPN connects to
- **Type**: the VPN type. **IPsec** is selected by default.
- **Billed By**: There are two options available, bandwidth, and traffic.
  - **Bandwidth**: You specify a bandwidth and pay the bill based on the amount of time you use the bandwidth.
  - **Traffic**: You specify a bandwidth and pay for the total traffic you generate.
- **Bandwidth (Mbit/s)**:
  - The bandwidth (Mbit/s) of the VPN gateway. The bandwidth size is shared by all VPN connections created for the VPN gateway. The total bandwidth size used by all VPN connections created for a VPN gateway cannot exceed the VPN gateway bandwidth size.
  - If the network traffic exceeds the VPN gateway bandwidth, the network may get congested and VPN connections may be interrupted. Make sure you configure enough bandwidth.
  - You can configure alarm rules on Cloud Eye to monitor the bandwidth.

VPN Configuration: VPN Connection

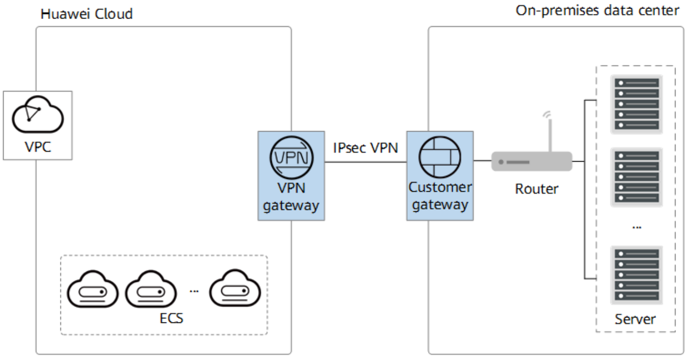
- To connect your ECSs in a VPC to your private network, after the VPN gateway is obtained, you also have to create a VPN connection.



- **VPN Connection**:
  - A VPN connection is an encrypted communications channel established between the VPN gateway in your VPC and that in your on-premises data center. The VPN connection can be modified later.
  - You can delete a VPN connection to release network resources if it is no longer required. When you delete the last VPN connection for a pay-per-use VPN gateway, the associated gateway will be deleted along with it.
- **VPN Gateway**: the name of the VPN gateway used by the VPN connection
- **Local Subnet**: the VPC subnets that will access your on-premises network through VPN. Possible values are **Select subnet** and **Specify CIDR block**.
- **Remote Gateway**: the public IP address of the VPN device translated by the VPN gateway in your on-premises private network. This IP address is used for communications with your VPC.
- **Remote Subnet**: the subnets of your on-premises network that will access the VPC through a VPN. The local subnet cannot include the CIDR block of the remote subnet.
- **PSK**: Enter 6 to 128 characters. The PSK at both ends of a VPN connection must be the same.

Application Scenario – Hybrid Cloud Deployment

- You can use a VPN to connect your on-premises data center to a VPC on the cloud and use the elastic and fast scaling capabilities of the cloud to expand application computing capabilities.



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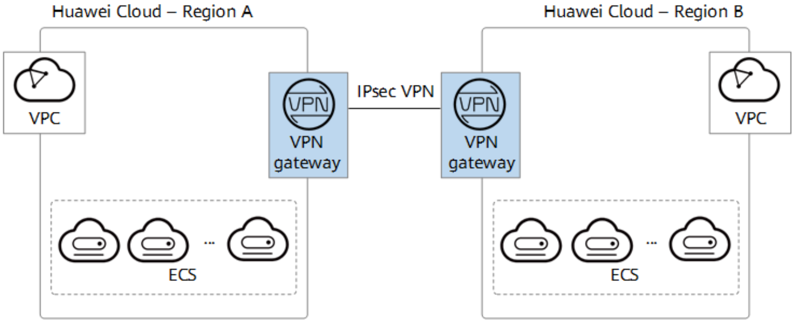
Application Scenario – Cross-Region Interconnection Between VPCs

- With VPNs, you can connect VPCs in different regions to enable connectivity between user services in these regions.

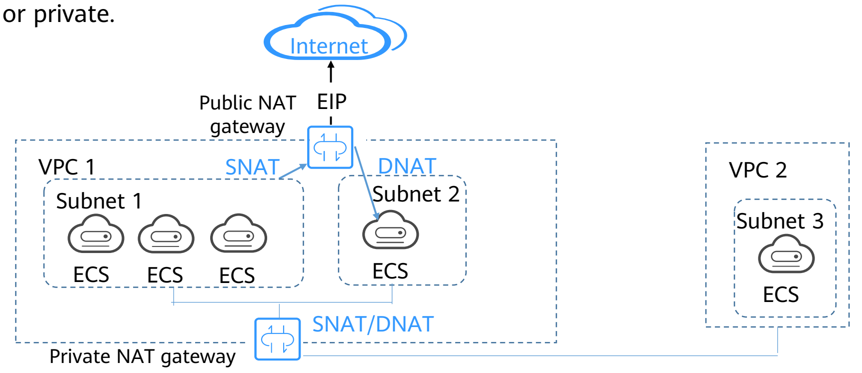


- **Enterprise Branch Interconnection**
    - A VPN gateway functions as a VPN hub to connect enterprise branches. This eliminates the need to configure VPN connections between every two branches.
- **Backup Between VPN and Direct Connect**
    - For high reliability purposes, you can connect your on-premises data center to a VPC on the cloud through Direct Connect and VPN that back up each other.

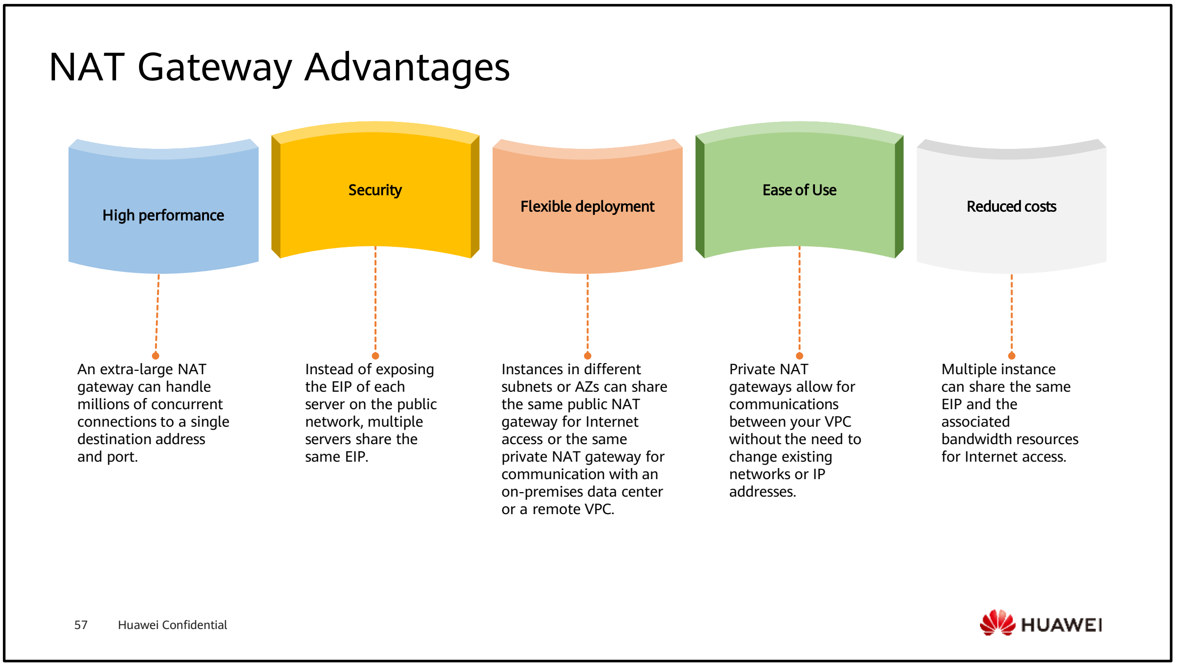
Contents

NAT Gateway

- The NAT Gateway service provides network address translation (NAT) service for servers in a VPC and enables servers to share an EIP to access the Internet. NAT gateways can be either public or private.



- With the increase of network applications, the problem of IPv4 address exhaustion becomes more and more serious. Although IPv4 can solve the problem of insufficient IPv4 address space, many network devices and network applications are based on IPv4. Therefore, before IPv6 is widely used, some transition technologies (such as CIDR and private network addresses) are the main solutions to this problem. NAT is one of these transition technologies.
- If the NAT function is enabled on the gateway, the device translates the IP address and port number in the header of the received packet to another IP address and port number, and then forwards the packet to the public network. In this process, the device can use the same public IP address to translate the packets sent by multiple private network users and distinguish different private network users by port number. In this way, the address reuse is achieved.
- A public NAT gateway enables cloud and on-premises servers in a private subnet to share an EIP to access the Internet or provide services accessible from the Internet.
- A private NAT gateway provides NAT service for servers in a VPC, so that multiple servers can share a private IP address to access or provide services accessible from an on-premises data center or other VPCs.



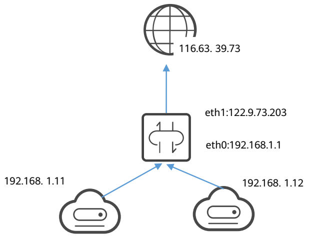
- **Advantages of Public NAT Gateways**
  - **Flexible deployment:** A NAT gateway can be shared across subnets and AZs, so that even if an AZ fails, the public NAT gateway can still run normally in another AZ. The specifications and EIP of a public NAT gateway can be changed at any time.
  - **Ease of use:** Multiple NAT gateway specifications are available. Public NAT gateway configuration is simple, the operation & maintenance is easy, and they can be provisioned quickly. Once provisioned, they can run stably.
  - **Cost-effectiveness:** Servers can share one EIP to connect to the Internet. You no longer need to configure one EIP for each server, which saves money on EIPs and bandwidth.

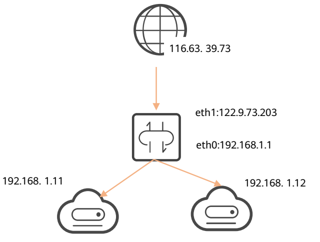
- Advantages of Private NAT Gateways
  - **Easier network planning:** Different departments in a large enterprise may have overlapping CIDR blocks, so the enterprise has to replan its network before migrating their workloads to the cloud. The replanning is time-consuming and stressful. The private NAT gateway eliminates the need to replan the network so that customers can retain their original network while migrating to the cloud.
  - **Easy operation & maintenance:** Departments of a large enterprise usually have hierarchical networks for hierarchical organizations, rights- and domain-based management, and security isolation. Such hierarchical networks need to be mapped to a large-scale network for enabling communication between them. A private NAT gateway can map the CIDR block of each department to the same VPC CIDR block, which simplifies the management of complex networks.
  - **Strong security:** Departments of an enterprise may need different levels of security. Private NAT gateways can expose the IP addresses and ports of only specified CIDR blocks to meet high security requirements. An industry regulation agency may require other organizations to use a specified IP address to access their regulation system. Private NAT gateways can help meet this requirement by mapping private IP addresses to that specified IP address.
  - **Zero IP conflicts:** Isolated services of multiple departments usually use IP addresses from the same private CIDR block. After the enterprise migrates

workloads to the cloud, IP address conflicts occur. Thanks to IP address mapping, the private NAT gateways allow for communication between overlapping CIDR blocks.

SNAT and DNAT

- Source network address translation (SNAT): During NAT, only the source address in packets is translated. This NAT mode applies to the scenario where private network users access the public network.
- Destination network address translation (DNAT): During NAT, only the destination address and port number in packets are translated. DNAT applies to the scenario where public network users access private network services.

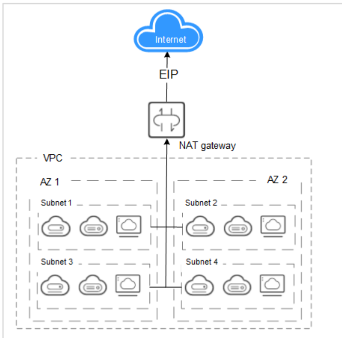


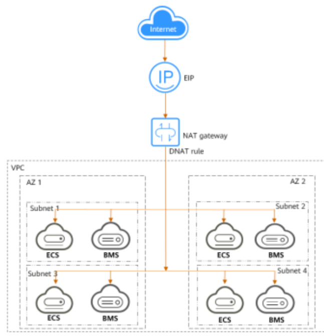


- NAT Gateway provides both source NAT (SNAT) and destination NAT (DNAT) for your resources in a VPC and allows servers in your VPC to access or provide services accessible from the Internet.

NAT Gateway Architecture (Public NAT Gateway)

- A public NAT gateway enables cloud and on-premises servers in a private subnet to share an EIP to access the Internet or provide services accessible from the Internet. Cloud servers are ECSs and BMSs in a VPC. On-premises servers are servers in on-premises data centers that connect to a VPC through Direct Connect or Virtual Private Network (VPN). A public NAT gateway supports up to 20 Gbit/s of bandwidth.



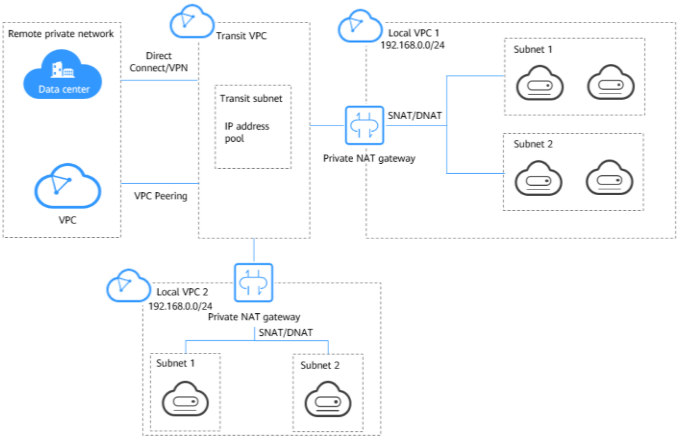


HUAWEI

- Public NAT gateways support SNAT and DNAT.
  - SNAT translates private IP addresses into EIPs, allowing servers in a VPC to share an EIP to access the Internet in a secure and efficient way.
  - DNAT enables servers in a VPC to share an EIP to provide services accessible from the Internet through IP address mapping or port mapping.

NAT Gateway Architecture (Private NAT Gateway)

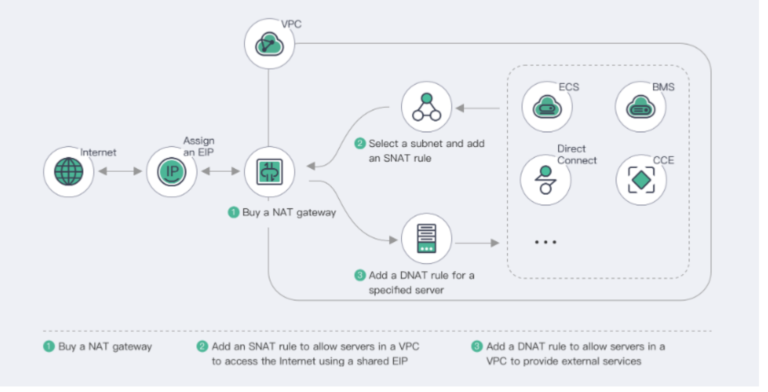
- A private NAT gateway provides NAT service for servers in a VPC, so that multiple servers can share a private IP address to access or provide services accessible from an on-premises data center or other VPCs.



- You can configure SNAT and DNAT rules for a NAT gateway to translate the source and destination IP addresses of originating packets into a transit IP address.
  - SNAT enables servers within one AZ or across AZs in a VPC to share a transit IP address to access on-premises data centers or other VPCs.
  - DNAT enables servers that share the same transit IP address in a VPC to provide services accessible from on-premises data centers or other VPCs.
- Transit subnet: A transit subnet is a transit network and is the subnet to which the transit IP address belongs.
- Transit IP Address: A transit IP address is a private IP address that can be assigned from a transit subnet. Cloud servers in your VPC can share a transit IP address to access on-premises networks or other VPCs.
- Transit VPC: A transit VPC is the VPC to which the transit subnet belongs.

Process for Buying a NAT Gateway

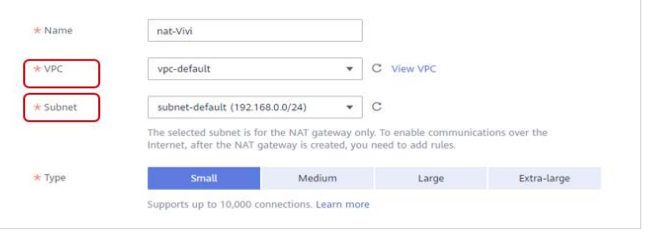
Public NAT gateway:



- Before you use public NAT gateway, buy an EIP.
- SNAT translates private IP addresses into EIPs, allowing servers in a VPC to share an EIP to access the Internet in a secure and efficient way.
- DNAT enables servers in a VPC to share an EIP to provide services accessible from the Internet through IP address mapping or port mapping.
- SNAT and DNAT rules are designed for different functions. If an SNAT rule and a DNAT rule use the same EIP, there may be service conflicts.
- An SNAT rule cannot share an EIP with a DNAT rule with **Port Type** set to **All ports**.

Buying a NAT Gateway

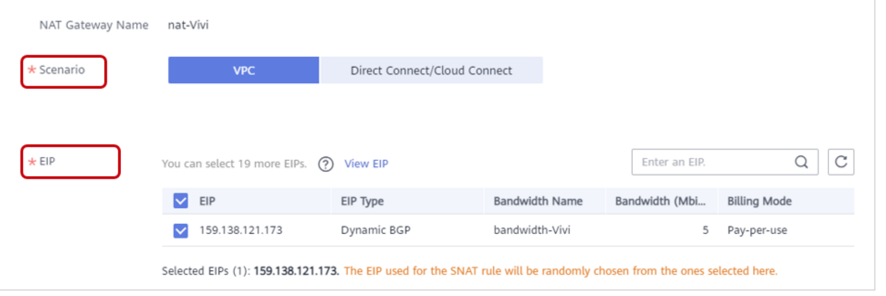
- When you buy a public NAT gateway, you must specify its VPC, subnet, and type.
- Check whether the default route (0.0.0.0/0) of the VPC is in use by any other gateways. If yes, add another route for the gateway you purchased or add the default route to a new route table that you will associate with the gateway.



- **Subnet**:
  - This is the subnet where the public NAT gateway is deployed.
  - The subnet must have at least one available IP address.
  - The selected subnet cannot be changed after the public NAT gateway is created.
- **Type**:
  - The type can be **Small**, **Medium**, **Large**, and **Extra-large**. You can click **Learn more** on the page to view details about each type.

SNAT Rule Configuration

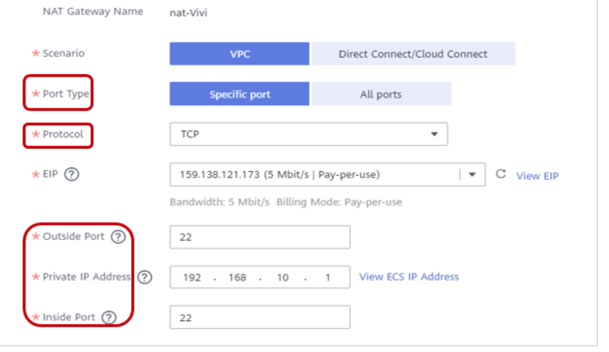
- If your servers are in a VPC and need to access the Internet, select VPC.
- If your on-premises servers access a VPC over a Direct Connect or VPN connection need to access the Internet, select Direct Connect/Cloud Connect.



- **Scenario**:
  - After the public NAT gateway is created, add SNAT rules to enable your cloud or on-premises servers to access the Internet by sharing an EIP.
  - Each SNAT rule is configured for one subnet. If there are multiple subnets in a VPC, you can create several SNAT rules to allow them to share EIPs.
- **Elastic IP**:
  - This is the EIP used for accessing the Internet.
  - You can select only an EIP that is not bound to any resource, an EIP that is bound to a DNAT rule whose **Port Type** is not set to **All ports**, or an EIP that is bound to an SNAT rule of the current NAT gateway.
  - You can select multiple EIPs at once. Up to 20 EIPs can be selected for each SNAT rule. If you have selected multiple EIPs for an SNAT rule, an EIP will be chosen from your selection at random.

DNAT Rule Configuration

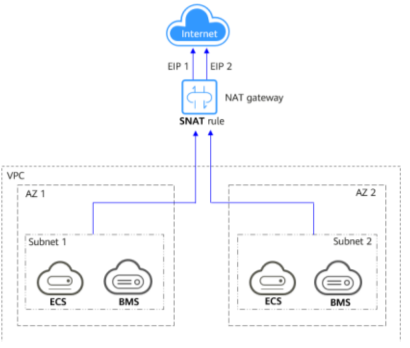
- VPC: A DNAT rule allows servers in a VPC to share an EIP and provide services accessible from the Internet.
- Direct Connect/Cloud Connect: A DNAT rule allows servers in an on-premises data center connected to a VPC through Direct Connect or Cloud Connect to provide services accessible from the Internet.



- **Scenario**:
  - After a public NAT gateway is created, you can add DNAT rules to allow servers in your VPC to provide services accessible from the Internet.
  - You can configure a DNAT rule for each port on a server. If multiple servers need to provide services accessible from the Internet, create multiple DNAT rules.
- **Outside Port**:
  - This is the port bound to the EIP. This parameter is available if you select **Specific port** for **Port Type**. Ports 1 to 65535 can all be selected.
- **Inside Port**:
  - This is the port of the server that provides services accessible from the Internet using the DNAT rule. This parameter is available if you select **Specific port** for **Port Type**. The value ranges from 1 to 65535.
- **Port Type**:
  - **Specific port**: The NAT gateway forwards requests to your servers only from the outside port and to the inside port configured here, and only if they use the right protocol.
  - **All ports**: This is effectively like having a regular EIP bound to your servers. Any requests received by the gateway will be forwarded to your servers, regardless of what port or protocol was used.

Application Scenario - Using SNAT to Access the Public Network (Public Network NAT)

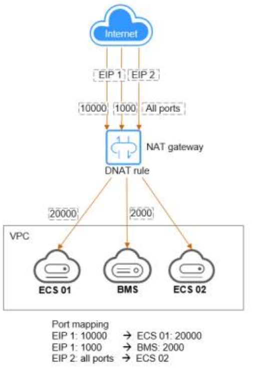
- When the ECSs in a VPC need to access the public network and a large number of requests are sent, the NAT gateway can provide different number of connections to save EIP resources and prevent the ECS IP addresses from being exposed to the public network. Based on the service plan, you can create multiple SNAT rules to share EIP resources.



- Application scenarios of other public network NAT:
  - If a large number of servers need to access the Internet securely, reliably, and at a high speed, or provide services for the Internet, the SNAT or DNAT function of the public network NAT gateway can be used.
  - In the IT system, the bound EIP may be blocked by attacks. To improve system reliability, you can add multiple EIPs when configuring an SNAT rule. When one EIP is blocked by an attack, services using other EIPs can be properly running. If multiple EIPs are bound to the SNAT rule, the system randomly selects an EIP to access the public network.
  - When the performance of a single gateway reaches the bottleneck, for example, the maximum number of connections supported by SNAT is 1 million or the maximum bandwidth conversion capability of 20 Gbit/s cannot meet service requirements, you are advised to use multiple gateways to expand the capacity horizontally. If you want to expand the capacity of multiple gateways, you only need to associate the routing table associated with the VPC subnet with the public network NAT gateway instance.

Application Scenario - Using DNAT to Provide Services for Cloud Hosts to the Public Network (Public Network NAT)

- When the cloud hosts in a VPC need to provide services for the public network, the DNAT function of the NAT gateway can be used.

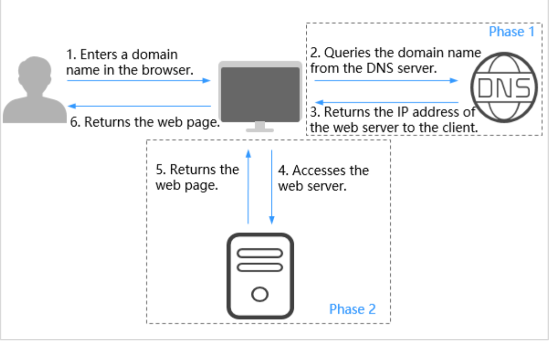


- If the DNAT function is bound to an EIP, the NAT gateway forwards the requests for accessing the EIP using the specified protocol and port to the specified port of the target ECS instance. You can also configure an EIP for the ECS through IP address mapping. Any request for accessing the EIP will be forwarded to the target ECS instance. This feature enables multiple ECSs to share EIPs and bandwidth, precisely controlling bandwidth resources.
- One DNAT rule is configured for a cloud host. If multiple cloud hosts need to provide services for the public network, you can configure multiple DNAT rules to share one or more EIP resources.

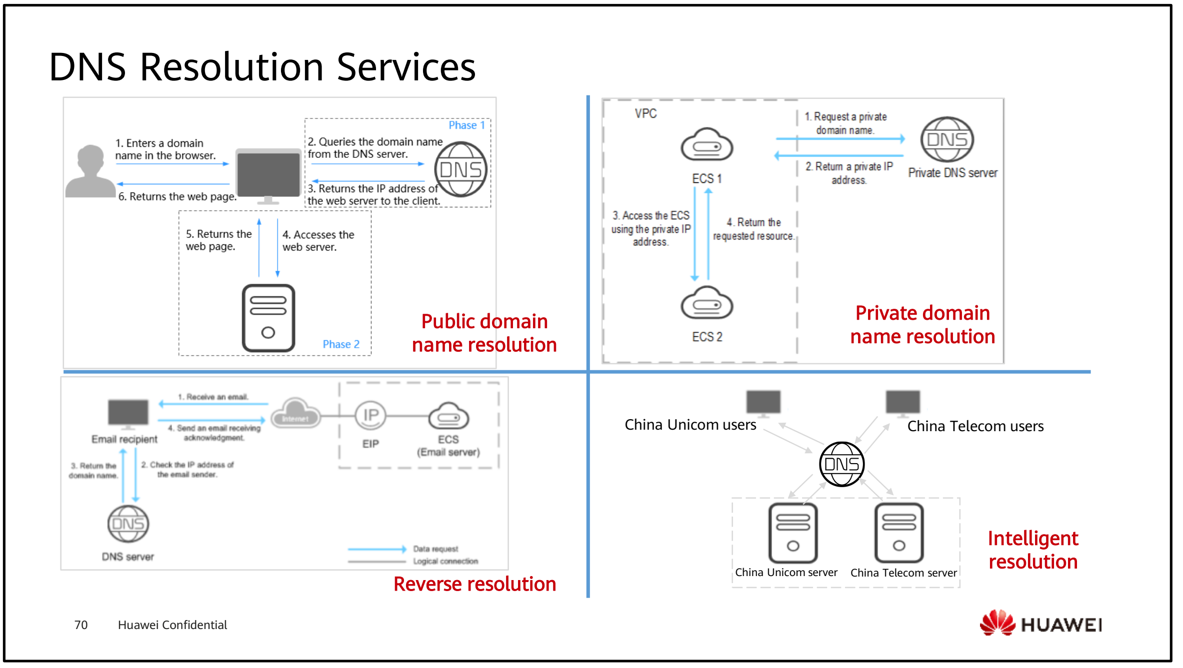
Contents

What Is DNS?

- Domain Name Service (DNS) provides highly available and scalable authoritative DNS services that translate domain names into IP addresses required for network connection, reliably directing end users to your applications.



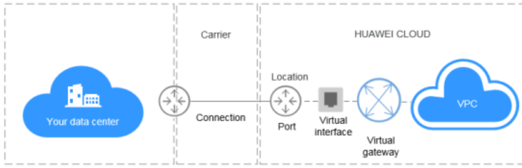
- DNS provides highly available and scalable authoritative DNS services that translate domain names (such as www.example.com) into IP addresses (such as 192.1.2.3) required for network connection, allowing users to visit your website or web application using your domain name.



- Public domain name resolution: DNS translates domain names like www.example.com to public IP addresses like 1.2.3.4, so that users can access your website or web application over the Internet by entering your domain name in the address box of their browser.
- Private domain name resolution: DNS translates domain names like ecs.com to private IP addresses like 192.168.1.1 that are used in associated VPCs. With private domain names, your ECSs can communicate with each other within the VPCs without having to connect to the Internet. You can also access cloud services, such as OBS and SMN, over a private network.
- Reverse resolution: DNS obtains a domain name based on an IP address. Reverse resolution, or reverse DNS lookup, is typically used to affirm the credibility of email servers. After a recipient server receives an email, it checks whether the IP address and domain name of the sender server are trustworthy and determines whether the email is spam. If the recipient server cannot obtain the domain name mapped to the IP address of the sender server, it concludes that the email was sent by a malicious host and rejects it. It is necessary to configure pointer records (PTR) to point the IP addresses of your email servers to domain names. If no PTR records are configured, the recipient server will treat emails from the email server as spam or malicious and discard them. If you want to build an email server, it is necessary to configure a PTR record to map the email server's IP address to your domain name.
- Intelligent resolution: DNS allows you to configure resolution lines. With these resolution lines, you can specify the DNS server that returns different resolution results for the same domain name based on the networks or geographic locations of visitors' IP addresses. For example, if the visitor is a China Unicom user, the DNS server will return an IP address of China Unicom. With this function, you can improve DNS resolution efficiency and speed up cross-network access. You can also create more fine-grained resolution lines based on source IP addresses.

What are DC and CC?

- Direct Connect allows you to establish a stable, high-speed, low-latency, secure dedicated network connection that connects your on-premises data center to Huawei Cloud. Direct Connect allows you to maximize legacy IT facilities and leverage cloud services to build a flexible, scalable hybrid cloud computing environment.
- Cloud Connect allows you to connect Virtual Private Clouds (VPCs) in different regions to allow instances in these VPCs to communicate over a private network as if they were within the same network.

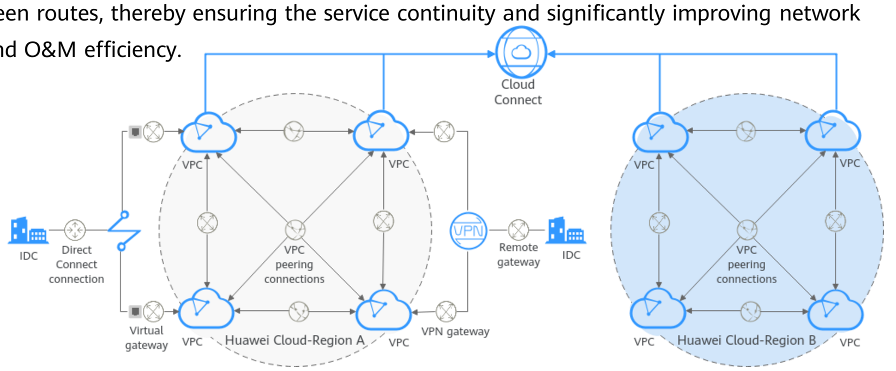


- In Direct Connect, a connection is an abstraction between a user's local data center and a VPC on the cloud. A virtual interface is the entry for the local data center to access the VPC. After the VGW associates the virtual interface with the VPC, the local data center can access the VPC. VGW is the access router of Direct Connect.
- Differences Between Direct Connect and VPN
  - Cost: VPN costs less than private lines and is easy to provision.
  - Public network quality; VPN has high latency, lower security than cloud private lines, and lower service stability.
  - Provisioning period: Direct Connect requires longer deployment period than VPN because it is limited by the deployment of Direct Connect and the line resources of the carrier. In this case, there is an order of magnitude. If both parties have Internet resources, the VPN is basically ready-to-use. After the configuration and negotiation, the two parties can communicate with each other.

- Application scenarios of the cloud connection:
  - Cross-region VPC private network communication: When VPCs in multiple regions on the cloud need to communicate with each other over a private network, Cloud Connect can easily connect multiple VPCs across regions based on your network plan, improving network topology flexibility. It also provides secure and reliable private network communication for users.
  - Interconnection between multiple data centers and VPCs in multiple regions: When multiple local data centers need to communicate with VPCs in multiple regions on the cloud, you can connect the local data centers to the VPCs on the cloud through Direct Connect. Then, the cloud connection is used to load the VPCs that need to communicate with each other and the VGWs connected to the data center to implement private network communication between the local data center and VPCs in multiple regions, implementing multi-point full-frequency communication.

What's an Enterprise Router?

- An enterprise router (ER) connects virtual private clouds (VPCs) and on-premises networks to build a central hub network. It has high specifications, provides high bandwidth, and delivers high performance. Enterprise routers use the Border Gateway Protocol (BGP) to learn, dynamically select, or switch between routes, thereby ensuring the service continuity and significantly improving network scalability and O&M efficiency.



- Enterprise routers have the following advantages:
- **High Performance:** Enterprise routers use exclusive resources and are deployed in clusters to deliver the highest possible performance for workloads on large-scale networks.
- **High Availability:** Enterprise routers can be deployed in multiple availability zones to work in active-active or multi-active mode, thereby ensuring service continuity and real-time seamless switchovers.
- **Simplified Management:** Enterprise routers can connect to multiple VPCs, Direct Connect connections, or enterprise routers in different regions and route traffic among them. The network topology is simpler and the network is easier to manage and maintain. For cross-VPC communications, you only need to maintain the route tables on the VPCs without requiring so many VPC peering connections. For communications between VPCs and an on-premises data center, multiple VPCs can connect to an enterprise router and then communicate with the data center over a shared Direct Connect or VPN connection. You do not need to establish Direct Connect or VPN connections between the data center and each of the VPCs. Enterprise routers can automatically learn, update, and synchronize routes, eliminating the need to manually configure or update routes whenever the network topology changes.
- **Seamless Failover Between Lines:** Enterprise routers use the Border Gateway Protocol (BGP) to select the best path from multiple lines working in load-sharing or active/standby mode. If a single line fails, services can be failed over to another functioning line within seconds to ensure service continuity.

Quiz

1. (Single choice) Which of the following is not a component of ELB?
   A. Backend server group
   B. Listener
   C. Load balancer
   D. NAT Gateway
2. (Single choice) Can resources in a subnet of one VPC communicate with those in a subnet of another VPC in the same region?
   A. Yes
   B. No
   C. Yes, they can communicate with each other by default
   D. Yes, but VPN is required

- D
- A

Summary

- This chapter described basic network knowledge and common network cloud services. After completing this course, you will be able to understand the functions of networks as well as how network cloud services work and where you can use these services. For example, a VPC is like the internal network used by an enterprise, and applications can provide Internet-accessible services using EIPs. Mastering these concepts can help you better prepare for cloud migration of legacy systems.

Recommendations

- Huawei Talent
  - https://e.huawei.com/en/talent/cert/#/careerCert
- Huawei Technical Support Website
  - https://support.huaweicloud.com/intl/en-us/help-novicedocument.html
- HUAWEI CLOUD Academy
  - https://edu.huaweicloud.com/intl/en-us/

Acronyms and Abbreviations

- ACL: access control list
- AS: autonomous system
- BGP: Border Gateway Protocol
- CC: Cloud Connect
- DHCP: Dynamic Host Configuration Protocol
- DNAT: destination network address translation
- DNS: Domain Name System/Domain Name Service
- ECS: Elastic Cloud Server

Acronyms and Abbreviations

- EIP: Elastic IP
- ELB: Elastic Load Balance
- HTTP: Hypertext Transfer Protocol
- HTTPS: Hypertext Transfer Protocol Secure
- ICT: information and communications technology
- IDC: Internet data center
- IPsec: IP security
- NAT: network address translation

Acronyms and Abbreviations

- SNAT: source network address translation
- TCP: Transmission Control Protocol
- UDP: User Datagram Protocol
- VPC: Virtual Private Cloud
- VPCEP: VPC Endpoint
- VPN: Virtual Private Network
- Web: World Wide Web (WWW)

Thank Users.

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每个组织，构建万物互联的智能世界。

Bring digital to every person, home, and organization for a fully connected, intelligent world.

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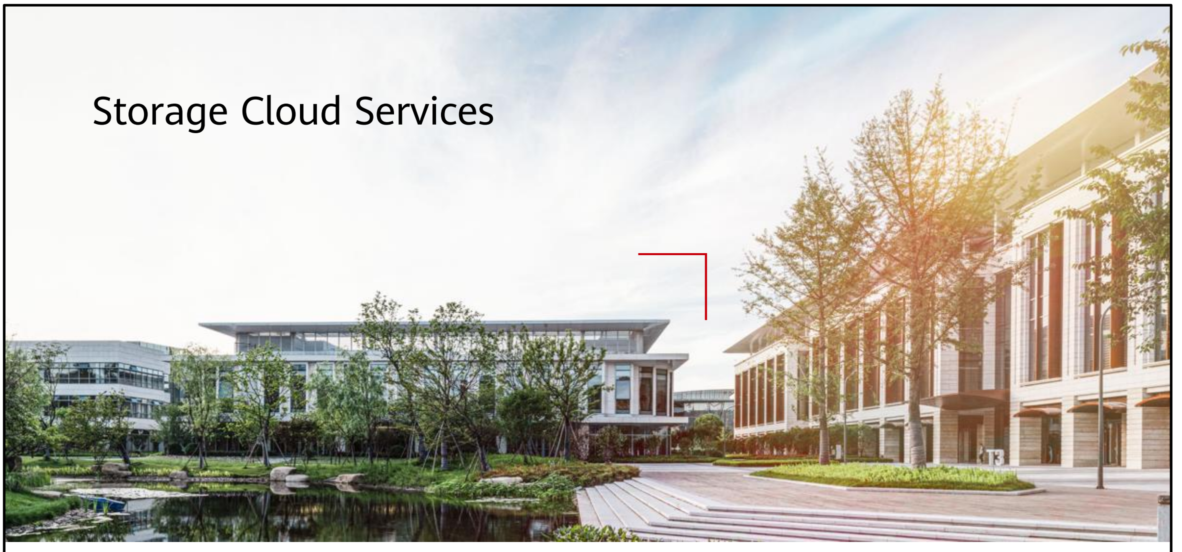
Storage Cloud Services



Foreword

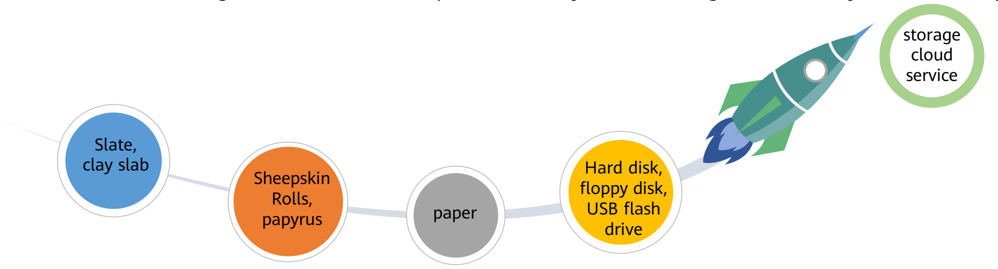
- Data is everywhere. We use USB flash drives and cloud disks to store data, and these devices are called storage devices. That is enough for most of us, but what do you use if you are an enterprise? In today's age of cloud computing, what are the most common storage cloud services?
- In this section, we will cover some common storage services on HUAWEI CLOUD.

Objectives

- On completion of this course, you will be able to:
  - Acquire a basic understanding of cloud storage.
  - Understand the principles behind and uses of common storage services on HUAWEI CLOUD.

What is a storage cloud service?

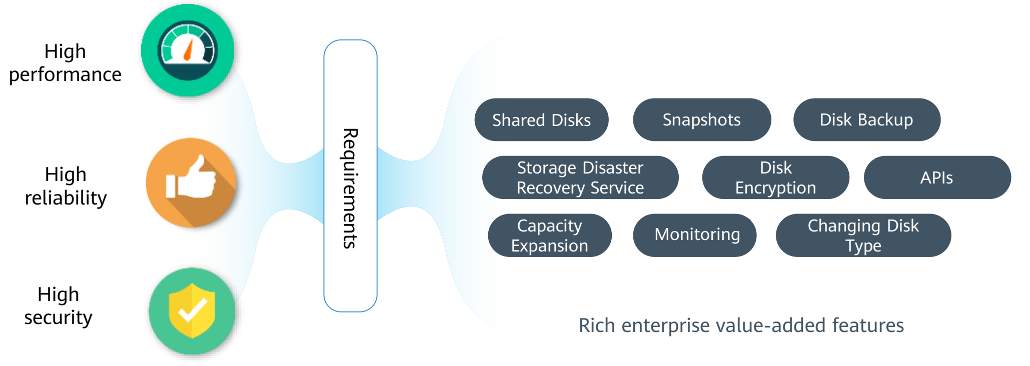
- Storage refers to the process of storing data or information in a computer or other electronic device. Storage can be temporary or long-term, such as keeping files on hard drives or cloud storage.
- Storage Services is a service that stores data on the Internet. It allows users to upload data to remote servers over the Internet, and access and manage that data anytime, anywhere. Cloud storage is usually provided by cloud service providers and can be used by users through subscription services. The benefits of cloud storage include data backup and recovery, data sharing, data security, and reliability.



- Storage is an important part of a computer system. It can not only store data, but also implement disaster recovery through storage.

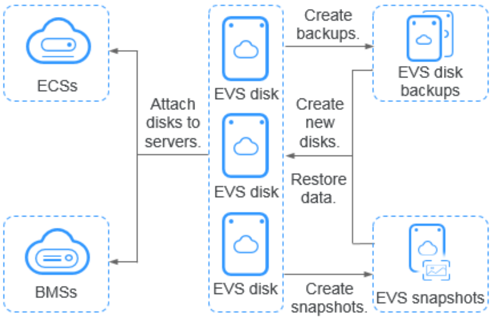
Contents

Enterprise requirements for block storage



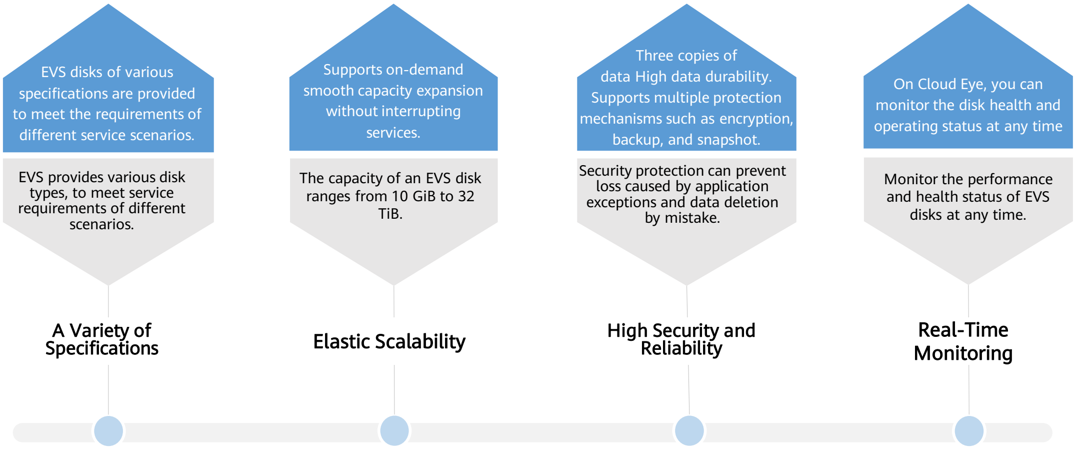
What Is EVS

- Elastic Volume Service (EVS) offers scalable block storage for cloud servers. With high reliability, high performance, and a variety of specifications, EVS disks can be used for distributed file systems, development and test environments, data warehouses, and high-performance computing (HPC) applications. Cloud servers that EVS supports include Elastic Cloud Servers (ECSs) and Bare Metal Servers (BMSs).



- An EVS disk is similar to a hard disk on a PC. It must be attached to an ECS and cannot be used independently. You can initialize attached EVS disks, create file systems, and store data persistently on EVS disks.

EVS Features and Benefits



Disk Types and Performance

Metric	Extreme SSD	General Purpose SSD V2	Ultra-high I/O	General Purpose SSD	High I/O
Max. capacity (GiB)	System disk: 1,024 Data disk: 32,768				
Max. IOPS （reference）	128000	128,000	50,000	20000	5000
Max. Throughput （reference）	1000 MiB/s	1,000 MiB/s	350 MiB/s	250 MiB/s	150 MiB/s
Single-queue access latencyd （reference）	Sub-millisecond	1 ms	1 ms	1 ms	1 ms ~ 3 ms
Typical workloads	Database workloads Oracle、SQL Server、ClickHouse、AI workloads	Enterprise OA and virtual desktops、Large-scale development and testing、Transcoding services、System disks	Transcoding services、I/O-intensive workloads NoSQL、Oracle、SQL Server、PostgreSQL	Enterprise OA、Medium-scale development and test environments、Small- and medium-sized databases、Web applications、System disks	Common development and test environments

IOPS: Number of read/write operations performed by an EVS disk per second.

Throughput: Amount of data read from and written into an EVS disk per second.

Read/write I/O latenc: Minimum interval between two consecutive read/write operations on an EVS disk

EVS Device Types

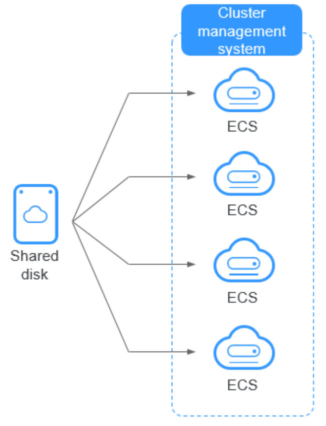
- There are two EVS device types: Virtual Block Device (VBD) and Small Computer System Interface (SCSI).

VBD (Virtual Block Device)	• VBD is the default EVS device type. • VBD EVS disks support only basic read/write SCSI commands.
SCSI (Small Computer System Interface)	• SCSI EVS disks support transparent SCSI command transmission and the cloud server OS can directly access the underlying storage media. • Besides basic read/write SCSI commands, SCSI EVS disks also support more advanced SCSI commands.

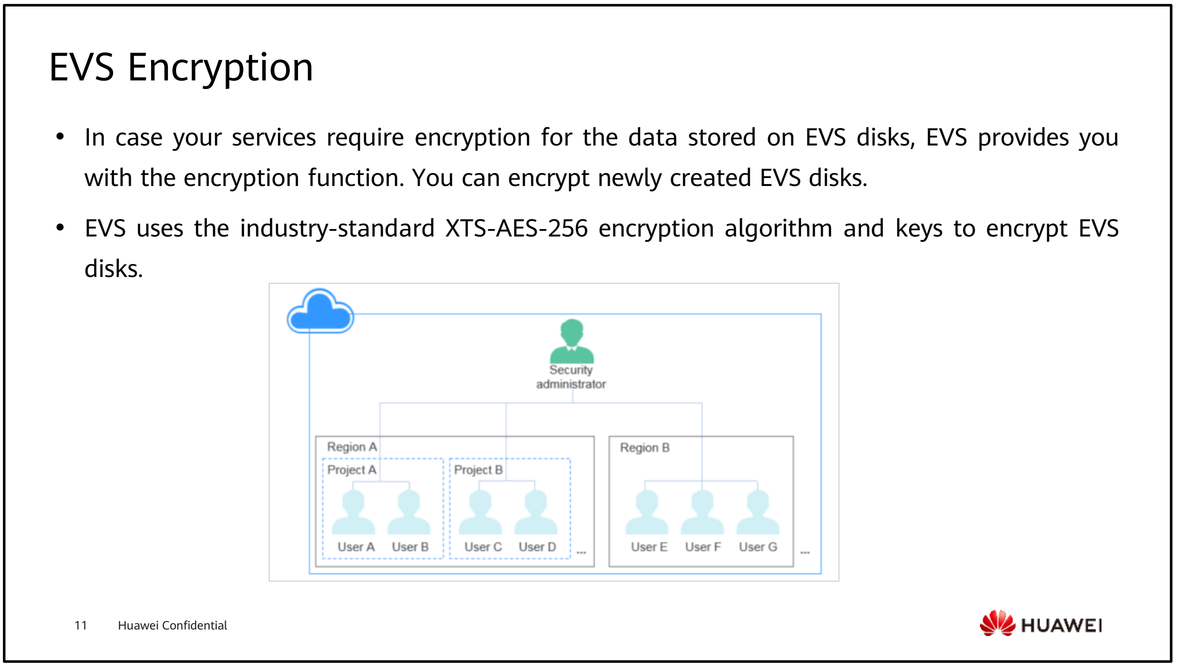
- The disk mode is configured when a disk is purchased and cannot be changed after the disk is purchased.

Shared EVS Disks

- Shared EVS disks are block storage devices that support concurrent read/write operations and can be attached to multiple servers. Shared EVS disks feature multiple attachments, high-concurrency, high-performance, and high-reliability. They are usually used for enterprise business-critical applications that require cluster deployment for high availability (HA). Multiple servers can access the same shared EVS disk at the same time.
- You can create shared EVS disks with device type VBD and SCSI.



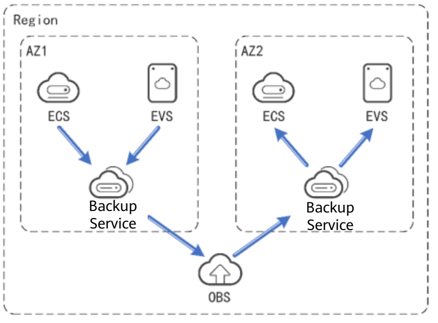
- You must set up a shared file system or cluster management system before using shared EVS disks. If you directly attach a disk to multiple servers, the sharing function will not work and data may be overwritten.
- A shared EVS disk can be attached to a maximum of 16 servers. Servers that EVS supports include ECSs and BMSs. To share files, you need to deploy a shared file system or a cluster management system, such as Windows MSCS, Veritas VCS, or CFS.



- A security administrator (with the Security Administrator permission) can directly authorize EVS to access Key Management Service (KMS) and use the encryption function.
- When an encrypted disk is mounted, EVS accesses KMS and KMS sends the data key (DK) to the memory of the host machine for storage. EVS encrypted disks use the DK plaintext in the memory of the host machine to encrypt and decrypt disk I/Os. The DK plaintext is used only in the memory of the host machine where the ECS instance is located and is not stored in plaintext on the media.
- For a tenant, common users in the same region can directly use the encryption function as long as the security administrator successfully authorizes EVS to access KMS.
- When a common user (without the Security Administrator permission) uses the encryption function, whether the common user is the first user in the current region or project to use the encryption function is classified as follows:
  - If yes, the common user is the first user in the current region or project to use the encryption function. You need to contact the security administrator for authorization before using the encryption function.
  - No: indicates that other users in the region or project have used the encryption function. The common user can use the encryption function directly.

EVS Backup

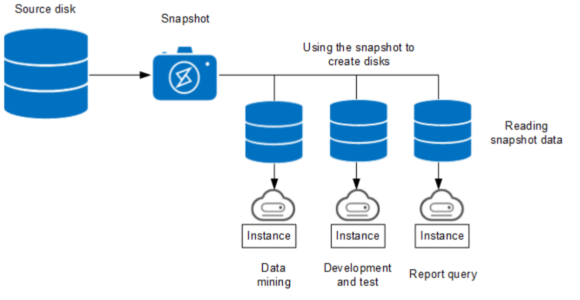
- If the data in an EVS disk is important, you can use the VBS backup function to back up the existing data.
- Cloud Disk Backup supports online backup without stopping the ECS. In addition, data can be restored using a backup copy at any time to ensure data correctness and security.



- Configure a backup policy to automatically back up EVS disk data based on the backup policy. Periodic backups are used as baseline data for creating EVS disks or restoring data to EVS disks.
- VBS backup data can be shared between users. You can use the shared backup data to create EVS disks.

EVS Snapshot

- An EVS snapshot is a complete copy or image of the disk data taken at a specific point in time. They are used for disaster recovery. If anything happens, you can completely restore the disk data to the state from when the snapshot was taken.
- You can create snapshots to rapidly save the disk data at specified time points. In addition, you can use snapshots to create new disks so that the created disks will contain the snapshot data in the beginning.



- The snapshot function helps address your following needs:
  - Routine data backup:You can create snapshots for disks on a timely basis and use snapshots to recover your data in case that data loss or data inconsistency occurred due to unintended operations, viruses, or attacks.
  - Rapid data restoration:You can create a snapshot or multiple snapshots before an application software upgrade or a service data migration. If an exception occurs during the upgrade or migration, service data can be rapidly restored to the time point when the snapshot was created.

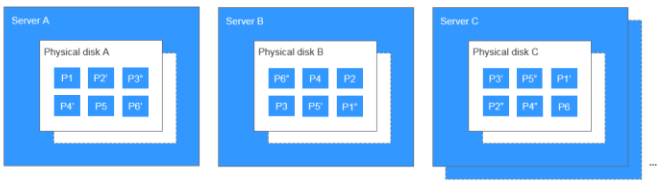
Differences Between EVS Backups and EVS Snapshots

- Both EVS backups and EVS snapshots provide redundancies for improved disk data reliability.

Metric	Storage Solution	Data Synchronization	DR Range	Service Recovery
Backup	Backups are stored in OBS, instead of disks. This ensures data restoration upon disk damage or corruption.	A backup is a copy of a disk taken at a given point of time and is stored in a different location. Automatic backup can be performed based on backup policies. Deleting a disk will not delete its backups.	A backup and its source disk reside in different AZs.	To restore data and recover services, you can restore the backups to their original disks or create new disks from the backups.
Snapshot	Snapshots are stored on the same disk as the original data.	A snapshot is the state of a disk at a specific point in time and is stored on the same disk. If the disk is deleted, all its snapshots will also be deleted.	A snapshot and its source disk reside in the same AZ.	You can use a snapshot to roll back its original disk or create a disk from the snapshot.

EVS Three-Copy Redundancy

- The backend storage system of EVS employs three-copy redundancy to guarantee data reliability. With this mechanism, one piece of data is by default divided into multiple 1 MiB data blocks. Each data block is saved in three copies, and these copies are stored on different nodes in the system according to the distributed algorithms.
- Three-copy redundancy has the following characteristics:
  - The storage system saves the data copies on different disks of different servers across cabinets, ensuring that services are not interrupted if a physical device fails.
  - The storage system guarantees strong consistency between the data copies.



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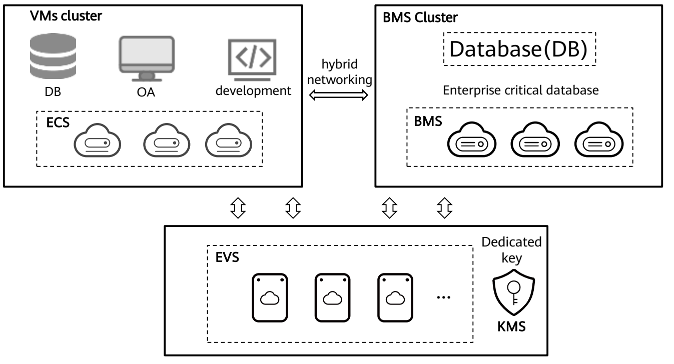
- The storage system ensures strong data consistency between the three copies of data: for example, for block P1 on physical disk A of server A, The system backs up its data as P1' on physical disk B of server B and P1' on physical disk C of server C. P1, P1', and P1' form three copies of the same data block. If the physical disk where P1 is located is faulty, P1' and P1'' can continue to provide storage services to ensure that services are not affected.
- When the storage system detects a hardware (server or physical disk) fault, it automatically starts data repair. Because copies of data blocks are stored on different nodes, data reconstruction is started on different nodes at the same time during data recovery. Only a small part of data needs to be reconstructed on each node, and multiple nodes work concurrently, thereby effectively avoiding a performance bottleneck generated when a single node rebuilds a large amount of data. Minimize the impact on upper-layer services.

EVS Application Scenario

- EVS disk encryption is used to improve Huawei's business security compliance.
  - The VM cluster runs development and testing, OA, and database services. A bare metal cluster runs a critical enterprise database.



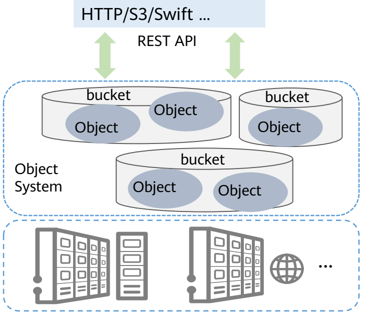
Contents

What Is Object storage

- OBS is an object-based massive storage service. It provides data storage capabilities that are easy to expand, secure, reliable, and cost-effective.
- OBS is a service oriented to Internet access. It provides HTTP/HTTPS-based web service interfaces for users to access the Internet anytime and anywhere.

Features:

- Flattened structure and data isolation between tenants
- Users can create buckets (like folders), upload or download objects, and share data by forwarding links.



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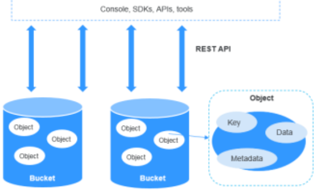
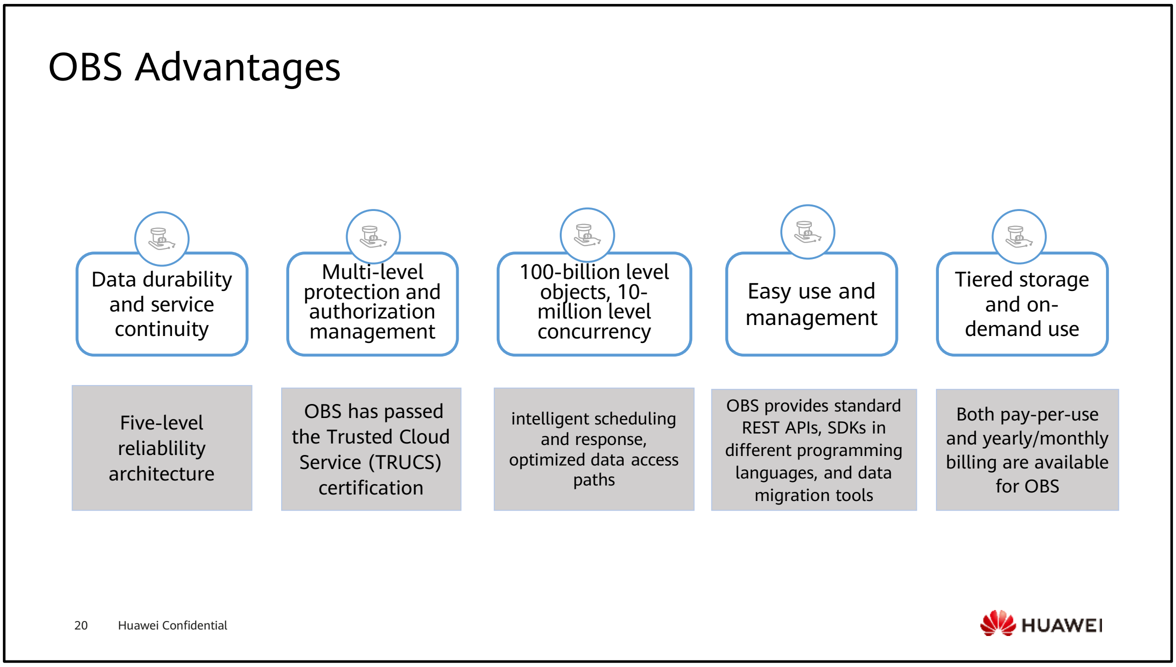
- It is suitable for storing files of any type. It is generally used in large-scale data storage scenarios, such as massive content on the Internet. (Videos, pictures, photos, books, audio-visual, magazines, etc.) , web disk, digital media, backup, and archiving.

Object Storage Service

- Object Storage Service (OBS) is a scalable service that provides secure, reliable, and cost-effective cloud storage for massive amounts of data.
- OBS basically consists of buckets and objects.
    - Buckets are containers for storing objects. OBS provides flat storage in the form of buckets and objects. Unlike the conventional multi-layer directory structure of file systems, all objects in a bucket are stored at the same logical layer..
    - Objects are basic units stored in OBS. An object contains both data and the metadata that describes data attributes. Data uploaded to OBS is stored in buckets as objects.



- An object consists of the following:
    - A key that specifies the name of an object. An object key is a UTF-8 string up to 1,024 characters long. Each object is uniquely identified by a key within a bucket.
    - Metadata that describes an object. The metadata is a set of key-value pairs that are assigned to objects stored in OBS. There are two types of metadata: system-defined metadata and custom metadata.
        - System-defined metadata is automatically assigned by OBS for processing objects. Such metadata includes Date, Content-Length, Last-Modified, ETag, and more.
        - You can specify custom metadata to describe the object when you upload an object to OBS.
    - Data that refers to the content of an object.



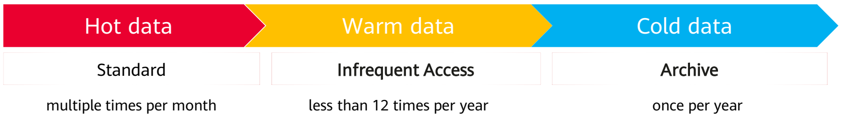
- Level5: region--Cross-region replication,Level4: data center--Multi-AZ,Level3: cabinets--Cabinet redundancy,Level2: servers--Erasure Code,Level1: storage media--Detection of slow disks and bad sectors.
- Measures, including versioning, server-side encryption, URL validation, virtual private cloud (VPC)-based network isolation, access log audit, and fine-grained access control are provided to keep data secure and trusted.
- Measures, including versioning, server-side encryption, URL validation, virtual private cloud (VPC)-based network isolation, access log audit, and fine-grained access control are provided to keep data secure and trusted.
- Online upgrade, online capacity expansion, and upgrade capacity expansion are supported without customer perception.
- Supports independent metering and billing of standard, infrequent access, and archived, reducing storage costs.

OBS Access Modes

Compared Item	Description	How to Use
OBS Console	OBS Console is a web-based GUI for you to easily manage OBS resources	Accessing the Cloud Service Console Using a Web Page
OBS Browser+	OBS Browser+ is a Windows client that lets you easily manage OBS resources from your desktop	Download OBS Browser+ and log in using the access key (AK/SK) or account and password
API	OBS offers the REST API for you to access it from web applications with ease. By making API calls, you can upload and download data anytime, anywhere over the Internet	Use the key (AK/SK) to perform operations
SDK	OBS SDKs encapsulate the REST API provided by OBS to simplify development. You can call API functions provided by the OBS SDKs to enjoy OBS capabilities	You can set the access key (AK/SK) and directly invoke the API functions provided by the SDK
obsutil	obsutil is a command line tool for you to perform common configuration and management operations on OBS. If you are comfortable using the command line interface (CLI), obsutil is recommended for batch processing and automated tasks	Download the obsutil tool, configure the server address, and use the access key (AK/SK) for identity authentication
obsfs	obsfs is an OBS tool based on Filesystem in Userspace (FUSE). It helps you mount parallel file systems to Linux, so that you can easily access virtually unlimited storage space of OBS the same way as you would use a regular local file system	Download the obsfs tool, configure the server address, and use the access key (AK/SK) for identity authentication

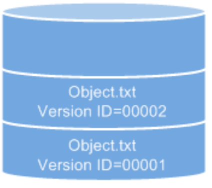
Storage Classes

- OBS offers the storage classes below to meet your requirements for storage performance and cost:
  - Standard: The Standard storage class features low latency and high throughput. It is therefore good for storing frequently (multiple times per month) accessed files or small files (less than 1 MB). Its application scenarios include big data analytics, mobile apps, hot videos, and social apps.
  - Infrequent Access: The Infrequent Access storage class is for storing data that is infrequently (less than 12 times per year) accessed, but when needed, the access has to be fast. It can be used for file synchronization, file sharing and many other scenarios. This storage class has the same durability, low latency, and high throughput as the Standard storage class, with a lower cost, but its availability is slightly lower than the Standard storage class.
  - Archive: The Archive storage class is ideal for storing data that is rarely (once per year) accessed. Its application scenarios include data archive and long-term backups. This storage class is secure, durable, and inexpensive, so it can be used to replace tape libraries. To keep cost low, it may take hours to restore data from the Archive storage class



Versioning

- OBS can store multiple versions of an object. You can quickly search for and restore different versions or restore data in the event of accidental deletions or application faults.
- By default, versioning is disabled for new OBS buckets. New objects will overwrite existing objects in case they have the same names.

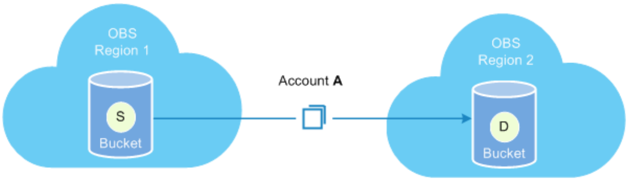


Versioning enabled

With versioning enabled, OBS automatically allocates a unique version ID to a newly uploaded object.
When an object with the same name as an existing object is uploaded again, both objects are stored in OBS with the same name but different version IDs.

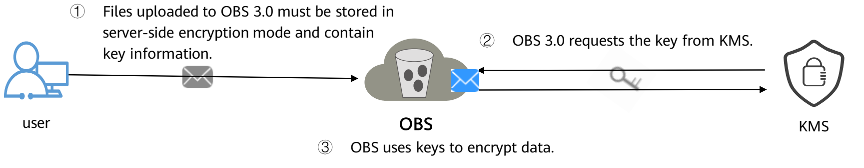
Cross-Region Replication

- Cross-region replication provides the capability for disaster recovery across regions, allowing you to set up a remote backup solution.
- Cross-region replication refers to the process of automatically and asynchronously replicating data from a bucket (source bucket) to another bucket (destination bucket) across regions by creating a cross-region replication rule. The source bucket and destination bucket must belong to the same account. Replication across accounts is not supported.



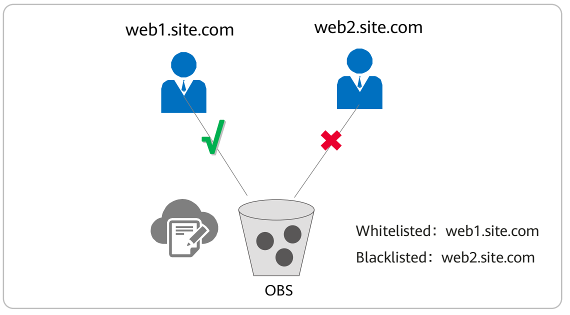
Encryption

- OBS allows you to configure default encryption for a bucket. After the configuration, objects uploaded to this bucket are automatically encrypted using the specified key, making data storage more secure.
- You can choose SSE-KMS or SSE-OBS for encryption when creating a bucket (see Creating a Bucket). You can also enable or disable encryption for a bucket after it is created.

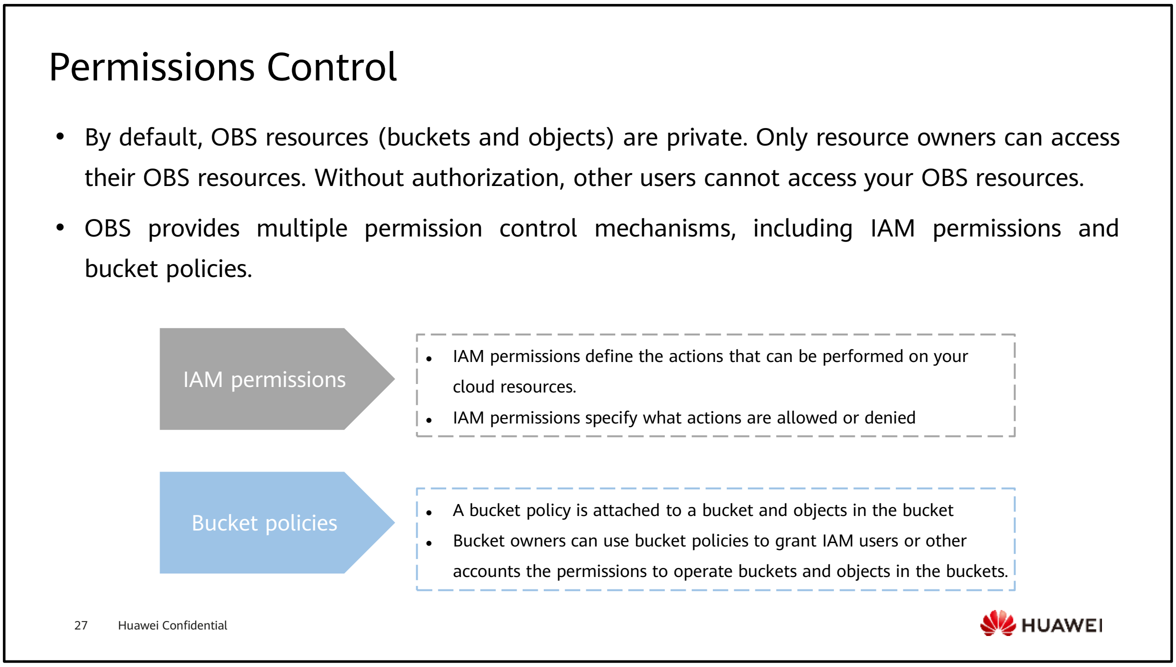


URL Validation

- Some rogue sites may steal links from other sites to enrich their content without any costs. Link stealing hurts the interests of the original websites and it is also a strain on their servers. OBS provides URL validation to solve this problem.
- Such authorization is controlled using a whitelist and a blacklist.



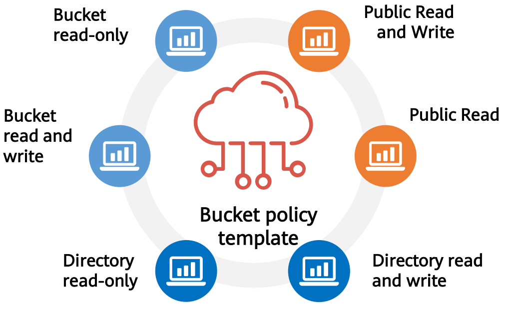
- In HTTP, the Referer field allows websites and web servers to identify where people are visiting them from. URL validation of OBS utilizes this Referer field. The idea is that once you find that a request to your resource is not originated from an authorized source (for example, a URL), you can have the request blocked or redirected to a specific web page. This way, OBS prevents unauthorized access to data stored in buckets.



- OBS permission control means to grant permissions to other accounts or IAM users by editing access policies. For example, if you have a bucket, you can authorize another IAM user to upload objects to your bucket. You can also open buckets to the public, so that anyone can access your buckets over the Internet. OBS offers multiple methods to help you to assign resource permissions to others. Resource owners can formulate different permissions control policies based on service requirements to ensure data security.
- Scenario:
  - IAM permissions: Controlling access to cloud resources as a whole under an account; Controlling access to all OBS buckets and objects under an account.
  - Bucket policies: Granting other Huawei Cloud accounts the permissions to access OBS resources; Configuring bucket policies to grant IAM users various access permissions to different buckets.

Bucket policies

- A bucket owner can configure a bucket policy to manage access to the bucket.
- Bucket policies centrally control access to buckets and objects based on a variety of request elements, such as actions, principals, resources, and others (like IP addresses).

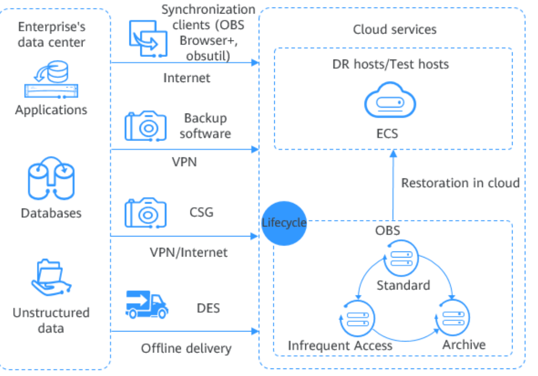


- If the resource is set to *, the permission applies to all objects in a bucket. For example, an account can create a policy to:
    - Grant users the write permission for a specific bucket.
    - Grant users in a specific network the write permission.

Scenario: Backup and Archiving

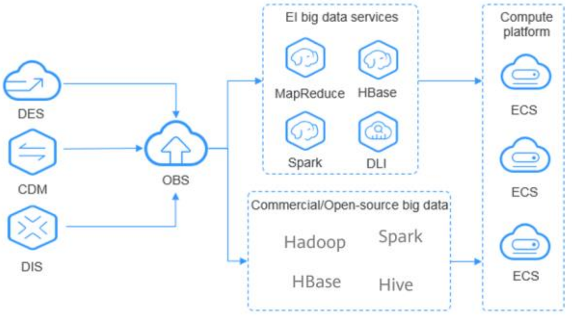
- OBS offers a highly reliable, inexpensive storage system featuring high concurrency and low latency. It can hold massive amounts of data, meeting the archive needs for unstructured data of applications and databases.

You can use the synchronization clients (such as OBS Browser+ and obsutil), Cloud Storage Gateway (CSG), DES, or mainstream backup software to back up your on-premises data to OBS.
OBS also provides lifecycle rules to automatically transition objects between storage classes to save your money on storage. You can restore data from OBS to a DR or test host on the cloud.



Scenario: Big Data Analytics

- OBS enables inexpensive big data solutions that feature high performance with zero service interruptions. It eliminates the need for capacity expansion. Such solutions are designed for scenarios that involve mass data storage and analysis, query of historical data details, analysis of numerous behavior logs, and statistical analysis of public transactions.

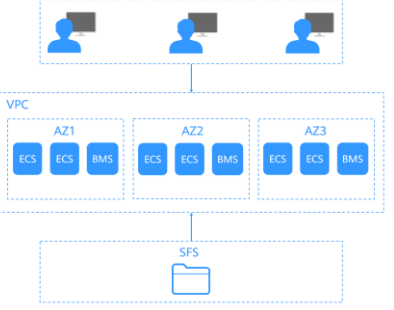


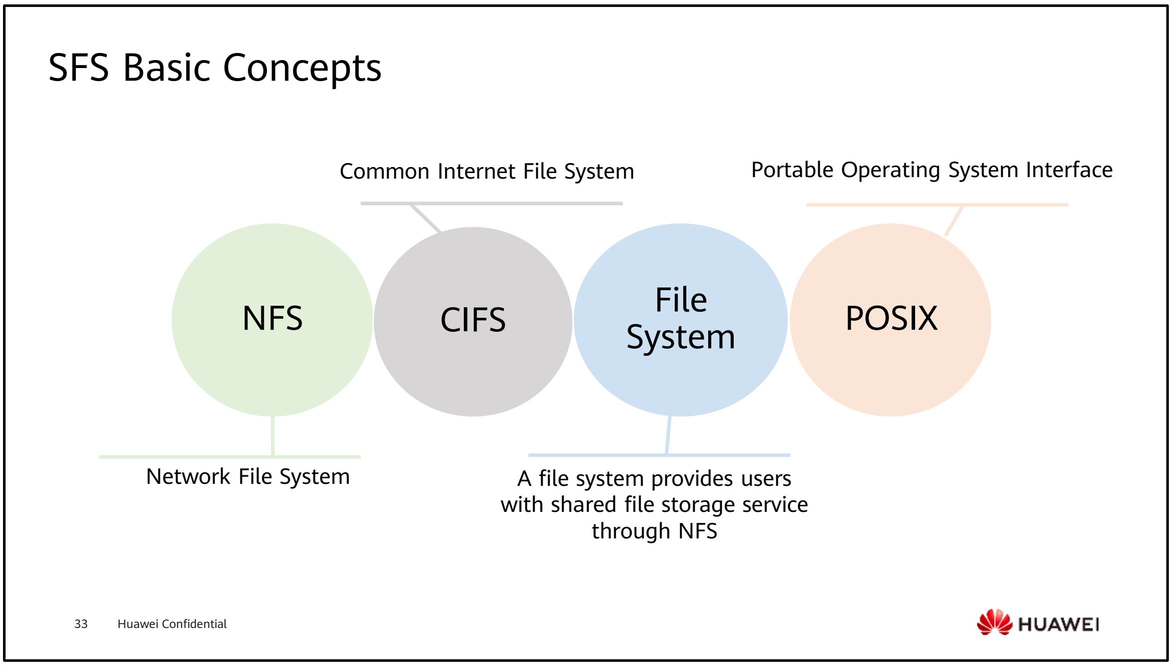
- You can migrate data to OBS with Data Express Service (DES), and then use Huawei Cloud big data services like MapReduce Service (MRS) or open-source computing frameworks such as Hadoop and Spark to analyze data stored in OBS. Such analysis results will be returned to your programs or applications on Elastic Cloud Servers (ECSs).

Contents

What Is SFS

- Scalable File Service (SFS) provides scalable, high-performance (NAS) file storage. With SFS, you can enjoy shared file access spanning multiple Elastic Cloud Servers (ECSs), Bare Metal Servers (BMSs), and containers created on Cloud Container Engine (CCE).
- You can access SFS on the management console or via APIs by sending HTTPS requests.

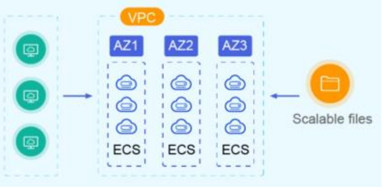




- NFS: Network File System (NFS) is a distributed file system protocol that allows different computers and operating systems to share data over a network. After the NFS client is installed on multiple ECSs, mount the file system to implement file sharing between ECSs. The NFS protocol is recommended for Linux clients.
- CIFS: Common Internet File System (CIFS) is a protocol used for network file access. Using the CIFS protocol, network files can be shared between hosts running Windows. The CIFS protocol is recommended for Windows clients.
- File System: A file system provides users with shared file storage service through NFS. It is used for accessing network files remotely. After a user creates a mount point on the management console, the file system can be mounted to multiple servers and is accessible through the standard POSIX.
- POSIX: Portable Operating System Interface (POSIX) is a set of interrelated standards specified by Institute of Electrical and Electronics Engineers (IEEE) to define the application programming interface (API) for software compatible with variants of the UNIX operating system. POSIX is intended to achieve software portability at the source code level.

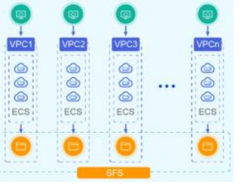
SFS functions

- Compatible with NFSv3, SFS meets your demands in various system environments.



- ECS instances can access file shares among AZs within the same region.

- Storage can be scaled up or down on demand to dynamically adapt to service changes without interrupting application services.



- A file system can be exclusively shared by ECSs within a designated VPC.

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SFS advantages

- Compared with traditional file sharing storage, SFS has the following advantages:

Elastic scaling

Storage can be scaled up or down on demand to dynamically adapt to service changes without interrupting applications.

Superior performance and reliability

The service enables file system performance to increase as capacity grows, and delivers a high data durability to support rapid service growth.

File sharing

Servers in multiple availability zones (AZs) of a same region can access the same file system concurrently and share files.

Seamless integration

SFS supports Network File System (NFS). With this standard protocol, a broad range of mainstream applications can read and write data in the file system.

Easy operation and low costs

In an intuitive graphical user interface (GUI), you can create and manage file systems with ease. SFS slashes the cost as it is charged on a pay-per-use basis.

File System Types

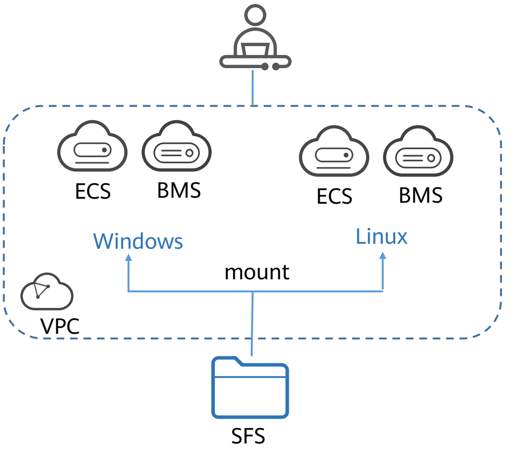
- SFS provides three types of file systems: SFS Capacity-Oriented, General Purpose File System and SFS Turbo.

Type	description	Highlight	Application Scenario	Remarks
SFS Capacity-Oriented	Powerful support for massive data and high-bandwidth applications	Large capacity, high bandwidth, and low cost	Cost-sensitive workloads which require large-capacity scalability, such as media processing, file sharing, HPC, and data backup	Not applicable to latency-sensitive services, such as massive small files and random small I/Os. Only internal network access is supported. Public network access is not supported. SFS Capacity-Oriented file systems is not support CBR
General Purpose File System	Cost-sensitive workloads which require large-capacity scalability, such as media processing, file sharing and data backup	Large capacity, high bandwidth, and low cost	Cost-sensitive workloads which require large-capacity scalability, such as media processing, file sharing, high-performance computing, and data backup	Compared with the traditional SFS Capacity-Oriented file system, the universal file system has higher performance
SFS Turbo	Supports massive small files, low latency, and high IOPS applications	Low latency and tenant exclusive Large capacity and low cost	Workloads dealing with massive small files, such as code storage, log storage, web services, and virtual desktop Autonomous driving, AI generated content, and EDA in chip design	SFS Turbo file systems can be backed up using CBR Off-cloud access (via VPN, Direct Connect, or other methods)

SFS configuration process

1. create SFS file system

2. mount the file system to a Linux or Windows host



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Mounting an NFS File System to ECSs

- After creating a file system, you need to mount the file system to servers so that they can share the file system。CentOS mounting is used as an example：
  - Log in to the ECS as user root. Run the following command to install the NFS software package.

    sudo yum -y install nfs-utils

  - （optional）Run the following command to check whether the domain name in the file system mount point can be resolved. SFS Turbo file systems do not require domain name resolution.

    nslookup File system domain name

  - Run the following command to create a local path for mounting the file system.

    mkdir Local path

  - Run the following command to mount the file system to the ECS that belongs to the same VPC as the file system

    mount -t nfs -o vers=3,timeo=600,noresvport,nolock Mount point Local path

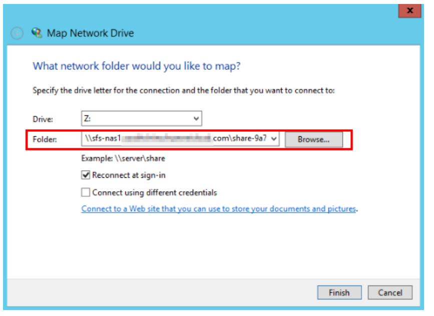
  - Run the following command to view the mounted file system.

    mount -l

- CIFS file systems cannot be mounted to Linux servers.
- An SFS Capacity-Oriented file system can use either NFS or CIFS. It cannot use both protocols.
- In this section, ECSs are used as example servers. Operations on BMSs and containers (CCE) are the same as those on ECSs.

Mounting a CIFS File System to ECSs

- After creating a file system, you need to mount the file system to ECSs so that they can share the file system.
- This section uses Windows Server 2012 as an example to describe how to mount a CIFS file system.
  - You have created a file system and have obtained the mount point of the file system.
  - At least one ECS that belongs to the same VPC as the file system exists.
  - The IP address of the DNS server for resolving the domain names of the file systems has been configured on the ECSs. For details, see Configuring DNS.
  - You need to mount the file system as user Administrator. You cannot switch to another user to mount the file system.



Data Protection

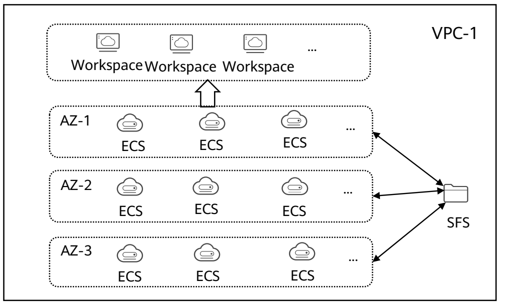
- Encryption:
  - SFS supports server-side encryption, which allows you to encrypt the data stored in SFS file systems. When data is accessed, SFS automatically decrypts the data and then returns it to you.
  - You can create a file system that is encrypted or not, but you cannot change the encryption settings of an existing file system.
- Backup:
  - A backup is a complete copy of an SFS Turbo file system at a specific time. It records all configuration data and service data at that time. For example, if a file system is faulty or encounters a logical error (accidental deletion, hacker attacks, and virus infection), you can use data backups to restore data quickly.

SFS vs OBS vs EVS

Dimension	SFS	OBS	EVS
Concept	SFS provides on-demand high-performance file storage, which can be shared by multiple ECSs	OBS provides massive, secure, reliable, and cost-effective data storage for users to store data of any type and size	EVS provides scalable block storage that features high reliability and high performance to meet various service requirements.
Data storage logic	Stores files. Data is sorted and displayed in files and folders	Stores objects. Files can be stored directly to OBS. The files automatically generate corresponding system metadata	Stores binary data and cannot directly store files. To store files, you need to format the file system first
Access method	SFS file systems can be accessed only after being mounted to ECSs or BMSs through NFS or CIFS	OBS buckets can be accessed through the Internet or Direct Connect. transmission protocols HTTP and HTTPS are used	EVS disks can be used and accessed from applications only after being attached to ECSs or BMSs and initialized
Capacity	PB-scale	EB-scale	TB-scale
Latency	3~10 ms	10 ms	Sub-millisecond level
Bandwidth	GB/s	TB/s	MB/s
Application Scenario	Gene sequencing, image rendering, media processing, file sharing, content management, and web services	Big data analysis, static website hosting, online video on demand (VoD), gene sequencing, and intelligent video surveillance	Industrial design, energy exploration, critical clustered applications, enterprise application systems, and development and testing

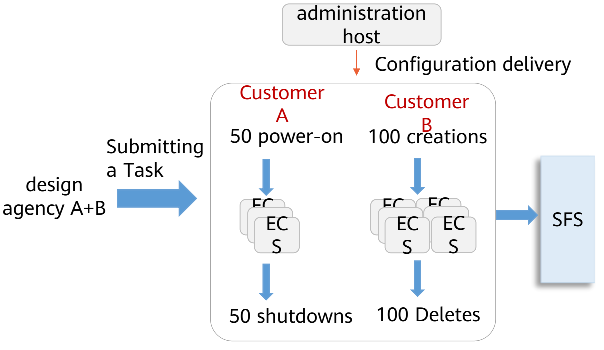
Scenario - File Sharing

- SFS applies to scenarios where there are a large number of departments or employees in an enterprise and the same documents need to be shared and accessed. The enterprise file storage hosted by SFS provides multiple file storage services, featuring high reliability, low latency, and high bandwidth. Users do not need to care about the underlying hardware infrastructure, avoiding the complexity of hardware deployment and maintenance.



Scenario - video rendering

- The high-bandwidth and large-capacity SFS file service meets the shared file storage requirements of video editing, transcoding, synthesis, HD video, and 4K video on demand scenarios. It supports multi-layer HD video editing and 4K video editing.

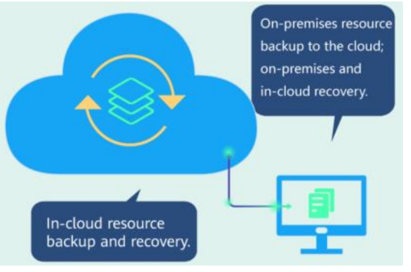


- **Shared storage performance:**
The rendering cluster accesses the shared storage to read and write rendering materials at the same time. This is a high-bandwidth scenario. The storage bandwidth of the shared storage must be greater than 10 GB/s.
- **The rendering performance of cloud hosts is stable:**
Customers are charged based on the rendering duration. Therefore, cloud host rendering processing must be stable and fluctuate slightly between different hosts so that customers can trust the cloud host.
- **Short batch operation time:**
The time for batch operations must be as short as possible to reduce the waiting time.

Contents

What Is CBR

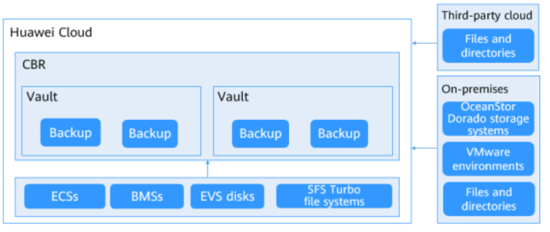
- Cloud Backup and Recovery (CBR) enables you to easily back up Elastic Cloud Servers (ECSs), Bare Metal Servers (BMSs), Elastic Volume Service (EVS) disks, SFS Turbo file systems, local files and directories, and on-premises VMware virtual environments.
- In case of a virus attack, accidental deletion, or software or hardware fault, you can use the backup to restore data to any point when the data was backed up.



CBR Architecture

- A backup is a copy of a specific block of data.
- CBR stores backups in vaults. Before creating a backup, you need to create at least one vault and associate the resources you want to back up with the vaults. Then the resources can be backed up to the associated vaults
- There are backup policies and replication policies.
  - A backup policy defines when you want to take a backup and for how long you would retain each backup.
  - A replication policy defines when you want to replicate from backup vaults and for how long you would retain each replica. Backup replicas are stored in replication vaults.

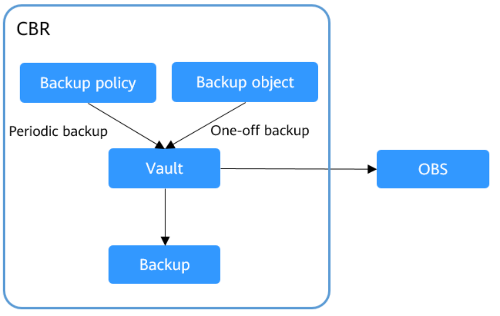
CBR involves backups, vaults, and policies



HUAWEI

Backup Options

- CBR supports one-off backup and periodic backup.
  - A one-off backup task is manually created and is executed only once.
  - Periodic backup tasks are automatically executed based on a user-defined backup policy.



- You can also use the two backup options together if needed. For example, you can associate resources with a vault and apply a backup policy to the vault to execute periodic backup for all the resources in the vault. Additionally, you can perform a one-off backup for the most important resources to enhance data security.

Advantages

Reliable

CBR offers crash-consistent backup for multiple disks on a server and application-consistent backup for database servers.

Efficient

Incremental forever backups shorten the time required for backup by 95%. With Instant Restore, CBR offers an RPO of as low as 1 hour and an RTO of only several minutes.

Easy to Use

CBR is easier to use than conventional backup systems. You can complete backup in just three steps, and no professional backup skills are required

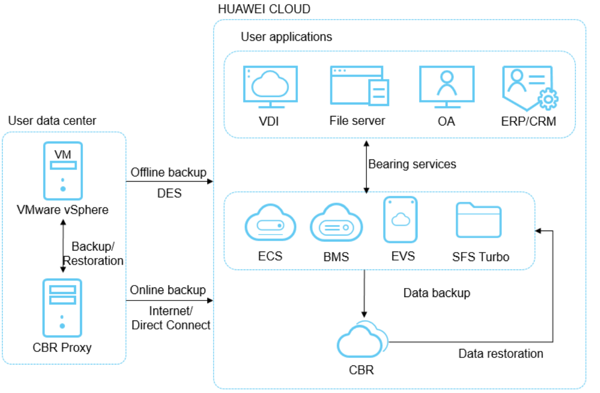
Secure

If the disks are encrypted, their backups are also encrypted to ensure data security. You can also replicate backups across regions to implement remote disaster recovery.

- Recovery Point Objective (RPO) specifies the maximum acceptable period in which data might be lost.
- Recovery Time Objective (RTO) specifies the maximum acceptable amount of time for restoring the entire system after a disaster occurs.

Scenarios: Data Backup and Restoration

- You can use CBR to quickly restore data to the latest backup point if any of the following incidents occur:
  - Hacker or virus attacks
  - Accidental deletion
  - Application update errors
  - System breakdown



Quiz

1. (True or false) Before attaching an EVS disk to an ECS, you must stop the ECS.

   True

   False

2. (Multiple-choice) Which of the following is not an OBS function?

   A. Cross-region replication

   B. Versioning

   C. URL validation

   D. Mounting in block storage mode

- False. You do not need to stop the ECS when attaching an EVS disk.
- D. EVS disks are attached to ECSs for use.

Summary

- Where there is data, there is a need for data storage. After studying the content presented here, we should have a new understanding of storage types and we should understand HUAWEI CLOUD storage services a little better. As more and more enterprises migrate to the cloud, we are more able to better meet their storage requirements if we understand the positioning, principles, and usages of various storage services, for example, which storage services are suitable for video cloud and which are the best for databases.

More Information

- Huawei iLearning
  - https://e.huawei.com/cn/talent/cert/#/careerCert
- HUAWEI CLOUD Help Center
  - https://support.huaweicloud.com/help-novice.html
- HUAWEI CLOUD Academy
  - https://edu.huaweicloud.com/

Acronyms and Abbreviations

- AK/SK: Access Key ID/Secret Access Key
- API: Application Programming Interface
- AZ: Availability Zone
- BMS: Bare Metal Server
- CAD/CAE: Computer Aided Design/Computer Aided Engineering
- CIFS: Common Internet File System
- DES: Data Express Service
- DHCP: Dynamic Host Configuration Protocol
- ECS: Elastic Cloud Server
- EVS: Elastic Volume Service
- HA: High Available

Acronyms and Abbreviations

- HPC: High Performance Computing
- HTTP: Hypertext Transfer Protocol
- HTTPS: Hypertext Transfer Protocol over Secure Sockets Layer
- IAM: Identity and Access Management
- IOPS: Input/Output Operations per Second
- NAS: Network Attached Storage
- NFS: Network File System
- OBS: Object Storage Service
- POSIX: Portable Operating System Interface
- SCSI: Small Computer System Interface
- SDK: Software Development Kit

Acronyms and Abbreviations

- SFS: Scalable File Service
- SSD: Solid-State Drive
- VBD: Virtual Block Device
- VPC: Virtual Private Cloud

Thank you.

把数字世界带入每个人、每个家庭、
每个组织，构建万物互联的智能世界。

Bring digital to every person, home, and organization for a fully connected, intelligent world.

HUAWEI

HUAWEI CLOUD O&M Basics

Foreword

- HUAWEI CLOUD not only provides resource services to meet enterprise needs to migrate their service systems to the cloud, but also ensures the normal running of the service systems on the cloud to meet the enterprise governance requirements.
- This section will help you understand HUAWEI CLOUD O&M.

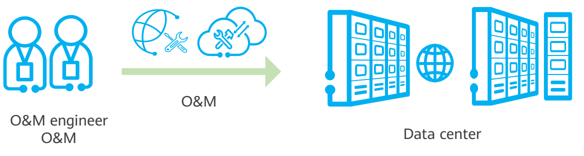
Objectives

- On completion of this course, you will be able to:
  - Gain basic knowledge about O&M, monitoring, and auditing.
  - Understand the positioning, principles, and usage of common governance services on HUAWEI CLOUD.

Contents

What Is O&M?

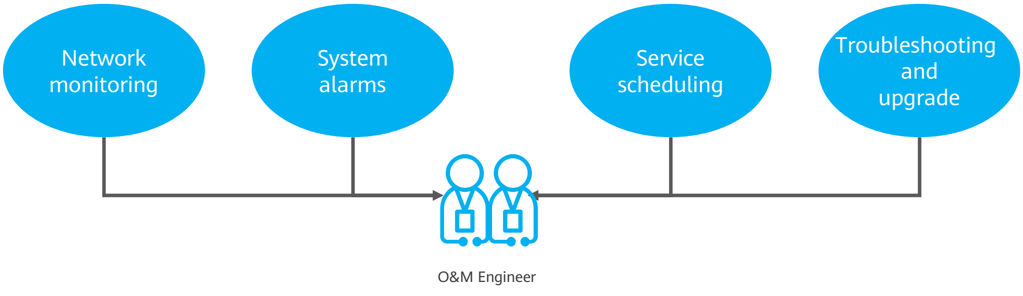
- O&M refers to operations and maintenance. It includes monitoring and managing devices and service systems to ensure services run normally. O&M also includes handling various problems and summarizing maintenance experiences to improve O&M efficiency and quality.
- O&M is essentially the operations and maintenance of devices and services such as servers and networks in each phase of their lifecycles, to achieve an optimum level of cost, stability, and efficiency.



- O&M focuses on various environments where the service system runs. It does not focus on programming, but on the use and management of these system platforms.
- In the ICT industry, those who perform O&M operations are typically referred to as O&M engineers.

Responsibilities of O&M Personnel

- O&M personnel are responsible for planning information, networks, and services based on service requirements and ensuring the long-term stability and availability of services by using various means, including but not limited to the following:



- O&M personnel stabilize the infrastructure, basic services, and online services that the enterprise Internet services rely on, perform routine inspection to detect potential risks, optimize the overall architecture to prevent common operation failures, and connect multiple data centers to improve the DR capability of services. By using technical means such as monitoring and log analysis, O&M personnel can detect and respond to service faults in a timely manner to minimize service interruption and meet enterprise availability requirements for Internet services.
- To be an excellent O&M engineer, one needs to have comprehensive technical knowledge and troubleshooting experience, and have a strong sense of responsibility for their work.
- In the common organizational structure of the Internet industry, O&M, development, and testing are basic technical positions. In terms of the phase, development and testing personnel are engaged in the work before software or services are launched, while O&M (except O&M development) personnel are engaged in the work after the software or services are launched. O&M can be further classified into IT operations, network O&M, service O&M, and O&M development.

Era of Automated O&M

- Traditional manual O&M is being gradually replaced by automated O&M platforms. Responsibilities of O&M personnel and development personnel are converging. The concept of integrated O&M and development (DevOps) is becoming more and more popular and is being used by most enterprises.

Traditional O&M

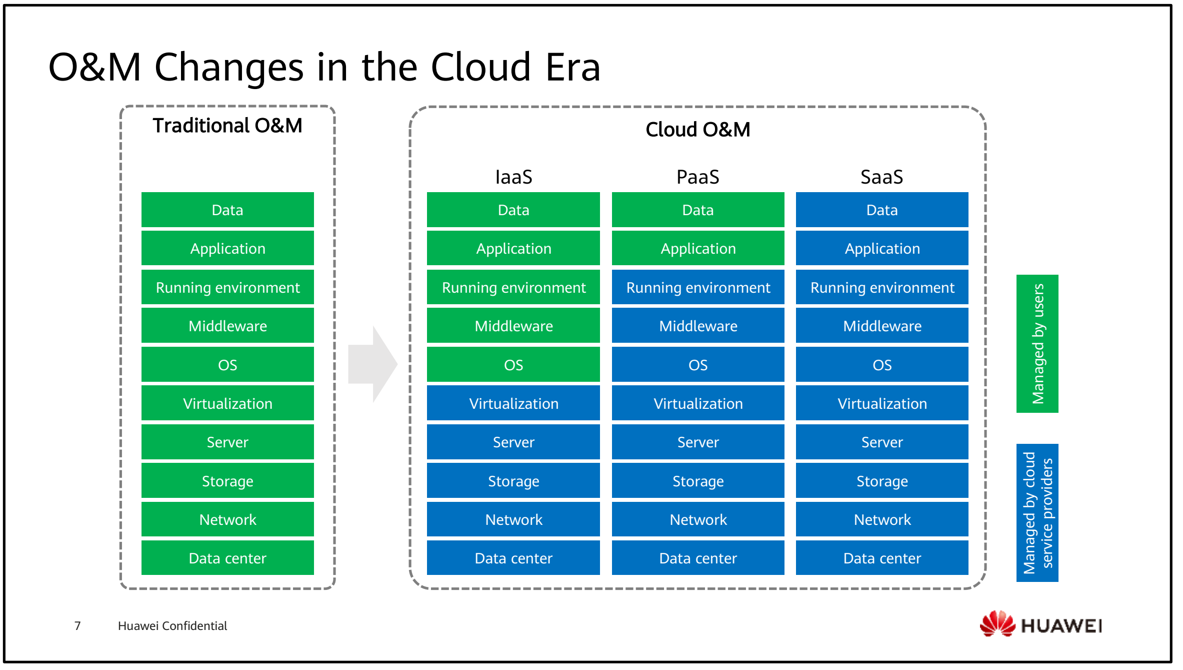
- Traditional O&M requires manual operations or processing.
- Because the problem needs to be handled manually, it takes some time to respond and solve the problem.
- Traditional O&M can be performed flexibly based on actual conditions, but human errors are prone to occur.
- It takes a lot of manpower.

VS

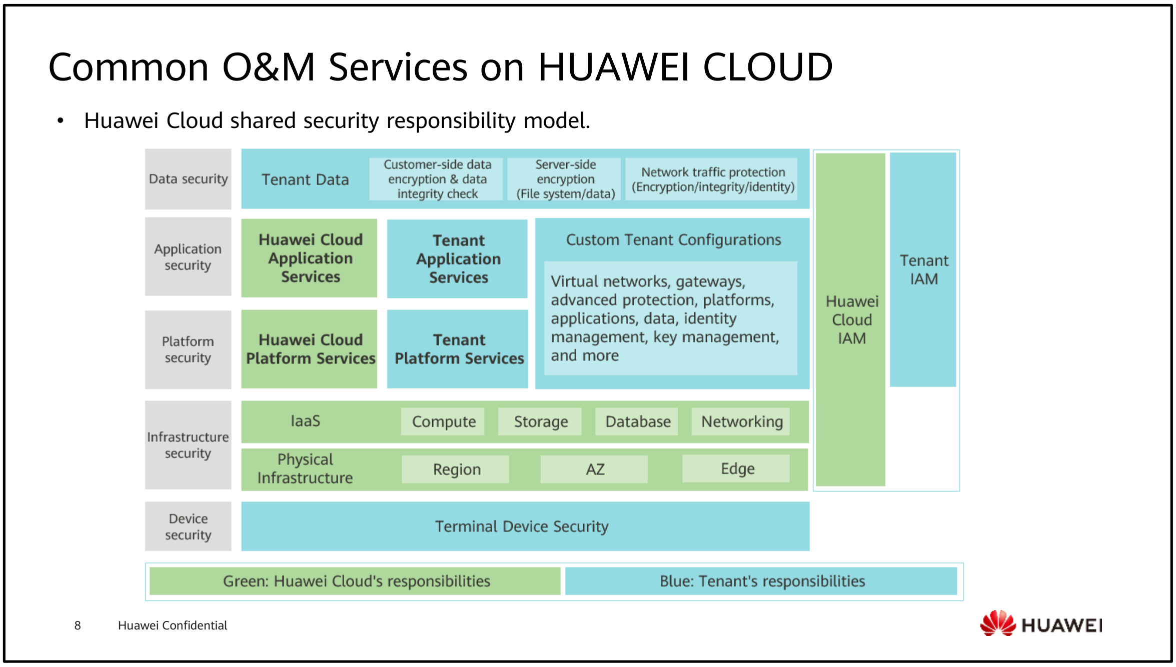
Automated O&M

- Manual operations are changed to automatic operations (by running scripts or using automatic tools). When a fault occurs, the fault can be automatically detected and quickly responded, reducing the troubleshooting time and error rate and improving O&M efficiency.
- This ensures the consistency of operations and reduces the occurrence of human errors.
- It requires certain technical costs, but it can reduce labor costs and improve efficiency.

- After more than a decade of development, IT operations is now facing a new direction: automation, which is an inevitable result of IT technology development. Nowadays, the complexity of IT systems requires digital and automated O&M. Automated O&M refers to the automation of daily and repeated work in IT operations and the transformation from manual work to automation. Automation is the sublimation of IT operations. IT operations automation is not only a maintenance process, but also a management improvement process. It is the highest level of IT operations and also the development trend in the future.
- DevOps is a group of processes, methods, and systems that are used to promote communication, collaboration, and integration between development, technical operation (O&M), and quality assurance (QA) departments. DevOps greatly reduces the gap between O&M and development and the delivery time.



- Compared with traditional O&M, cloud O&M greatly reduces the enterprise O&M costs. The O&M management services provided on the public cloud enable users to complete routine O&M at little or no cost. All these services are based on automatic O&M technologies.

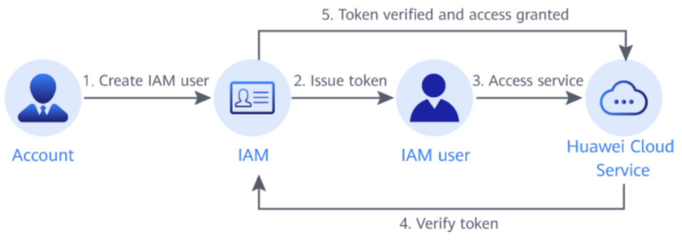


- Huawei Cloud: Ensure the security of cloud services and provide secure clouds. Huawei Cloud's security responsibilities include ensuring the security of our IaaS, PaaS, and SaaS services, as well as the physical environments of the Huawei Cloud data centers where our IaaS, PaaS, and SaaS services operate. Huawei Cloud is responsible for not only the security functions and performance of our infrastructure, cloud services, and technologies, but also for the overall cloud O&M security and, in the broader sense, the security and compliance of our infrastructure and services.
- Tenant: Use the cloud securely. Tenants of Huawei Cloud are responsible for the secure and effective management of the tenant-customized configurations of cloud services including IaaS, PaaS, and SaaS. This includes but is not limited to virtual networks, the OS of virtual machine hosts and guests, virtual firewalls, API Gateway, advanced security services, all types of cloud services, tenant data, identity accounts, and key management.

Contents

What Is IAM?

- Identity and Access Management (IAM) helps you manage your users and control their access to HUAWEI CLOUD services and resources.
- If you want to share resources with others but do not want to share your account and password, you can create an IAM user.



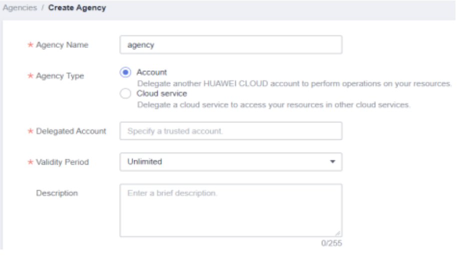
- A typical enterprise has multiple IT administrators, each responsible for managing different resources. It's more secure to not give every administrator super administrator permissions. Thanks to HUAWEI CLOUD IAM, an enterprise administrator can create multiple users with separate permissions.
- Credentials authenticate a user on the HUAWEI CLOUD console or APIs. Credentials include a password and access keys. The enterprise administrator manages both their own credentials and the credentials of IAM users they create.

IAM Permissions

- IAM is a global service deployed for all regions. When you set the authorization scope to Global services, users have permission to access IAM in all regions.
- You can grant permissions by using roles and policies.
    - Roles: A coarse-grained authorization strategy provided by IAM to assign permissions based on users' job responsibilities. Only a limited number of service-level roles are available for authorization. Cloud services depend on each other. When you grant permissions using roles, you also need to attach any existing role dependencies. Roles are not ideal for fine-grained authorization and least privilege access.
    - Policies: A fine-grained authorization strategy that defines permissions required to perform operations on specific cloud resources under certain conditions. This type of authorization is more flexible and is ideal for least privilege access. For example, you can grant users only permission to manage ECSs of a certain type. A majority of fine-grained policies contain permissions for specific APIs, and permissions are defined using API actions.

Agency

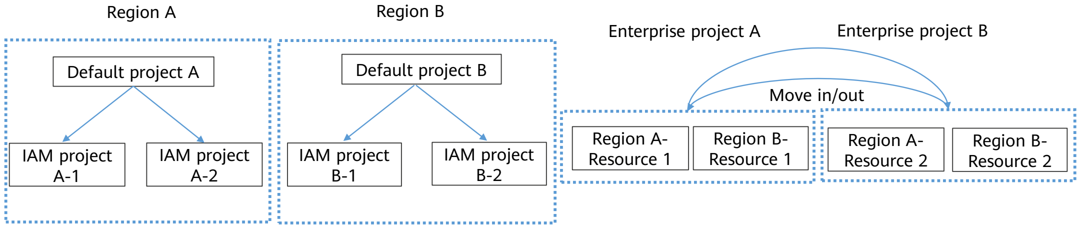
- A trust relationship that you can establish between your account and another account or a cloud service to delegate resource access.
  - Account delegation: You can delegate another account to implement O&M on your resources based on assigned permissions.
  - Cloud service delegation: Huawei Cloud services interwork with each other, and some cloud services are dependent on other services. You can create an agency to delegate a cloud service to access other services.



- You can delegate more professional, efficient accounts or other cloud services to manage specific resources in your account

Differences Between IAM Projects and Enterprise Projects

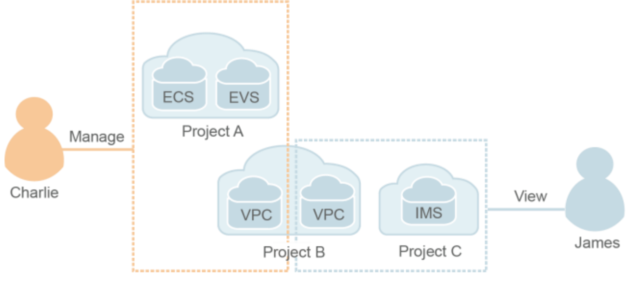
- IAM projects group and physically isolate resources in the same region. Resources cannot be transferred between IAM projects. They can only be deleted and then provisioned again.
- Enterprise projects group and logically isolate resources. An enterprise project can contain resources from multiple regions, and resources can be added to or removed from enterprise projects. Enterprise projects can be used to grant permissions to use specific cloud resources.



- Project: A region corresponds to a project. Default projects are defined to group and physically isolate resources (including computing, storage, and network resources) across regions. You can grant users permissions in a default project to access all resources in the region associated with the project.
- Enterprise Project: Enterprise projects allow you to group and manage resources across regions. Resources in enterprise projects are logically isolated from each other. An enterprise project can contain resources of multiple regions, and you can easily add resources to or remove resources from enterprise projects.

Refined Permissions Management

- You can grant IAM users permissions to manage different resources in your account. As shown in the following figure, you can grant Charlie permission to manage Virtual Private Cloud (VPC) resources in project B, and only grant James permission to view VPC resources in project B.

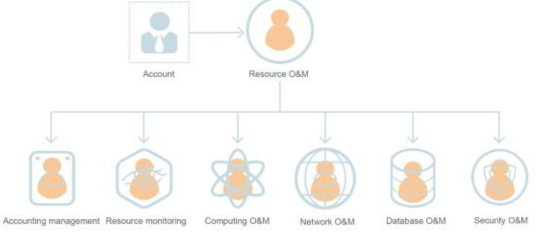


Federated Identity Authentication

- Enterprises with identity authentication systems can access Huawei Cloud through single sign-on (SSO), eliminating the need to create users on Huawei Cloud.
- If your enterprise has an identity system, you can create an identity provider (IdP) in IAM to provide single sign-on (SSO) access to Huawei Cloud for employees in your enterprise. The identity provider establishes a trust relationship between your enterprise and Huawei Cloud, allowing the employees to access Huawei Cloud using their existing accounts.

Application Scenario - IAM

- Assume that company A has purchased different resources on HUAWEI CLOUD, and has multiple functional teams that need to use one or more types of resources. Company A can use IAM to assign the permissions required for the O&M personnel to use resources.

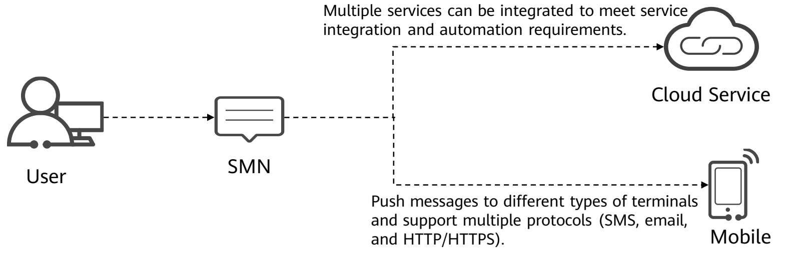


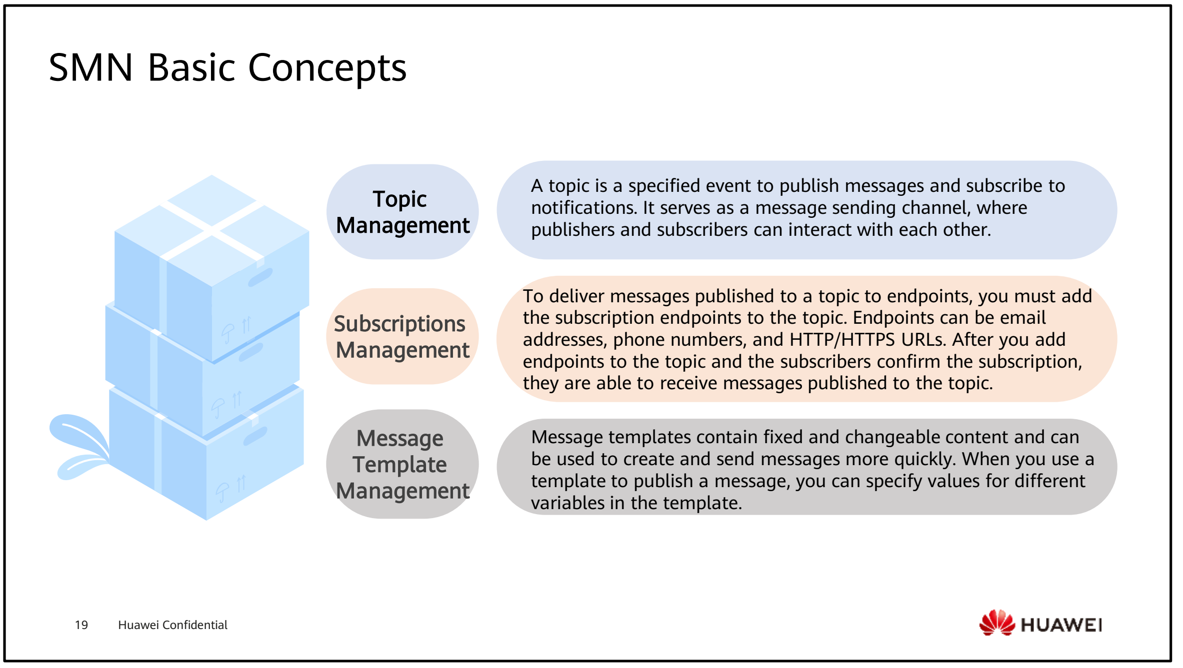
Functional Team	Policy	Permissions Description
Resource O&M team	Tenant Administrator	Full permissions for all cloud services
Accounting management team	BSS Administrator	Full permissions for Billing Center, Resource Center, and My Account
Resource monitoring team	Tenant Guest	Read-only permissions
Computing O&M team	ECS FullAccess CCE FullAccess AutoScaling FullAccess	Full permissions for ECS, CCE, AS
Network O&M team	VPC FullAccess EIP FullAccess ELB FullAccess	Full permissions for VPC, ELB,
Database O&M team	RDS FullAccess DDS FullAccess	Full permissions for RDS, DDS
Security O&M team	Anti-DDoS Administrator CAD Administrator KMS Administrator	Full permissions for Anti-DDoS, DDoS, DEW

Contents

What Is SMN?

- Simple Message Notification (SMN) is a reliable and flexible large-scale message notification service. It enables you to efficiently send messages to various endpoints, such as phone numbers, and email addresses.
- SMN offers a publish/subscribe model to achieve one-to-multiple message subscriptions and notifications in a variety of message types.

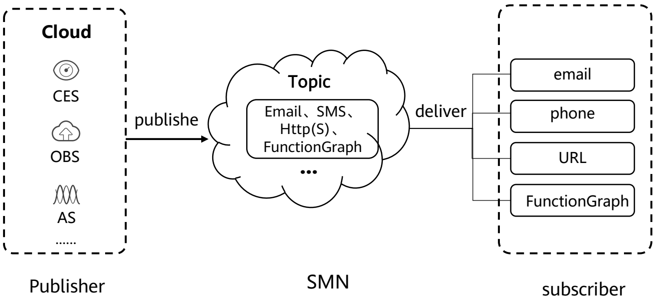




- Topic Management: After a topic is created, the system generates a topic URN, which uniquely identifies the topic and cannot be changed. The topic you created is displayed in the topic list.
- Subscriptions Management: You can add multiple subscriptions to each topic. After you add a subscription, SMN sends a confirmation message to the subscription endpoint. The message contains a link for confirming the subscription. The subscription confirmation link is valid within 48 hours. Confirm the subscription on your mobile phone, mailbox, or other endpoints in time.
- Message Template Management: Message templates are identified by name, but you can create different templates with the same name as long as they are configured for different protocols. All template messages must include a Default template or they cannot be sent out. The Default template is used anytime a template has not been configured for a given protocol, but as long as there is a template for the protocol, then any subscriber who selected that protocol when they subscribed will receive a message using the corresponding template.

SMN Architecture

- SMN involves two roles: publisher and subscriber.
  - A publisher publishes messages to a topic, and SMN then delivers the messages to subscribers in the topic.
  - The subscribers can be email addresses, phone numbers, and URLs.



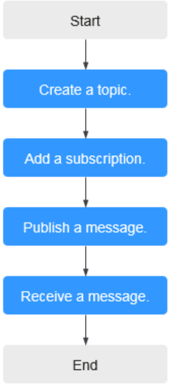
- A topic is a collection of messages and a logical access point, through which the publisher and the subscriber can interact with each other. Each topic has a unique name. The topic creator can configure topic policies to grant other users or cloud services permissions to perform certain operations to a topic, for example, querying subscriptions or publishing messages.

SMN Advantages

Item	Advantages
Simplicity	SMN provides three basic APIs to create topics, add subscriptions, and publish messages and can be quickly integrated with your services.
Stability and reliability	SMN stores messages in multiple data centers and supports transparent topic migration. Once a message fails to deliver, SMN saves it in a message queue and tries to deliver it again. If one service node is faulty, your requests are automatically processed by another available node.
Multiple message types	You publish a message once, and SMN delivers it to endpoints in various message types.
Security	SMN isolates data based on topics and does not allow any unauthorized users to access message queues, thereby protecting your service data.

Application Scenario - SMN

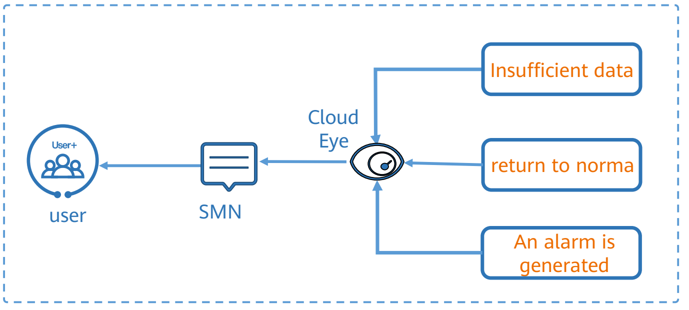
- System notifications
  - After events or alarms are triggered, SMN can send notifications to specified users by email, SMS message, or HTTP/HTTPS message. For example, Cloud Trace Service (CTS) detects key cloud service operations and uses SMN to notify you and other users.
- Integrating with cloud services
  - SMN can function as a message middleware to directly connect cloud services, improving service efficiency. For example, Cloud Eye does not have to be integrated with Object Storage Service (OBS) to interact with each other. Instead, they can be connected by SMN, so faults in one service will not affect the other.
- Off-peak traffic control
  - If there is a discrepancy between processing capabilities of the upstream and downstream systems, SMN can cache data to reduce downstream pressure to reduce breakdowns, enhance availability, and mitigate complexity in the system。



Contents

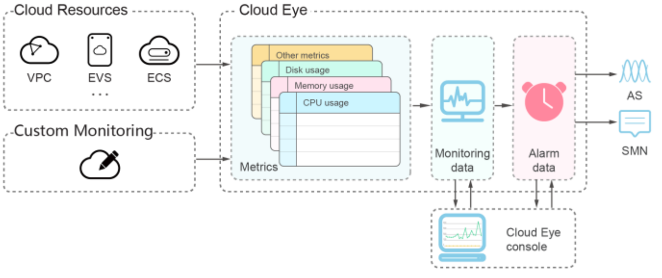
What Is Cloud Eye

- Cloud Eye is a multi-dimensional resource monitoring service. You can use Cloud Eye to monitor resources, set alarm rules, identify resource exceptions, and quickly respond to resource changes.



Cloud Eye Architecture

- Cloud Eye is a multi-dimensional resource monitoring service.
- Cloud Eye Provides cloud monitoring services for users in terms of computing, storage, network, and security. provides multiple cloud services, including Elastic Cloud Server (ECS), cloud database, cloud storage, cloud network, and cloud security, to meet enterprise requirements in different scenarios..



- Cloud Eye provides the following functions:
- Automatic monitoring: Monitoring starts automatically after you created resources such as Elastic Cloud Servers (ECSs). On the Cloud Eye console, you can view the service status and set alarm rules for these resources.
- Server monitoring: After you install the Agent (Telescope) on an ECS and Bare Metal Server (BMS), you can collect 60-second granularity ECS and BMS monitoring data in real-time. Cloud Eye provides 40 metrics, such as CPU, memory, and disk metrics.
- Flexible alarm rule configuration: You can create alarm rules for multiple resources at the same time. After you create an alarm rule, you can modify, enable, disable, or delete it at any time.
- Real-time notification: You can enable Alarm Notification when creating alarm rules. When the cloud service status changes and metrics reach the thresholds specified in alarm rules, Cloud Eye notifies you by emails, or by sending messages to server addresses, allowing you to monitor the cloud resource status and changes in real time.
- Monitoring panel: The panel enables you to view cross-service and cross-dimension monitoring data. It displays key metrics, providing an overview of the service status and monitoring details that you can use for troubleshooting.
- Resource group: A resource group allows you to add and monitor correlated resources and provides a collective health status for all resources that it contains.

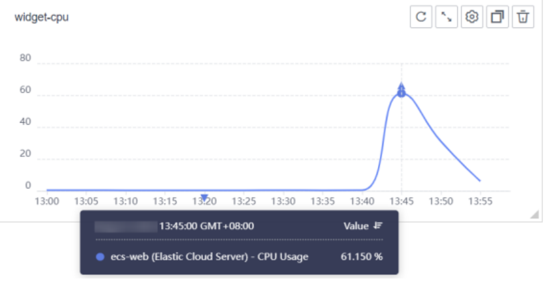
Cloud Eye Advantages

- Cloud Eye provides the following Advantages functions:

- Automatic Provisioning: Cloud Eye is automatically provisioned for all users. You can use the Cloud Eye console or APIs to view cloud service statuses and set alarm rules.
- Reliable Real-time Monitoring: Raw data is reported to Cloud Eye in real time for monitoring of cloud services.Alarms are generated and notifications are sent to you in real time.
- Visualized Monitoring: You can create monitoring panels and graphs to compare multiple metrics. The graphs automatically refresh to display the latest data.
- Multiple Notification Types: You can enable Alarm Notification when creating alarm rules. When the metric reaches the threshold specified in an alarm rule, Cloud Eye notifies you by emails, or by sending HTTP/HTTPS messages to an IP address of your choice, allowing you to keep track of the statuses of cloud services and enabling you to build smart alarm handling programs.
- Batch Creation of Alarm Rules: Alarm templates allow you to create alarm rules in batches for multiple cloud services.

Monitoring Panels

- Panels serve as custom monitoring platforms and allow you to view core metrics and compare the performance data of different services.
- After you create a panel, you can add graphs to the panel to monitor cloud services.



- Each panel supports up to 24 graphs. You can add up to 50 metrics to one graph. Monitoring comparison between different services, dimensions, and metrics is supported.

Server Monitoring

- Server monitoring includes basic monitoring, process monitoring, and OS monitoring for servers.
  - Basic monitoring covers metrics automatically reported by ECSs.
  - OS monitoring provides proactive and fine-grained OS monitoring for ECSs or BMSs, and it requires the Agent to be installed on all servers that will be monitored. OS monitoring supports metrics such as CPU usage and memory usage (Linux).
  - Process monitoring provides monitoring of active processes on hosts. By default, Cloud Eye collects CPU usage, memory usage, and number of opened files of active processes.

Name/ID	IP Address	ECS Stat...	Agent Status	CP...	Me...	Di...	Perma...	Enterprise...	Tag	Operation
ecs-web2 887ed36b-ecb5-4806-a7...	192.168....	Run...	Running	4.97%	25.61%	6.63%	--	default	--	View Metric
ecs-web be8f8f0c-6467-4033-ab9...	192.168....	Run...	Running	100%	16.04%	6.1%	--	default	--	View Metric

Event Monitoring

- In event monitoring, you can query system events that are automatically reported to Cloud Eye and custom events reported to Cloud Eye through the API.
- Events are key operations on cloud service resources that are stored and monitored by Cloud Eye. You can view events to see operations performed by specific users on specific resources, such as deleting or rebooting an ECS.

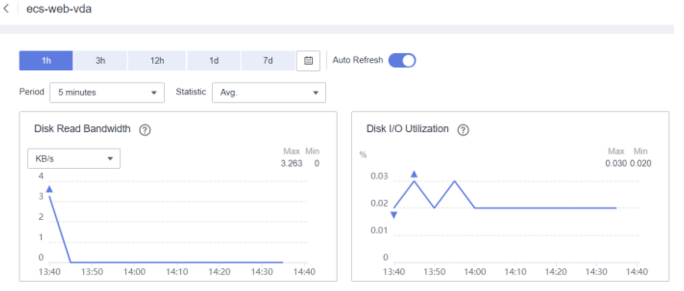
Event Type	Event Name	Event Source	Quantity	Last Occurred At	Operation
System Event	Login	Identity and Access Manageme...	1	10:21:53 GMT+08:00	View Graph | Create Alarm Rule

Filter

Monitored Object/ID	Event Severity	Event Status	Operator	Occurred At	Operation
097904e769000f811fb9c016f78869d9	Minor	Normal		10:21:53 GMT+08:00	View Event

Cloud Service Monitoring

- Cloud Eye provides multiple built-in metrics based on the attribute of each service. After you enabled one cloud service on the cloud platform, the system automatically associates its metrics based on the service type. Monitoring of these metrics helps you accurately grasp the service running status.



Alarm Function

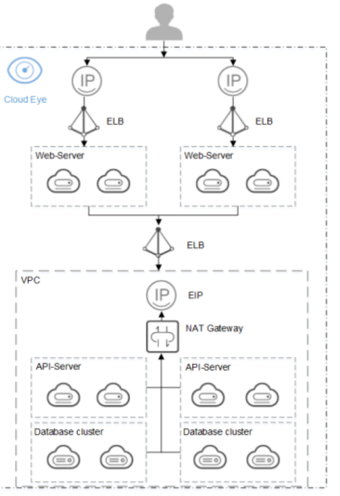
- Alarm rules allow you to monitor the performance of resources and their running status. You can set alarm rules for key metrics of cloud services.
- When the conditions in the alarm rule are met, Cloud Eye sends emailsor SMS messages, or sends HTTP/HTTPS messages, enabling you to quickly respond to resource changes.
- Cloud Eye invokes SMN APIs to send notifications. This requires you to create a topic and add subscriptions to this topic on the SMN console.

Name/ID	Resource Type	Monitored Obj...	Alarm Policy	Status	Notification G...	Alarm ...	Alarm Maskin...	Operation
alarm-5mb7 al16929462060...	Elastic Cloud S...	ECSs All resources	Associate template: ECS Alarm Template	Enabled	smn	Not...	--	View Details
alarm-5mb7 al16929462063...	Elastic Cloud S...	ECSs All resources	Associate template: ECS Alarm Template	Enabled	smn	Not...	--	View Details

- If no alarm notification topic is created, alarm notifications will be sent to the default email address of the login account.

Application Scenario – Cloud Eye

- Crowdsourcing platforms, as knowledge worker sharing platforms, use the Internet to allocate jobs and connect employers with service providers. Many service providers provide customized solutions for enterprises, public institutions, and individuals to transform ideas, wisdom, and skills into business value and social value.
- The core databases use the BMS clusters to deploy the database clusters. Web-Servers and API-Servers are deployed on ECSs. Web-Servers provide website search, category, store, and transaction services, and API-Servers are basic interfaces for connecting services with databases. The running statuses of BMSs and ECSs are critical to the entire service. CPU, memory, and disk usages affect the overall service status. Therefore, you need to use the server monitoring and event monitoring functions to monitor the running statuses of ECSs and BMSs at any time.



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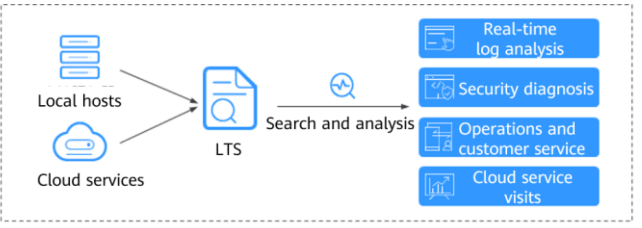
- Services like VPC, NAT Gateway, and ELB provide basic network support. The network status affects the connectivity between services. Therefore, you need to use the cloud service monitoring function to monitor the running status of each service system at any time.

Contents

What Is Log Tank Service?

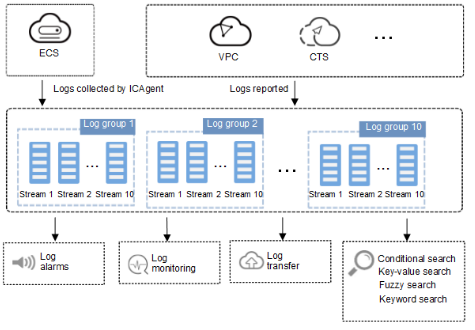
- Log Tank Service (LTS) enables you to collect logs from hosts and cloud services for centralized management, and analyze large volumes of logs efficiently, securely, and in real time.
- LTS provides you with the insights for optimizing the availability and performance of cloud services and applications. It allows you to make faster data-driven decisions, perform device O&M with ease, and analyze service trend.



- Real-time log ingestion: You can ingest logs from hosts and cloud services using ICAgent, APIs, or SDKs.
- Log transfer: Log transfer is to create log copies in destination cloud services. You can transfer logs to Object Storage Service (OBS), or Data Ingestion Service (DIS) for long-term storage.

LTS Architecture

- LTS collects logs from hosts and cloud services, and displays them on the LTS console in an intuitive and orderly manner. You can transfer logs for long-term storage. Collected logs can be quickly queried by keyword or fuzzy match. You can analyze real-time logs for security diagnosis and analysis, or obtain operations statistics, such as cloud service visits and clicks.



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LTS Basic Concepts

Log groups: A log group is the basic unit in LTS for log management. You can set log retention duration for a log group.

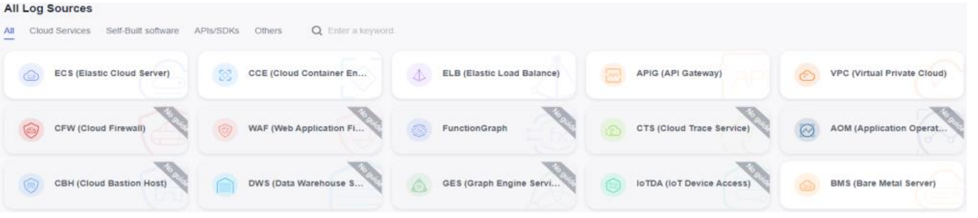
Log streams: A log stream is the basic unit for reading and writing logs. You can create log streams in a log group for finer log management.

ICAgent: ICAgent is the log collection tool of LTS. Install ICAgent on a host from which you want to collect logs.

- Log groups: Log Tank Service (LTS) collects log data from hosts and cloud services. By processing massive amounts of logs efficiently, securely, and in real time, LTS provides useful insights for you to optimize the availability and performance of cloud services and applications. It also helps you efficiently perform real-time decision-making, device O&M, and service trend analysis. Log groups can be created in two ways. They are automatically created when other services are connected to LTS, or you can create one manually by following the steps described here.
- Log streams: A log stream is the basic unit for reading and writing logs. You can separate different types of logs (such as operation logs and access logs) into different log streams for easier management. Sorting logs into different log streams makes it easier to find specific logs when you need them. Up to 100 log streams can be created in a log group.
- ICAgent: Before installing ICAgent, ensure that the time and time zone of your local browser are consistent with those of the host.

Log Ingestion

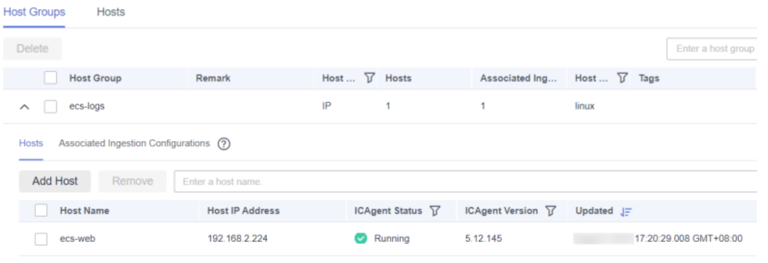
- LTS enables you to ingest logs from cloud services in real time using multiple means such as ICAgent, APIs, or SDKs. Ingested logs are displayed on the LTS console in an intuitive and orderly manner. You can query logs that you need quickly and with ease.
  - Cloud service: LTS supports log ingestion from cloud services. Click a cloud service to configure access to it.
  - Self-built software: Configure the paths of the host logs to be collected in a log stream.
  - API: You can use LTS APIs to report logs to LTS.
  - Other: You can use an agency to map the log stream of a delegator account to that of a delegated account.



- Self-built software: ICAgent will collect logs based on the ingestion configurations and send the logs to LTS.

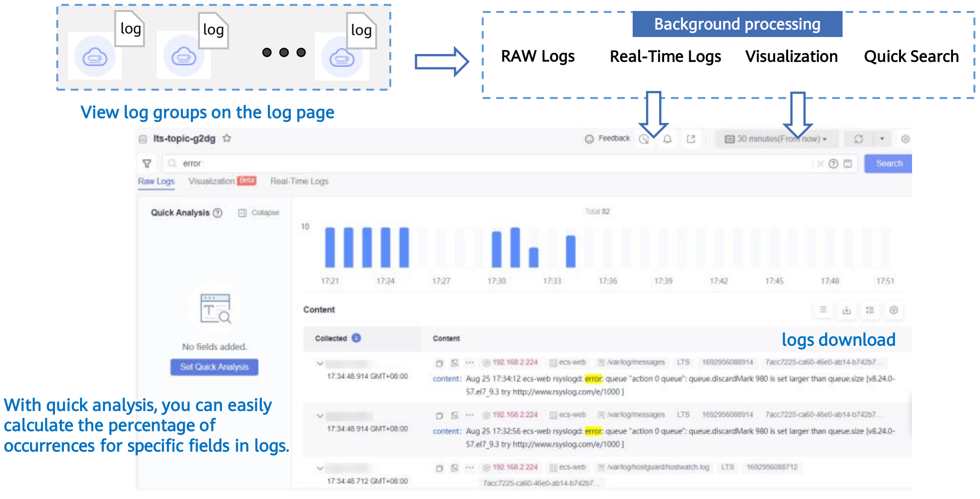
Host Management and Host Groups

- Host groups allow you to configure host log ingestion efficiently. You can sort multiple hosts to a host group and associate the host group with log ingestion configurations. The ingestion configurations will be applied to all the hosts in the host group, saving you the trouble of configuring the hosts individually.



- When there is a new host, simply add it to a host group and the host will automatically inherit the log ingestion configurations associated with the host group.
- You can also use host groups to modify the log collection paths for multiple hosts at one go.

Log Search and View



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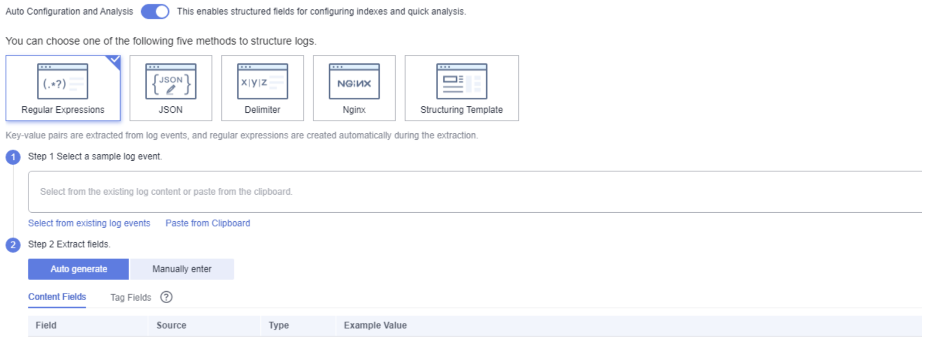
Log Structuring

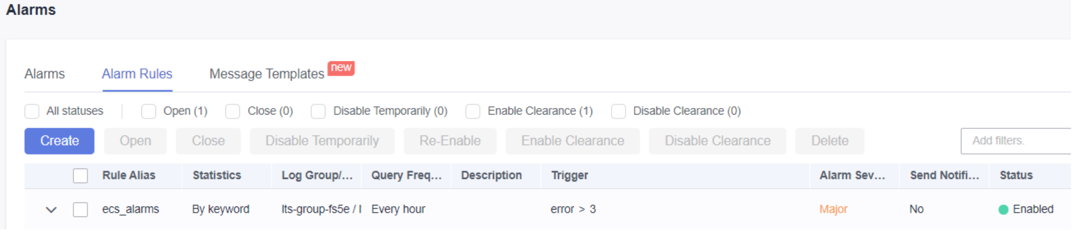
- Log data can be structured or unstructured. Structured data is quantitative data or can be defined by unified data models. It has a fixed length and format. Unstructured data has no pre-defined data models and cannot be fit into two-dimensional tables of databases.



- During log structuring, logs with fixed or similar formats are extracted from a log stream based on your defined structuring method and irrelevant logs are filtered out. You can then use SQL syntax to query and analyze the structured logs.

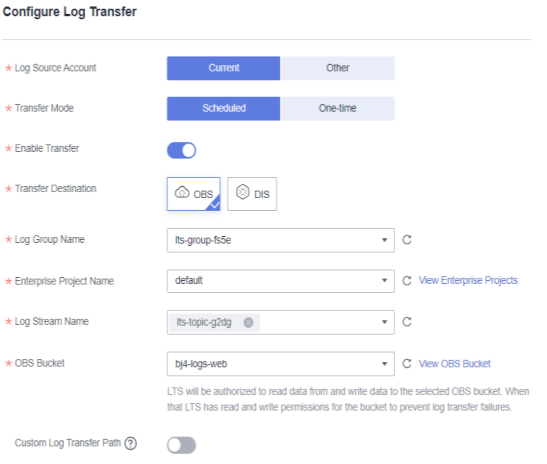
Log Alarms

- LTS allows you to collect statistics on log keywords and set alarm rules to monitor them. By checking the number of keyword occurrences in a specified period, you can have a real-time view of the service running.
- You can configure keyword alarm rules to query and monitor log data. When alarm rules are met, alarms will be triggered. You can view the alarms on the LTS console.



Log Transfer

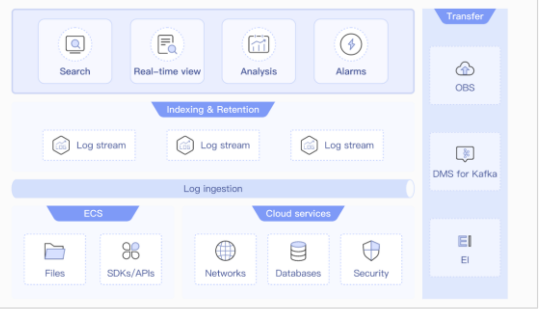
- Logs reported from hosts and cloud services are retained in LTS for seven days by default. You can set the retention period to be 1 to 365 days. Retained logs are deleted once the retention period is over. For long-term storage, you can transfer logs to other cloud services.
- You can transfer logs to OBS, DIS, or DMS based on your service scenario.



- Log transfer refers to when logs are replicated to other cloud services. Retained logs are deleted once the retention period is over, but the logs that have been transferred to other services are not affected.
- Transferring Logs to OBS: OBS is suitable for long-term storage.
- Transferring Logs to DIS: DIS provides both log storage and big data analysis.DIS can perform offline analysis, and transmit a large number of log files to the cloud for backup, query, and machine learning. You can also use it for data recovery and fault analysis after data loss or exceptions. In addition, a large number of small text files can be combined and transferred into large files to improve data processing performance.
- Transferring Logs to DMS: You can use DMS APIs to process logs in real time.

Application Scenario – LTS

- O&M logs of enterprise applications are distributed on different VMs, including application run logs and middleware logs. The logs are scattered and large in scale, providing a centralized management platform for enterprise logs.
- Advantages:
  - Fully managed: provides log collection, storage, search, and dumping for multiple cloud services.
  - Massive log management: 100 TB logs can be accessed every day, and billions of logs can be searched in seconds.
  - High cost-effectiveness: Low maintenance costs and on-demand charging, easily coping with peak log traffic.



Contents

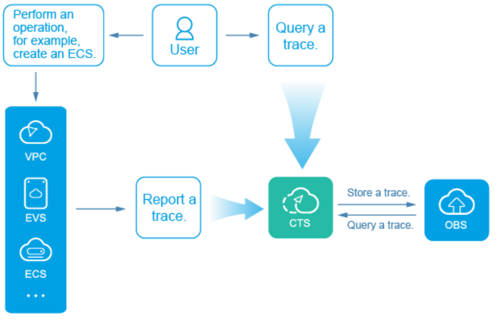
What Is Audit?

- The log audit module is a core component necessary for information security audit and an important part for the information systems of enterprises and public institutions to provide security risk management and control.
- Through auditing the financial statements and actual operation of the enterprise, the authenticity, legality and efficiency of the financial revenue and expenditure of the enterprise can be ensured, and the purpose of ensuring the healthy operation of the enterprise and promoting the long-term development of the enterprise can be achieved. In the ICT industry, audits are also used to ensure the healthy operation of the entire information system.

Generally speaking, the audit of institutions, organizations, and enterprises is mainly based on the following two things:

What Is Cloud Trace Service?

- Cloud Trace Service (CTS) is a log audit service for Huawei Cloud security. It allows you to collect, store, and query resource operation records. You can use these records to perform security analysis, track resource changes, audit compliance, and locate faults.



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- CTS provides the following functions:
  - Trace recording: CTS records operations performed on the management console or by calling APIs, as well as operations triggered by each interconnected service.
  - Trace query: Operation records of the last seven days can be queried on the management console from multiple dimensions, such as the trace type, trace source, resource type, filter, operator and trace status.
  - Trace transfer: Traces are transferred to Object Storage Service (OBS) buckets on a regular basis for long-term storage. In this process, traces are compressed into trace files by service.
  - Trace file encryption: Trace files are encrypted using keys provided by Data Encryption Workshop (DEW) during transfer.

CTS Advantages

Traditional Audit

- Traditional IT environments cannot systematically record operations and API records in real time, such as server, database, and operating system violations.
- System configuration changes need to be manually collected by IT personnel.
- Traditional audit content is manually recorded and stored without multiple copies. Therefore, it is not suitable for long-term storage.

VS

CTS

- Real-time recording: Quickly collects operation events and views them on the management console after resource changes are complete.
- Entire records: Records operations performed by the management console, open APIs, and internal operations triggered by the system.
- Reliable and low-cost: Event files are periodically generated and stored for a long time (for example, dumped to OBS).

CTS Basic Concepts

Trackers

- When you enable CTS for the first time, a management tracker named system is created automatically. You can also manually create multiple data trackers on the Tracker List page.
- Th management tracker identifies and associates with all cloud services your tenant account is using, and records all operations of your tenant account.

Traces

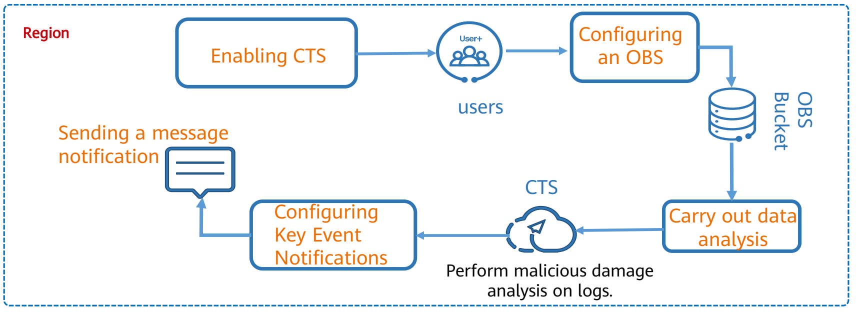
- Traces are operation logs of cloud service resources and are captured and stored by CTS. You can view traces to get to know details of operations performed on specific resources.
- here are two types of traces:
  - Management traces: Traces reported by cloud services.
  - Data traces: Traces of read and write operations reported by OBS.

Trace List

- The trace list displays traces generated in the last seven days. These traces record operations on cloud service resources, including creation, modification, and deletion, but query operations are not recorded. There are two types of traces。
  - Management traces: record details about creating, configuring, and deleting cloud service resources in your tenant account.
  - Data traces: record operations on data in OBS buckets, such as data upload and download.

CTS security analysis

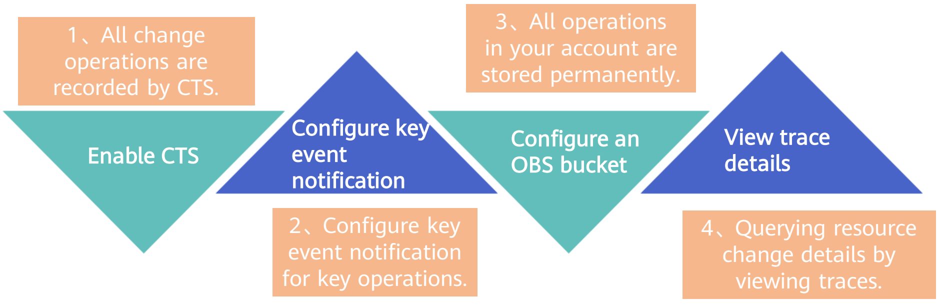
- Each trace generated by CTS records the user, time, and IP address of an operation request. You can perform security analysis and detect users' behavior patterns to determine whether to configure Key Event Notification.



- How security analysis works:
  - Log in to the system as a user and enable CTS. CTS records all operation logs of the account.
  - Operation logs are stored in OBS buckets.
  - The Analytics can download logs from buckets for analysis.
  - The final result can be reported to the SMN service.

Resource Change

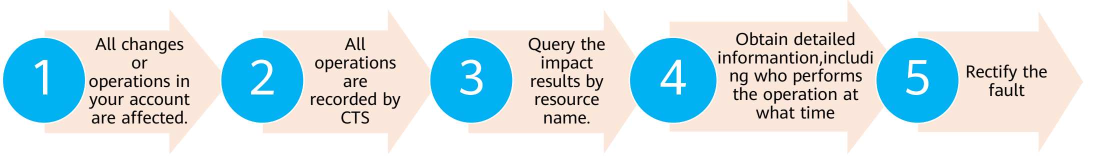
- Each trace generated by CTS records a resource change and change results. You can collect statistics on resource usage and perform backtracking operations based on these records.



- The working principle of resource change is as follows:
  - All change operations performed by users on cloud services have been recorded by the audit service.
  - You can also configure key operation message notifications to obtain the latest service status in a timely manner.
  - All changes under the account will be stored for a long time.
  - Users can query detailed information about resource changes based on logs.

Fault Locating

- Traces generated by CTS record the cause of a fault. You can easily rectify the fault based on the cause. For example, you may delete a system disk when expanding an ECS, causing an expansion failure.



- Working principle of fault locating:
  - Users perform changes or misoperations under their own accounts that may cause related faults.
  - The audit service records all these operations.
  - In this case, you can search for the resource name to query the impact result.
  - Users can obtain detailed information, including the operator and specific time.
  - Users can correct the incorrect operation based on the information.

Application Scenario - Using CTS to Monitor Resources

- With CTS, you can record operations performed by public cloud accounts on cloud service resources and their results in real time.
  - You can create a CTS trigger in FunctionGraph to obtain subscribed resource operation information, use user-defined functions to analyze and process the resource operation information, and generate alarm logs.
  - SMN sends alarms to service personnel through SMS messages and emails.
- Case:
  - With CTS, you can quickly analyze logs and filter logs based on specified IP addresses.
  - Function computing based on the serverless non-service architecture provides data processing, analysis, event triggering, and elastic scaling, requiring no O&M and paying on demand.
  - Provides log and alarm functions together with SMN.

- CTS mines data in audit logs for service health analysis, risk analysis, resource tracing, and cost analysis. It also opens audit data to customers so that they can explore data value.
- CTS allows you to set search criteria to accurately search for operations and details when an issue occurs, reducing the time and labor costs for detecting, locating, and resolving the issue.

Quiz

1. (True or false) Cloud Eye is a free cloud service.

   True

   False

2. (Multiple-choice) Which of the following are scenarios of CTS?

   A. Resource tracking

   B. Compliance auditing

   C. Fault locating

   D. Security analysis

- True. Cloud Eye is free. You can use Cloud Eye to monitor and manage your purchased cloud services.
- ABCD

Summary

- O&M services play an important role in ensuring that platforms are secure and operate normally. We can use CTS to better manage platforms, and use Cloud Eye to monitor platforms in real time. With LTS, we can obtain logs in real time and evaluate and eliminate potential risks.

More Information

Huawei iLearning

- https://e.huawei.com/cn/talent/cert/#/careerCert

HUAWEI CLOUD Help Center

- https://support.huaweicloud.com/help-novice.html

HUAWEI CLOUD Academy

- https://edu.huaweicloud.com/

Acronyms and Abbreviations

- API：Application Programming Interface
- AZ：Availability Zone
- AS：Auto Scaling
- BMS：Bare Metal Server
- BSS：Business Support System
- CBR：Cloud Backup and Recovery
- CTS：Cloud Trace Service
- CES：Cloud Eye Service
- CAD：Cloud Anti-DDoS
- DES：Data Express Service
- DSS：Dedicated Distributed Storage Service

Acronyms and Abbreviations

- DHCP：Dynamic Host Configuration Protocol
- ECS：Elastic Cloud Server
- EVS：Elastic Volume Service
- HA：High Available
- HTTP：Hypertext Transfer Protocol
- HTTPS：Hypertext Transfer Protocol over Secure Sockets Layer
- IAM：Identity and Access Management
- IDP：Identity Provider
- IOPS：Input/Output Per Second
- KMS：Key Management Service
- LTS： Log Tank Service

Acronyms and Abbreviations

- OBS：Object Storage Service
- O&M： Operations and Maintenance
- POSIX：Portable Operating System Interface
- RTO：Recovery time objective
- RPO：Recovery Point Objective
- SDK：Software Development Kit
- SFS：Scalable File Service
- SSD：Solid-State Drive
- SDRS：Storage Disaster Recovery Service
- SP：Service Provider
- SAML：Sectuirty Assertion Markup Language

Thank you.

把数字世界带入每个人、每个家庭、
每个组织，构建万物互联的智能世界。

Bring digital to every person, home, and organization for a fully connected, intelligent world.

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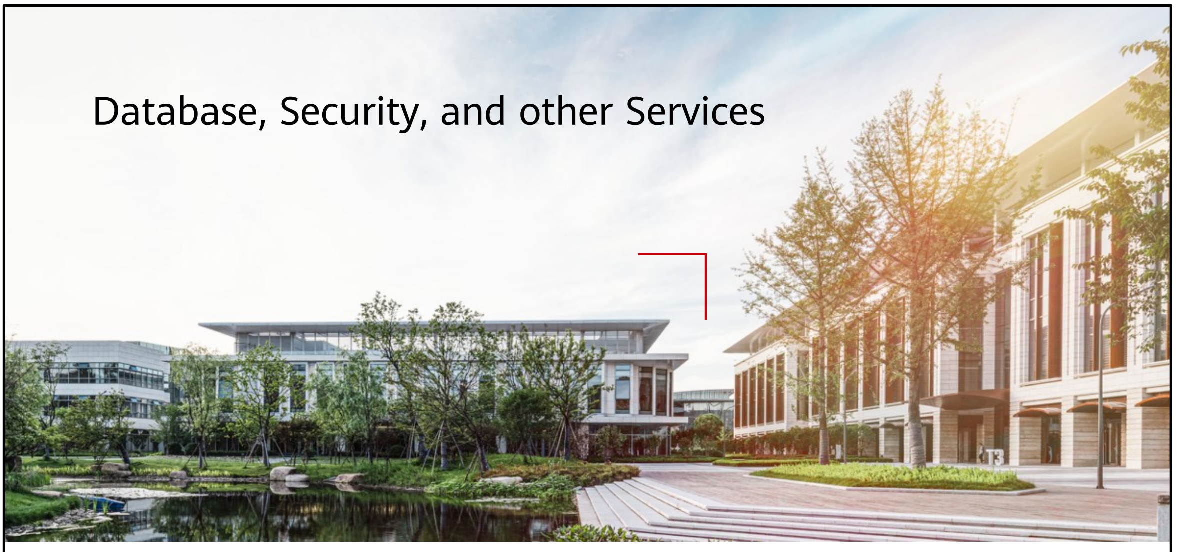
Database, Security, and other Services



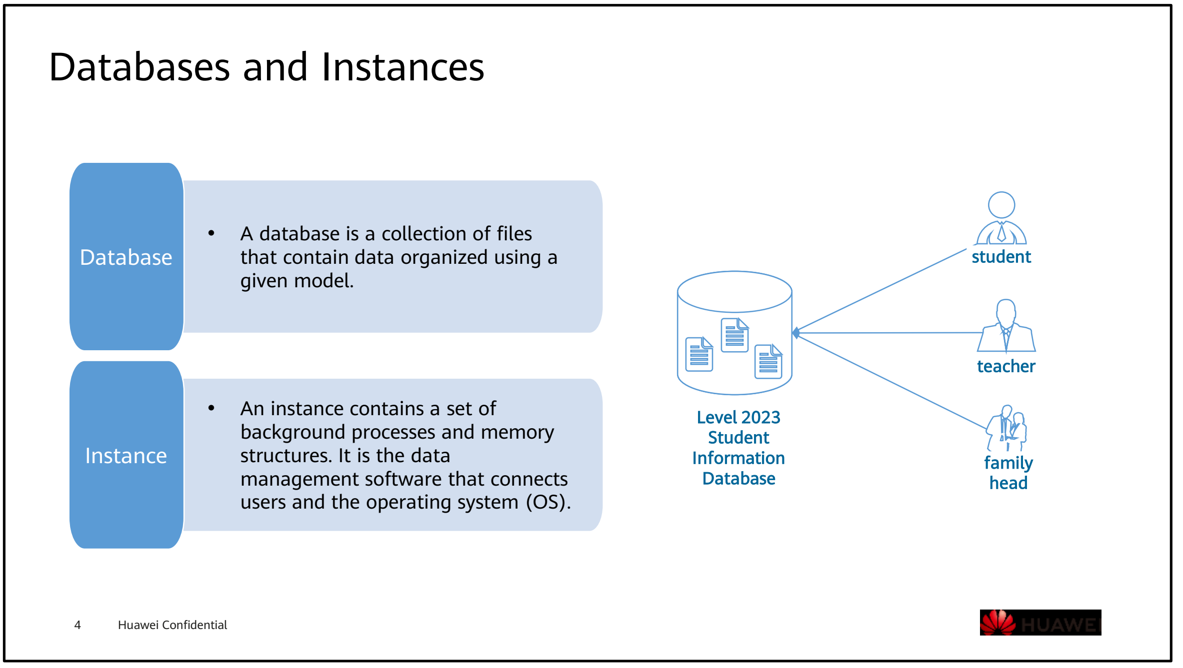
Foreword

- In addition to compute, storage, and networking services, enterprises need database services, security services, Content Delivery Network (CDN), API services, and EI services. These services help users quickly deploy services on the cloud to meet the requirements of various service scenarios, simplify service deployment, and facilitate O&M.
- This chapter describes the database service, security service, CDN, API, and EI services.

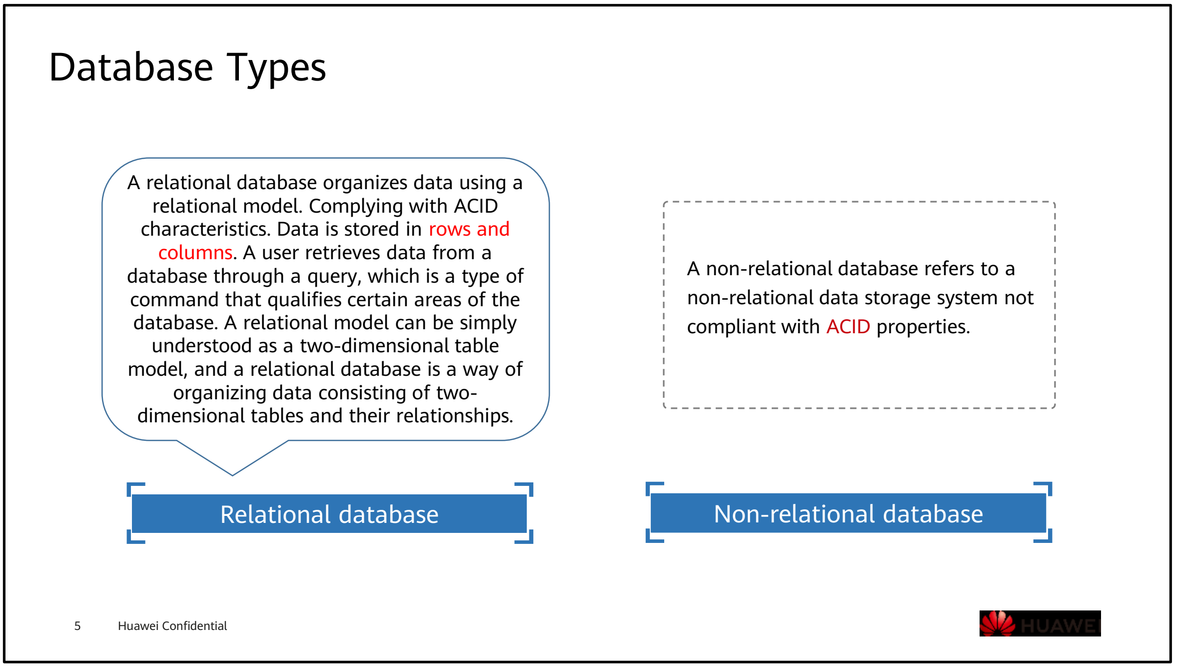
Objectives

- On completion of this course, you will be able to:
  - Understand the basic concepts.
  - Understand the service positioning, principles, and functions.

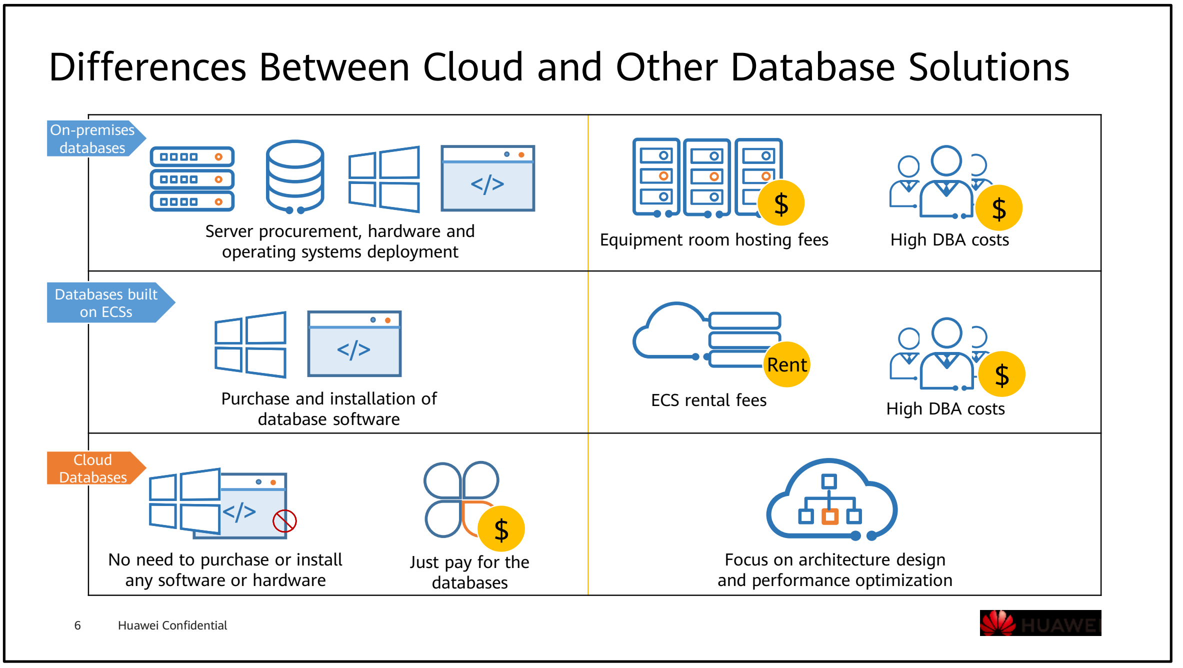
Contents



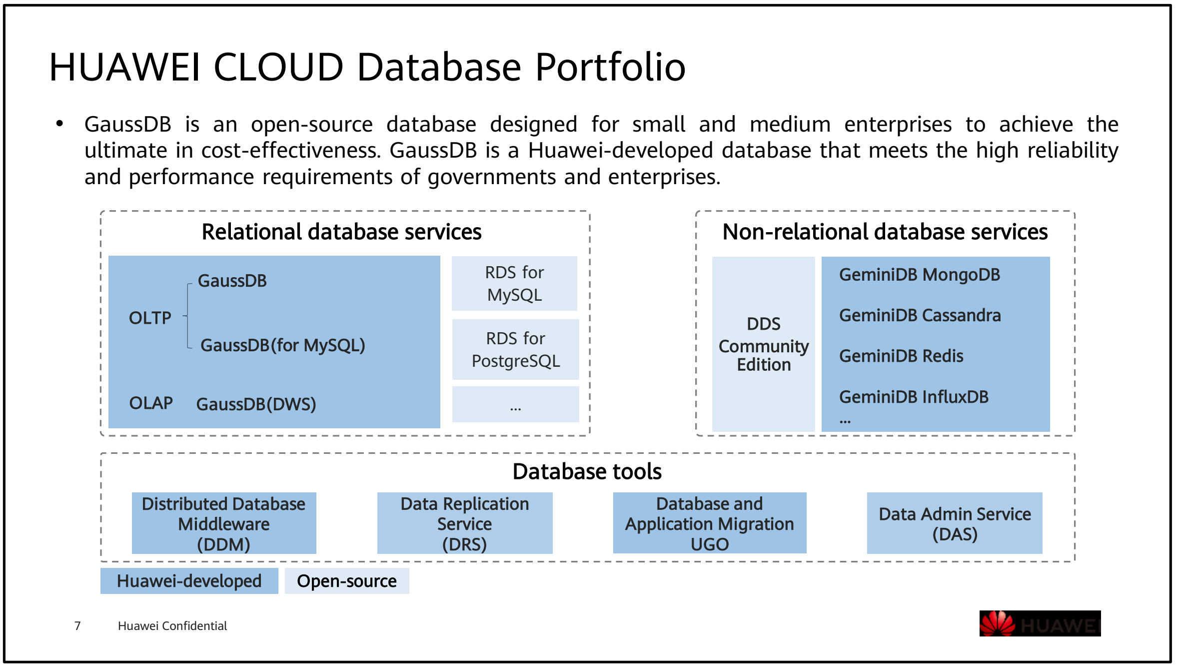
- We all know that data can be stored in multiple media formats, such as in memory or saved to disks. In fact, a database is also a medium for storing data.
- To be more specific, a database stores electronic documents, and you can add, intercept, update, and delete data in these documents.
- All operations on the data of databases, such as defining data, querying data, maintaining data, and managing database operations, are performed using database instances. Your applications only interact with the databases only through the instances.
- The smallest management unit of RDS is a database instance. A database instance is an isolated database environment running in the cloud. You can use RDS to create and manage database instances running various DB engines.



- ACID stands for atomicity, consistency, isolation, and durability.
  - Atomicity: Atomicity is the guarantee that series of database operations in an atomic transaction will either all occur or none will occur. If an error occurs during transaction execution, the transaction will be rolled back to the state from before it was committed.
  - Consistency: A consistent transaction will not violate integrity constraints placed on the data by the database rules. That is, executing a transaction cannot destroy the integrity or consistency of database data.
  - Isolation: Isolation means that concurrent transactions are executed sequentially. It guarantees the individuality of each transaction and prevents them from being affected by other transactions.
  - Durability: Once a transaction is committed, it will remain in the system even in the event of a system failure.



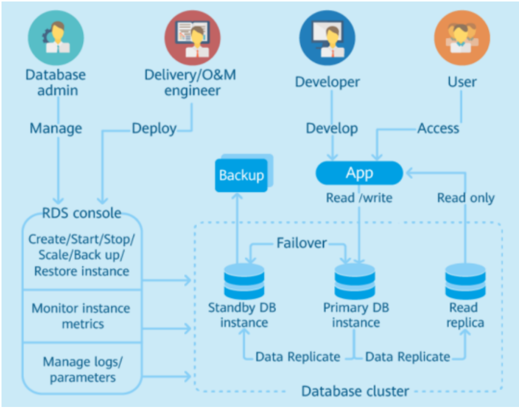
- If customers want to build their own databases, they need to purchase hardware such as database servers and switches. If the hardware is damaged or replaced, the cost of repairing or replacing it is typically at least 30% of the project budget. It costs at least 3000 CNY per year to host 1U of cabinet space. If there are two 1U servers and a 1U intranet switch required for databases, the total hosting fee would be 9,000 CNY (3,000 x 3) in a year. The monthly salary of a junior DBA engineer is at least 5,000 CNY per month. If building the databases occupies 30% of the engineer's workload, the yearly labor cost is 18,000 CNY (5,000 x 12 x 30%). The sunk cost of this project is considerable. Open-source databases cannot be optimized. To ensure database reliability, customers have to prepare backup resources, which means more money. Public network traffic and domain name transfer are not free either.
- If customers want to deploy databases on ECSs, they need to purchase primary/standby ECS instances. Physical devices are provided by the service provider. Customers do not need to pay for the equipment room. They only need to hire DBA engineers to operate and maintain the database services. Elastic resources are provided. But open-source databases cannot be optimized, and backup represents a separate cost, along with traffic over a public network.
- Using cloud databases, customers only need to pay for the DB instances. The service provider provides the physical devices and maintains databases at its own cost. Resources are elastic and there is no charge for any public network traffic. Even the domain name generated for the DB instance is free, and regular updates help keep your instances updated to the latest MySQL version.



- Introduction to HUAWEI CLOUD database services:
  - PostgreSQL is an object-relational database management system (ORDBMS) derived from the POSTGRES package based on the 4.2 version written at the University of California, Berkeley. Many leading POSTGRES concepts were not around until fairly late in the development of business databases.
  - NoSQL refers to non-relational databases. Traditional relational databases are unable to keep up with the ultra-large-scale processing and massive concurrent SNS website requests involved with Internet Web 2.0 websites. NoSQL databases are designed to address the challenges of handling the multiple data types involved in large-scale data collections, especially where big data applications are concerned. NoSQL databases come in a variety of types based on different data models. The main types are key-value pair, wide column, document, and graph.
  - Distributed Database Middleware (DDM) works with the RDS service to remove a single node's dependency on hardware, facilitate capacity expansion to address data growth challenges, and ensure fast response to query requests. DDM eliminates the bottlenecks in capacity and performance and ensures that concurrent access is possible for a massive amount of data.
  - Data Replication Service (DRS) is a stable, secure, and efficient cloud service for online database migration and real-time database synchronization. DRS simplifies data transmission between databases and reduces data transfer costs.
  - Data Admin Service (DAS) enables you to manage DB instances on a web-based console, simplifying database management and improving efficiency and security.

Architecture and Advantages of RDS

- RDS is a reliable and scalable cloud database service that is easy to manage.



Advantages

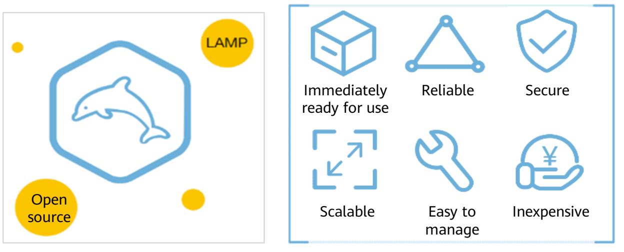
- Relax with High Availability and Durability
- More Flexibility and Lower Upfront Costs
- Optimized for Performance and Security
- Simple Deployment and Maintenance

- Relax with High Availability and Durability :
    - Ensure business continuity with quick failover to a standby instance and automated data backups to a different availability zone.
- Simple Deployment and Maintenance :
    - Launch a production-ready MySQL database in minutes Eliminate time-consuming operational tasks with automated patch updates, backups, monitoring, and scaling
- More Flexibility and Lower Upfront Costs :
    - Cost-effective pricing, with no hardware investment needed. Fully utilize storage with pay-per-use pricing, or save money with pre-allocated packages.
- Optimized for Performance and Security :
    - Scale quickly to handle increasing concurrency and demand. Protect data with end-to-end server security and data encryption.

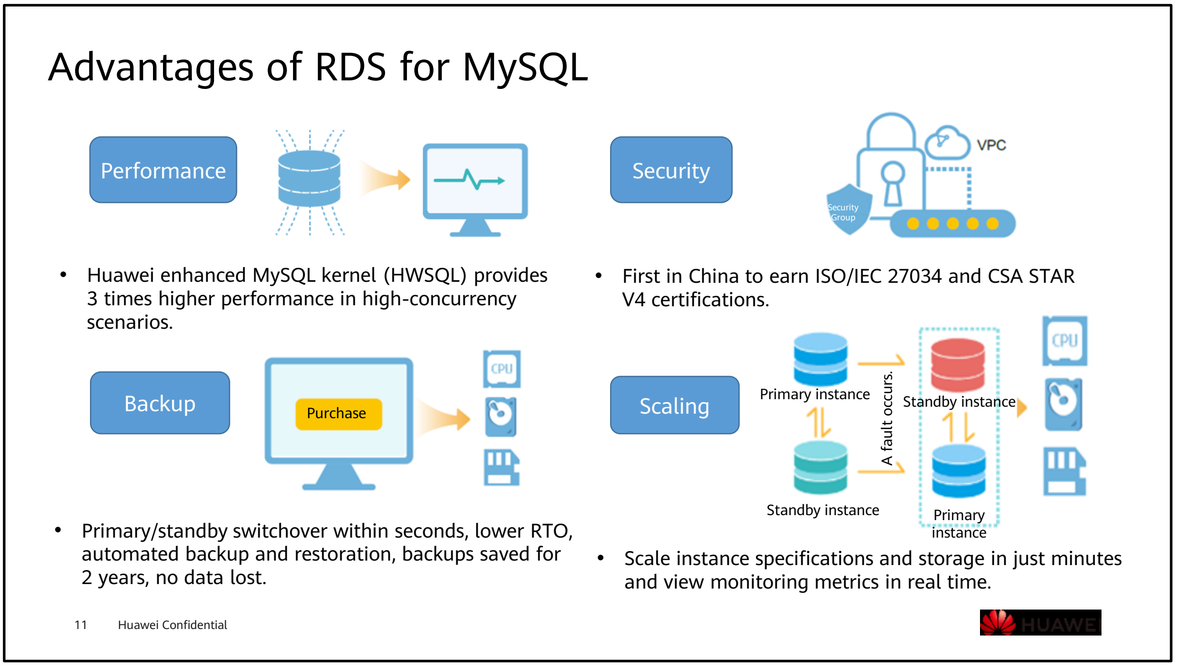
Contents

RDS for MySQL

- MySQL is one of the world's most popular open-source relational databases. It works with the Linux, Apache, and Perl/PHP/Python to establish a LAMP model for efficient web solutions.
- RDS for MySQL is reliable, secure, scalable, inexpensive, and easy to manage.



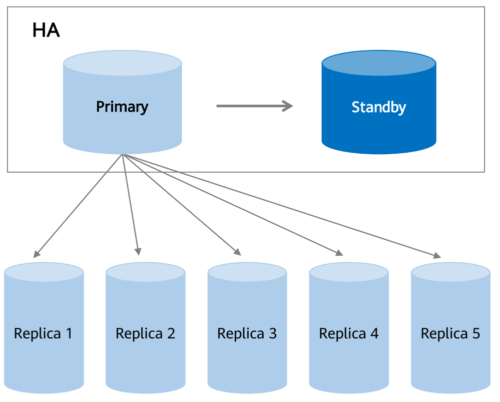
- RDS for MySQL includes a comprehensive performance monitoring system, multi-level security protection measures, and a professional database management platform that allow you to easily set up and scale up databases. On the RDS for MySQL console, you can perform necessary tasks and no programming is required. The console simplifies operations and reduces routine O&M workloads, so you can stay focused on application and service development.
  - It uses a stable architecture and supports a range of web applications. It is also a cost-effective solution for small and medium enterprises.
  - A web-based console is available for you to monitor a comprehensive range of performance metrics.
  - You can flexibly scale resources based on your service requirements but you still pay only for what you use.



- Huawei Cloud RDS presents a significant edge over traditional databases. With RDS, you can deploy enterprise-grade MySQL databases without any worries about setup, configuration, maintenance, backups, or uptime.

RDS for MySQL Features - Cross-AZ HA

Primary/standby instances across AZs

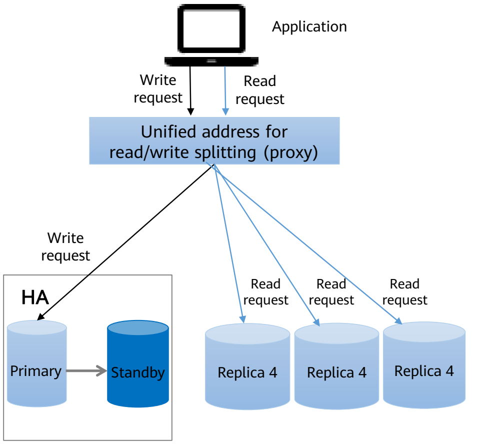


Functions

- Cross-AZ HA supports switchover in seconds.
- Up to 5 read replicas can be created for offloading read traffic.
- Standby DB instances are invisible to users. Users can access DB instances through virtual IP addresses.
- Read replicas cannot exist alone and must come with single or primary/standby DB instances.

- Cross-AZ HA is an effective disaster recovery mechanism. When your database services have high reliability requirements, you can deploy the primary and standby databases across AZs to achieve AZ-level DR and ensure system reliability.

RDS for MySQL Features - Read/Write Splitting



Functions

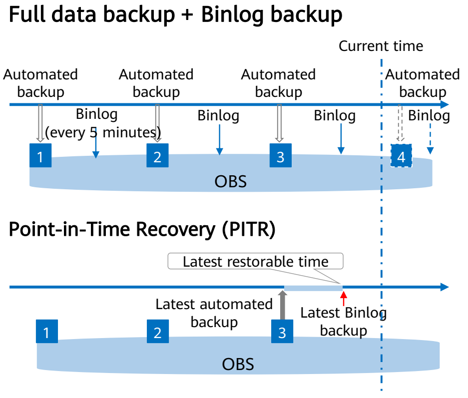
- A single read/write splitting address is provided, transparent to applications.
- Read-only permissions can be configured for each node.
- Instance health check is performed. If a DB instance breaks down or the latency exceeds what is supported, read requests are no longer allocated to the instance.

Advantages

- A single read/write splitting address is provided, and read/write splitting does not require application reconstruction.
- The read weight assigned to a read replica is configurable.

- Read/write splitting enables read and write requests to be automatically routed through a read/write splitting address. You can enable read/write splitting after read replicas are created. Write requests are automatically routed to the primary DB instance and read requests are routed to read replicas by user-defined weights.

RDS for MySQL Feature - Point-In-Time Recovery (PITR)



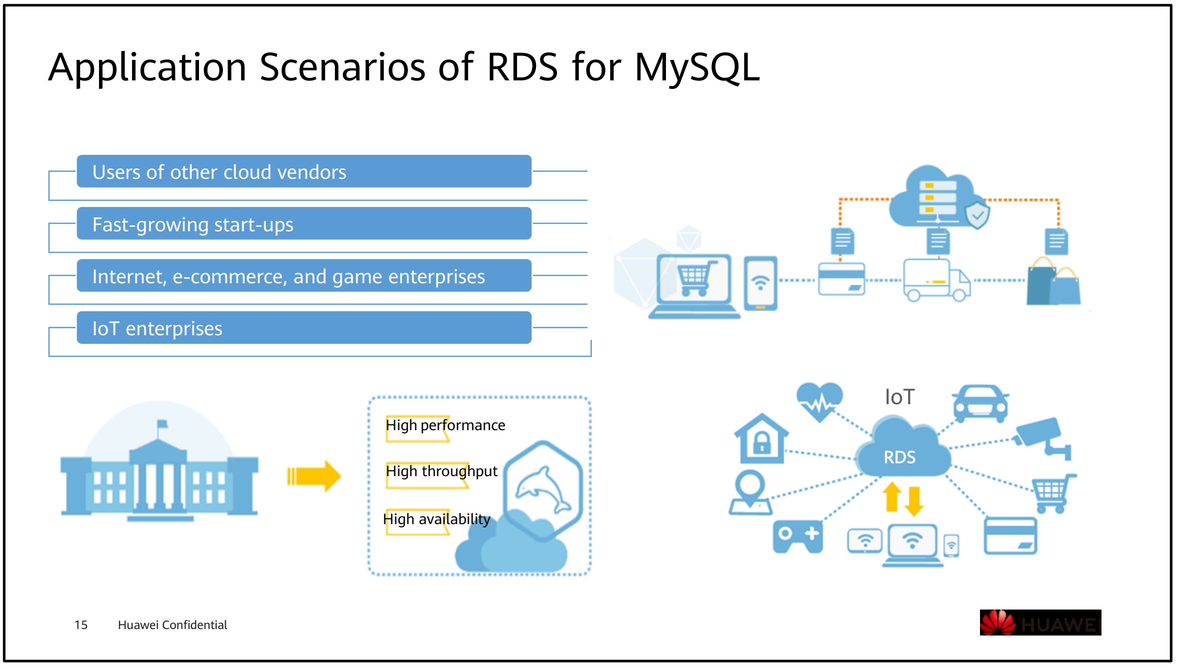
Functions

- Instance-level restoration in seconds is supported.
- Automated backups can be configured to be saved for up to 732 days (approximately 2 years).
- You can restore data to any point in time at least 5 minutes ago and restore the data to a new DB instance or to the original DB instance.

Advantages

- The backup retention period is up to 732 days.
- RDS provides free backup space approximately equal to your purchased storage space.

- You can use backups to restore data to any point in time. Binlog is a binary log used to record MySQL DB table structure changes and table data changes.

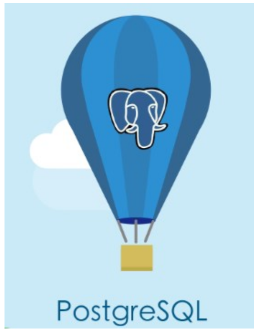


- RDS for MySQL is mainly used in the following scenarios:
  - Users of public cloud platforms other than HUAWEI CLOUD generally use RDS for MySQL.
  - Start-ups choose RDS for MySQL in the early stages because they need ways to support fast growth on a limited budget.
  - MySQL is used widely by Internet, e-commerce, and game enterprises. When migrating databases to the cloud, these types of enterprises choose RDS for MySQL.
  - IoT applications tend to be very large scale and they need to be extremely reliable. RDS for MySQL is the first choice for IoT enterprises because it allows for a large number of concurrent connections and does not require customers to reconstruct their applications.

Contents

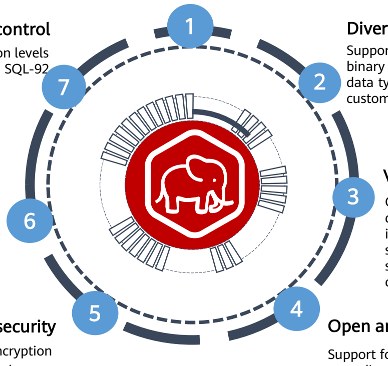
What Is RDS for PostgreSQL?

- RDS for PostgreSQL is a typical open-source relational database that excels in data reliability and integrity. It supports Internet e-commerce, geographic location application systems, financial insurance systems, complex data object processing, and other applications.



- PostgreSQL is based on Postgres, which was developed at the University of California, Berkeley. After more than 30 years of development, PostgreSQL has become the most powerful open-source database in the world. It has earned a reputation for reliability, stability, and data consistency, and has become the preferred open-source relational database for many enterprises.
- PostgreSQL is an open source object-relational database management system focused on extensibility and standards compliance. It is known as the most advanced open source database. RDS for PostgreSQL is designed for enterprise-oriented OLTP scenarios and supports NoSQL (JSON, XML, or hstore) and GIS data types. It has earned a reputation for reliability and data integrity, and is suitable for websites, location-based applications, and complex data object processing.
- RDS for PostgreSQL supports the Postgres plugin, which provides excellent spatial performance.
- RDS for PostgreSQL is a cost-effective solution for a range of different scenarios. You can flexibly scale resources based on service requirements and you only pay for what you use.

RDS for PostgreSQL Features



Reliability, stability, data consistency, and integrity
Always the most important features for database products

Diversified data types
Support for not only traditional data types, but also binary large objects, arrays, spatial data types, network data types, JSON, and XML, all of which can be customized

Various third-party plugins
Geographic information, space, heterogeneous database access, machine learning, text retrieval, image, time series, multi-dimension, word segmentation, blockchain, column storage, and similarity algorithms, all of which can be customized

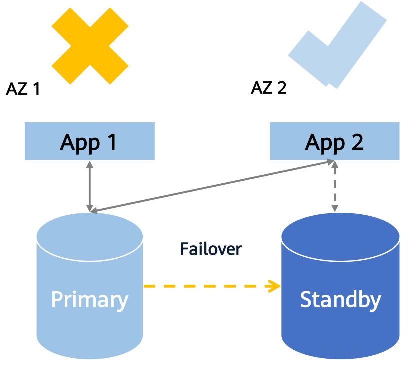
Open architecture
Support for functions, operators, indexes, and languages, extending functions and maintaining kernel stability

High security
Field encryption
Permission control based on databases, schemas, tables, and columns
Identity authentication and certificate

Excellent performance and abundant indexes
Query optimizers, parallel computing, complex query optimization, hash aggregation, hash JOIN, subquery, analysis query, function processing, and other analysis functions
Indexes: B-tree, GiST, GIN, SP-GiST, Hash, Brin, rum, bloom, and bitmap

Powerful concurrency control
Support for four transaction isolation levels defined in ANSI SQL-92

RDS for PostgreSQL Features - High Availability

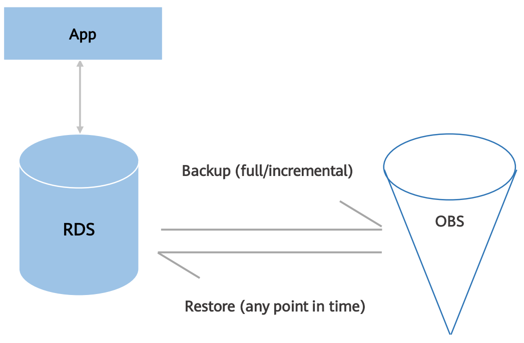


Benefits of the HA cluster architecture:

- You can choose a failover policy to prioritize reliability or availability.
- DB instances can be deployed in one AZ or across AZs and can automatically fail over within a cluster.
- You can manually switch a primary instance to standby to simulate a fault.
- A read replica can automatically associate itself with a new primary node.
- A switchover can be completed in seconds.
- The standby database does not handle traffic. It only ensures RTO.
- A Huawei-developed HA Monitor module is used.
- Virtual IP addresses can be switched completely invisibly to the applications.
- Multiple primary/standby switchovers can be performed.
- Automatic fault detection is provided.

- RTO stands for Recovery Time Objective. It is the length of time from when an IT system breaks down and services stop to when the system recovers.
- HA Monitor is an HA monitoring module.

RDS for PostgreSQL Features - Point-In-Time Recovery(PITR)



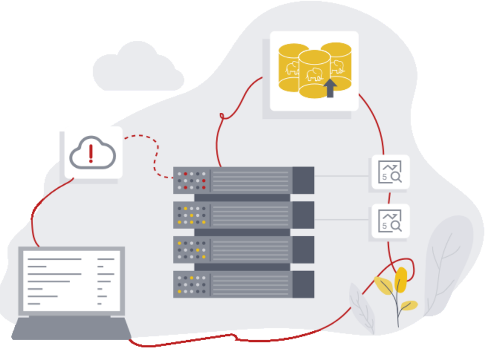
- Backup cycle: 7 to 732 days
- Pay-per-use: Free EVS storage space equal to the requested storage and virtually limitlessly expandable
- Reliability: Up to 11 nines of data reliability
- Security encryption: KMS encryption and multiple protections

Data archived in OBS can be restored to any point in time.

- RDS stands for Relational Database Service.
- EVS stands for Elastic Volume Service.
- OBS stands for Object Storage Service.

RDS for PostgreSQL Features - Enhanced Features

- Read Replica
  - Up to five read replicas can be added to an existing DB instance to offload read-heavy database work
- Logical Replication Subscription
  - All logical slots can be replicated from the prir to the standby instance.
- Scheduled Tasks
  - Tasks can be run automatically during off-pea based on a schedule that you control.



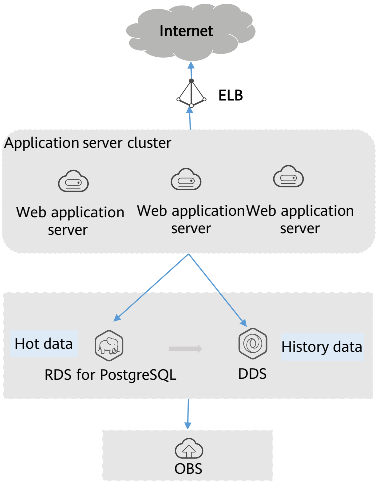
RDS for PostgreSQL Features - Support for 90+ Plugins

- Multiple Types of Geospatial Data:
  - Support PostGIS for 2D and 3D models, with spa
    objects, indexes, operation functions, and operat
- Time Series Data:
  - Support the TimescaleDB time-series database plugin, partition tables, and BRIN indexes.
- Search Indexes:
  - Provide a wide range of indexes, including function- and condition-based indexes, for faster full-text search.

RDS for PostgreSQL Features - Professional Database O&M Platform

- Instance Management:
  - Manage your instances with ease using flexible console-based capabilities.
- Real-Time Monitoring:
  - View key operational metrics of your instances, including vCPU/storage utilization, I/O activity, and instance connections, define custom alarm rules as needed.
- Backup and Restoration:
  - Restore data from backups to any point in time. Backups can b saved for up to 732 days.
- Automated Failover:
  - Maintain the uptime of your workloads with automated failove

Application Scenarios of RDS for PostgreSQL



- **Utilize Spatial and Geographic Objects**

RDS for PostgreSQL supports the PostGIS extender plugin for location-based applications providing excellent spatial performance. It provides spatial features including space objects, indexes, operation functions, and operators.

- **Advantages**
  - Extra types: Supports various types of spatial data including points, lines, planes, grids, and three-dimensional data.
  - Efficient Spatial Analysis: Provides efficient spatial analysis functions and works with OBS for unlimited object storage.
  - Simple Spatial Operations: Reduces the code complexity for location applications, making spatial operations easier.

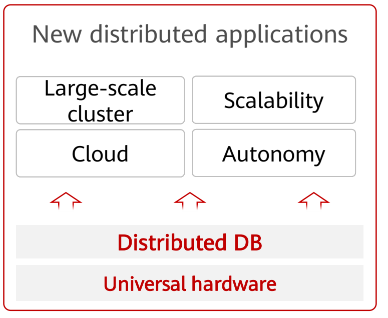
- Mixed-mode operations combining OLTP and OLAP are supported.
- Multiple data models are applicable to spatiotemporal, geographic, heterogeneous, image, text retrieval, time series, stream computing, and multi-dimensional scenarios.
- Huawei provides you with a reliable database service and keeps your data consistent.
- To replace Oracle databases, there are two solutions available:
  - Use RDS for PostgreSQL Enhanced Edition.
  - Use RDS for PostgreSQL Community Edition and Oracle plug-ins.

Contents

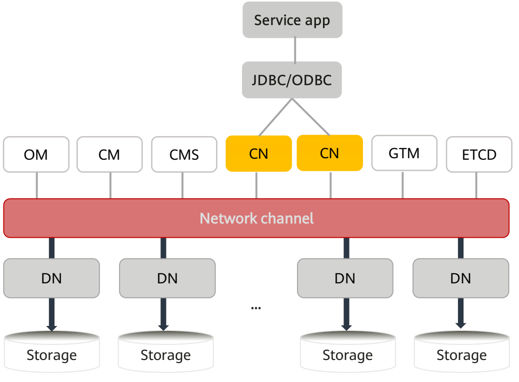
What Is GaussDB

- GaussDB is a distributed relational database from Huawei. It supports intra-city cross-AZ deployment with zero data loss. With a distributed architecture, GaussDB supports petabytes of storage and contains more than 1,000 nodes per DB instance. It is highly available, secure, and scalable and provides services including quick deployment, backup, restoration, monitoring, and alarm reporting for enterprises.

- Migrating Data from Centralized DB to Distributed DB Is Driving Digital Transformation.



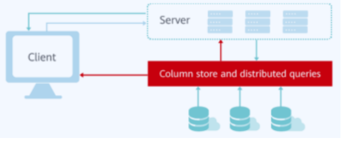
Key Components of the GaussDB Distributed Architecture



Component	Description
OM	OM(Operation Manager): Provides management APIs and tools for routine O&M and configurations
CM	CM(Cluster Manager): Manages and monitors the running status of functional units and physical resources in a distributed system, ensuring the stable running of the entire system
GTM	GTM(Global Transaction Manager): Generates and maintains globally unique information based on global transaction IDs
CN	CN(Coordinator Node): Receives access requests from applications and returns execution results to clients. A CN also splits tasks and schedules task shards on each data node
DN	DN(Data Node): Stores data, executes data queries, and returns the results to a CN
ETCD	ETCD(Editable Text Configuration Daemon): Ensures the consistency of replicas as an arbitration compo
Storage	Storage resource, used for persistent data storage

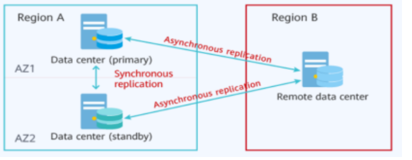
Advantages of GaussDB –(1)

Performace



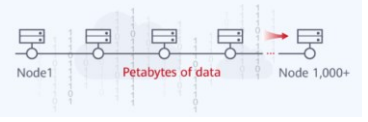
- A DB instance with 32 nodes can reach up to 15 million tpmC. According to the TPC-H benchmark test for performance.

Availability



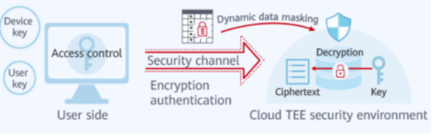
- You can deploy a DB instance within a single AZ, across AZs, or across regions as required. GaussDB supports RPO=0，RTO<10 s.

Scalability



- Global consistency of distributed transactions breaks through the performance bottlenecks of traditional databases. Compute and storage can be scaled separately and flexibly.

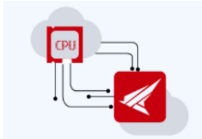
Security



- GaussDB provides end-to-end data security with access control, encryption authentication, database audit, d dynamic data masking, and the Always Encrypted feature, all ensuring you data remains safe and secure.

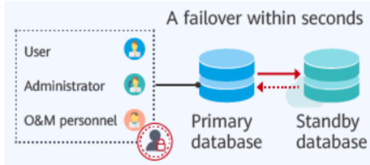
Advantages of GaussDB –(2)

Powerful Comput



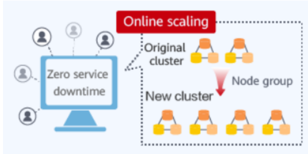
- Powerful Compute 6th Gen Intel Core processors and Kunpeng processors

Online Scaling



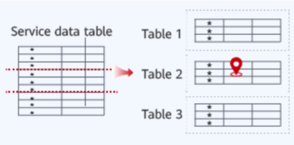
- Excellent linear performance scaling and online redistribution of new shards .

Security and Reliabilit



- Service faults are automatically monitored and recovered from, ensuring service continuity and zero data loss.

Powerful Comput

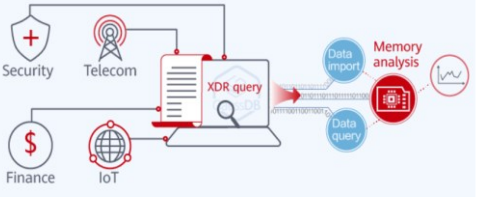


- Based on a shared-nothing architecture, data is sharded automatically. GTM-lite technology is used to ensure strong consistency for transactions and to eliminate performance bottlenecks on the central node.

Application Scenarios of GaussDB

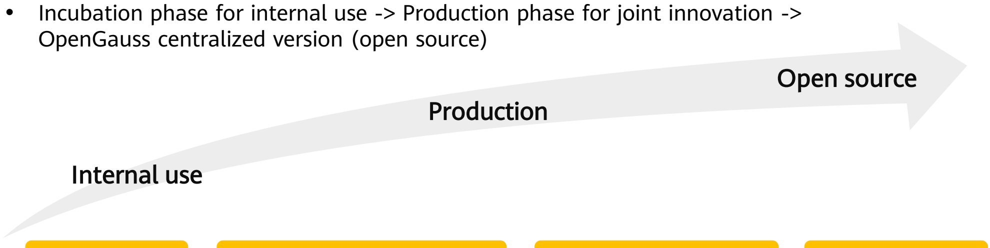
- **Transaction applications**
  - The distributed, highly scalable architecture of GaussDB makes it an ideal fit for highly concurrent online transactions containing a large volume of data from government, finance, e-commerce, O2O, telecom customer relationship management (CRM), and billing. GaussDB supports different deployment models.

- **CDR query**
  - GaussDB can process petabytes of data and use the memory analysis technology to query massive volumes of data when data is being written to databases. Therefore, it is suitable for the Call Detail Record (CDR) query service in the security, telecom, finance, and Internet of things (IoT) sector



Centralized openGauss Kernel Completely Open-Source

- Incubation phase for internal use -> Production phase for joint innovation -> OpenGauss centralized version (open source)



2001–2011

- Introduced enterprise-class in-memory databases.

2011–2019

- Bank G's core data warehouse and GaussDB(DWS) on HUAWEI CLOUD were deployed for commercial use.
- Z Bank's core service system replaced the commercial database.
- 30,000+ sets of Huawei's 40+ flagship products deployed commercially for 70+ carriers worldwide, and a user base of over 2 billion.

2019–2020

- GaussDB launched globally on May 15, 2019.
- Partner ecosystem established.
- Compatible with mainstream ecosystems in the industry and completed interconnection with finance and other industries.

2020–Now

- OpenGauss centralized version (open source) used.

Contents

Customer Requirements on Cloud Security

CSA Top Threats	
• Data Leakage	• Advanced persistent threat (APT)
• Insufficient identity, credential, and access management	• Data loss
• Insecure ports and APIs	• Insufficient due diligence
• System vulnerabilities	• Abuse and nefarious use of cloud services
• Account hijacking	• Denial of service (DoS)
• Malicious insiders	• Shared technology vulnerabilities

Key Security Requirements for Enterprise Cloudification

Continuous services	Controllable O&M	Data confidentiality
• Defend against network attackers and hackers. Comply with laws and regulations.	• Configure security policies. Detect and eliminate risks. Audit and trace operations.	• Prevent data breach. Data is accessible only to authorized staff.

HUAWEI CLOUD Security Services

- Build a series of top-quality security services for ensuring data security.

Data Security
- Data Encryption Workshop (DEW)
- Database Security Service (DBSS)

App security
- Web Application Firewall (WAF)
- Vulnerability Scan Service (VSS)

Compute security
- Host Security Service (HSS)
- Container Guard Service (CGS)

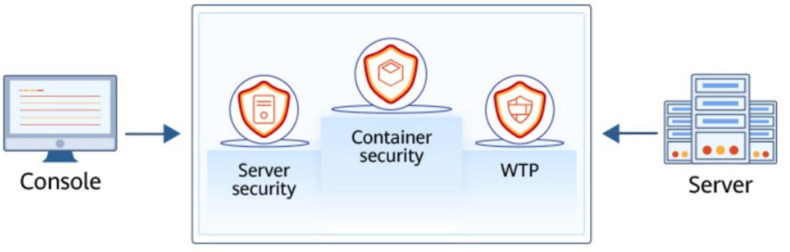
Cyber security
- Cloud Firewall (CFW)
- Advanced Anti-DDoS (AAD)

Management security
- Cloud Trace Service (CTS)
- Log Tank Service (LTS)
- Cloud Certificate Manager (CCM)
- Cloud Bastion Host (CBH)
- Situation Awareness (SA)

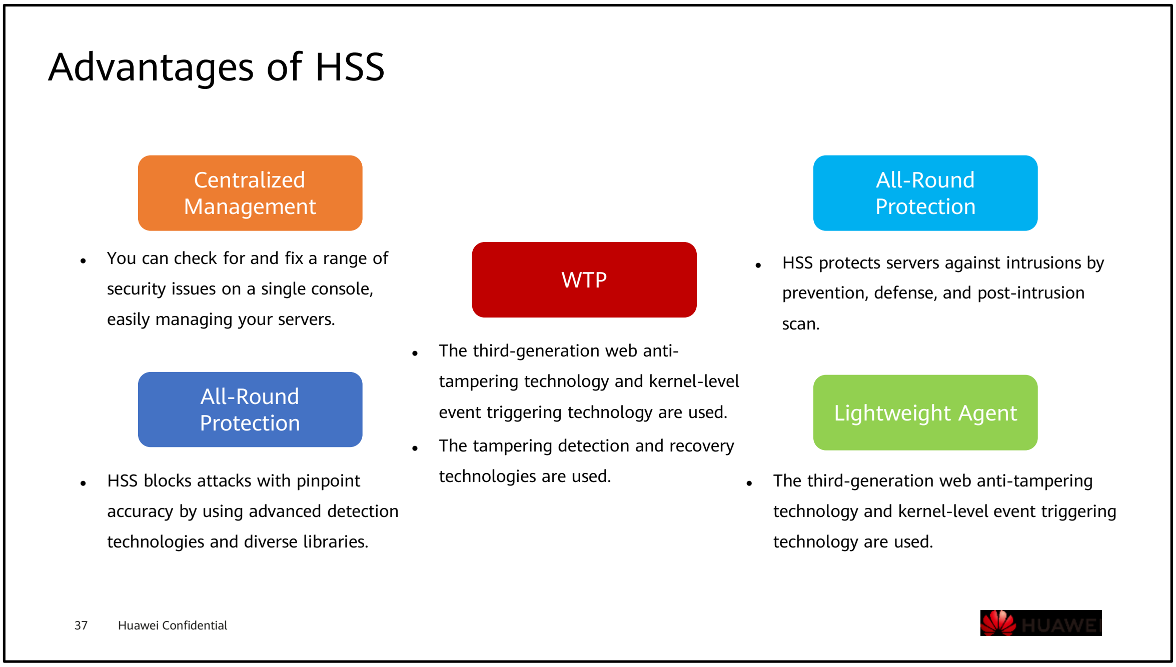
Contents

What Is HSS

- Host Security Service (HSS) is designed to protect server workloads in hybrid clouds and multi-cloud data centers. It provides host security functions and Web Tamper Protection (WTP).
- HSS can help you remotely check and manage your servers and containers in a unified manner.



- HSS protects your system integrity, enhances application security, monitors user operations, and detects intrusion
- Host Security Service (HSS) helps you identify and manage the assets on your servers, eliminate risks, and defend against intrusions and web page tampering. There are also advanced protection and security operations functions available to help you easily detect and handle threats.
- Install the HSS agent on your servers, and you will be able to check the server protection status and risks in a region on the HSS console.



- You can install the agent on Huawei Cloud ECSs, BMSs, offline servers, and third-party cloud servers in the same region to manage them all on a single console.
- On the security console, you can view the sources of server risks in a region, handle them according to displayed suggestions, and use filter, search, and batch processing functions to quickly analyze the risks of all servers in the region.
- The tampering detection and recovery technologies are used. Files modified only by authorized users are backed up on local and remote servers in real time, and will be used to recover tampered websites (if any) detected by HSS.

What Is HSS

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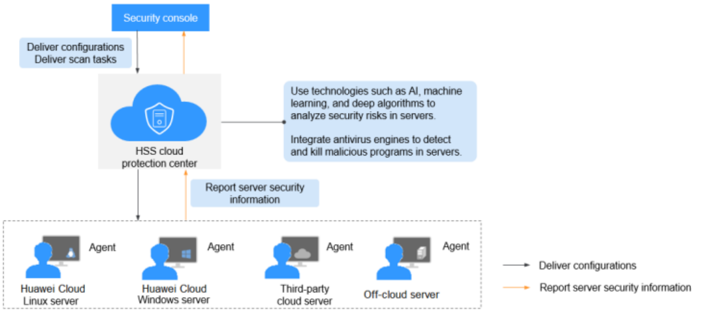

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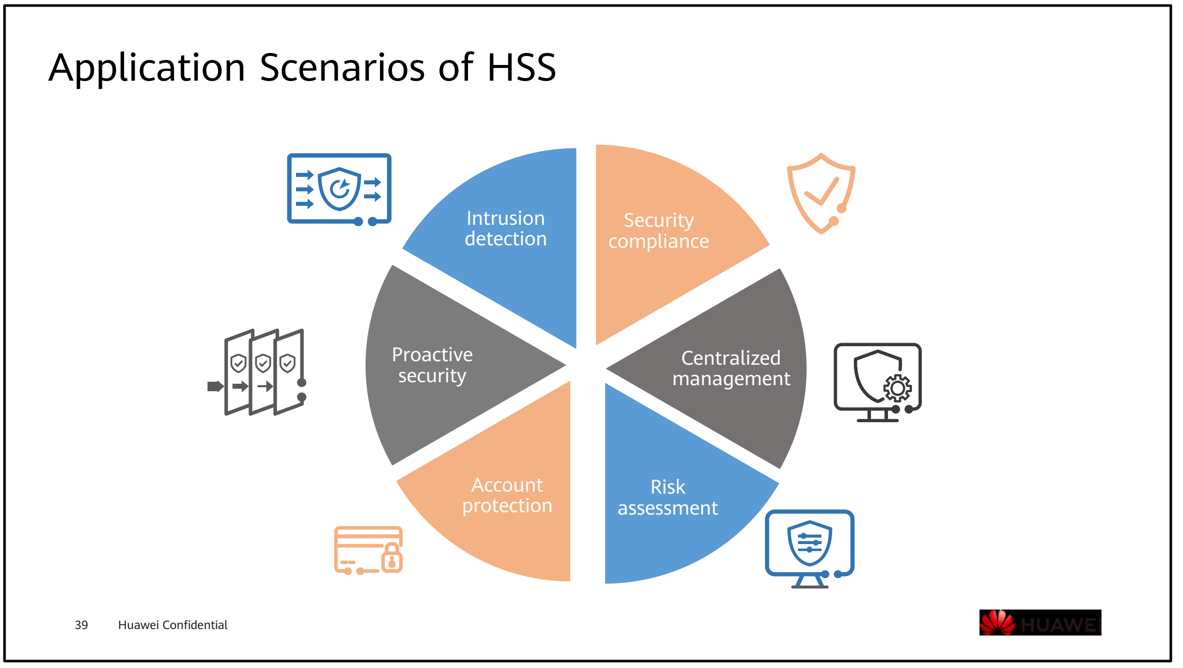
- Host Security Service (HSS) helps you identify and manage the assets on your servers, eliminate risks, and defend against intrusions and web page tampering. There are also advanced protection and security operations functions available to help you easily detect and handle threats.
- Install the HSS agent on your servers, and you will be able to check the server protection status and risks in a region on the HSS console.

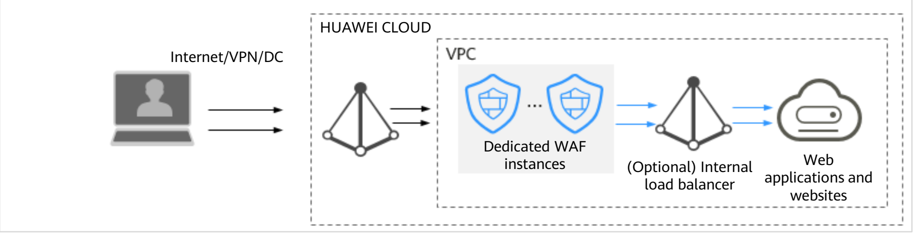


- HSS applications:
  - DJCP Multi-level Protection Scheme (MLPS) compliance, The intrusion detection function of HSS protects accounts and systems on cloud servers, helping companies meet compliance standards.
  - Centralized management: You can manage servers, security configurations, and security events all from the HSS console. You can reduce security risks from a single convenient portal and keep management costs down.
  - Risk assessment: HSS scans your servers for risks, including unsafe accounts, ports, software vulnerabilities, and weak passwords, and prompts you to eliminate any security risks identified and harden the system in a timely manner.
  - Account protection: Accounts are protected before, during, and after a security event. You can also use 2FA to block brute-force attacks on accounts, enhancing the security of your cloud servers.
  - Proactive defense: You can count and scan your server assets, check and fix vulnerabilities and unsafe settings, and proactively protect your network, applications, and files from attacks.
  - Intrusion detection: You can scan all possible attack vectors to detect and fight APTs and other threats in real time, protecting your system from their impacts.

Contents

What Is WAF

- Web Application Firewall (WAF) keeps web services stable and secure. It examines all HTTP and HTTPS requests to detect and block the following attacks: Structured Query Language (SQL) injection, cross-site scripting (XSS), web shells, command and code injections, file inclusion, sensitive file access, third-party vulnerability exploits, Challenge Collapsar (CC) attacks, malicious crawlers, and cross-site request forgery (CSRF) .



- After you purchase a WAF instance, add your website domain to the WAF instance on the WAF console. All public network traffic for your website then goes to WAF first. WAF identifies and filters out the illegitimate traffic, and routes only the legitimate traffic to your origin server to ensure site security.

Advantages of WAF

- WAF examines web traffic from multiple dimensions to accurately identify malicious requests and filter attacks, reducing the risks of data being tampered with or stolen.

Precisely and Efficiently Identify Threats

- WAF uses rule and AI dual engines and integrates our latest security rules and best practices.

Zero-Day Vulnerabilities Patched Fast

- A specialized security team provides 24/7 service support to fix zero-day vulnerabilities within 2 hours.

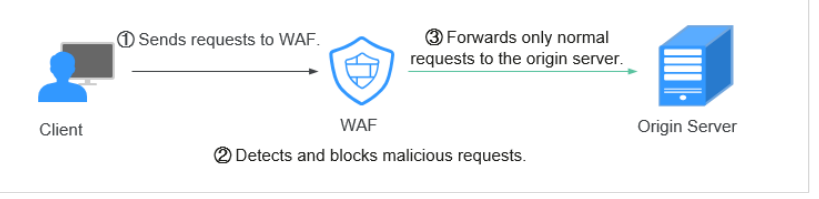
Strong Protection for User Data Privacy

- Sensitive information, such as accounts and passwords, in attack logs can be anonymized.
- PCI-DSS checks for SSL encryption are available.
- The minimum TLS protocol version and cipher suite can be configured.

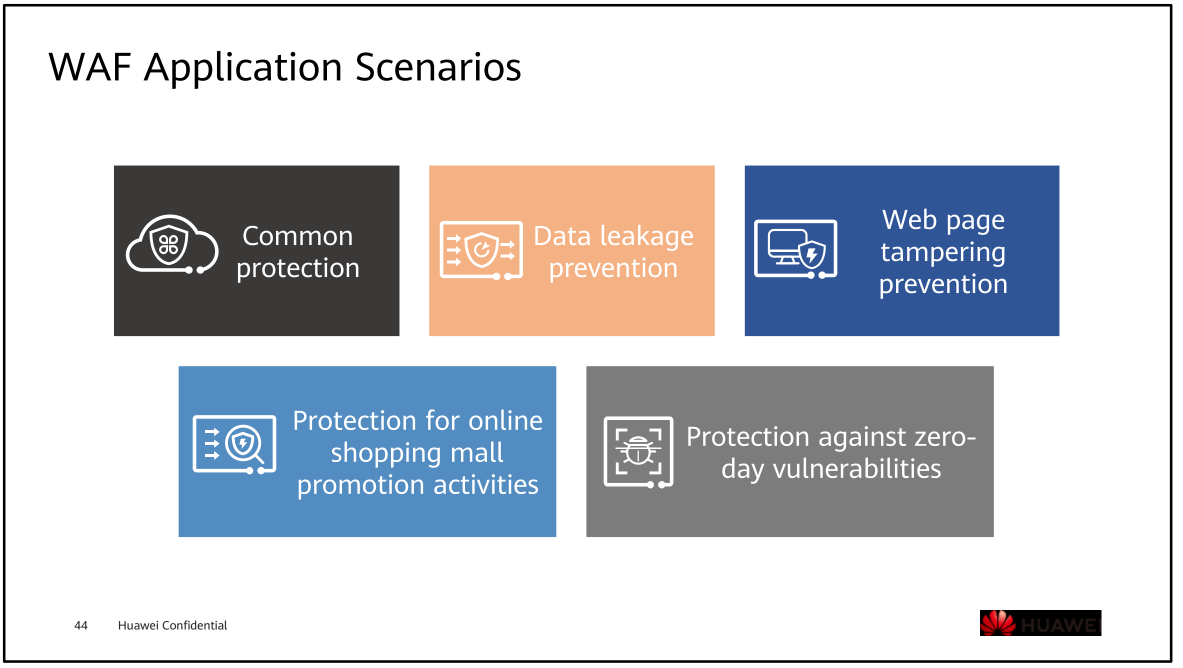
- Basic web protection:
  - Backed by an extensive preset reputation database, WAF can defend against the Open Web Application Security Project (OWASP) top 10 threats, vulnerability exploits, web shells, and other threats.
  - WAF detects and blocks varied attacks, such as SQL injection, XSS attacks, remote overflow vulnerabilities, file inclusion, Bash vulnerabilities, remote command execution, directory traversal, sensitive file access, and command/code injection.
  - WAF provides web shell detection, protecting web applications from web shells.
- Precise identification:
  - WAF uses a wide range of techniques to identify attacks. For example, WAF uses a dual-engine architecture, combining built-in semantic analysis engine and regex engine. WAF enables users to configure blacklist and whitelist rules, so WAF has a low false positive rate.
  - WAF can automatically decode common codes no matter how many times they are encoded. WAF can decode a wide range of code types, including url_encode, Unicode, XML, C-OCT, hexadecimal, HTML escape, and base64 code, case confusion, JavaScript, shell, and PHP concatenation confusion.
  - WAF deep inspection identifies and blocks evasion attacks, including those that use homomorphic character obfuscation, command injection with deformed wildcard characters, UTF7, data URI scheme, and other techniques.
  - WAF header detection inspects all header fields in received requests.

How WAF Works

- After a website is connected to WAF, all website access requests are forwarded to WAF first. Then, WAF inspects the traffic, filters out malicious traffic, and routes only normal traffic to the origin server, keeping the origin server secure, stable, and available.



- To enable WAF, after purchasing a WAF instance, go to the WAF console and connect the website to be protected to the WAF instance. After that, all website access requests go to WAF first. Then, WAF inspects the traffic, filters out attacks, and routes only normal traffic to the origin server, keeping the origin server secure, stable, and available.
- The process of forwarding website traffic to the origin server through WAF is called back-to-source. WAF inspects traffic originating from the client and uses WAF back-to-source IP addresses to forward normal traffic to the origin server. To the origin server, source IP addresses of all requests are the WAF back-to-source IP addresses. In this way, the IP address of the origin server is hidden from the client.

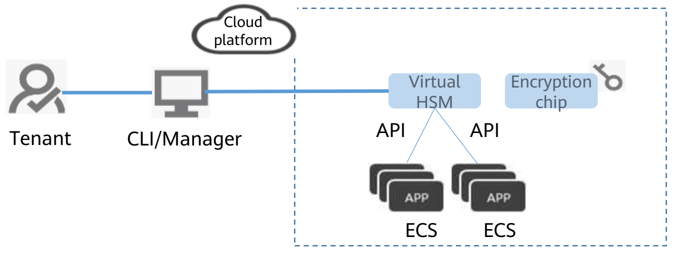


- WAF application scenarios:
  - Common protection: WAF helps users defend against common web attacks, such as command injection and sensitive file access.
  - Protection for online shopping mall promotion activities: Countless malicious requests may be sent to service interfaces during online promotions. WAF allows configurable rate limiting policies to defend against CC attacks. This prevents services from breaking down due to many concurrent requests, ensuring response to legitimate requests.
  - Protection against zero-day vulnerabilities: Services cannot recover quickly from impact of zero-day vulnerabilities in third-party web frameworks and plug-ins.
  - Data leakage prevention: WAF prevents malicious actors from using methods such as SQL injection and web shells to bypass application security and gain remote access to web databases.
  - Web page tampering prevention: WAF ensures that attackers cannot leave backdoors on your web servers or tamper with your web page content, preventing damage to your credibility.

Contents

What Is DEW?

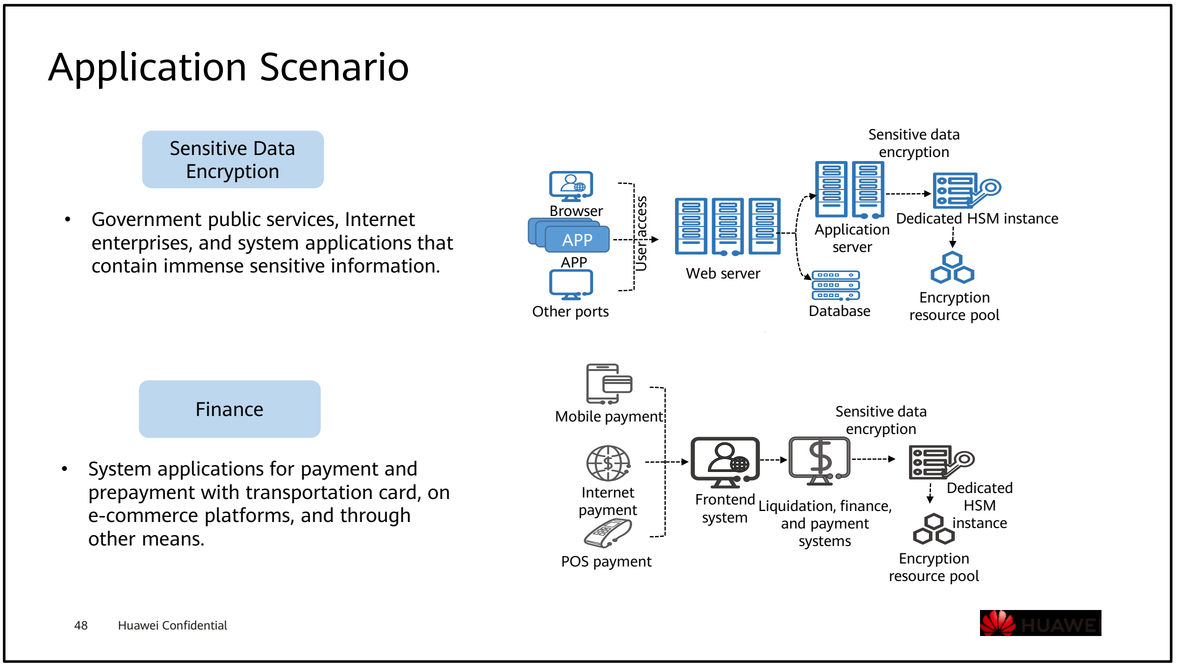
- Data Encryption Workshop (DEW) is a cloud data encryption service. It consists of the following services: Key Management Service (KMS), Cloud Secret Management Service (CSMS), Key Pair Service (KPS), and Dedicated Hardware Security Module (Dedicated HSM).
- It helps you secure your data and keys, simplifying key management. DEW uses HSMs to protect the security of your keys, and can be integrated with other Huawei Cloud services to address data security, key security, and key management issues.



- Data is a core enterprise asset, and data breaches can result in immeasurable losses. DEW can encrypt customer data and protect it from data leaks.
- DEW uses HSMs to protect your keys, and can be integrated with other HUAWEI CLOUD services to address data security, key security, and key management issues. You can also develop your own encryption applications based on DEW.

DEW Services

- DEW consists of the following services: Key Management Service (KMS), Cloud Secret Management Service (CSMS), Key Pair Service (KPS), and Dedicated Hardware Security Module (Dedicated HSM).



- After a Dedicated HSM instance is purchased, you can use the UKey provided by Dedicated HSM to initialize and manage the instance. You can fully control the key generation, storage, and access authentication.
- Applications:
  - Sensitive data encryption: government public services, Internet enterprises, and system applications that contain immense volumes of sensitive information
  - Payment: payment and prepayment applications, such as transportation cards and e-commerce platforms
  - Verification: Dedicated HSM can ensure the confidentiality and integrity of electronic contracts, invoices, insurance policies, and medical records during transmission and storage.

Contents

Pain Points

Poor user experience

Slow access
Download or access failures
Video freezing

High concurrency

Traffic bursts and concurrent downloads when e-commerce promotions, popular games, or hit TV series go online

Large bandwidth consumption and high costs

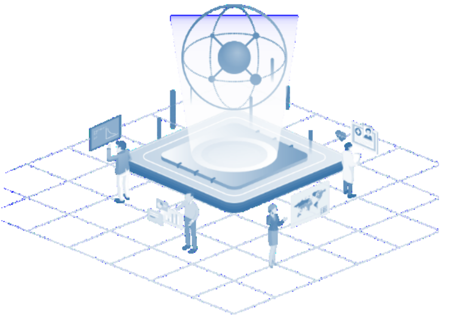
Without CDN, everything must be fetched from origin servers, which wastes bandwidth and costs money.

Heavy O&M workload

Limited bandwidth
A huge number of concurrent requests
Inefficient O&M

What Is CDN

- Content Delivery Network (CDN) is a smart virtual network on the Internet infrastructure. CDN can cache origin content on nodes closer to users, so content can load faster. CDN speeds up site response and improves site availability. It breaks through the bottlenecks caused by low bandwidth, heavy access traffic, and uneven distribution of edge nodes.



- Huawei Cloud CDN caches origin content on edge nodes across the globe. Users can get content from the nearest nodes instead of from the origin server far way from them. This reduces latency and improves user experience. Using preset policies (including content types, geological locations, and network loads), CDN provides users with the IP address of a node that responds the fastest. Users get the requested content faster than would have otherwise been possible.

HUAWEI CLOUD Global CND Node Information

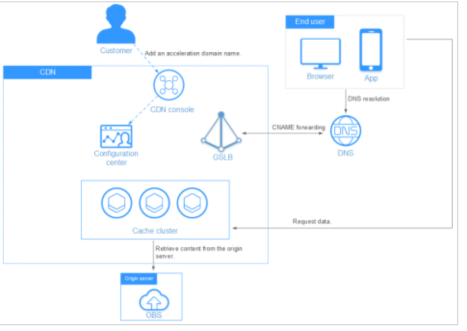
- Huawei Cloud CDN has over 2000 edge nodes in the Chinese mainland and over 800 edge nodes outside the Chinese mainland. The network-wide bandwidth reaches 150 Tbit/s. The edge nodes are connected to the networks of top carriers in China such as China Telecom, China Unicom, China Mobile, and China Education and Research Network (CERNET), as well as many small- and medium-sized carriers. CDN covers more than 130 countries and regions. It deploys nodes on networks of over 1600 carriers. CDN schedules user requests to the most appropriate nodes, accelerating content delivery.

Advantages of CDN

Global Network	Precise Scheduling	Ease of Use	High-Performance Cache	Security
Get content delivered from 2800+ edge nodes from popular carriers in six continents. Bandwidth reaches 150 Tbit/s network-wide.	Enhance user access experience with accurate, evolving IP geolocation database and dynamic node adjustment.	Configure domains with CDN in several steps, customize them on the console, and call open APIs for app integration and cross-cloud management.	Improve cache hit ratio and shorten access queues with proprietary AICache, multi-level cache scheduling, and fast, massive SSD storage.	Protect your resources with network-wide HTTPS transmission, anti-leeching, referer validation, URL validation, and access control.

Application Scenarios of CDN

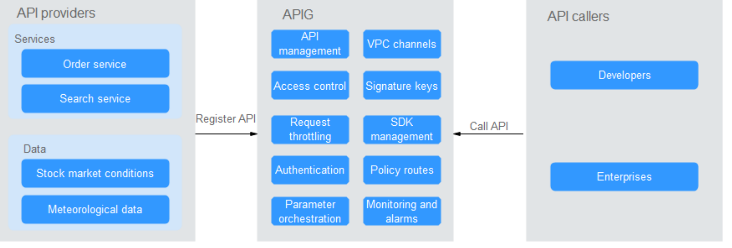
- CDN is useful for download clients, game clients, app stores, and websites that provide download services based on HTTP or HTTPS. An increasing number of new services need to update software in real time. Conventional download services need to provide even more and larger downloads. If origin servers have to handle all these requests, it places tremendous strain on these servers and results in bottlenecks. CDN can distribute content to edge nodes, ease the pressure on origin servers, and speed up downloads.



Contents

What Is APIG

- API Gateway (APIG) is your cloud native gateway service. With APIG, you can build, manage, and deploy APIs at any scale to package your capabilities. With just a few clicks, you can integrate internal systems, monetize service capabilities, and selectively expose capabilities with minimal costs and risks.
- APIG helps you monetize service capabilities and reduce R&D investment, and enables you to focus on core enterprise services to improve operational efficiency.



- To monetize your capabilities (VM clusters, data, and microservice clusters), you can open them up by creating APIs in APIG. Then you can provide the APIs for API callers using offline channels.
- You can also obtain open APIs from APIG to reduce your development time and costs.

Product Functions

API Lifecycle Management

APIG provides full-lifecycle API management, including design, development, test, publish, and O&M.

Available Out-of-the-Box

You can quickly create APIs by configuring the required settings on the APIG console.

Refined Request Throttling

APIG combines synchronous and asynchronous traffic control and multiple algorithms to throttle requests at the second level.

Higher Performance

APIG integrates the nodes for security, load balancing, inbound traffic governance, and microservice governance, improving performance while reducing deployment and O&M costs.

Operation Quality Assurance

Hosting open APIs of all Huawei Cloud services, APIG helps improve the quality process system with ensured reliability and stability.

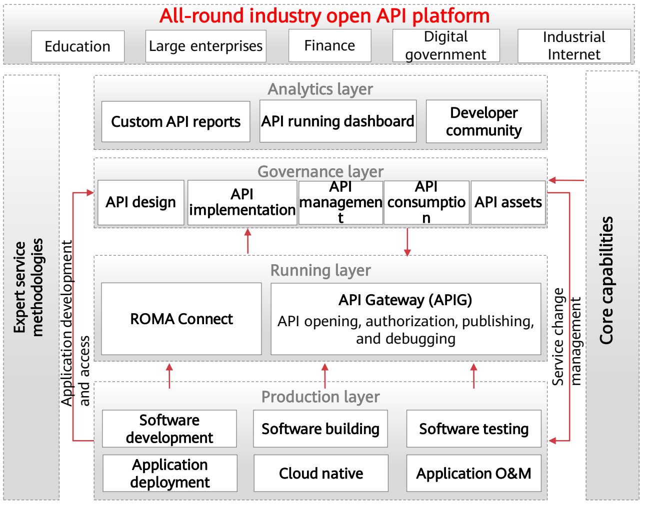
Function Invocation

APIG seamlessly works with FunctionGraph, enabling you to selectively expose FunctionGraph functions in the form of APIs

Security Protection

APIG support SSL transfer, strict access control, IP address blacklist/whitelist, authentication, anti-replay, anti-attack, and multiple audit rules

API open platform

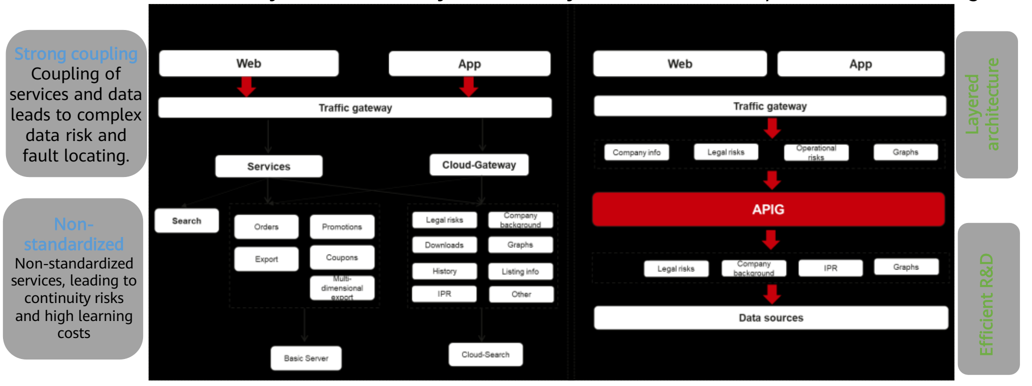


- API Arts is an integrated solution platform for API lifecycle management.
- It enables developers to efficiently implement one-stop experience in API design, API development, API testing, API hosting, API O&M, and API monetization. With API contracts as anchors, API Arts ensures high data consistency in API phases and provides developers with a user-friendly end-to-end solution for the entire API process.
- With API Arts, developers can efficiently, standardize, and accurately cultivate and protect their APIs, and easily participate in the API economy.

- API O&M personnel:
  - Provides cloud-native API gateways to flexibly connect to any backend, meeting unified microservice governance requirements.
  - Provides SLA assurance for ultra-high concurrency.
  - Provides full-stack security assurance to ensure service security at the protocol layer, access layer, audit layer, and backend integration layer.
- API O&M personnel:
  - Provides cloud-native API gateways to flexibly connect to any backend, meeting unified microservice governance requirements.
  - Provides SLA assurance for ultra-high concurrency.
  - Provides full-stack security assurance to ensure service security at the protocol layer, access layer, audit layer, and backend integration layer.
- API Manager:
  - One-stop overview of enterprise API asset subscription, usage, and evaluation.
- API customers and partners:
  - Quickly access API assets through the Marketplace.
  - Reuse API assets on the Developer Portal.

Application Scenarios of API

- With the rapid development of enterprises and rapid service changes, internal systems of enterprises need to change along with service requirements. However, internal systems of enterprises depend on each other. To maintain system universality and stability, it is difficult to cope with service changes.

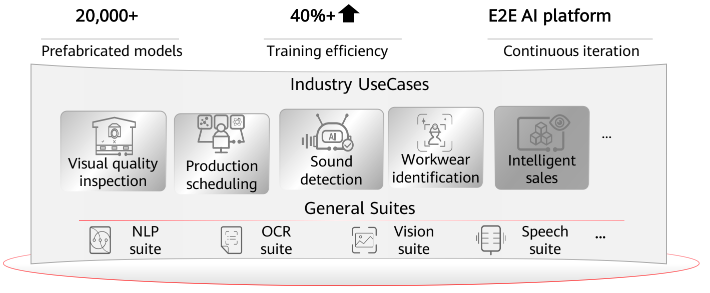


- API Gateway uses RESTful APIs to simplify the service architecture. Standardized APIs are used to quickly decouple internal systems and separate front-end and back-end systems. In addition, existing capabilities are reused to avoid resource waste caused by repeated development.

Contents

One-Stop AI Development Platform ModelArts

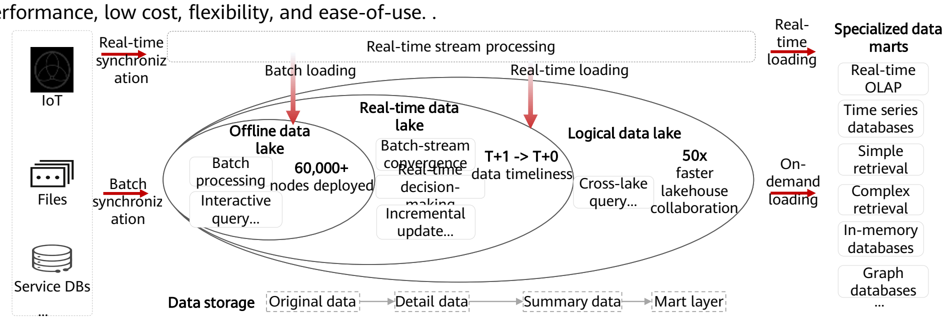
- ModelArts is a one-stop AI development platform geared toward developers and data scientists of all skill levels. It enables you to rapidly build, train, and deploy models anywhere (from the cloud to the edge), and manage full-lifecycle AI workflows. ModelArts accelerates AI development and fosters AI innovation with key capabilities, including data preprocessing and auto labeling, distributed training, automated model building, and one-click workflow execution.



- ModelArts covers all stages of AI development, including data processing, algorithm development, and model training and deployment. The underlying technologies of ModelArts support various heterogeneous computing resources, allowing developers to flexibly select and use resources. In addition, ModelArts supports popular open-source AI development frameworks such as TensorFlow, PyTorch, and MindSpore. ModelArts also allows you to use customized algorithm frameworks tailored to your needs.
- AI engineers face challenges in the installation and configuration of various AI tools, data preparation, and model training. To address these challenges, the one-stop AI development platform ModelArts is provided. The platform integrates data preparation, algorithm development, model training, and model deployment into the production environment, allowing AI engineers to perform one-stop AI development.
- ModelArts has the following features:
  - Data governance: Manages data preparation, such as data filtering and labeling, and dataset versions.
  - Rapid and simplified model training: Enables high-performance distributed training and simplifies coding with the self-developed MoXing deep learning framework.
  - Cloud-edge-device synergy: Deploys models in various production environments such as devices, the edge, and the cloud, and supports real-time and batch inference.
  - Auto learning: Enables model building without coding and supports image classification, object detection, and predictive analytic.

Introduction to MRS

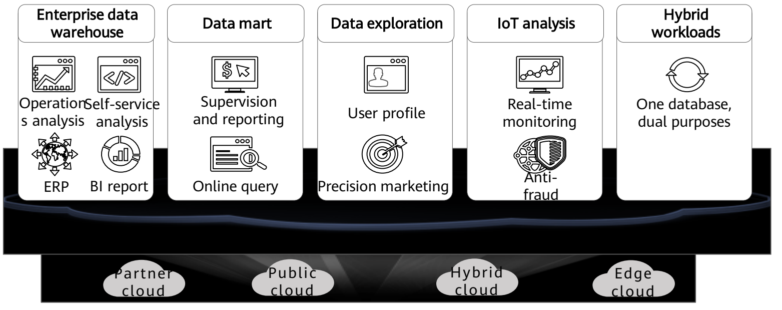
- MapReduce Service (MRS) is provided on Huawei Cloud for you to manage Hadoop-based components. With MRS, you can deploy a Hadoop cluster with a few clicks. MRS provides enterprise-level big data clusters on the cloud. Tenants can fully control clusters and easily run big data components such as Storm, Hadoop, Spark, HBase, and Kafka. MRS is fully compatible with open source APIs, and incorporates advantages of Huawei Cloud computing and storage and big data industry experience to provide customers with a full-stack big data platform featuring high performance, low cost, flexibility, and ease-of-use. .



- Big data is a huge challenge facing the Internet era as the data volume and types increase rapidly. Conventional data processing technologies, such as single-node storage and relational databases, are unable to solve the emerging big data problems. In this case, the Apache Software Foundation (ASF) has launched an open source Hadoop big data processing solution. Hadoop is an open source distributed computing platform that can fully utilize computing and storage capabilities of clusters to process massive amounts of data. If enterprises deploy Hadoop systems by themselves, the disadvantages include high costs, long deployment period, difficult maintenance, and inflexible use.

Introduction to GaussDB(DWS) Data Warehouse

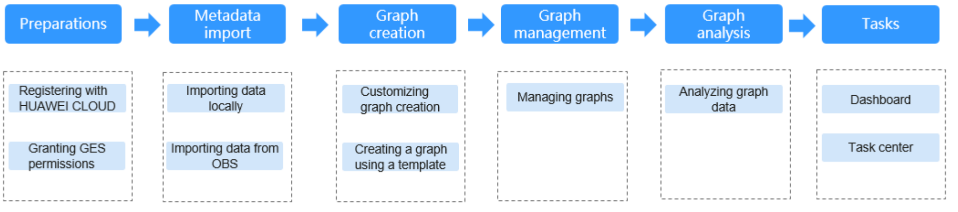
- GaussDB(DWS) is an online data processing database that runs on the Huawei Cloud infrastructure to provide scalable, fully-managed, and out-of-the-box analytic database service, freeing you from complex database management and monitoring. It is a native cloud service based on the Huawei converged data warehouse GaussDB, and is fully compatible with the standard ANSI SQL 99 and SQL 2003, as well as the PostgreSQL and Oracle ecosystems. GaussDB(DWS) provides competitive solutions for PB-level big data analysis in various industries



- Advantages:
  - Seamless migration: GaussDB(DWS) provides various migration tools to ensure seamless migration of popular data analysis systems such as Teradata, Oracle, MySQL, SQL Server, PostgreSQL, Greenplum, and Impala.
  - Compatible with conventional data warehouses: GaussDB(DWS) supports the SQL 2003 standard and stored procedures. It is compatible with some Oracle syntax and data structures, and can be seamlessly interconnected with common BI tools, saving service migration efforts.
  - Secure and reliable: GaussDB(DWS) supports data encryption and connects to DBSS to ensure data security on the cloud. In addition, GaussDB(DWS) supports automatic full and incremental backup of data, improving data reliability.

Introduction to GES

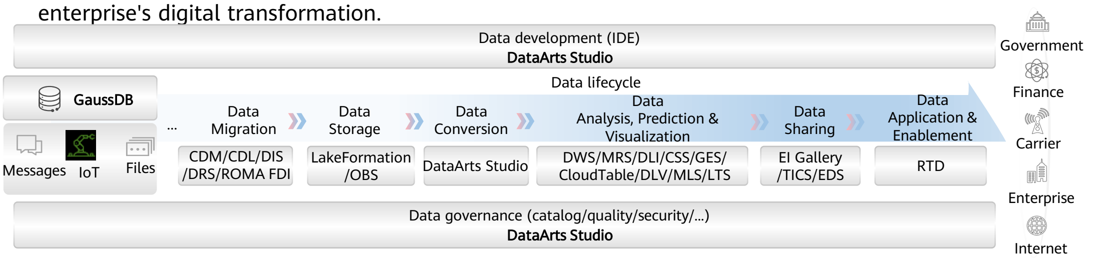
- Graph Engine Service (GES) uses the Huawei-developed EYWA kernel to facilitate query and analysis of multi-relational graph data structures. It is specifically suited for scenarios requiring analysis of rich relationship data, including social relationship analysis, marketing and recommendations, public opinions and social listening, information communication, and anti-fraud.



- GES has the following functions:
  - Extensive Algorithms: Algorithms such as PageRank, K-core, Shortest Path, Label Propagation, Triangle Count, and Link Prediction are all supported.
  - Visualized Graph Analysis: A wizard-based exploration environment is included, along with visualized query results.
  - Query/Analysis APIs: GES provides APIs for graph query, metrics statistics, Gremlin query, Cypher query, graph algorithms, and graph and backup management.
  - Good Compatibility: Compatible with open source Apache TinkerPop Gremlin 3.4.
  - Graph Management: GES provides graph overview, graph management, graph backup, and metadata management functions.

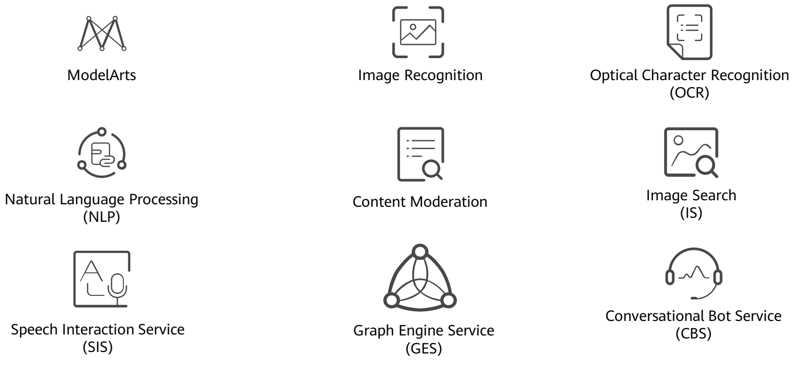
Introduction to DataArts Studio

- DataArts Studio is a one-stop data operations platform that drives digital transformation. It allows you to perform many operations, such as integrating and developing data, designing data architecture, controlling data quality, managing data assets, creating data services, and ensuring data security. Incorporating big data storage, computing and analytical engines, it can also construct industry knowledge bases and help your enterprise build an intelligent end-to-end data system. This system can eliminate data silos, unify data standards, accelerate data monetization, and accelerate your enterprise's digital transformation.



Huawei EI Service Panorama - Artificial Intelligence

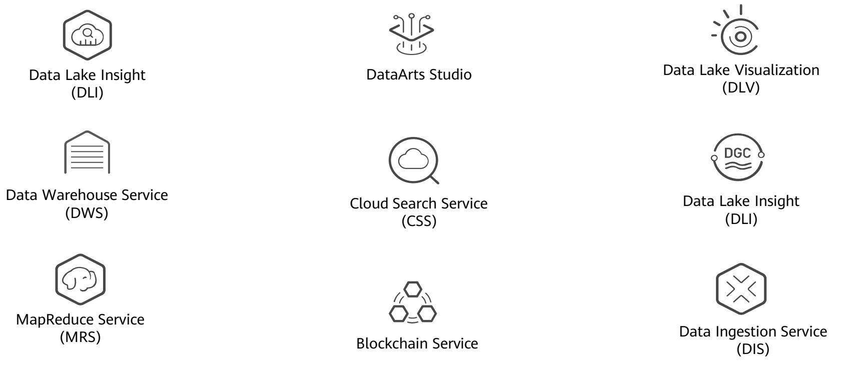
- HUAWEI CLOUD provides comprehensive AI and big data cloud services to facilitate the intelligent upgrades of governments and enterprises and build ubiquitous and pervasive AI.



- HUAWEI CLOUD EI includes AI services and big data services. This slide introduces the former.

HUAWEI CLOUD EI Service Panorama - Big Data

- HUAWEI CLOUD provides comprehensive AI and big data cloud services to facilitate the intelligent upgrades of governments and enterprises and build ubiquitous and pervasive AI.



Quiz

1. (True or false) Read replicas of RDS for MySQL can exist independently only when you have purchased a single-node system or active/standby.

   True

   False

2. (Multiple-Answer Question)Which of the following are the application scenarios for HUAWEI CLOUD CDN?

   A. Website acceleration

   B. File download acceleration

   C. VOD acceleration

   D. ECS running acceleration

- False. The read replicas of RDS for MySQL cannot exist independently.
- ABC. CDN mainly accelerates applications. It cannot accelerate cloud servers.

Summary

This course introduces database services, security services, CDN, API,and EI services of HUAWEI CLOUD, including:

- Relational and non-relational database types, and the application scenarios and key features of different databases.
- Basic concepts and importance of security services.
- Functions and working rules of theAPI,CDN and Enterprise Intelligence (EI) services.

After completing this course, you will have a comprehensive understanding of HUAWEI CLOUD and can better help enterprises accelerate cloud migration and business innovation.

More Information

Huawei iLearning

- https://e.huawei.com/cn/talent/cert/#/careerCert

HUAWEI CLOUD Help Center

- https://support.huaweicloud.com/help-novice.html

HUAWEI CLOUD Academy

- https://edu.huaweicloud.com/

Acronyms and Abbreviations

- AZ: availability zone
- APP: application
- API: application programming interface
- APT: advanced persistent threat
- CDN: content delivery network
- CPU: central processing unit
- CSA: cloud security alliance
- DDoS attack: distributed denial-of-service attack
- DDS: document database service
- DDM: distributed database middleware

Acronyms and Abbreviations

- DAS: data admin service
- DWS: data warehouse service
- DEW: data encryption workshop
- EI: enterprise intelligence
- ELB: elastic load balance
- HA: highly available
- HSS: host security service
- IT: Internet technology
- IAM: identity and access management
- KMS: key management system

Acronyms and Abbreviations

- LAMP: Linux+Apache+PHP+MySQL (a set of open-source software usually used to build dynamic websites)
- OLAP: online analytical processing
- OLTP: online transaction processing
- OBS: object storage service
- PITR: point-in-time recovery
- RTO: recovery time object
- UGC: user generated content
- VIP: virtual IP address
- WAF: web application firewall

Thank you.

把数字世界带入每个人、每个家庭、
每个组织，构建万物互联的智能世界。

Bring digital to every person, home, and organization for a fully connected, intelligent world.

HUAWEI